

# US Patent & Trademark Office

## Patent Public Search | Text View

United States Patent  
Kind Code  
Date of Patent  
Inventor(s)

12412140  
B2  
September 09, 2025  
Beaumont; Richard A. et al.

### Data processing systems and methods for bundled privacy policies

#### Abstract

Data processing systems and methods, according to various embodiments, are adapted for determining an applicable privacy policy based on various criteria associated with a user and the associated product or service. User and product criteria may be obtained automatically and/or based on user input and analyzed by a privacy policy rules engine to determine the applicable policy. Text from the applicable policy can then be presented to the user. A default policy can be used when no particular applicable policy can be identified using by the rules engine. Policies may be ranked or prioritized so that a policy can be selected in the event the rules engine identifies two, conflicting policies based on the criteria.

**Inventors:** Beaumont; Richard A. (London, GB), Brannon; Jonathan Blake (Smyrna, GA)  
**Applicant:** OneTrust, LLC (Atlanta, GA)  
**Family ID:** 78292192  
**Assignee:** OneTrust, LLC (Atlanta, GA)  
**Appl. No.:** 17/962470  
**Filed:** October 08, 2022

#### Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20230047653 A1   | Feb. 16, 2023    |

#### Related U.S. Application Data

continuation parent-doc US 17573999 20220112 US 11468386 child-doc US 17962470  
continuation parent-doc US 17334909 20210531 US 11227247 20220118 child-doc US 17573999  
continuation parent-doc US 16403358 20190503 US 10510031 20191217 child-doc US 16714355  
continuation parent-doc US 16159634 20181013 US 10282692 20190507 child-doc US 16403358  
continuation-in-part parent-doc US 17086732 20201102 US 11023842 20210601 child-doc US 17334909  
continuation-in-part parent-doc US 16808503 20200304 US 10885485 20210105 child-doc US 17086732  
continuation-in-part parent-doc US 16714355 20191213 US 10692033 20200623 child-doc US 16808503  
continuation-in-part parent-doc US 16055083 20180804 US 10289870 20190514 child-doc US 16159634  
continuation-in-part parent-doc US 15996208 20180601 US 10181051 20190115 child-doc US 16055083  
continuation-in-part parent-doc US 15853674 20171222 US 10019597 20180710 child-doc US 15996208  
continuation-in-part parent-doc US 15619455 20170610 US 9851966 20171226 child-doc US 15853674  
continuation-in-part parent-doc US 15254901 20160901 US 9729583 20170808 child-doc US 15619455  
us-provisional-application US 62929583 20191101  
us-provisional-application US 62813584 20190304  
us-provisional-application US 62360123 20160708  
us-provisional-application US 62353802 20160623  
us-provisional-application US 62348695 20160610  
us-provisional-application US 62541613 20170804  
us-provisional-application US 62537839 20170727  
us-provisional-application US 62547530 20170818  
us-provisional-application US 62572096 20171013  
us-provisional-application US 62728435 20180907

#### Publication Classification

**Int. Cl.:** G06Q10/0635 (20230101); G06F15/76 (20060101); G06F21/55 (20130101); G06F21/57 (20130101); G06F21/62 (20130101); G06Q10/067 (20230101); G06F16/95 (20190101)

#### U.S. Cl.:

**CPC** G06Q10/0635 (20130101); G06F15/76 (20130101); G06F21/552 (20130101); G06F21/577 (20130101); G06F21/6245 (20130101); G06Q10/067 (20130101); G06F16/95 (20190101)

Field of Classification Search

CPC: G06Q (10/0635); G06Q (10/067); G06F (15/76); G06F (21/552); G06F (21/577); G06F (21/6245); G06F (16/95)

References Cited

U.S. PATENT DOCUMENTS

| Patent No. | Issued Date | Patentee Name       | U.S. Cl. | CPC |
|------------|-------------|---------------------|----------|-----|
| 4536866    | 12/1984     | Jerome et al.       | N/A      | N/A |
| 4574350    | 12/1985     | Starr               | N/A      | N/A |
| 5193162    | 12/1992     | Bordsen et al.      | N/A      | N/A |
| 5276735    | 12/1993     | Boebert et al.      | N/A      | N/A |
| 5329447    | 12/1993     | Leedom, Jr.         | N/A      | N/A |
| 5404299    | 12/1994     | Tsurubayashi et al. | N/A      | N/A |
| 5535393    | 12/1995     | Reeve et al.        | N/A      | N/A |
| 5560005    | 12/1995     | Hoover et al.       | N/A      | N/A |
| 5668986    | 12/1996     | Nilsen et al.       | N/A      | N/A |
| 5710917    | 12/1997     | Musa et al.         | N/A      | N/A |
| 5761529    | 12/1997     | Raji                | N/A      | N/A |
| 5764906    | 12/1997     | Edelstein et al.    | N/A      | N/A |
| 5872973    | 12/1998     | Mitchell et al.     | N/A      | N/A |
| 5913041    | 12/1998     | Ramanathan et al.   | N/A      | N/A |
| 5913214    | 12/1998     | Madnick et al.      | N/A      | N/A |
| 6009429    | 12/1998     | Greer et al.        | N/A      | N/A |
| 6016394    | 12/1999     | Walker              | N/A      | N/A |
| 6122627    | 12/1999     | Carey et al.        | N/A      | N/A |
| 6148297    | 12/1999     | Swor et al.         | N/A      | N/A |
| 6148342    | 12/1999     | Ho                  | N/A      | N/A |
| 6240416    | 12/2000     | Immon et al.        | N/A      | N/A |
| 6240422    | 12/2000     | Atkins et al.       | N/A      | N/A |
| 6243816    | 12/2000     | Fang et al.         | N/A      | N/A |
| 6253203    | 12/2000     | Oflaherty et al.    | N/A      | N/A |
| 6263335    | 12/2000     | Paik et al.         | N/A      | N/A |
| 6272631    | 12/2000     | Thomlinson et al.   | N/A      | N/A |
| 6275824    | 12/2000     | Oflaherty et al.    | N/A      | N/A |
| 6282548    | 12/2000     | Burner et al.       | N/A      | N/A |
| 6330562    | 12/2000     | Boden et al.        | N/A      | N/A |
| 6363488    | 12/2001     | Ginter et al.       | N/A      | N/A |
| 6374237    | 12/2001     | Reese               | N/A      | N/A |
| 6374252    | 12/2001     | Althoff et al.      | N/A      | N/A |
| 6408336    | 12/2001     | Schneider et al.    | N/A      | N/A |
| 6427230    | 12/2001     | Goiffon et al.      | N/A      | N/A |
| 6442688    | 12/2001     | Moses et al.        | N/A      | N/A |
| 6446120    | 12/2001     | Dantressangle       | N/A      | N/A |
| 6463488    | 12/2001     | San Juan            | N/A      | N/A |
| 6484149    | 12/2001     | Jammes et al.       | N/A      | N/A |
| 6484180    | 12/2001     | Lyons et al.        | N/A      | N/A |
| 6516314    | 12/2002     | Birkler et al.      | N/A      | N/A |
| 6516337    | 12/2002     | Tripp et al.        | N/A      | N/A |
| 6519571    | 12/2002     | Guheen et al.       | N/A      | N/A |
| 6574631    | 12/2002     | Subramanian et al.  | N/A      | N/A |
| 6591272    | 12/2002     | Williams            | N/A      | N/A |
| 6601233    | 12/2002     | Underwood           | N/A      | N/A |
| 6606744    | 12/2002     | Mikurak             | N/A      | N/A |
| 6611812    | 12/2002     | Hurtado et al.      | N/A      | N/A |
| 6625602    | 12/2002     | Meredith et al.     | N/A      | N/A |
| 6629081    | 12/2002     | Cornelius et al.    | N/A      | N/A |
| 6633878    | 12/2002     | Underwood           | N/A      | N/A |
| 6662192    | 12/2002     | Rebane              | N/A      | N/A |
| 6662357    | 12/2002     | Bowman-Amuah        | N/A      | N/A |
| 6697824    | 12/2003     | Bowman-Amuah        | N/A      | N/A |
| 6699042    | 12/2003     | Smith et al.        | N/A      | N/A |
| 6701314    | 12/2003     | Conover et al.      | N/A      | N/A |
| 6721713    | 12/2003     | Guheen et al.       | N/A      | N/A |
| 6725200    | 12/2003     | Rost                | N/A      | N/A |
| 6732109    | 12/2003     | Lindberg et al.     | N/A      | N/A |
| 6754665    | 12/2003     | Futagami et al.     | N/A      | N/A |
| 6755344    | 12/2003     | Mollett et al.      | N/A      | N/A |
| 6757685    | 12/2003     | Raffaele et al.     | N/A      | N/A |
| 6757888    | 12/2003     | Knutson et al.      | N/A      | N/A |
| 6816944    | 12/2003     | Peng                | N/A      | N/A |
| 6826693    | 12/2003     | Yoshida et al.      | N/A      | N/A |
| 6850252    | 12/2004     | Hoffberg            | N/A      | N/A |

|         |         |                      |     |     |
|---------|---------|----------------------|-----|-----|
| 6886101 | 12/2004 | Glazer et al.        | N/A | N/A |
| 6901346 | 12/2004 | Tracy et al.         | N/A | N/A |
| 6904417 | 12/2004 | Clayton et al.       | N/A | N/A |
| 6909897 | 12/2004 | Kikuchi              | N/A | N/A |
| 6925443 | 12/2004 | Baggett, Jr. et al.  | N/A | N/A |
| 6938041 | 12/2004 | Brandow et al.       | N/A | N/A |
| 6956845 | 12/2004 | Baker et al.         | N/A | N/A |
| 6978270 | 12/2004 | Carty et al.         | N/A | N/A |
| 6980927 | 12/2004 | Tracy et al.         | N/A | N/A |
| 6980987 | 12/2004 | Kaminer              | N/A | N/A |
| 6983221 | 12/2005 | Tracy et al.         | N/A | N/A |
| 6985887 | 12/2005 | Sunstein et al.      | N/A | N/A |
| 6990454 | 12/2005 | McIntosh             | N/A | N/A |
| 6993448 | 12/2005 | Tracy et al.         | N/A | N/A |
| 6993495 | 12/2005 | Smith, Jr. et al.    | N/A | N/A |
| 6996807 | 12/2005 | Vardi et al.         | N/A | N/A |
| 7003560 | 12/2005 | Mullen et al.        | N/A | N/A |
| 7003662 | 12/2005 | Genty et al.         | N/A | N/A |
| 7013290 | 12/2005 | Ananian              | N/A | N/A |
| 7017105 | 12/2005 | Flanagin et al.      | N/A | N/A |
| 7023979 | 12/2005 | Wu et al.            | N/A | N/A |
| 7039594 | 12/2005 | Gersting             | N/A | N/A |
| 7039654 | 12/2005 | Eder                 | N/A | N/A |
| 7047517 | 12/2005 | Brown et al.         | N/A | N/A |
| 7051036 | 12/2005 | Rosnow et al.        | N/A | N/A |
| 7051038 | 12/2005 | Yeh et al.           | N/A | N/A |
| 7058970 | 12/2005 | Shaw                 | N/A | N/A |
| 7069427 | 12/2005 | Adler et al.         | N/A | N/A |
| 7076558 | 12/2005 | Dunn                 | N/A | N/A |
| 7093200 | 12/2005 | Schreiber et al.     | N/A | N/A |
| 7095854 | 12/2005 | Ginter et al.        | N/A | N/A |
| 7100195 | 12/2005 | Underwood            | N/A | N/A |
| 7120800 | 12/2005 | Ginter et al.        | N/A | N/A |
| 7124101 | 12/2005 | Mikurak              | N/A | N/A |
| 7124107 | 12/2005 | Pishevar et al.      | N/A | N/A |
| 7127705 | 12/2005 | Christfort et al.    | N/A | N/A |
| 7127741 | 12/2005 | Bandini et al.       | N/A | N/A |
| 7133845 | 12/2005 | Ginter et al.        | N/A | N/A |
| 7139999 | 12/2005 | Bowman-Amuah         | N/A | N/A |
| 7143091 | 12/2005 | Charnock et al.      | N/A | N/A |
| 7149698 | 12/2005 | Guheen et al.        | N/A | N/A |
| 7165041 | 12/2006 | Guheen et al.        | N/A | N/A |
| 7167842 | 12/2006 | Josephson, II et al. | N/A | N/A |
| 7167844 | 12/2006 | Leong et al.         | N/A | N/A |
| 7171379 | 12/2006 | Menninger et al.     | N/A | N/A |
| 7181438 | 12/2006 | Szabo                | N/A | N/A |
| 7203929 | 12/2006 | Vinodkrishnan et al. | N/A | N/A |
| 7213233 | 12/2006 | Vinodkrish           | N/A | N/A |
| 7216155 | 12/2006 | Stuckman et al.      | N/A | N/A |
| 7216340 | 12/2006 | Vinodkrishnan et al. | N/A | N/A |
| 7219066 | 12/2006 | Parks et al.         | N/A | N/A |
| 7223234 | 12/2006 | Stupp et al.         | N/A | N/A |
| 7225460 | 12/2006 | Barzilai et al.      | N/A | N/A |
| 7234065 | 12/2006 | Breslin et al.       | N/A | N/A |
| 7247625 | 12/2006 | Zhang et al.         | N/A | N/A |
| 7251624 | 12/2006 | Lee et al.           | N/A | N/A |
| 7260830 | 12/2006 | Sugimoto             | N/A | N/A |
| 7266566 | 12/2006 | Kennaley et al.      | N/A | N/A |
| 7272818 | 12/2006 | Ishimitsu et al.     | N/A | N/A |
| 7275063 | 12/2006 | Horn                 | N/A | N/A |
| 7281020 | 12/2006 | Fine                 | N/A | N/A |
| 7284232 | 12/2006 | Bates et al.         | N/A | N/A |
| 7284271 | 12/2006 | Lucovsky et al.      | N/A | N/A |
| 7287280 | 12/2006 | Young                | N/A | N/A |
| 7290275 | 12/2006 | Baudoin et al.       | N/A | N/A |
| 7293119 | 12/2006 | Beale                | N/A | N/A |
| 7299299 | 12/2006 | Hollenbeck et al.    | N/A | N/A |
| 7302569 | 12/2006 | Betz et al.          | N/A | N/A |
| 7313575 | 12/2006 | Carr et al.          | N/A | N/A |
| 7313699 | 12/2006 | Koga                 | N/A | N/A |
| 7313825 | 12/2006 | Redlich et al.       | N/A | N/A |
| 7315826 | 12/2007 | Guheen et al.        | N/A | N/A |
| 7315849 | 12/2007 | Bakalash et al.      | N/A | N/A |
| 7322047 | 12/2007 | Redlich et al.       | N/A | N/A |
| 7330850 | 12/2007 | Seibel et al.        | N/A | N/A |

|         |         |                       |     |     |
|---------|---------|-----------------------|-----|-----|
| 7340447 | 12/2007 | Ghatore               | N/A | N/A |
| 7340776 | 12/2007 | Zobel et al.          | N/A | N/A |
| 7343434 | 12/2007 | Kapoor et al.         | N/A | N/A |
| 7346518 | 12/2007 | Frank et al.          | N/A | N/A |
| 7353204 | 12/2007 | Liu                   | N/A | N/A |
| 7353283 | 12/2007 | Henaff et al.         | N/A | N/A |
| 7356559 | 12/2007 | Jacobs et al.         | N/A | N/A |
| 7367014 | 12/2007 | Griffin               | N/A | N/A |
| 7370025 | 12/2007 | Pandit                | N/A | N/A |
| 7376835 | 12/2007 | Olkin et al.          | N/A | N/A |
| 7380120 | 12/2007 | Garcia                | N/A | N/A |
| 7382903 | 12/2007 | Ray                   | N/A | N/A |
| 7383570 | 12/2007 | Pinkas et al.         | N/A | N/A |
| 7391854 | 12/2007 | Salonen et al.        | N/A | N/A |
| 7398393 | 12/2007 | Mont et al.           | N/A | N/A |
| 7401235 | 12/2007 | Mowers et al.         | N/A | N/A |
| 7403942 | 12/2007 | Bayliss               | N/A | N/A |
| 7409354 | 12/2007 | Putnam et al.         | N/A | N/A |
| 7412402 | 12/2007 | Cooper                | N/A | N/A |
| 7424680 | 12/2007 | Carpenter             | N/A | N/A |
| 7428546 | 12/2007 | Nori et al.           | N/A | N/A |
| 7428707 | 12/2007 | Quimby                | N/A | N/A |
| 7430585 | 12/2007 | Sibert                | N/A | N/A |
| 7454457 | 12/2007 | Lowery et al.         | N/A | N/A |
| 7454508 | 12/2007 | Mathew et al.         | N/A | N/A |
| 7478157 | 12/2008 | Bohrer et al.         | N/A | N/A |
| 7480694 | 12/2008 | Blennerhassett et al. | N/A | N/A |
| 7480755 | 12/2008 | Herrell et al.        | N/A | N/A |
| 7487170 | 12/2008 | Stevens               | N/A | N/A |
| 7493282 | 12/2008 | Manly et al.          | N/A | N/A |
| 7500607 | 12/2008 | Williams              | N/A | N/A |
| 7512987 | 12/2008 | Williams              | N/A | N/A |
| 7516882 | 12/2008 | Cucinotta             | N/A | N/A |
| 7523053 | 12/2008 | Pudhukottai et al.    | N/A | N/A |
| 7529836 | 12/2008 | Bolen                 | N/A | N/A |
| 7548968 | 12/2008 | Bura et al.           | N/A | N/A |
| 7552480 | 12/2008 | Voss                  | N/A | N/A |
| 7562339 | 12/2008 | Racca et al.          | N/A | N/A |
| 7565685 | 12/2008 | Ross et al.           | N/A | N/A |
| 7567541 | 12/2008 | Karimi et al.         | N/A | N/A |
| 7584505 | 12/2008 | Mondri et al.         | N/A | N/A |
| 7584508 | 12/2008 | Kashchenko et al.     | N/A | N/A |
| 7587749 | 12/2008 | Leser et al.          | N/A | N/A |
| 7590705 | 12/2008 | Mathew et al.         | N/A | N/A |
| 7590972 | 12/2008 | Axelrod et al.        | N/A | N/A |
| 7603356 | 12/2008 | Schran et al.         | N/A | N/A |
| 7606783 | 12/2008 | Carter                | N/A | N/A |
| 7606790 | 12/2008 | Levy                  | N/A | N/A |
| 7607120 | 12/2008 | Sanyal et al.         | N/A | N/A |
| 7613700 | 12/2008 | Lobo et al.           | N/A | N/A |
| 7617136 | 12/2008 | Lessing et al.        | N/A | N/A |
| 7617167 | 12/2008 | Griffis et al.        | N/A | N/A |
| 7620644 | 12/2008 | Cote et al.           | N/A | N/A |
| 7627666 | 12/2008 | Degiulio et al.       | N/A | N/A |
| 7630874 | 12/2008 | Fables et al.         | N/A | N/A |
| 7630998 | 12/2008 | Zhou et al.           | N/A | N/A |
| 7636742 | 12/2008 | Olavarrieta et al.    | N/A | N/A |
| 7640322 | 12/2008 | Wendkos et al.        | N/A | N/A |
| 7650497 | 12/2009 | Thornton et al.       | N/A | N/A |
| 7653592 | 12/2009 | Flaxman et al.        | N/A | N/A |
| 7657476 | 12/2009 | Barney                | N/A | N/A |
| 7657694 | 12/2009 | Mansell et al.        | N/A | N/A |
| 7665073 | 12/2009 | Meijer et al.         | N/A | N/A |
| 7665125 | 12/2009 | Heard et al.          | N/A | N/A |
| 7668947 | 12/2009 | Hutchinson et al.     | N/A | N/A |
| 7673282 | 12/2009 | Amaru et al.          | N/A | N/A |
| 7676034 | 12/2009 | Wu et al.             | N/A | N/A |
| 7681034 | 12/2009 | Lee et al.            | N/A | N/A |
| 7681140 | 12/2009 | Ebert                 | N/A | N/A |
| 7685561 | 12/2009 | Deem et al.           | N/A | N/A |
| 7685577 | 12/2009 | Pace et al.           | N/A | N/A |
| 7693593 | 12/2009 | Ishibashi et al.      | N/A | N/A |
| 7698398 | 12/2009 | Lai                   | N/A | N/A |
| 7702639 | 12/2009 | Stanley et al.        | N/A | N/A |
| 7707224 | 12/2009 | Chastagnol et al.     | N/A | N/A |

|         |         |                        |     |     |
|---------|---------|------------------------|-----|-----|
| 7712029 | 12/2009 | Ferreira et al.        | N/A | N/A |
| 7716242 | 12/2009 | Pae et al.             | N/A | N/A |
| 7725474 | 12/2009 | Tamai et al.           | N/A | N/A |
| 7725875 | 12/2009 | Waldrep                | N/A | N/A |
| 7729940 | 12/2009 | Harvey et al.          | N/A | N/A |
| 7730142 | 12/2009 | Levasseur et al.       | N/A | N/A |
| 7752124 | 12/2009 | Green et al.           | N/A | N/A |
| 7756826 | 12/2009 | Bots et al.            | N/A | N/A |
| 7756987 | 12/2009 | Wang et al.            | N/A | N/A |
| 7761586 | 12/2009 | Olenick et al.         | N/A | N/A |
| 7774745 | 12/2009 | Fildebrandt et al.     | N/A | N/A |
| 7788212 | 12/2009 | Beckmann et al.        | N/A | N/A |
| 7788222 | 12/2009 | Shah et al.            | N/A | N/A |
| 7788632 | 12/2009 | Kuester et al.         | N/A | N/A |
| 7788726 | 12/2009 | Teixeira               | N/A | N/A |
| 7797726 | 12/2009 | Ashley et al.          | N/A | N/A |
| 7801758 | 12/2009 | Gracie et al.          | N/A | N/A |
| 7801826 | 12/2009 | Labrou et al.          | N/A | N/A |
| 7801912 | 12/2009 | Ransil et al.          | N/A | N/A |
| 7802305 | 12/2009 | Leeds                  | N/A | N/A |
| 7805349 | 12/2009 | Yu et al.              | N/A | N/A |
| 7805451 | 12/2009 | Hosokawa               | N/A | N/A |
| 7813947 | 12/2009 | DeAngelis et al.       | N/A | N/A |
| 7822620 | 12/2009 | Dixon et al.           | N/A | N/A |
| 7827523 | 12/2009 | Ahmed et al.           | N/A | N/A |
| 7836078 | 12/2009 | Dettinger et al.       | N/A | N/A |
| 7844640 | 12/2009 | Bender et al.          | N/A | N/A |
| 7849143 | 12/2009 | Vuong                  | N/A | N/A |
| 7853468 | 12/2009 | Callahan et al.        | N/A | N/A |
| 7853470 | 12/2009 | Sonnleithner et al.    | N/A | N/A |
| 7853925 | 12/2009 | Kemmler                | N/A | N/A |
| 7860816 | 12/2009 | Fokoue-Nkoutche et al. | N/A | N/A |
| 7870540 | 12/2010 | Zare et al.            | N/A | N/A |
| 7870608 | 12/2010 | Shraim et al.          | N/A | N/A |
| 7873541 | 12/2010 | Klar et al.            | N/A | N/A |
| 7877327 | 12/2010 | Gwiazda et al.         | N/A | N/A |
| 7877812 | 12/2010 | Koved et al.           | N/A | N/A |
| 7885841 | 12/2010 | King                   | N/A | N/A |
| 7890461 | 12/2010 | Oeda et al.            | N/A | N/A |
| 7895260 | 12/2010 | Archer et al.          | N/A | N/A |
| 7904478 | 12/2010 | Yu et al.              | N/A | N/A |
| 7904487 | 12/2010 | Ghatore                | N/A | N/A |
| 7917888 | 12/2010 | Chong et al.           | N/A | N/A |
| 7917963 | 12/2010 | Goyal et al.           | N/A | N/A |
| 7921152 | 12/2010 | Ashley et al.          | N/A | N/A |
| 7930197 | 12/2010 | Ozzie et al.           | N/A | N/A |
| 7930753 | 12/2010 | Mellinger et al.       | N/A | N/A |
| 7953725 | 12/2010 | Burris et al.          | N/A | N/A |
| 7954150 | 12/2010 | Croft et al.           | N/A | N/A |
| 7958087 | 12/2010 | Blumenau               | N/A | N/A |
| 7958494 | 12/2010 | Chaar et al.           | N/A | N/A |
| 7962900 | 12/2010 | Barraclough et al.     | N/A | N/A |
| 7966310 | 12/2010 | Sullivan et al.        | N/A | N/A |
| 7966599 | 12/2010 | Malasky et al.         | N/A | N/A |
| 7966663 | 12/2010 | Strickland et al.      | N/A | N/A |
| 7974992 | 12/2010 | Fastabend et al.       | N/A | N/A |
| 7975000 | 12/2010 | Dixon et al.           | N/A | N/A |
| 7991559 | 12/2010 | Dzekunov et al.        | N/A | N/A |
| 7991747 | 12/2010 | Upadhyay et al.        | N/A | N/A |
| 7996372 | 12/2010 | Rubel, Jr.             | N/A | N/A |
| 8005891 | 12/2010 | Knowles et al.         | N/A | N/A |
| 8010612 | 12/2010 | Costea et al.          | N/A | N/A |
| 8010720 | 12/2010 | Iwaoka et al.          | N/A | N/A |
| 8019881 | 12/2010 | Sandhu et al.          | N/A | N/A |
| 8020206 | 12/2010 | Hubbard et al.         | N/A | N/A |
| 8024384 | 12/2010 | Prabhakar et al.       | N/A | N/A |
| 8032721 | 12/2010 | Murai                  | N/A | N/A |
| 8036374 | 12/2010 | Noble, Jr.             | N/A | N/A |
| 8037409 | 12/2010 | Jacob et al.           | N/A | N/A |
| 8041749 | 12/2010 | Beck                   | N/A | N/A |
| 8041763 | 12/2010 | Kordun et al.          | N/A | N/A |
| 8041913 | 12/2010 | Wang                   | N/A | N/A |
| 8069161 | 12/2010 | Bugir et al.           | N/A | N/A |
| 8069471 | 12/2010 | Boren                  | N/A | N/A |
| 8082539 | 12/2010 | Schelkogonov           | N/A | N/A |

|         |         |                    |     |     |
|---------|---------|--------------------|-----|-----|
| 8090754 | 12/2011 | Schmidt et al.     | N/A | N/A |
| 8095923 | 12/2011 | Harvey et al.      | N/A | N/A |
| 8099709 | 12/2011 | Baikov et al.      | N/A | N/A |
| 8103962 | 12/2011 | Embley et al.      | N/A | N/A |
| 8117441 | 12/2011 | Kurien et al.      | N/A | N/A |
| 8135815 | 12/2011 | Mayer              | N/A | N/A |
| 8146054 | 12/2011 | Baker et al.       | N/A | N/A |
| 8146074 | 12/2011 | Ito et al.         | N/A | N/A |
| 8150717 | 12/2011 | Whitmore           | N/A | N/A |
| 8156105 | 12/2011 | Altounian et al.   | N/A | N/A |
| 8156158 | 12/2011 | Rolls et al.       | N/A | N/A |
| 8156159 | 12/2011 | Ebrahimi et al.    | N/A | N/A |
| 8166406 | 12/2011 | Goldfeder et al.   | N/A | N/A |
| 8176061 | 12/2011 | Swanbeck et al.    | N/A | N/A |
| 8176177 | 12/2011 | Sussman et al.     | N/A | N/A |
| 8176334 | 12/2011 | Vainstein          | N/A | N/A |
| 8176470 | 12/2011 | Klumpp et al.      | N/A | N/A |
| 8180759 | 12/2011 | Hamzy              | N/A | N/A |
| 8181151 | 12/2011 | Sedukhin et al.    | N/A | N/A |
| 8185409 | 12/2011 | Putnam et al.      | N/A | N/A |
| 8196176 | 12/2011 | Berteau et al.     | N/A | N/A |
| 8205093 | 12/2011 | Argott             | N/A | N/A |
| 8205140 | 12/2011 | Hafeez et al.      | N/A | N/A |
| 8214362 | 12/2011 | Djabarov           | N/A | N/A |
| 8214803 | 12/2011 | Horii et al.       | N/A | N/A |
| 8234377 | 12/2011 | Cohn               | N/A | N/A |
| 8239244 | 12/2011 | Ginsberg et al.    | N/A | N/A |
| 8250051 | 12/2011 | Bugir et al.       | N/A | N/A |
| 8255468 | 12/2011 | Vitaldevara et al. | N/A | N/A |
| 8260262 | 12/2011 | Ben Ayed           | N/A | N/A |
| 8261362 | 12/2011 | Goodwin et al.     | N/A | N/A |
| 8266231 | 12/2011 | Golovin et al.     | N/A | N/A |
| 8275632 | 12/2011 | Awaraji et al.     | N/A | N/A |
| 8275793 | 12/2011 | Ahmad et al.       | N/A | N/A |
| 8286239 | 12/2011 | Sutton             | N/A | N/A |
| 8312549 | 12/2011 | Goldberg et al.    | N/A | N/A |
| 8316237 | 12/2011 | Felsher et al.     | N/A | N/A |
| 8332908 | 12/2011 | Hatakeyama et al.  | N/A | N/A |
| 8340999 | 12/2011 | Kumaran et al.     | N/A | N/A |
| 8341405 | 12/2011 | Meijer et al.      | N/A | N/A |
| 8346929 | 12/2012 | Lai                | N/A | N/A |
| 8364713 | 12/2012 | Pollard            | N/A | N/A |
| 8370224 | 12/2012 | Grewal             | N/A | N/A |
| 8370794 | 12/2012 | Moosmann et al.    | N/A | N/A |
| 8380630 | 12/2012 | Felsher            | N/A | N/A |
| 8380743 | 12/2012 | Convertino et al.  | N/A | N/A |
| 8381180 | 12/2012 | Rostoker           | N/A | N/A |
| 8381297 | 12/2012 | Touboul            | N/A | N/A |
| 8386314 | 12/2012 | Kirkby et al.      | N/A | N/A |
| 8392982 | 12/2012 | Harris et al.      | N/A | N/A |
| 8418226 | 12/2012 | Gardner            | N/A | N/A |
| 8423954 | 12/2012 | Ronen et al.       | N/A | N/A |
| 8429179 | 12/2012 | Mirhaji            | N/A | N/A |
| 8429597 | 12/2012 | Prigge             | N/A | N/A |
| 8429630 | 12/2012 | Nickolov et al.    | N/A | N/A |
| 8429758 | 12/2012 | Chen et al.        | N/A | N/A |
| 8438644 | 12/2012 | Watters et al.     | N/A | N/A |
| 8452693 | 12/2012 | Shah et al.        | N/A | N/A |
| 8463247 | 12/2012 | Misiag             | N/A | N/A |
| 8464311 | 12/2012 | Ashley et al.      | N/A | N/A |
| 8468244 | 12/2012 | Redlich et al.     | N/A | N/A |
| 8473324 | 12/2012 | Alvarez et al.     | N/A | N/A |
| 8474012 | 12/2012 | Ahmed et al.       | N/A | N/A |
| 8494894 | 12/2012 | Jaster et al.      | N/A | N/A |
| 8504481 | 12/2012 | Motahari et al.    | N/A | N/A |
| 8510199 | 12/2012 | Erlanger           | N/A | N/A |
| 8515988 | 12/2012 | Jones et al.       | N/A | N/A |
| 8516076 | 12/2012 | Thomas             | N/A | N/A |
| 8527337 | 12/2012 | Lim et al.         | N/A | N/A |
| 8533746 | 12/2012 | Nolan et al.       | N/A | N/A |
| 8533844 | 12/2012 | Mahaffey et al.    | N/A | N/A |
| 8538817 | 12/2012 | Wilson             | N/A | N/A |
| 8539359 | 12/2012 | Rapaport et al.    | N/A | N/A |
| 8539437 | 12/2012 | Finlayson et al.   | N/A | N/A |
| 8560645 | 12/2012 | Linden et al.      | N/A | N/A |

|         |         |                      |     |     |
|---------|---------|----------------------|-----|-----|
| 8560841 | 12/2012 | Chin et al.          | N/A | N/A |
| 8560956 | 12/2012 | Curtis et al.        | N/A | N/A |
| 8561100 | 12/2012 | Hu et al.            | N/A | N/A |
| 8561153 | 12/2012 | Grason et al.        | N/A | N/A |
| 8565729 | 12/2012 | Moseler et al.       | N/A | N/A |
| 8566726 | 12/2012 | Dixon et al.         | N/A | N/A |
| 8566938 | 12/2012 | Prakash et al.       | N/A | N/A |
| 8571909 | 12/2012 | Miller et al.        | N/A | N/A |
| 8572717 | 12/2012 | Narayanaswamy        | N/A | N/A |
| 8578036 | 12/2012 | Holfelder et al.     | N/A | N/A |
| 8578166 | 12/2012 | De Monseignat et al. | N/A | N/A |
| 8578481 | 12/2012 | Rowley               | N/A | N/A |
| 8578501 | 12/2012 | Ogilvie              | N/A | N/A |
| 8583694 | 12/2012 | Siegel et al.        | N/A | N/A |
| 8583766 | 12/2012 | Dixon et al.         | N/A | N/A |
| 8589183 | 12/2012 | Awaraji et al.       | N/A | N/A |
| 8601467 | 12/2012 | Hofhansl et al.      | N/A | N/A |
| 8601591 | 12/2012 | Krishnamurthy et al. | N/A | N/A |
| 8606746 | 12/2012 | Yeap et al.          | N/A | N/A |
| 8612420 | 12/2012 | Sun et al.           | N/A | N/A |
| 8612993 | 12/2012 | Grant et al.         | N/A | N/A |
| 8615549 | 12/2012 | Knowles et al.       | N/A | N/A |
| 8615731 | 12/2012 | Doshi                | N/A | N/A |
| 8620952 | 12/2012 | Bennett et al.       | N/A | N/A |
| 8621637 | 12/2012 | Al-Harbi et al.      | N/A | N/A |
| 8626671 | 12/2013 | Federgreen           | N/A | N/A |
| 8627114 | 12/2013 | Resch et al.         | N/A | N/A |
| 8630961 | 12/2013 | Beilby et al.        | N/A | N/A |
| 8631048 | 12/2013 | Davis et al.         | N/A | N/A |
| 8640110 | 12/2013 | Kopp et al.          | N/A | N/A |
| 8646072 | 12/2013 | Savant               | N/A | N/A |
| 8650399 | 12/2013 | Le Bihan et al.      | N/A | N/A |
| 8655939 | 12/2013 | Redlich et al.       | N/A | N/A |
| 8656265 | 12/2013 | Paulin et al.        | N/A | N/A |
| 8656456 | 12/2013 | Maxson et al.        | N/A | N/A |
| 8661036 | 12/2013 | Turski et al.        | N/A | N/A |
| 8667074 | 12/2013 | Farkas               | N/A | N/A |
| 8667487 | 12/2013 | Boodman et al.       | N/A | N/A |
| 8677472 | 12/2013 | Dotan et al.         | N/A | N/A |
| 8681984 | 12/2013 | Lee et al.           | N/A | N/A |
| 8682698 | 12/2013 | Cashman et al.       | N/A | N/A |
| 8683502 | 12/2013 | Shkedi et al.        | N/A | N/A |
| 8688601 | 12/2013 | Jaiswal              | N/A | N/A |
| 8689292 | 12/2013 | Williams et al.      | N/A | N/A |
| 8693689 | 12/2013 | Belenkiy et al.      | N/A | N/A |
| 8700524 | 12/2013 | Williams et al.      | N/A | N/A |
| 8700699 | 12/2013 | Shen et al.          | N/A | N/A |
| 8706742 | 12/2013 | Ravid et al.         | N/A | N/A |
| 8707451 | 12/2013 | Ture et al.          | N/A | N/A |
| 8712813 | 12/2013 | King                 | N/A | N/A |
| 8713098 | 12/2013 | Adya et al.          | N/A | N/A |
| 8713638 | 12/2013 | Hu et al.            | N/A | N/A |
| 8719366 | 12/2013 | Mathew et al.        | N/A | N/A |
| 8732839 | 12/2013 | Hohl                 | N/A | N/A |
| 8744894 | 12/2013 | Christiansen et al.  | N/A | N/A |
| 8751285 | 12/2013 | Deb et al.           | N/A | N/A |
| 8762406 | 12/2013 | Ho et al.            | N/A | N/A |
| 8762413 | 12/2013 | Graham, Jr. et al.   | N/A | N/A |
| 8763071 | 12/2013 | Sinha et al.         | N/A | N/A |
| 8763082 | 12/2013 | Huber et al.         | N/A | N/A |
| 8763131 | 12/2013 | Archer et al.        | N/A | N/A |
| 8767947 | 12/2013 | Ristock et al.       | N/A | N/A |
| 8769242 | 12/2013 | Tkac et al.          | N/A | N/A |
| 8769412 | 12/2013 | Gill et al.          | N/A | N/A |
| 8769671 | 12/2013 | Shraim et al.        | N/A | N/A |
| 8776241 | 12/2013 | Zaitsev              | N/A | N/A |
| 8788935 | 12/2013 | Hirsch et al.        | N/A | N/A |
| 8793614 | 12/2013 | Wilson et al.        | N/A | N/A |
| 8793650 | 12/2013 | Hilerio et al.       | N/A | N/A |
| 8793781 | 12/2013 | Grossi et al.        | N/A | N/A |
| 8793809 | 12/2013 | Falkenburg et al.    | N/A | N/A |
| 8799984 | 12/2013 | Ahn                  | N/A | N/A |
| 8805707 | 12/2013 | Schumann, Jr. et al. | N/A | N/A |
| 8805806 | 12/2013 | Amarendran et al.    | N/A | N/A |
| 8805925 | 12/2013 | Price et al.         | N/A | N/A |

|         |         |                       |     |     |
|---------|---------|-----------------------|-----|-----|
| 8812342 | 12/2013 | Barcelo et al.        | N/A | N/A |
| 8812752 | 12/2013 | Shih et al.           | N/A | N/A |
| 8812766 | 12/2013 | Kranendonk et al.     | N/A | N/A |
| 8813028 | 12/2013 | Farooqi               | N/A | N/A |
| 8813214 | 12/2013 | McNair et al.         | N/A | N/A |
| 8819253 | 12/2013 | Simeloff et al.       | N/A | N/A |
| 8819617 | 12/2013 | Koenig et al.         | N/A | N/A |
| 8819800 | 12/2013 | Gao et al.            | N/A | N/A |
| 8826446 | 12/2013 | Liu et al.            | N/A | N/A |
| 8832649 | 12/2013 | Bishop et al.         | N/A | N/A |
| 8832854 | 12/2013 | Staddon et al.        | N/A | N/A |
| 8839232 | 12/2013 | Taylor et al.         | N/A | N/A |
| 8843487 | 12/2013 | McGraw et al.         | N/A | N/A |
| 8843745 | 12/2013 | Roberts, Jr.          | N/A | N/A |
| 8849757 | 12/2013 | Kruglick              | N/A | N/A |
| 8856534 | 12/2013 | Khosravi et al.       | N/A | N/A |
| 8856936 | 12/2013 | Datta Ray et al.      | N/A | N/A |
| 8862507 | 12/2013 | Sandhu et al.         | N/A | N/A |
| 8863261 | 12/2013 | Yang                  | N/A | N/A |
| 8875232 | 12/2013 | Blom et al.           | N/A | N/A |
| 8887047 | 12/2013 | Digiantomaso et al.   | N/A | N/A |
| 8893078 | 12/2013 | Schaude et al.        | N/A | N/A |
| 8893286 | 12/2013 | Oliver                | N/A | N/A |
| 8893297 | 12/2013 | Eversoll et al.       | N/A | N/A |
| 8904494 | 12/2013 | Kindler et al.        | N/A | N/A |
| 8914263 | 12/2013 | Shimada et al.        | N/A | N/A |
| 8914299 | 12/2013 | Pesci-Anderson et al. | N/A | N/A |
| 8914342 | 12/2013 | Kalaboukis et al.     | N/A | N/A |
| 8914902 | 12/2013 | Moritz et al.         | N/A | N/A |
| 8918306 | 12/2013 | Cashman et al.        | N/A | N/A |
| 8918392 | 12/2013 | Brooker et al.        | N/A | N/A |
| 8918632 | 12/2013 | Sartor                | N/A | N/A |
| 8924388 | 12/2013 | Elliot et al.         | N/A | N/A |
| 8930896 | 12/2014 | Wiggins               | N/A | N/A |
| 8930897 | 12/2014 | Nassar                | N/A | N/A |
| 8935198 | 12/2014 | Phillips et al.       | N/A | N/A |
| 8935266 | 12/2014 | Wu                    | N/A | N/A |
| 8935342 | 12/2014 | Patel                 | N/A | N/A |
| 8935804 | 12/2014 | Clark et al.          | N/A | N/A |
| 8938221 | 12/2014 | Brazier et al.        | N/A | N/A |
| 8943076 | 12/2014 | Stewart et al.        | N/A | N/A |
| 8943548 | 12/2014 | Drokov et al.         | N/A | N/A |
| 8943602 | 12/2014 | Roy et al.            | N/A | N/A |
| 8949137 | 12/2014 | Crapo et al.          | N/A | N/A |
| 8955038 | 12/2014 | Nicodemus et al.      | N/A | N/A |
| 8959568 | 12/2014 | Hudis et al.          | N/A | N/A |
| 8959584 | 12/2014 | Piliouras             | N/A | N/A |
| 8966575 | 12/2014 | McQuay et al.         | N/A | N/A |
| 8966597 | 12/2014 | Saylor et al.         | N/A | N/A |
| 8973108 | 12/2014 | Roth et al.           | N/A | N/A |
| 8977234 | 12/2014 | Chava                 | N/A | N/A |
| 8977643 | 12/2014 | Schindlauer et al.    | N/A | N/A |
| 8978158 | 12/2014 | Rajkumar et al.       | N/A | N/A |
| 8983972 | 12/2014 | Kriebel et al.        | N/A | N/A |
| 8984031 | 12/2014 | Todd                  | N/A | N/A |
| 8990933 | 12/2014 | Magdalin              | N/A | N/A |
| 8996417 | 12/2014 | Channakeshava         | N/A | N/A |
| 8996480 | 12/2014 | Agarwala et al.       | N/A | N/A |
| 8997213 | 12/2014 | Papakipos et al.      | N/A | N/A |
| 9001673 | 12/2014 | Birdsall et al.       | N/A | N/A |
| 9003295 | 12/2014 | Baschy                | N/A | N/A |
| 9003552 | 12/2014 | Goodwin et al.        | N/A | N/A |
| 9009851 | 12/2014 | Droste et al.         | N/A | N/A |
| 9014661 | 12/2014 | Decharms              | N/A | N/A |
| 9015796 | 12/2014 | Fujioka               | N/A | N/A |
| 9021469 | 12/2014 | Hilerio et al.        | N/A | N/A |
| 9026526 | 12/2014 | Bau et al.            | N/A | N/A |
| 9030987 | 12/2014 | Bianchetti et al.     | N/A | N/A |
| 9032067 | 12/2014 | Prasad et al.         | N/A | N/A |
| 9043217 | 12/2014 | Cashman et al.        | N/A | N/A |
| 9043480 | 12/2014 | Barton et al.         | N/A | N/A |
| 9047463 | 12/2014 | Porras                | N/A | N/A |
| 9047582 | 12/2014 | Hutchinson et al.     | N/A | N/A |
| 9047583 | 12/2014 | Patton et al.         | N/A | N/A |
| 9047639 | 12/2014 | Quintiliani et al.    | N/A | N/A |



|         |         |                       |     |     |
|---------|---------|-----------------------|-----|-----|
| 9049244 | 12/2014 | Prince et al.         | N/A | N/A |
| 9049314 | 12/2014 | Pugh et al.           | N/A | N/A |
| 9055071 | 12/2014 | Gates et al.          | N/A | N/A |
| 9058590 | 12/2014 | Criddle et al.        | N/A | N/A |
| 9064033 | 12/2014 | Jin et al.            | N/A | N/A |
| 9069940 | 12/2014 | Hars                  | N/A | N/A |
| 9076231 | 12/2014 | Hill et al.           | N/A | N/A |
| 9077736 | 12/2014 | Werth et al.          | N/A | N/A |
| 9081952 | 12/2014 | Sagi et al.           | N/A | N/A |
| 9087090 | 12/2014 | Cormier et al.        | N/A | N/A |
| 9092796 | 12/2014 | Eversoll et al.       | N/A | N/A |
| 9094434 | 12/2014 | Williams et al.       | N/A | N/A |
| 9098515 | 12/2014 | Richter et al.        | N/A | N/A |
| 9100337 | 12/2014 | Battré et al.         | N/A | N/A |
| 9100778 | 12/2014 | Stogaitis et al.      | N/A | N/A |
| 9106691 | 12/2014 | Burger et al.         | N/A | N/A |
| 9106710 | 12/2014 | Feimster              | N/A | N/A |
| 9110918 | 12/2014 | Rajaa et al.          | N/A | N/A |
| 9111105 | 12/2014 | Barton et al.         | N/A | N/A |
| 9111295 | 12/2014 | Tietzen et al.        | N/A | N/A |
| 9123330 | 12/2014 | Sharifi et al.        | N/A | N/A |
| 9123339 | 12/2014 | Shaw et al.           | N/A | N/A |
| 9129311 | 12/2014 | Schoen et al.         | N/A | N/A |
| 9135261 | 12/2014 | Maunder et al.        | N/A | N/A |
| 9135444 | 12/2014 | Carter et al.         | N/A | N/A |
| 9141823 | 12/2014 | Dawson                | N/A | N/A |
| 9141911 | 12/2014 | Zhao et al.           | N/A | N/A |
| 9152818 | 12/2014 | Hathaway et al.       | N/A | N/A |
| 9152820 | 12/2014 | Pauley, Jr. et al.    | N/A | N/A |
| 9154514 | 12/2014 | Prakash               | N/A | N/A |
| 9154556 | 12/2014 | Dotan et al.          | N/A | N/A |
| 9158655 | 12/2014 | Wadhwani et al.       | N/A | N/A |
| 9165036 | 12/2014 | Mehra                 | N/A | N/A |
| 9170996 | 12/2014 | Lovric et al.         | N/A | N/A |
| 9172706 | 12/2014 | Krishnamurthy et al.  | N/A | N/A |
| 9177293 | 12/2014 | Gagnon et al.         | N/A | N/A |
| 9178901 | 12/2014 | Xue et al.            | N/A | N/A |
| 9183100 | 12/2014 | Gventer et al.        | N/A | N/A |
| 9189642 | 12/2014 | Perlman               | N/A | N/A |
| 9201572 | 12/2014 | Lyon et al.           | N/A | N/A |
| 9201770 | 12/2014 | Duerk                 | N/A | N/A |
| 9202026 | 12/2014 | Reeves                | N/A | N/A |
| 9202085 | 12/2014 | Mawdsley et al.       | N/A | N/A |
| 9215076 | 12/2014 | Roth et al.           | N/A | N/A |
| 9215252 | 12/2014 | Smith et al.          | N/A | N/A |
| 9218596 | 12/2014 | Ronca et al.          | N/A | N/A |
| 9224009 | 12/2014 | Liu et al.            | N/A | N/A |
| 9230036 | 12/2015 | Davis                 | N/A | N/A |
| 9231935 | 12/2015 | Bridge et al.         | N/A | N/A |
| 9232040 | 12/2015 | Barash et al.         | N/A | N/A |
| 9235476 | 12/2015 | McHugh et al.         | N/A | N/A |
| 9240987 | 12/2015 | Barrett-Bowen et al.  | N/A | N/A |
| 9241259 | 12/2015 | Daniela et al.        | N/A | N/A |
| 9245126 | 12/2015 | Christodorescu et al. | N/A | N/A |
| 9245266 | 12/2015 | Hardt                 | N/A | N/A |
| 9253609 | 12/2015 | Hosier, Jr.           | N/A | N/A |
| 9264443 | 12/2015 | Weisman               | N/A | N/A |
| 9274858 | 12/2015 | Milliron et al.       | N/A | N/A |
| 9280581 | 12/2015 | Grimes et al.         | N/A | N/A |
| 9286149 | 12/2015 | Sampson et al.        | N/A | N/A |
| 9286282 | 12/2015 | Ling, III et al.      | N/A | N/A |
| 9288118 | 12/2015 | Pattan                | N/A | N/A |
| 9288556 | 12/2015 | Kim et al.            | N/A | N/A |
| 9294498 | 12/2015 | Yampolskiy et al.     | N/A | N/A |
| 9299050 | 12/2015 | Stiffler et al.       | N/A | N/A |
| 9306939 | 12/2015 | Chan et al.           | N/A | N/A |
| 9317697 | 12/2015 | Maier et al.          | N/A | N/A |
| 9317715 | 12/2015 | Schuette et al.       | N/A | N/A |
| 9325731 | 12/2015 | McGeehan              | N/A | N/A |
| 9336184 | 12/2015 | Mital et al.          | N/A | N/A |
| 9336220 | 12/2015 | Li et al.             | N/A | N/A |
| 9336324 | 12/2015 | Lomme et al.          | N/A | N/A |
| 9336332 | 12/2015 | Davis et al.          | N/A | N/A |
| 9336400 | 12/2015 | Milman et al.         | N/A | N/A |
| 9338188 | 12/2015 | Ahn                   | N/A | N/A |

|         |         |                    |     |     |
|---------|---------|--------------------|-----|-----|
| 9342706 | 12/2015 | Chawla et al.      | N/A | N/A |
| 9344297 | 12/2015 | Shah et al.        | N/A | N/A |
| 9344424 | 12/2015 | Tenenboym et al.   | N/A | N/A |
| 9344484 | 12/2015 | Ferris             | N/A | N/A |
| 9348802 | 12/2015 | Massand            | N/A | N/A |
| 9348862 | 12/2015 | Kawecki, III       | N/A | N/A |
| 9348929 | 12/2015 | Eberlein           | N/A | N/A |
| 9349016 | 12/2015 | Brisebois et al.   | N/A | N/A |
| 9350718 | 12/2015 | Sondhi et al.      | N/A | N/A |
| 9355157 | 12/2015 | Mohammed et al.    | N/A | N/A |
| 9356961 | 12/2015 | Todd et al.        | N/A | N/A |
| 9361446 | 12/2015 | Demirjian et al.   | N/A | N/A |
| 9369488 | 12/2015 | Woods et al.       | N/A | N/A |
| 9372869 | 12/2015 | Joseph et al.      | N/A | N/A |
| 9374693 | 12/2015 | Olincy et al.      | N/A | N/A |
| 9384199 | 12/2015 | Thereska et al.    | N/A | N/A |
| 9384357 | 12/2015 | Patil et al.       | N/A | N/A |
| 9386078 | 12/2015 | Reno et al.        | N/A | N/A |
| 9386104 | 12/2015 | Adams et al.       | N/A | N/A |
| 9395959 | 12/2015 | Hatfield et al.    | N/A | N/A |
| 9396332 | 12/2015 | Abrams et al.      | N/A | N/A |
| 9401900 | 12/2015 | Levasseur et al.   | N/A | N/A |
| 9411967 | 12/2015 | Parecki et al.     | N/A | N/A |
| 9411982 | 12/2015 | Dippenaar et al.   | N/A | N/A |
| 9417859 | 12/2015 | Gounares et al.    | N/A | N/A |
| 9418221 | 12/2015 | Turgeman           | N/A | N/A |
| 9424021 | 12/2015 | Zamir              | N/A | N/A |
| 9424414 | 12/2015 | Demirjian et al.   | N/A | N/A |
| 9426177 | 12/2015 | Wang et al.        | N/A | N/A |
| 9450940 | 12/2015 | Belov et al.       | N/A | N/A |
| 9460136 | 12/2015 | Todd et al.        | N/A | N/A |
| 9460171 | 12/2015 | Marrelli et al.    | N/A | N/A |
| 9460307 | 12/2015 | Breslau et al.     | N/A | N/A |
| 9461876 | 12/2015 | Van Dusen et al.   | N/A | N/A |
| 9462009 | 12/2015 | Kolman et al.      | N/A | N/A |
| 9465702 | 12/2015 | Gventer et al.     | N/A | N/A |
| 9465800 | 12/2015 | Lacey              | N/A | N/A |
| 9473446 | 12/2015 | Vijay et al.       | N/A | N/A |
| 9473505 | 12/2015 | Asano et al.       | N/A | N/A |
| 9473535 | 12/2015 | Sartor             | N/A | N/A |
| 9477523 | 12/2015 | Warman et al.      | N/A | N/A |
| 9477660 | 12/2015 | Scott et al.       | N/A | N/A |
| 9477685 | 12/2015 | Leung et al.       | N/A | N/A |
| 9477942 | 12/2015 | Adachi et al.      | N/A | N/A |
| 9483659 | 12/2015 | Bao et al.         | N/A | N/A |
| 9489366 | 12/2015 | Scott et al.       | N/A | N/A |
| 9495547 | 12/2015 | Schepis et al.     | N/A | N/A |
| 9501523 | 12/2015 | Hyatt et al.       | N/A | N/A |
| 9507960 | 12/2015 | Bell et al.        | N/A | N/A |
| 9509674 | 12/2015 | Nasserbakht et al. | N/A | N/A |
| 9509702 | 12/2015 | Grigg et al.       | N/A | N/A |
| 9514231 | 12/2015 | Eden               | N/A | N/A |
| 9516012 | 12/2015 | Chochois et al.    | N/A | N/A |
| 9521166 | 12/2015 | Wilson             | N/A | N/A |
| 9524500 | 12/2015 | Dave et al.        | N/A | N/A |
| 9529989 | 12/2015 | Kling et al.       | N/A | N/A |
| 9536108 | 12/2016 | Powell et al.      | N/A | N/A |
| 9537546 | 12/2016 | Cordeiro et al.    | N/A | N/A |
| 9542568 | 12/2016 | Francis et al.     | N/A | N/A |
| 9549047 | 12/2016 | Fredinburg et al.  | N/A | N/A |
| 9552395 | 12/2016 | Bayer et al.       | N/A | N/A |
| 9552470 | 12/2016 | Turgeman et al.    | N/A | N/A |
| 9553918 | 12/2016 | Manion et al.      | N/A | N/A |
| 9558497 | 12/2016 | Carvalho           | N/A | N/A |
| 9569752 | 12/2016 | Deering et al.     | N/A | N/A |
| 9571506 | 12/2016 | Boss et al.        | N/A | N/A |
| 9571509 | 12/2016 | Satish et al.      | N/A | N/A |
| 9571526 | 12/2016 | Sartor             | N/A | N/A |
| 9571559 | 12/2016 | Raleigh et al.     | N/A | N/A |
| 9571991 | 12/2016 | Brizendine et al.  | N/A | N/A |
| 9576289 | 12/2016 | Henderson et al.   | N/A | N/A |
| 9578060 | 12/2016 | Brisebois et al.   | N/A | N/A |
| 9578173 | 12/2016 | Sanghavi et al.    | N/A | N/A |
| 9582681 | 12/2016 | Mishra             | N/A | N/A |
| 9584964 | 12/2016 | Pelkey             | N/A | N/A |

|         |         |                      |     |     |
|---------|---------|----------------------|-----|-----|
| 9589110 | 12/2016 | Carey et al.         | N/A | N/A |
| 9600181 | 12/2016 | Patel et al.         | N/A | N/A |
| 9602529 | 12/2016 | Jones et al.         | N/A | N/A |
| 9606971 | 12/2016 | Seolas et al.        | N/A | N/A |
| 9607041 | 12/2016 | Himmelstein          | N/A | N/A |
| 9619652 | 12/2016 | Slater               | N/A | N/A |
| 9619661 | 12/2016 | Finkelstein          | N/A | N/A |
| 9621357 | 12/2016 | Williams et al.      | N/A | N/A |
| 9621566 | 12/2016 | Gupta et al.         | N/A | N/A |
| 9626124 | 12/2016 | Lipinski et al.      | N/A | N/A |
| 9626680 | 12/2016 | Ryan et al.          | N/A | N/A |
| 9629064 | 12/2016 | Graves et al.        | N/A | N/A |
| 9642008 | 12/2016 | Wyatt et al.         | N/A | N/A |
| 9646095 | 12/2016 | Gottlieb et al.      | N/A | N/A |
| 9647949 | 12/2016 | Varki et al.         | N/A | N/A |
| 9648036 | 12/2016 | Seiver et al.        | N/A | N/A |
| 9652314 | 12/2016 | Mahiddini            | N/A | N/A |
| 9654506 | 12/2016 | Barrett              | N/A | N/A |
| 9654541 | 12/2016 | Kapczynski et al.    | N/A | N/A |
| 9665722 | 12/2016 | Nagasundaram et al.  | N/A | N/A |
| 9665733 | 12/2016 | Sills et al.         | N/A | N/A |
| 9665883 | 12/2016 | Roullier et al.      | N/A | N/A |
| 9672053 | 12/2016 | Tang et al.          | N/A | N/A |
| 9672355 | 12/2016 | Titonis et al.       | N/A | N/A |
| 9678794 | 12/2016 | Barrett et al.       | N/A | N/A |
| 9691090 | 12/2016 | Barday               | N/A | N/A |
| 9697368 | 12/2016 | Dharawat             | N/A | N/A |
| 9699209 | 12/2016 | Ng et al.            | N/A | N/A |
| 9703549 | 12/2016 | Dufresne             | N/A | N/A |
| 9704103 | 12/2016 | Suskind et al.       | N/A | N/A |
| 9705840 | 12/2016 | Pujare et al.        | N/A | N/A |
| 9705880 | 12/2016 | Siris                | N/A | N/A |
| 9721078 | 12/2016 | Cornick et al.       | N/A | N/A |
| 9721108 | 12/2016 | Krishnamurthy et al. | N/A | N/A |
| 9727751 | 12/2016 | Oliver et al.        | N/A | N/A |
| 9729583 | 12/2016 | Barday               | N/A | N/A |
| 9734148 | 12/2016 | Bendersky et al.     | N/A | N/A |
| 9734255 | 12/2016 | Jiang                | N/A | N/A |
| 9736004 | 12/2016 | Jung et al.          | N/A | N/A |
| 9740985 | 12/2016 | Byron et al.         | N/A | N/A |
| 9740987 | 12/2016 | Dolan                | N/A | N/A |
| 9749408 | 12/2016 | Subramani et al.     | N/A | N/A |
| 9753703 | 12/2016 | Jemiolo              | N/A | N/A |
| 9754091 | 12/2016 | Kode et al.          | N/A | N/A |
| 9756059 | 12/2016 | Demirjian et al.     | N/A | N/A |
| 9760620 | 12/2016 | Nachnani et al.      | N/A | N/A |
| 9760635 | 12/2016 | Bliss et al.         | N/A | N/A |
| 9760697 | 12/2016 | Walker               | N/A | N/A |
| 9760849 | 12/2016 | Vinnakota et al.     | N/A | N/A |
| 9762553 | 12/2016 | Ford et al.          | N/A | N/A |
| 9767202 | 12/2016 | Darby et al.         | N/A | N/A |
| 9767309 | 12/2016 | Patel et al.         | N/A | N/A |
| 9769124 | 12/2016 | Yan                  | N/A | N/A |
| 9773269 | 12/2016 | Lazarus              | N/A | N/A |
| 9785795 | 12/2016 | Grondin et al.       | N/A | N/A |
| 9787671 | 12/2016 | Bogrett              | N/A | N/A |
| 9798749 | 12/2016 | Saner                | N/A | N/A |
| 9798826 | 12/2016 | Wilson et al.        | N/A | N/A |
| 9798896 | 12/2016 | Jakobsson            | N/A | N/A |
| 9800605 | 12/2016 | Baikalov et al.      | N/A | N/A |
| 9800606 | 12/2016 | Yumer                | N/A | N/A |
| 9804649 | 12/2016 | Cohen et al.         | N/A | N/A |
| 9804928 | 12/2016 | Davis et al.         | N/A | N/A |
| 9805381 | 12/2016 | Frank et al.         | N/A | N/A |
| 9811532 | 12/2016 | Parkison et al.      | N/A | N/A |
| 9817850 | 12/2016 | Dubbels et al.       | N/A | N/A |
| 9817978 | 12/2016 | Marsh et al.         | N/A | N/A |
| 9819684 | 12/2016 | Cernoch et al.       | N/A | N/A |
| 9825928 | 12/2016 | Lelcuk et al.        | N/A | N/A |
| 9830563 | 12/2016 | Paknad               | N/A | N/A |
| 9832633 | 12/2016 | Gerber, Jr. et al.   | N/A | N/A |
| 9836598 | 12/2016 | Yer et al.           | N/A | N/A |
| 9838407 | 12/2016 | Oprea et al.         | N/A | N/A |
| 9838839 | 12/2016 | Vudali et al.        | N/A | N/A |
| 9841969 | 12/2016 | Seibert, Jr. et al.  | N/A | N/A |

|          |         |                      |     |     |
|----------|---------|----------------------|-----|-----|
| 9842042  | 12/2016 | Chhatwal et al.      | N/A | N/A |
| 9842349  | 12/2016 | Sawczuk et al.       | N/A | N/A |
| 9848005  | 12/2016 | Ardeli et al.        | N/A | N/A |
| 9848061  | 12/2016 | Jain et al.          | N/A | N/A |
| 9852150  | 12/2016 | Sharpe et al.        | N/A | N/A |
| 9853959  | 12/2016 | Kapczynski et al.    | N/A | N/A |
| 9860226  | 12/2017 | Thormaehlen          | N/A | N/A |
| 9864735  | 12/2017 | Lamprecht            | N/A | N/A |
| 9876825  | 12/2017 | Amar et al.          | N/A | N/A |
| 9877138  | 12/2017 | Franklin             | N/A | N/A |
| 9880157  | 12/2017 | Levak et al.         | N/A | N/A |
| 9882935  | 12/2017 | Barday               | N/A | N/A |
| 9887965  | 12/2017 | Kay et al.           | N/A | N/A |
| 9888377  | 12/2017 | McCorkendale et al.  | N/A | N/A |
| 9892441  | 12/2017 | Barday               | N/A | N/A |
| 9892442  | 12/2017 | Barday               | N/A | N/A |
| 9892443  | 12/2017 | Barday               | N/A | N/A |
| 9892444  | 12/2017 | Barday               | N/A | N/A |
| 9894076  | 12/2017 | Li et al.            | N/A | N/A |
| 9898613  | 12/2017 | Swerdlow et al.      | N/A | N/A |
| 9898739  | 12/2017 | Monastyrsky et al.   | N/A | N/A |
| 9898769  | 12/2017 | Barday               | N/A | N/A |
| 9912625  | 12/2017 | Mutha et al.         | N/A | N/A |
| 9912677  | 12/2017 | Chien                | N/A | N/A |
| 9912810  | 12/2017 | Segre et al.         | N/A | N/A |
| 9916703  | 12/2017 | Levinson et al.      | N/A | N/A |
| 9922124  | 12/2017 | Rathod               | N/A | N/A |
| 9923927  | 12/2017 | McClintock et al.    | N/A | N/A |
| 9928379  | 12/2017 | Hoffer               | N/A | N/A |
| 9934493  | 12/2017 | Castinado et al.     | N/A | N/A |
| 9934544  | 12/2017 | Whitfield et al.     | N/A | N/A |
| 9936127  | 12/2017 | Todasco              | N/A | N/A |
| 9942214  | 12/2017 | Burciu et al.        | N/A | N/A |
| 9942244  | 12/2017 | Lahoz et al.         | N/A | N/A |
| 9942276  | 12/2017 | Sartor               | N/A | N/A |
| 9946897  | 12/2017 | Lovin                | N/A | N/A |
| 9948652  | 12/2017 | Yu et al.            | N/A | N/A |
| 9948663  | 12/2017 | Wang et al.          | N/A | N/A |
| 9953189  | 12/2017 | Cook et al.          | N/A | N/A |
| 9954879  | 12/2017 | Sadaghiani et al.    | N/A | N/A |
| 9954883  | 12/2017 | Ahuja et al.         | N/A | N/A |
| 9959551  | 12/2017 | Schermerhorn et al.  | N/A | N/A |
| 9959582  | 12/2017 | Sukman et al.        | N/A | N/A |
| 9961070  | 12/2017 | Tang                 | N/A | N/A |
| 9973518  | 12/2017 | Lee et al.           | N/A | N/A |
| 9973585  | 12/2017 | Ruback et al.        | N/A | N/A |
| 9977904  | 12/2017 | Khan et al.          | N/A | N/A |
| 9977920  | 12/2017 | Danielson et al.     | N/A | N/A |
| 9983936  | 12/2017 | Dornemann et al.     | N/A | N/A |
| 9984252  | 12/2017 | Pollard              | N/A | N/A |
| 9990499  | 12/2017 | Chan et al.          | N/A | N/A |
| 9992213  | 12/2017 | Sinnema              | N/A | N/A |
| 10001975 | 12/2017 | Bharthulwar          | N/A | N/A |
| 10002064 | 12/2017 | Muske                | N/A | N/A |
| 10007895 | 12/2017 | Vanasco              | N/A | N/A |
| 10013577 | 12/2017 | Beaumont et al.      | N/A | N/A |
| 10015164 | 12/2017 | Hamburg et al.       | N/A | N/A |
| 10019339 | 12/2017 | Von Hanxleden et al. | N/A | N/A |
| 10019588 | 12/2017 | Garcia et al.        | N/A | N/A |
| 10019591 | 12/2017 | Beguin               | N/A | N/A |
| 10019741 | 12/2017 | Hesselink            | N/A | N/A |
| 10021143 | 12/2017 | Cabrera et al.       | N/A | N/A |
| 10025804 | 12/2017 | Vranyes et al.       | N/A | N/A |
| 10025836 | 12/2017 | Batchu et al.        | N/A | N/A |
| 10028226 | 12/2017 | Ayyagari et al.      | N/A | N/A |
| 10032172 | 12/2017 | Barday               | N/A | N/A |
| 10044761 | 12/2017 | Ducatel et al.       | N/A | N/A |
| 10055426 | 12/2017 | Arasan et al.        | N/A | N/A |
| 10055869 | 12/2017 | Borrelli et al.      | N/A | N/A |
| 10061847 | 12/2017 | Mohammed et al.      | N/A | N/A |
| 10069858 | 12/2017 | Robinson et al.      | N/A | N/A |
| 10069914 | 12/2017 | Smith                | N/A | N/A |
| 10073924 | 12/2017 | Karp et al.          | N/A | N/A |
| 10075437 | 12/2017 | Costigan et al.      | N/A | N/A |
| 10075451 | 12/2017 | Hall et al.          | N/A | N/A |

|          |         |                    |     |     |
|----------|---------|--------------------|-----|-----|
| 10084817 | 12/2017 | Saher et al.       | N/A | N/A |
| 10091214 | 12/2017 | Godlewski et al.   | N/A | N/A |
| 10091312 | 12/2017 | Khanwalkar et al.  | N/A | N/A |
| 10097551 | 12/2017 | Chan et al.        | N/A | N/A |
| 10102533 | 12/2017 | Barday             | N/A | N/A |
| 10108409 | 12/2017 | Pirzadeh et al.    | N/A | N/A |
| 10122663 | 12/2017 | Hu et al.          | N/A | N/A |
| 10122760 | 12/2017 | Terrill et al.     | N/A | N/A |
| 10127403 | 12/2017 | Kong et al.        | N/A | N/A |
| 10129211 | 12/2017 | Heath              | N/A | N/A |
| 10140666 | 12/2017 | Wang et al.        | N/A | N/A |
| 10142113 | 12/2017 | Zaidi et al.       | N/A | N/A |
| 10152560 | 12/2017 | Potiagalov et al.  | N/A | N/A |
| 10158676 | 12/2017 | Barday             | N/A | N/A |
| 10165011 | 12/2017 | Barday             | N/A | N/A |
| 10169762 | 12/2018 | Ogawa              | N/A | N/A |
| 10176503 | 12/2018 | Barday et al.      | N/A | N/A |
| 10181043 | 12/2018 | Pauley, Jr. et al. | N/A | N/A |
| 10181051 | 12/2018 | Barday et al.      | N/A | N/A |
| 10187363 | 12/2018 | Smirnoff et al.    | N/A | N/A |
| 10187394 | 12/2018 | Bar et al.         | N/A | N/A |
| 10188950 | 12/2018 | Biswas et al.      | N/A | N/A |
| 10204154 | 12/2018 | Barday et al.      | N/A | N/A |
| 10205994 | 12/2018 | Splaine et al.     | N/A | N/A |
| 10212134 | 12/2018 | Rai                | N/A | N/A |
| 10212175 | 12/2018 | Seul et al.        | N/A | N/A |
| 10223533 | 12/2018 | Dawson             | N/A | N/A |
| 10230571 | 12/2018 | Rangasamy et al.   | N/A | N/A |
| 10242228 | 12/2018 | Barday et al.      | N/A | N/A |
| 10250594 | 12/2018 | Chathoth et al.    | N/A | N/A |
| 10255602 | 12/2018 | Wang               | N/A | N/A |
| 10257127 | 12/2018 | Dotan-Cohen et al. | N/A | N/A |
| 10257181 | 12/2018 | Sherif et al.      | N/A | N/A |
| 10268838 | 12/2018 | Yadgiri et al.     | N/A | N/A |
| 10275221 | 12/2018 | Thattai et al.     | N/A | N/A |
| 10275614 | 12/2018 | Barday et al.      | N/A | N/A |
| 10282370 | 12/2018 | Barday et al.      | N/A | N/A |
| 10282559 | 12/2018 | Barday et al.      | N/A | N/A |
| 10284604 | 12/2018 | Barday et al.      | N/A | N/A |
| 10289584 | 12/2018 | Chiba              | N/A | N/A |
| 10289857 | 12/2018 | Brinskelle         | N/A | N/A |
| 10289866 | 12/2018 | Barday et al.      | N/A | N/A |
| 10289867 | 12/2018 | Barday et al.      | N/A | N/A |
| 10289870 | 12/2018 | Barday et al.      | N/A | N/A |
| 10291731 | 12/2018 | Huang              | N/A | N/A |
| 10296504 | 12/2018 | Hock et al.        | N/A | N/A |
| 10304442 | 12/2018 | Rudden et al.      | N/A | N/A |
| 10310723 | 12/2018 | Rathod             | N/A | N/A |
| 10311042 | 12/2018 | Kumar              | N/A | N/A |
| 10311475 | 12/2018 | Yuasa              | N/A | N/A |
| 10311492 | 12/2018 | Gelfenbeyn et al.  | N/A | N/A |
| 10318761 | 12/2018 | Barday et al.      | N/A | N/A |
| 10320940 | 12/2018 | Brennan et al.     | N/A | N/A |
| 10324960 | 12/2018 | Skvortsov et al.   | N/A | N/A |
| 10326768 | 12/2018 | Verweyst et al.    | N/A | N/A |
| 10326798 | 12/2018 | Lambert            | N/A | N/A |
| 10326841 | 12/2018 | Bradley et al.     | N/A | N/A |
| 10327100 | 12/2018 | Davis et al.       | N/A | N/A |
| 10331689 | 12/2018 | Sorrentino et al.  | N/A | N/A |
| 10331904 | 12/2018 | Sher-Jan et al.    | N/A | N/A |
| 10333975 | 12/2018 | Soman et al.       | N/A | N/A |
| 10339470 | 12/2018 | Dutta et al.       | N/A | N/A |
| 10346186 | 12/2018 | Kalyanpur          | N/A | N/A |
| 10346635 | 12/2018 | Kumar et al.       | N/A | N/A |
| 10346637 | 12/2018 | Barday et al.      | N/A | N/A |
| 10346638 | 12/2018 | Barday et al.      | N/A | N/A |
| 10346849 | 12/2018 | Ionescu et al.     | N/A | N/A |
| 10348726 | 12/2018 | Caluwaert          | N/A | N/A |
| 10348775 | 12/2018 | Barday             | N/A | N/A |
| 10353673 | 12/2018 | Barday et al.      | N/A | N/A |
| 10361857 | 12/2018 | Woo                | N/A | N/A |
| 10366241 | 12/2018 | Sartor             | N/A | N/A |
| 10373119 | 12/2018 | Driscoll et al.    | N/A | N/A |
| 10373409 | 12/2018 | White et al.       | N/A | N/A |
| 10375115 | 12/2018 | Mallya             | N/A | N/A |

|          |         |                      |     |     |
|----------|---------|----------------------|-----|-----|
| 10387559 | 12/2018 | Wendt et al.         | N/A | N/A |
| 10387577 | 12/2018 | Hill et al.          | N/A | N/A |
| 10387657 | 12/2018 | Belfiore, Jr. et al. | N/A | N/A |
| 10387952 | 12/2018 | Sandhu et al.        | N/A | N/A |
| 10395201 | 12/2018 | Vescio               | N/A | N/A |
| 10402545 | 12/2018 | Gorfein et al.       | N/A | N/A |
| 10404729 | 12/2018 | Turgeman             | N/A | N/A |
| 10417401 | 12/2018 | Votaw et al.         | N/A | N/A |
| 10417621 | 12/2018 | Cassel et al.        | N/A | N/A |
| 10419476 | 12/2018 | Parekh               | N/A | N/A |
| 10423985 | 12/2018 | Dutta et al.         | N/A | N/A |
| 10425492 | 12/2018 | Comstock et al.      | N/A | N/A |
| 10430608 | 12/2018 | Peri et al.          | N/A | N/A |
| 10435350 | 12/2018 | Ito et al.           | N/A | N/A |
| 10437412 | 12/2018 | Barday et al.        | N/A | N/A |
| 10437860 | 12/2018 | Barday et al.        | N/A | N/A |
| 10438016 | 12/2018 | Barday et al.        | N/A | N/A |
| 10438273 | 12/2018 | Burns et al.         | N/A | N/A |
| 10440062 | 12/2018 | Barday et al.        | N/A | N/A |
| 10445508 | 12/2018 | Sher-Jan et al.      | N/A | N/A |
| 10445526 | 12/2018 | Barday et al.        | N/A | N/A |
| 10452864 | 12/2018 | Barday et al.        | N/A | N/A |
| 10452866 | 12/2018 | Barday et al.        | N/A | N/A |
| 10453076 | 12/2018 | Parekh et al.        | N/A | N/A |
| 10453092 | 12/2018 | Wang et al.          | N/A | N/A |
| 10454934 | 12/2018 | Parimi et al.        | N/A | N/A |
| 10481763 | 12/2018 | Bartkiewicz et al.   | N/A | N/A |
| 10489454 | 12/2018 | Chen                 | N/A | N/A |
| 10503926 | 12/2018 | Barday et al.        | N/A | N/A |
| 10510031 | 12/2018 | Barday et al.        | N/A | N/A |
| 10521623 | 12/2018 | Rodriguez et al.     | N/A | N/A |
| 10534851 | 12/2019 | Chan et al.          | N/A | N/A |
| 10535081 | 12/2019 | Ferreira et al.      | N/A | N/A |
| 10536475 | 12/2019 | McCorkle, Jr. et al. | N/A | N/A |
| 10536478 | 12/2019 | Kirti et al.         | N/A | N/A |
| 10540212 | 12/2019 | Feng et al.          | N/A | N/A |
| 10541938 | 12/2019 | Timmerman et al.     | N/A | N/A |
| 10546135 | 12/2019 | Kassoumeh et al.     | N/A | N/A |
| 10552462 | 12/2019 | Hart                 | N/A | N/A |
| 10558809 | 12/2019 | Joyce et al.         | N/A | N/A |
| 10558821 | 12/2019 | Barday et al.        | N/A | N/A |
| 10564815 | 12/2019 | Soon-Shiong          | N/A | N/A |
| 10564935 | 12/2019 | Barday et al.        | N/A | N/A |
| 10564936 | 12/2019 | Barday et al.        | N/A | N/A |
| 10565161 | 12/2019 | Barday et al.        | N/A | N/A |
| 10565236 | 12/2019 | Barday et al.        | N/A | N/A |
| 10567439 | 12/2019 | Barday               | N/A | N/A |
| 10567517 | 12/2019 | Weinig et al.        | N/A | N/A |
| 10572684 | 12/2019 | Lafever et al.       | N/A | N/A |
| 10572686 | 12/2019 | Barday et al.        | N/A | N/A |
| 10574705 | 12/2019 | Barday et al.        | N/A | N/A |
| 10581825 | 12/2019 | Poschel et al.       | N/A | N/A |
| 10592648 | 12/2019 | Barday et al.        | N/A | N/A |
| 10592692 | 12/2019 | Brannon              | N/A | N/A |
| 10599456 | 12/2019 | Lissack              | N/A | N/A |
| 10606916 | 12/2019 | Brannon et al.       | N/A | N/A |
| 10613971 | 12/2019 | Vasikarla            | N/A | N/A |
| 10614365 | 12/2019 | Sathish et al.       | N/A | N/A |
| 10628553 | 12/2019 | Murrish et al.       | N/A | N/A |
| 10645102 | 12/2019 | Hamdi                | N/A | N/A |
| 10645548 | 12/2019 | Reynolds et al.      | N/A | N/A |
| 10649630 | 12/2019 | Vora et al.          | N/A | N/A |
| 10650408 | 12/2019 | Andersen et al.      | N/A | N/A |
| 10657469 | 12/2019 | Bade et al.          | N/A | N/A |
| 10657504 | 12/2019 | Zimmerman et al.     | N/A | N/A |
| 10659566 | 12/2019 | Luah et al.          | N/A | N/A |
| 10671749 | 12/2019 | Felice-Steele et al. | N/A | N/A |
| 10671760 | 12/2019 | Esmailzadeh et al.   | N/A | N/A |
| 10678945 | 12/2019 | Barday et al.        | N/A | N/A |
| 10685140 | 12/2019 | Barday et al.        | N/A | N/A |
| 10706176 | 12/2019 | Brannon et al.       | N/A | N/A |
| 10706226 | 12/2019 | Byun et al.          | N/A | N/A |
| 10708305 | 12/2019 | Barday et al.        | N/A | N/A |
| 10713387 | 12/2019 | Brannon et al.       | N/A | N/A |
| 10726145 | 12/2019 | Duminy et al.        | N/A | N/A |

|          |         |                           |     |     |
|----------|---------|---------------------------|-----|-----|
| 10726153 | 12/2019 | Nerurkar et al.           | N/A | N/A |
| 10726158 | 12/2019 | Brannon et al.            | N/A | N/A |
| 10732865 | 12/2019 | Jain et al.               | N/A | N/A |
| 10735388 | 12/2019 | Rose et al.               | N/A | N/A |
| 10740487 | 12/2019 | Barday et al.             | N/A | N/A |
| 10747893 | 12/2019 | Kiriyama et al.           | N/A | N/A |
| 10747897 | 12/2019 | Cook                      | N/A | N/A |
| 10749870 | 12/2019 | Brouillette et al.        | N/A | N/A |
| 10762213 | 12/2019 | Rudek et al.              | N/A | N/A |
| 10762230 | 12/2019 | Ancin et al.              | N/A | N/A |
| 10762236 | 12/2019 | Brannon et al.            | N/A | N/A |
| 10769302 | 12/2019 | Barday et al.             | N/A | N/A |
| 10769303 | 12/2019 | Brannon et al.            | N/A | N/A |
| 10776510 | 12/2019 | Antonelli et al.          | N/A | N/A |
| 10776518 | 12/2019 | Barday et al.             | N/A | N/A |
| 10778792 | 12/2019 | Handy Bosma et al.        | N/A | N/A |
| 10783256 | 12/2019 | Brannon et al.            | N/A | N/A |
| 10785173 | 12/2019 | Willett et al.            | N/A | N/A |
| 10785299 | 12/2019 | Gupta et al.              | N/A | N/A |
| 10791150 | 12/2019 | Barday et al.             | N/A | N/A |
| 10795527 | 12/2019 | Legge et al.              | N/A | N/A |
| 10796020 | 12/2019 | Barday et al.             | N/A | N/A |
| 10796260 | 12/2019 | Brannon et al.            | N/A | N/A |
| 10798133 | 12/2019 | Barday et al.             | N/A | N/A |
| 10803196 | 12/2019 | Bodegas Martinez et al.   | N/A | N/A |
| 10805331 | 12/2019 | Boyer et al.              | N/A | N/A |
| 10831831 | 12/2019 | Greene                    | N/A | N/A |
| 10834590 | 12/2019 | Turgeman et al.           | N/A | N/A |
| 10839100 | 12/2019 | Hollinger et al.          | N/A | N/A |
| 10846433 | 12/2019 | Brannon et al.            | N/A | N/A |
| 10853356 | 12/2019 | McPherson et al.          | N/A | N/A |
| 10853501 | 12/2019 | Brannon                   | N/A | N/A |
| 10860721 | 12/2019 | Gentile                   | N/A | N/A |
| 10860742 | 12/2019 | Joseph et al.             | N/A | N/A |
| 10860979 | 12/2019 | Geffen et al.             | N/A | N/A |
| 10878127 | 12/2019 | Brannon et al.            | N/A | N/A |
| 10885485 | 12/2020 | Brannon et al.            | N/A | N/A |
| 10891393 | 12/2020 | Currier et al.            | N/A | N/A |
| 10893074 | 12/2020 | Sartor                    | N/A | N/A |
| 10896394 | 12/2020 | Brannon et al.            | N/A | N/A |
| 10902490 | 12/2020 | He et al.                 | N/A | N/A |
| 10909488 | 12/2020 | Hecht et al.              | N/A | N/A |
| 10924514 | 12/2020 | Altman et al.             | N/A | N/A |
| 10929557 | 12/2020 | Chavez                    | N/A | N/A |
| 10949555 | 12/2020 | Rattan et al.             | N/A | N/A |
| 10949565 | 12/2020 | Barday et al.             | N/A | N/A |
| 10956213 | 12/2020 | Chambers et al.           | N/A | N/A |
| 10957326 | 12/2020 | Bhaya et al.              | N/A | N/A |
| 10963571 | 12/2020 | Bar Joseph et al.         | N/A | N/A |
| 10963572 | 12/2020 | Belfiore, Jr. et al.      | N/A | N/A |
| 10965547 | 12/2020 | Esposito et al.           | N/A | N/A |
| 10970418 | 12/2020 | Durvasula et al.          | N/A | N/A |
| 10972509 | 12/2020 | Barday et al.             | N/A | N/A |
| 10976950 | 12/2020 | Trezza et al.             | N/A | N/A |
| 10983963 | 12/2020 | Venkatasubramanian et al. | N/A | N/A |
| 10984458 | 12/2020 | Gutierrez                 | N/A | N/A |
| 10997318 | 12/2020 | Barday et al.             | N/A | N/A |
| 11003748 | 12/2020 | Oliker et al.             | N/A | N/A |
| 11012475 | 12/2020 | Patnala et al.            | N/A | N/A |
| 11023528 | 12/2020 | Lee et al.                | N/A | N/A |
| 11023921 | 12/2020 | Wang et al.               | N/A | N/A |
| 11037168 | 12/2020 | Lee et al.                | N/A | N/A |
| 11057356 | 12/2020 | Malhotra et al.           | N/A | N/A |
| 11057427 | 12/2020 | Wright et al.             | N/A | N/A |
| 11062051 | 12/2020 | Barday et al.             | N/A | N/A |
| 11068318 | 12/2020 | Kuesel et al.             | N/A | N/A |
| 11068584 | 12/2020 | Burriesci et al.          | N/A | N/A |
| 11068618 | 12/2020 | Brannon et al.            | N/A | N/A |
| 11068797 | 12/2020 | Bhide et al.              | N/A | N/A |
| 11068847 | 12/2020 | Boutros et al.            | N/A | N/A |
| 11093950 | 12/2020 | Hersh et al.              | N/A | N/A |
| 11138299 | 12/2020 | Brannon et al.            | N/A | N/A |
| 11144622 | 12/2020 | Brannon et al.            | N/A | N/A |
| 11144678 | 12/2020 | Dondini et al.            | N/A | N/A |
| 11144862 | 12/2020 | Jackson et al.            | N/A | N/A |

|              |         |                  |         |            |
|--------------|---------|------------------|---------|------------|
| 11195134     | 12/2020 | Brannon et al.   | N/A     | N/A        |
| 11201929     | 12/2020 | Dudmesh et al.   | N/A     | N/A        |
| 11210420     | 12/2020 | Brannon et al.   | N/A     | N/A        |
| 11238390     | 12/2021 | Brannon et al.   | N/A     | N/A        |
| 11240273     | 12/2021 | Barday et al.    | N/A     | N/A        |
| 11245951     | 12/2021 | Lim et al.       | N/A     | N/A        |
| 11246520     | 12/2021 | Clifford et al.  | N/A     | N/A        |
| 11252159     | 12/2021 | Kannan et al.    | N/A     | N/A        |
| 11256777     | 12/2021 | Brannon et al.   | N/A     | N/A        |
| 11263262     | 12/2021 | Chen             | N/A     | N/A        |
| 11327996     | 12/2021 | Reynolds et al.  | N/A     | N/A        |
| 2002/0004736 | 12/2001 | Roundtree et al. | N/A     | N/A        |
| 2002/0049907 | 12/2001 | Woods et al.     | N/A     | N/A        |
| 2002/0055932 | 12/2001 | Wheeler et al.   | N/A     | N/A        |
| 2002/0077941 | 12/2001 | Halligan et al.  | N/A     | N/A        |
| 2002/0103854 | 12/2001 | Okita            | N/A     | N/A        |
| 2002/0129216 | 12/2001 | Collins          | N/A     | N/A        |
| 2002/0161594 | 12/2001 | Bryan et al.     | N/A     | N/A        |
| 2002/0161733 | 12/2001 | Grainger         | N/A     | N/A        |
| 2003/0041250 | 12/2002 | Proudlar         | N/A     | N/A        |
| 2003/0065641 | 12/2002 | Chaloux          | N/A     | N/A        |
| 2003/0093680 | 12/2002 | Astley et al.    | N/A     | N/A        |
| 2003/0097451 | 12/2002 | Bjorksten et al. | N/A     | N/A        |
| 2003/0097661 | 12/2002 | Li et al.        | N/A     | N/A        |
| 2003/0115142 | 12/2002 | Brickell et al.  | N/A     | N/A        |
| 2003/0130893 | 12/2002 | Farmer           | N/A     | N/A        |
| 2003/0131001 | 12/2002 | Matsuo           | N/A     | N/A        |
| 2003/0131093 | 12/2002 | Aschen et al.    | N/A     | N/A        |
| 2003/0140150 | 12/2002 | Kemp et al.      | N/A     | N/A        |
| 2003/0167216 | 12/2002 | Brown et al.     | N/A     | N/A        |
| 2003/0212604 | 12/2002 | Cullen           | N/A     | N/A        |
| 2004/0002818 | 12/2003 | Kulp et al.      | N/A     | N/A        |
| 2004/0025053 | 12/2003 | Hayward          | N/A     | N/A        |
| 2004/0088235 | 12/2003 | Ziekle et al.    | N/A     | N/A        |
| 2004/0098366 | 12/2003 | Sinclair et al.  | N/A     | N/A        |
| 2004/0098493 | 12/2003 | Rees             | N/A     | N/A        |
| 2004/0111359 | 12/2003 | Hudock           | N/A     | N/A        |
| 2004/0186912 | 12/2003 | Harlow et al.    | N/A     | N/A        |
| 2004/0193907 | 12/2003 | Patanella        | N/A     | N/A        |
| 2005/0022198 | 12/2004 | Olapurath et al. | N/A     | N/A        |
| 2005/0033616 | 12/2004 | Vavul et al.     | N/A     | N/A        |
| 2005/0076294 | 12/2004 | Dehamer et al.   | N/A     | N/A        |
| 2005/0114343 | 12/2004 | Wesinger et al.  | N/A     | N/A        |
| 2005/0144066 | 12/2004 | Cope et al.      | N/A     | N/A        |
| 2005/0197884 | 12/2004 | Mullen           | N/A     | N/A        |
| 2005/0198177 | 12/2004 | Black            | N/A     | N/A        |
| 2005/0198646 | 12/2004 | Kortela          | N/A     | N/A        |
| 2005/0246292 | 12/2004 | Sarcanin         | N/A     | N/A        |
| 2005/0278538 | 12/2004 | Fowler           | N/A     | N/A        |
| 2006/0031078 | 12/2005 | Pizzinger et al. | N/A     | N/A        |
| 2006/0035204 | 12/2005 | LaMarche et al.  | N/A     | N/A        |
| 2006/0075122 | 12/2005 | Lindskog et al.  | N/A     | N/A        |
| 2006/0149730 | 12/2005 | Curtis           | N/A     | N/A        |
| 2006/0156052 | 12/2005 | Bodnar et al.    | N/A     | N/A        |
| 2006/0190280 | 12/2005 | Hoebel et al.    | N/A     | N/A        |
| 2006/0206375 | 12/2005 | Scott et al.     | N/A     | N/A        |
| 2006/0224422 | 12/2005 | Cohen            | N/A     | N/A        |
| 2006/0253597 | 12/2005 | Mujica           | N/A     | N/A        |
| 2006/0259416 | 12/2005 | Johnson          | N/A     | N/A        |
| 2007/0011058 | 12/2006 | Dev              | N/A     | N/A        |
| 2007/0027715 | 12/2006 | Gropper et al.   | N/A     | N/A        |
| 2007/0061125 | 12/2006 | Bhatt et al.     | N/A     | N/A        |
| 2007/0061393 | 12/2006 | Moore            | N/A     | N/A        |
| 2007/0130101 | 12/2006 | Anderson et al.  | N/A     | N/A        |
| 2007/0130323 | 12/2006 | Landsman et al.  | N/A     | N/A        |
| 2007/0157311 | 12/2006 | Meier et al.     | N/A     | N/A        |
| 2007/0173355 | 12/2006 | Klein            | N/A     | N/A        |
| 2007/0179793 | 12/2006 | Bagchi et al.    | N/A     | N/A        |
| 2007/0180490 | 12/2006 | Renzi et al.     | N/A     | N/A        |
| 2007/0192438 | 12/2006 | Goei             | N/A     | N/A        |
| 2007/0266420 | 12/2006 | Hawkins et al.   | N/A     | N/A        |
| 2007/0283171 | 12/2006 | Breslin et al.   | N/A     | N/A        |
| 2008/0005194 | 12/2007 | Smolen et al.    | N/A     | N/A        |
| 2008/0015927 | 12/2007 | Ramirez          | N/A     | N/A        |
| 2008/0022093 | 12/2007 | Kurien           | 713/164 | G06F 21/50 |



|              |         |                    |         |              |
|--------------|---------|--------------------|---------|--------------|
| 2008/0028065 | 12/2007 | Caso et al.        | N/A     | N/A          |
| 2008/0028435 | 12/2007 | Strickland et al.  | N/A     | N/A          |
| 2008/0047016 | 12/2007 | Spoonamore         | N/A     | N/A          |
| 2008/0120699 | 12/2007 | Spear              | N/A     | N/A          |
| 2008/0140696 | 12/2007 | Mathuria           | N/A     | N/A          |
| 2008/0189306 | 12/2007 | Hewett et al.      | N/A     | N/A          |
| 2008/0195436 | 12/2007 | Whyte              | N/A     | N/A          |
| 2008/0222271 | 12/2007 | Spires             | N/A     | N/A          |
| 2008/0235177 | 12/2007 | Kim et al.         | N/A     | N/A          |
| 2008/0270203 | 12/2007 | Holmes et al.      | N/A     | N/A          |
| 2008/0270351 | 12/2007 | Thomsen            | N/A     | N/A          |
| 2008/0270381 | 12/2007 | Thomsen            | N/A     | N/A          |
| 2008/0270382 | 12/2007 | Thomsen et al.     | N/A     | N/A          |
| 2008/0270451 | 12/2007 | Thomsen et al.     | N/A     | N/A          |
| 2008/0270462 | 12/2007 | Thomsen            | N/A     | N/A          |
| 2008/0281649 | 12/2007 | Morris             | N/A     | N/A          |
| 2008/0282320 | 12/2007 | DeNovo et al.      | N/A     | N/A          |
| 2008/0288271 | 12/2007 | Faust              | N/A     | N/A          |
| 2008/0288299 | 12/2007 | Schultz            | N/A     | N/A          |
| 2009/0012896 | 12/2008 | Arnold             | N/A     | N/A          |
| 2009/0022301 | 12/2008 | Mudaliar           | N/A     | N/A          |
| 2009/0037975 | 12/2008 | Ishikawa et al.    | N/A     | N/A          |
| 2009/0100498 | 12/2008 | Grossi             | 726/1   | G06Q 10/10   |
| 2009/0119500 | 12/2008 | Roth et al.        | N/A     | N/A          |
| 2009/0132419 | 12/2008 | Grammer et al.     | N/A     | N/A          |
| 2009/0138276 | 12/2008 | Hayashida          | 705/325 | G06Q 10/10   |
| 2009/0140035 | 12/2008 | Miller             | N/A     | N/A          |
| 2009/0144702 | 12/2008 | Atkin et al.       | N/A     | N/A          |
| 2009/0158249 | 12/2008 | Tomkins et al.     | N/A     | N/A          |
| 2009/0172705 | 12/2008 | Cheong             | N/A     | N/A          |
| 2009/0182818 | 12/2008 | Krywaniuk          | N/A     | N/A          |
| 2009/0187764 | 12/2008 | Astakhov et al.    | N/A     | N/A          |
| 2009/0204452 | 12/2008 | Iskandar et al.    | N/A     | N/A          |
| 2009/0204820 | 12/2008 | Brandenburg et al. | N/A     | N/A          |
| 2009/0210347 | 12/2008 | Sarcanin           | N/A     | N/A          |
| 2009/0216610 | 12/2008 | Chorny             | N/A     | N/A          |
| 2009/0249076 | 12/2008 | Reed et al.        | N/A     | N/A          |
| 2009/0300716 | 12/2008 | Ahn                | 726/1   | H04L 63/0876 |
| 2009/0303237 | 12/2008 | Liu et al.         | N/A     | N/A          |
| 2010/0010912 | 12/2009 | Jones et al.       | N/A     | N/A          |
| 2010/0010968 | 12/2009 | Redlich et al.     | N/A     | N/A          |
| 2010/0077484 | 12/2009 | Paretti            | 726/1   | H04W 4/023   |
| 2010/0082533 | 12/2009 | Nakamura et al.    | N/A     | N/A          |
| 2010/0094650 | 12/2009 | Tran et al.        | N/A     | N/A          |
| 2010/0100398 | 12/2009 | Auker et al.       | N/A     | N/A          |
| 2010/0121773 | 12/2009 | Currier et al.     | N/A     | N/A          |
| 2010/0192201 | 12/2009 | Shimoni et al.     | N/A     | N/A          |
| 2010/0205057 | 12/2009 | Hook et al.        | N/A     | N/A          |
| 2010/0223349 | 12/2009 | Thorson            | N/A     | N/A          |
| 2010/0228786 | 12/2009 | Török              | N/A     | N/A          |
| 2010/0234987 | 12/2009 | Benschop et al.    | N/A     | N/A          |
| 2010/0235297 | 12/2009 | Mamorsky           | N/A     | N/A          |
| 2010/0235915 | 12/2009 | Memon et al.       | N/A     | N/A          |
| 2010/0262624 | 12/2009 | Pullikottil        | N/A     | N/A          |
| 2010/0268628 | 12/2009 | Pitkow et al.      | N/A     | N/A          |
| 2010/0268932 | 12/2009 | Bhattacharjee      | N/A     | N/A          |
| 2010/0281313 | 12/2009 | White et al.       | N/A     | N/A          |
| 2010/0287114 | 12/2009 | Bartko et al.      | N/A     | N/A          |
| 2010/0333012 | 12/2009 | Adachi et al.      | N/A     | N/A          |
| 2011/0006996 | 12/2010 | Smith et al.       | N/A     | N/A          |
| 2011/0010202 | 12/2010 | Neale              | N/A     | N/A          |
| 2011/0082794 | 12/2010 | Blechman           | N/A     | N/A          |
| 2011/0137696 | 12/2010 | Meyer et al.       | N/A     | N/A          |
| 2011/0145154 | 12/2010 | Rivers et al.      | N/A     | N/A          |
| 2011/0153396 | 12/2010 | Marcuvitz et al.   | N/A     | N/A          |
| 2011/0191664 | 12/2010 | Sheleheda et al.   | N/A     | N/A          |
| 2011/0208850 | 12/2010 | Sheleheda et al.   | N/A     | N/A          |
| 2011/0209067 | 12/2010 | Bogess et al.      | N/A     | N/A          |
| 2011/0231896 | 12/2010 | Tovar              | N/A     | N/A          |
| 2011/0238573 | 12/2010 | Varadarajan        | N/A     | N/A          |
| 2011/0252456 | 12/2010 | Hatakeyama         | N/A     | N/A          |
| 2011/0302643 | 12/2010 | Pichna et al.      | N/A     | N/A          |
| 2012/0041939 | 12/2011 | Amsterdamski       | N/A     | N/A          |
| 2012/0084151 | 12/2011 | Kozak et al.       | N/A     | N/A          |
| 2012/0084349 | 12/2011 | Lee et al.         | N/A     | N/A          |

|              |         |                       |         |           |
|--------------|---------|-----------------------|---------|-----------|
| 2012/0102411 | 12/2011 | Sathish               | N/A     | N/A       |
| 2012/0102543 | 12/2011 | Kohli et al.          | N/A     | N/A       |
| 2012/0106738 | 12/2011 | Belenkiy              | 380/270 | H04L 9/14 |
| 2012/0109830 | 12/2011 | Vogel                 | N/A     | N/A       |
| 2012/0110674 | 12/2011 | Belani et al.         | N/A     | N/A       |
| 2012/0116923 | 12/2011 | Irving et al.         | N/A     | N/A       |
| 2012/0131438 | 12/2011 | Li et al.             | N/A     | N/A       |
| 2012/0143650 | 12/2011 | Crowley et al.        | N/A     | N/A       |
| 2012/0144499 | 12/2011 | Tan et al.            | N/A     | N/A       |
| 2012/0191596 | 12/2011 | Kremen et al.         | N/A     | N/A       |
| 2012/0226621 | 12/2011 | Petran et al.         | N/A     | N/A       |
| 2012/0239557 | 12/2011 | Weinflash et al.      | N/A     | N/A       |
| 2012/0254320 | 12/2011 | Dove et al.           | N/A     | N/A       |
| 2012/0259752 | 12/2011 | Agee                  | N/A     | N/A       |
| 2012/0323700 | 12/2011 | Aleksandrovich et al. | N/A     | N/A       |
| 2012/0330769 | 12/2011 | Arceo                 | N/A     | N/A       |
| 2012/0330869 | 12/2011 | Durham                | N/A     | N/A       |
| 2013/0004933 | 12/2012 | Bhaskaran             | N/A     | N/A       |
| 2013/0018954 | 12/2012 | Cheng                 | N/A     | N/A       |
| 2013/0085801 | 12/2012 | Sharpe et al.         | N/A     | N/A       |
| 2013/0091156 | 12/2012 | Raiche et al.         | N/A     | N/A       |
| 2013/0103485 | 12/2012 | Postrel               | N/A     | N/A       |
| 2013/0111323 | 12/2012 | Taghaddos et al.      | N/A     | N/A       |
| 2013/0124257 | 12/2012 | Schubert              | N/A     | N/A       |
| 2013/0159351 | 12/2012 | Hamann et al.         | N/A     | N/A       |
| 2013/0171968 | 12/2012 | Wang                  | N/A     | N/A       |
| 2013/0179982 | 12/2012 | Bridges et al.        | N/A     | N/A       |
| 2013/0179988 | 12/2012 | Bekker et al.         | N/A     | N/A       |
| 2013/0185806 | 12/2012 | Hatakeyama            | N/A     | N/A       |
| 2013/0211872 | 12/2012 | Cherry et al.         | N/A     | N/A       |
| 2013/0218829 | 12/2012 | Martinez              | N/A     | N/A       |
| 2013/0219459 | 12/2012 | Bradley               | N/A     | N/A       |
| 2013/0254139 | 12/2012 | Lei                   | N/A     | N/A       |
| 2013/0254649 | 12/2012 | Michael               | N/A     | N/A       |
| 2013/0254699 | 12/2012 | Bashir et al.         | N/A     | N/A       |
| 2013/0262328 | 12/2012 | Federgreen            | N/A     | N/A       |
| 2013/0282438 | 12/2012 | Hunter et al.         | N/A     | N/A       |
| 2013/0282466 | 12/2012 | Hampton               | N/A     | N/A       |
| 2013/0290169 | 12/2012 | Bathula et al.        | N/A     | N/A       |
| 2013/0298071 | 12/2012 | Wine                  | N/A     | N/A       |
| 2013/0311224 | 12/2012 | Heroux et al.         | N/A     | N/A       |
| 2013/0318207 | 12/2012 | Dotter                | N/A     | N/A       |
| 2013/0326112 | 12/2012 | Park et al.           | N/A     | N/A       |
| 2013/0332362 | 12/2012 | Ciurea                | N/A     | N/A       |
| 2013/0340086 | 12/2012 | Blom                  | N/A     | N/A       |
| 2014/0006355 | 12/2013 | Kirihata              | N/A     | N/A       |
| 2014/0006616 | 12/2013 | Aad et al.            | N/A     | N/A       |
| 2014/0012833 | 12/2013 | Humprecht             | N/A     | N/A       |
| 2014/0019561 | 12/2013 | Belity et al.         | N/A     | N/A       |
| 2014/0032259 | 12/2013 | Lafever et al.        | N/A     | N/A       |
| 2014/0032265 | 12/2013 | Paprocki              | N/A     | N/A       |
| 2014/0040134 | 12/2013 | Peter                 | N/A     | N/A       |
| 2014/0040161 | 12/2013 | Jason                 | N/A     | N/A       |
| 2014/0040979 | 12/2013 | Barton et al.         | N/A     | N/A       |
| 2014/0041048 | 12/2013 | Goodwin et al.        | N/A     | N/A       |
| 2014/0047551 | 12/2013 | Nagasundaram et al.   | N/A     | N/A       |
| 2014/0052463 | 12/2013 | Cashman et al.        | N/A     | N/A       |
| 2014/0067973 | 12/2013 | Eden                  | N/A     | N/A       |
| 2014/0074550 | 12/2013 | Chourey               | N/A     | N/A       |
| 2014/0074645 | 12/2013 | Ingram                | N/A     | N/A       |
| 2014/0075493 | 12/2013 | Krishnan et al.       | N/A     | N/A       |
| 2014/0089027 | 12/2013 | Brown                 | N/A     | N/A       |
| 2014/0089039 | 12/2013 | McClellan             | N/A     | N/A       |
| 2014/0108173 | 12/2013 | Cooper et al.         | N/A     | N/A       |
| 2014/0108968 | 12/2013 | Vishria               | N/A     | N/A       |
| 2014/0137257 | 12/2013 | Martinez et al.       | N/A     | N/A       |
| 2014/0142988 | 12/2013 | Grosso et al.         | N/A     | N/A       |
| 2014/0143011 | 12/2013 | Mudugu et al.         | N/A     | N/A       |
| 2014/0143844 | 12/2013 | Goertzen              | N/A     | N/A       |
| 2014/0164476 | 12/2013 | Thomson               | N/A     | N/A       |
| 2014/0188956 | 12/2013 | Subba et al.          | N/A     | N/A       |
| 2014/0196143 | 12/2013 | Fliderman et al.      | N/A     | N/A       |
| 2014/0208418 | 12/2013 | Libin                 | N/A     | N/A       |
| 2014/0222468 | 12/2013 | Araya et al.          | N/A     | N/A       |
| 2014/0244309 | 12/2013 | Francois              | N/A     | N/A       |

|              |         |                           |       |             |
|--------------|---------|---------------------------|-------|-------------|
| 2014/0244325 | 12/2013 | Cartwright                | N/A   | N/A         |
| 2014/0244375 | 12/2013 | Kim                       | N/A   | N/A         |
| 2014/0244399 | 12/2013 | Orduna et al.             | N/A   | N/A         |
| 2014/0257917 | 12/2013 | Spencer et al.            | N/A   | N/A         |
| 2014/0258093 | 12/2013 | Gardiner et al.           | N/A   | N/A         |
| 2014/0278539 | 12/2013 | Edwards                   | N/A   | N/A         |
| 2014/0278663 | 12/2013 | Samuel et al.             | N/A   | N/A         |
| 2014/0278730 | 12/2013 | Muhart et al.             | N/A   | N/A         |
| 2014/0278802 | 12/2013 | MacPherson                | N/A   | N/A         |
| 2014/0283027 | 12/2013 | Orona et al.              | N/A   | N/A         |
| 2014/0283106 | 12/2013 | Stahura et al.            | N/A   | N/A         |
| 2014/0288971 | 12/2013 | Whibbs, III               | N/A   | N/A         |
| 2014/0289681 | 12/2013 | Wielgosz                  | N/A   | N/A         |
| 2014/0289862 | 12/2013 | Gorfein et al.            | N/A   | N/A         |
| 2014/0317171 | 12/2013 | Fox et al.                | N/A   | N/A         |
| 2014/0324480 | 12/2013 | Dufel et al.              | N/A   | N/A         |
| 2014/0337041 | 12/2013 | Madden et al.             | N/A   | N/A         |
| 2014/0337466 | 12/2013 | Li et al.                 | N/A   | N/A         |
| 2014/0344015 | 12/2013 | Puértolas-Montañés et al. | N/A   | N/A         |
| 2015/0006514 | 12/2014 | Hung                      | N/A   | N/A         |
| 2015/0012363 | 12/2014 | Grant et al.              | N/A   | N/A         |
| 2015/0019530 | 12/2014 | Felch                     | N/A   | N/A         |
| 2015/0026056 | 12/2014 | Calman et al.             | N/A   | N/A         |
| 2015/0026260 | 12/2014 | Worthley                  | N/A   | N/A         |
| 2015/0033112 | 12/2014 | Norwood et al.            | N/A   | N/A         |
| 2015/0066577 | 12/2014 | Christiansen et al.       | N/A   | N/A         |
| 2015/0066865 | 12/2014 | Yara et al.               | N/A   | N/A         |
| 2015/0088598 | 12/2014 | Acharyya et al.           | N/A   | N/A         |
| 2015/0088635 | 12/2014 | Maycotte et al.           | N/A   | N/A         |
| 2015/0101007 | 12/2014 | Fujioka                   | 726/1 | H04L 63/107 |
| 2015/0106264 | 12/2014 | Johnson                   | N/A   | N/A         |
| 2015/0106867 | 12/2014 | Liang                     | N/A   | N/A         |
| 2015/0106948 | 12/2014 | Holman et al.             | N/A   | N/A         |
| 2015/0106949 | 12/2014 | Holman et al.             | N/A   | N/A         |
| 2015/0121462 | 12/2014 | Courage et al.            | N/A   | N/A         |
| 2015/0143258 | 12/2014 | Carolan et al.            | N/A   | N/A         |
| 2015/0149362 | 12/2014 | Baum et al.               | N/A   | N/A         |
| 2015/0154520 | 12/2014 | Federgreen et al.         | N/A   | N/A         |
| 2015/0169318 | 12/2014 | Nash                      | N/A   | N/A         |
| 2015/0172296 | 12/2014 | Fujioka                   | N/A   | N/A         |
| 2015/0178740 | 12/2014 | Borawski et al.           | N/A   | N/A         |
| 2015/0199534 | 12/2014 | Francis et al.            | N/A   | N/A         |
| 2015/0199541 | 12/2014 | Koch et al.               | N/A   | N/A         |
| 2015/0199702 | 12/2014 | Singh                     | N/A   | N/A         |
| 2015/0229664 | 12/2014 | Hawthorn et al.           | N/A   | N/A         |
| 2015/0235049 | 12/2014 | Cohen et al.              | N/A   | N/A         |
| 2015/0235050 | 12/2014 | Wouhaybi et al.           | N/A   | N/A         |
| 2015/0235283 | 12/2014 | Nishikawa                 | N/A   | N/A         |
| 2015/0242638 | 12/2014 | Bitran et al.             | N/A   | N/A         |
| 2015/0242778 | 12/2014 | Wilcox et al.             | N/A   | N/A         |
| 2015/0242858 | 12/2014 | Smith et al.              | N/A   | N/A         |
| 2015/0248391 | 12/2014 | Watanabe                  | N/A   | N/A         |
| 2015/0254597 | 12/2014 | Jahagirdar                | N/A   | N/A         |
| 2015/0261887 | 12/2014 | Joukov                    | N/A   | N/A         |
| 2015/0262189 | 12/2014 | Vergeer                   | N/A   | N/A         |
| 2015/0264417 | 12/2014 | Spitz et al.              | N/A   | N/A         |
| 2015/0269384 | 12/2014 | Holman et al.             | N/A   | N/A         |
| 2015/0271167 | 12/2014 | Kalai                     | N/A   | N/A         |
| 2015/0288715 | 12/2014 | Hotchkiss                 | N/A   | N/A         |
| 2015/0309813 | 12/2014 | Patel                     | N/A   | N/A         |
| 2015/0310227 | 12/2014 | Ishida et al.             | N/A   | N/A         |
| 2015/0310575 | 12/2014 | Shelton                   | N/A   | N/A         |
| 2015/0348200 | 12/2014 | Fair et al.               | N/A   | N/A         |
| 2015/0356362 | 12/2014 | Demos                     | N/A   | N/A         |
| 2015/0379430 | 12/2014 | Dirac et al.              | N/A   | N/A         |
| 2016/0006760 | 12/2015 | Lala et al.               | N/A   | N/A         |
| 2016/0012465 | 12/2015 | Sharp                     | N/A   | N/A         |
| 2016/0026394 | 12/2015 | Goto                      | N/A   | N/A         |
| 2016/0034918 | 12/2015 | Bjelajac et al.           | N/A   | N/A         |
| 2016/0048700 | 12/2015 | Stransky-Heilkron         | N/A   | N/A         |
| 2016/0050213 | 12/2015 | Storr                     | N/A   | N/A         |
| 2016/0063523 | 12/2015 | Nistor et al.             | N/A   | N/A         |
| 2016/0063567 | 12/2015 | Srivastava                | N/A   | N/A         |
| 2016/0071112 | 12/2015 | Unser                     | N/A   | N/A         |
| 2016/0080405 | 12/2015 | Schler et al.             | N/A   | N/A         |

|              |         |                          |         |             |
|--------------|---------|--------------------------|---------|-------------|
| 2016/0099963 | 12/2015 | Mahaffey et al.          | N/A     | N/A         |
| 2016/0103963 | 12/2015 | Mishra                   | N/A     | N/A         |
| 2016/0104259 | 12/2015 | Menrad                   | N/A     | N/A         |
| 2016/0125550 | 12/2015 | Joao et al.              | N/A     | N/A         |
| 2016/0125749 | 12/2015 | Delacroix et al.         | N/A     | N/A         |
| 2016/0125751 | 12/2015 | Barker et al.            | N/A     | N/A         |
| 2016/0140466 | 12/2015 | Sidebottom et al.        | N/A     | N/A         |
| 2016/0143570 | 12/2015 | Valacich et al.          | N/A     | N/A         |
| 2016/0148143 | 12/2015 | Anderson et al.          | N/A     | N/A         |
| 2016/0148259 | 12/2015 | Baek et al.              | N/A     | N/A         |
| 2016/0162269 | 12/2015 | Pogorelik et al.         | N/A     | N/A         |
| 2016/0164915 | 12/2015 | Cook                     | N/A     | N/A         |
| 2016/0180386 | 12/2015 | Konig                    | N/A     | N/A         |
| 2016/0188450 | 12/2015 | Appusamy et al.          | N/A     | N/A         |
| 2016/0189156 | 12/2015 | Kim et al.               | N/A     | N/A         |
| 2016/0196189 | 12/2015 | Miyagi et al.            | N/A     | N/A         |
| 2016/0225000 | 12/2015 | Glasgow                  | N/A     | N/A         |
| 2016/0232465 | 12/2015 | Kurtz et al.             | N/A     | N/A         |
| 2016/0232534 | 12/2015 | Lacey et al.             | N/A     | N/A         |
| 2016/0234319 | 12/2015 | Griffin                  | N/A     | N/A         |
| 2016/0253497 | 12/2015 | Christodorescu et al.    | N/A     | N/A         |
| 2016/0255139 | 12/2015 | Rathod                   | 709/203 | H04L 51/046 |
| 2016/0261631 | 12/2015 | Vissamsetty et al.       | N/A     | N/A         |
| 2016/0262163 | 12/2015 | Gonzalez Garrido et al.  | N/A     | N/A         |
| 2016/0292453 | 12/2015 | Patterson et al.         | N/A     | N/A         |
| 2016/0292621 | 12/2015 | Ciccone et al.           | N/A     | N/A         |
| 2016/0321582 | 12/2015 | Broudou et al.           | N/A     | N/A         |
| 2016/0321748 | 12/2015 | Mahatma et al.           | N/A     | N/A         |
| 2016/0330237 | 12/2015 | Edlabadkar               | N/A     | N/A         |
| 2016/0335531 | 12/2015 | Mullen et al.            | N/A     | N/A         |
| 2016/0342811 | 12/2015 | Whitcomb et al.          | N/A     | N/A         |
| 2016/0359861 | 12/2015 | Manov et al.             | N/A     | N/A         |
| 2016/0364736 | 12/2015 | Maugans, III             | N/A     | N/A         |
| 2016/0370954 | 12/2015 | Burningham et al.        | N/A     | N/A         |
| 2016/0378762 | 12/2015 | Rohter                   | N/A     | N/A         |
| 2016/0381064 | 12/2015 | Chan et al.              | N/A     | N/A         |
| 2016/0381560 | 12/2015 | Margaliot                | N/A     | N/A         |
| 2017/0004055 | 12/2016 | Horan et al.             | N/A     | N/A         |
| 2017/0032395 | 12/2016 | Kaufman et al.           | N/A     | N/A         |
| 2017/0032408 | 12/2016 | Kumar et al.             | N/A     | N/A         |
| 2017/0034101 | 12/2016 | Kumar et al.             | N/A     | N/A         |
| 2017/0041324 | 12/2016 | Ionutescu et al.         | N/A     | N/A         |
| 2017/0046399 | 12/2016 | Sankaranarasimhan et al. | N/A     | N/A         |
| 2017/0046753 | 12/2016 | Deupree, IV              | N/A     | N/A         |
| 2017/0061501 | 12/2016 | Horwich                  | N/A     | N/A         |
| 2017/0068785 | 12/2016 | Experton et al.          | N/A     | N/A         |
| 2017/0070495 | 12/2016 | Cherry et al.            | N/A     | N/A         |
| 2017/0075513 | 12/2016 | Watson et al.            | N/A     | N/A         |
| 2017/0093917 | 12/2016 | Chandra et al.           | N/A     | N/A         |
| 2017/0115864 | 12/2016 | Thomas et al.            | N/A     | N/A         |
| 2017/0124570 | 12/2016 | Nidamanuri et al.        | N/A     | N/A         |
| 2017/0140174 | 12/2016 | Lacey et al.             | N/A     | N/A         |
| 2017/0140467 | 12/2016 | Neag et al.              | N/A     | N/A         |
| 2017/0142158 | 12/2016 | Laoutaris et al.         | N/A     | N/A         |
| 2017/0142177 | 12/2016 | Hu                       | N/A     | N/A         |
| 2017/0154188 | 12/2016 | Meier et al.             | N/A     | N/A         |
| 2017/0161520 | 12/2016 | Lockhart, III et al.     | N/A     | N/A         |
| 2017/0171235 | 12/2016 | Mulchandani et al.       | N/A     | N/A         |
| 2017/0171325 | 12/2016 | Perez                    | N/A     | N/A         |
| 2017/0177324 | 12/2016 | Frank et al.             | N/A     | N/A         |
| 2017/0180378 | 12/2016 | Tyler et al.             | N/A     | N/A         |
| 2017/0180505 | 12/2016 | Shaw et al.              | N/A     | N/A         |
| 2017/0193017 | 12/2016 | Migliori                 | N/A     | N/A         |
| 2017/0193624 | 12/2016 | Tsai                     | N/A     | N/A         |
| 2017/0201518 | 12/2016 | Holmqvist et al.         | N/A     | N/A         |
| 2017/0206707 | 12/2016 | Guay et al.              | N/A     | N/A         |
| 2017/0208084 | 12/2016 | Steelman et al.          | N/A     | N/A         |
| 2017/0213206 | 12/2016 | Shearer                  | N/A     | N/A         |
| 2017/0220685 | 12/2016 | Yan et al.               | N/A     | N/A         |
| 2017/0220964 | 12/2016 | Datta Ray                | N/A     | N/A         |
| 2017/0249394 | 12/2016 | Loeb et al.              | N/A     | N/A         |
| 2017/0249710 | 12/2016 | Guillama et al.          | N/A     | N/A         |
| 2017/0269791 | 12/2016 | Meyerzon et al.          | N/A     | N/A         |
| 2017/0270318 | 12/2016 | Ritchie                  | N/A     | N/A         |
| 2017/0278004 | 12/2016 | McElhinney et al.        | N/A     | N/A         |

|              |         |                        |     |            |
|--------------|---------|------------------------|-----|------------|
| 2017/0278117 | 12/2016 | Wallace et al.         | N/A | N/A        |
| 2017/0286719 | 12/2016 | Krishnamurthy et al.   | N/A | N/A        |
| 2017/0287031 | 12/2016 | Barday                 | N/A | N/A        |
| 2017/0289199 | 12/2016 | Barday                 | N/A | N/A        |
| 2017/0308875 | 12/2016 | O'Regan et al.         | N/A | N/A        |
| 2017/0316400 | 12/2016 | Venkatakrishnan et al. | N/A | N/A        |
| 2017/0330197 | 12/2016 | DiMaggio et al.        | N/A | N/A        |
| 2017/0353404 | 12/2016 | Hodge                  | N/A | N/A        |
| 2018/0032757 | 12/2017 | Michael                | N/A | N/A        |
| 2018/0039975 | 12/2017 | Hefetz                 | N/A | N/A        |
| 2018/0041498 | 12/2017 | Kikuchi                | N/A | N/A        |
| 2018/0046753 | 12/2017 | Shelton                | N/A | N/A        |
| 2018/0046939 | 12/2017 | Meron et al.           | N/A | N/A        |
| 2018/0063174 | 12/2017 | Grill et al.           | N/A | N/A        |
| 2018/0063190 | 12/2017 | Wright et al.          | N/A | N/A        |
| 2018/0082368 | 12/2017 | Weinflash et al.       | N/A | N/A        |
| 2018/0083843 | 12/2017 | Sambandam              | N/A | N/A        |
| 2018/0091476 | 12/2017 | Jakobsson et al.       | N/A | N/A        |
| 2018/0131574 | 12/2017 | Jacobs et al.          | N/A | N/A        |
| 2018/0131658 | 12/2017 | Bhagwan et al.         | N/A | N/A        |
| 2018/0165637 | 12/2017 | Romero et al.          | N/A | N/A        |
| 2018/0198614 | 12/2017 | Neumann                | N/A | N/A        |
| 2018/0204281 | 12/2017 | Painter et al.         | N/A | N/A        |
| 2018/0219917 | 12/2017 | Chiang                 | N/A | N/A        |
| 2018/0239500 | 12/2017 | Allen et al.           | N/A | N/A        |
| 2018/0248914 | 12/2017 | Sartor                 | N/A | N/A        |
| 2018/0285887 | 12/2017 | Maung                  | N/A | N/A        |
| 2018/0301222 | 12/2017 | Dew, Sr.               | N/A | G16H 15/00 |
| 2018/0307859 | 12/2017 | Lafever et al.         | N/A | N/A        |
| 2018/0336509 | 12/2017 | Guttmann               | N/A | N/A        |
| 2018/0343215 | 12/2017 | Ganapathi et al.       | N/A | N/A        |
| 2018/0349583 | 12/2017 | Turgeman et al.        | N/A | N/A        |
| 2018/0351888 | 12/2017 | Howard                 | N/A | N/A        |
| 2018/0352003 | 12/2017 | Winn et al.            | N/A | N/A        |
| 2018/0357243 | 12/2017 | Yoon                   | N/A | N/A        |
| 2018/0365720 | 12/2017 | Goldman et al.         | N/A | N/A        |
| 2018/0374030 | 12/2017 | Barday et al.          | N/A | N/A        |
| 2018/0375814 | 12/2017 | Hart                   | N/A | N/A        |
| 2019/0005210 | 12/2018 | Wiederspohn et al.     | N/A | N/A        |
| 2019/0012211 | 12/2018 | Selvaraj               | N/A | N/A        |
| 2019/0012672 | 12/2018 | Francesco              | N/A | N/A        |
| 2019/0019184 | 12/2018 | Lacey et al.           | N/A | N/A        |
| 2019/0050547 | 12/2018 | Welsh et al.           | N/A | N/A        |
| 2019/0087570 | 12/2018 | Sloane                 | N/A | N/A        |
| 2019/0096020 | 12/2018 | Barday et al.          | N/A | N/A        |
| 2019/0108353 | 12/2018 | Sadeh et al.           | N/A | N/A        |
| 2019/0130132 | 12/2018 | Barbas et al.          | N/A | N/A        |
| 2019/0132350 | 12/2018 | Smith et al.           | N/A | N/A        |
| 2019/0138496 | 12/2018 | Yamaguchi              | N/A | N/A        |
| 2019/0139087 | 12/2018 | Dabbs et al.           | N/A | N/A        |
| 2019/0148003 | 12/2018 | Van Hoe                | N/A | N/A        |
| 2019/0156053 | 12/2018 | Vogel et al.           | N/A | N/A        |
| 2019/0156058 | 12/2018 | Van Dyne et al.        | N/A | N/A        |
| 2019/0171801 | 12/2018 | Barday et al.          | N/A | N/A        |
| 2019/0179652 | 12/2018 | Hesener et al.         | N/A | N/A        |
| 2019/0180051 | 12/2018 | Barday et al.          | N/A | N/A        |
| 2019/0182294 | 12/2018 | Rieke et al.           | N/A | N/A        |
| 2019/0188402 | 12/2018 | Wang et al.            | N/A | N/A        |
| 2019/0266200 | 12/2018 | Francolla              | N/A | N/A        |
| 2019/0266201 | 12/2018 | Barday et al.          | N/A | N/A        |
| 2019/0266350 | 12/2018 | Barday et al.          | N/A | N/A        |
| 2019/0268343 | 12/2018 | Barday et al.          | N/A | N/A        |
| 2019/0268344 | 12/2018 | Barday et al.          | N/A | N/A        |
| 2019/0272492 | 12/2018 | Elledge et al.         | N/A | N/A        |
| 2019/0294818 | 12/2018 | Barday et al.          | N/A | N/A        |
| 2019/0332802 | 12/2018 | Barday et al.          | N/A | N/A        |
| 2019/0332807 | 12/2018 | Lafever et al.         | N/A | N/A        |
| 2019/0333118 | 12/2018 | Crimmins et al.        | N/A | N/A        |
| 2019/0354709 | 12/2018 | Brinskelle             | N/A | N/A        |
| 2019/0356684 | 12/2018 | Sinha et al.           | N/A | N/A        |
| 2019/0362169 | 12/2018 | Lin et al.             | N/A | N/A        |
| 2019/0362268 | 12/2018 | Fogarty et al.         | N/A | N/A        |
| 2019/0377901 | 12/2018 | Balzer et al.          | N/A | N/A        |
| 2019/0378073 | 12/2018 | Lopez et al.           | N/A | N/A        |
| 2019/0384934 | 12/2018 | Kim                    | N/A | N/A        |

|              |         |                       |     |     |
|--------------|---------|-----------------------|-----|-----|
| 2019/0392162 | 12/2018 | Stern et al.          | N/A | N/A |
| 2019/0392170 | 12/2018 | Barday et al.         | N/A | N/A |
| 2019/0392171 | 12/2018 | Barday et al.         | N/A | N/A |
| 2020/0004938 | 12/2019 | Brannon et al.        | N/A | N/A |
| 2020/0020454 | 12/2019 | McGarvey et al.       | N/A | N/A |
| 2020/0050966 | 12/2019 | Enuka et al.          | N/A | N/A |
| 2020/0051117 | 12/2019 | Mitchell              | N/A | N/A |
| 2020/0057781 | 12/2019 | McCormick             | N/A | N/A |
| 2020/0074471 | 12/2019 | Adjaoute              | N/A | N/A |
| 2020/0081865 | 12/2019 | Farrar et al.         | N/A | N/A |
| 2020/0082270 | 12/2019 | Gu et al.             | N/A | N/A |
| 2020/0090197 | 12/2019 | Rodriguez et al.      | N/A | N/A |
| 2020/0092179 | 12/2019 | Chieu et al.          | N/A | N/A |
| 2020/0110589 | 12/2019 | Bequet et al.         | N/A | N/A |
| 2020/0110904 | 12/2019 | Shinde et al.         | N/A | N/A |
| 2020/0117737 | 12/2019 | Gopalakrishnan et al. | N/A | N/A |
| 2020/0137097 | 12/2019 | Zimmermann et al.     | N/A | N/A |
| 2020/0143301 | 12/2019 | Bowers                | N/A | N/A |
| 2020/0143797 | 12/2019 | Manoharan et al.      | N/A | N/A |
| 2020/0159952 | 12/2019 | Dain et al.           | N/A | N/A |
| 2020/0159955 | 12/2019 | Barlik et al.         | N/A | N/A |
| 2020/0167653 | 12/2019 | Manjunath et al.      | N/A | N/A |
| 2020/0175424 | 12/2019 | Kursun                | N/A | N/A |
| 2020/0183655 | 12/2019 | Barday et al.         | N/A | N/A |
| 2020/0186355 | 12/2019 | Davies                | N/A | N/A |
| 2020/0193018 | 12/2019 | Van Dyke              | N/A | N/A |
| 2020/0193022 | 12/2019 | Lunsford et al.       | N/A | N/A |
| 2020/0210558 | 12/2019 | Barday et al.         | N/A | N/A |
| 2020/0210620 | 12/2019 | Haletky               | N/A | N/A |
| 2020/0211002 | 12/2019 | Steinberg             | N/A | N/A |
| 2020/0220901 | 12/2019 | Barday et al.         | N/A | N/A |
| 2020/0226156 | 12/2019 | Borra et al.          | N/A | N/A |
| 2020/0226196 | 12/2019 | Brannon et al.        | N/A | N/A |
| 2020/0242259 | 12/2019 | Chirravuri et al.     | N/A | N/A |
| 2020/0242719 | 12/2019 | Lee                   | N/A | N/A |
| 2020/0250342 | 12/2019 | Miller et al.         | N/A | N/A |
| 2020/0252413 | 12/2019 | Buzbee et al.         | N/A | N/A |
| 2020/0252817 | 12/2019 | Brouillette et al.    | N/A | N/A |
| 2020/0272764 | 12/2019 | Brannon et al.        | N/A | N/A |
| 2020/0285755 | 12/2019 | Kassoumeh et al.      | N/A | N/A |
| 2020/0293679 | 12/2019 | Handy Bosma et al.    | N/A | N/A |
| 2020/0296171 | 12/2019 | Mocanu et al.         | N/A | N/A |
| 2020/0302089 | 12/2019 | Barday et al.         | N/A | N/A |
| 2020/0310917 | 12/2019 | Tkachev et al.        | N/A | N/A |
| 2020/0311310 | 12/2019 | Barday et al.         | N/A | N/A |
| 2020/0344243 | 12/2019 | Brannon et al.        | N/A | N/A |
| 2020/0356695 | 12/2019 | Brannon et al.        | N/A | N/A |
| 2020/0364369 | 12/2019 | Brannon et al.        | N/A | N/A |
| 2020/0372178 | 12/2019 | Barday et al.         | N/A | N/A |
| 2020/0394327 | 12/2019 | Childress et al.      | N/A | N/A |
| 2020/0401380 | 12/2019 | Jacobs et al.         | N/A | N/A |
| 2020/0401962 | 12/2019 | Gottemukkala et al.   | N/A | N/A |
| 2020/0410117 | 12/2019 | Barday et al.         | N/A | N/A |
| 2020/0410131 | 12/2019 | Barday et al.         | N/A | N/A |
| 2020/0410132 | 12/2019 | Brannon et al.        | N/A | N/A |
| 2021/0012341 | 12/2020 | Garg et al.           | N/A | N/A |
| 2021/0056569 | 12/2020 | Silberman et al.      | N/A | N/A |
| 2021/0075775 | 12/2020 | Cheng et al.          | N/A | N/A |
| 2021/0081567 | 12/2020 | Park et al.           | N/A | N/A |
| 2021/0099449 | 12/2020 | Frederick et al.      | N/A | N/A |
| 2021/0110047 | 12/2020 | Victor                | N/A | N/A |
| 2021/0125089 | 12/2020 | Nickl et al.          | N/A | N/A |
| 2021/0152496 | 12/2020 | Kim et al.            | N/A | N/A |
| 2021/0182940 | 12/2020 | Gupta et al.          | N/A | N/A |
| 2021/0224402 | 12/2020 | Sher-Jan et al.       | N/A | N/A |
| 2021/0233157 | 12/2020 | Crutchfield, Jr.      | N/A | N/A |
| 2021/0243595 | 12/2020 | Buck et al.           | N/A | N/A |
| 2021/0248247 | 12/2020 | Poothokaran et al.    | N/A | N/A |
| 2021/0256163 | 12/2020 | Fleming et al.        | N/A | N/A |
| 2021/0279360 | 12/2020 | Jimenez Palop et al.  | N/A | N/A |
| 2021/0288995 | 12/2020 | Attar et al.          | N/A | N/A |
| 2021/0297441 | 12/2020 | Olalere               | N/A | N/A |
| 2021/0303828 | 12/2020 | Lafreniere et al.     | N/A | N/A |
| 2021/0312061 | 12/2020 | Schroeder et al.      | N/A | N/A |
| 2021/0326786 | 12/2020 | Sun et al.            | N/A | N/A |

|              |         |                       |     |     |
|--------------|---------|-----------------------|-----|-----|
| 2021/0328969 | 12/2020 | Gaddam et al.         | N/A | N/A |
| 2021/0382949 | 12/2020 | Yastrebenetsky et al. | N/A | N/A |
| 2021/0397735 | 12/2020 | Samatov et al.        | N/A | N/A |
| 2021/0400018 | 12/2020 | Vettaikaran et al.    | N/A | N/A |
| 2021/0406712 | 12/2020 | Bhide et al.          | N/A | N/A |
| 2022/0137850 | 12/2021 | Boddu et al.          | N/A | N/A |
| 2022/0171759 | 12/2021 | Jindal et al.         | N/A | N/A |
| 2022/0217045 | 12/2021 | Blau et al.           | N/A | N/A |

#### FOREIGN PATENT DOCUMENTS

| Patent No.  | Application Date | Country | CPC |
|-------------|------------------|---------|-----|
| 111496802   | 12/2019          | CN      | N/A |
| 112115859   | 12/2019          | CN      | N/A |
| 1394698     | 12/2003          | EP      | N/A |
| 2031540     | 12/2008          | EP      | N/A |
| 20130062500 | 12/2012          | KR      | N/A |
| 2001033430  | 12/2000          | WO      | N/A |
| 20020067158 | 12/2001          | WO      | N/A |
| 20030050773 | 12/2002          | WO      | N/A |
| 2005008411  | 12/2004          | WO      | N/A |
| 2007002412  | 12/2006          | WO      | N/A |
| 2008/134203 | 12/2007          | WO      | N/A |
| 2012174659  | 12/2011          | WO      | N/A |
| 2015116905  | 12/2014          | WO      | N/A |
| 2020/146028 | 12/2019          | WO      | N/A |
| 2022006421  | 12/2021          | WO      | N/A |

#### OTHER PUBLICATIONS

Notice of Allowance, dated Mar. 4, 2022, from corresponding U.S. Appl. No. 17/409,999. cited by applicant  
 Notice of Allowance, dated May 1, 2020, from corresponding U.S. Appl. No. 16/586,202. cited by applicant  
 Notice of Allowance, dated May 11, 2020, from corresponding U.S. Appl. No. 16/786,196. cited by applicant  
 Notice of Allowance, dated May 11, 2022, from corresponding U.S. Appl. No. 17/395,759. cited by applicant  
 Notice of Allowance, dated May 13, 2021, from corresponding U.S. Appl. No. 17/101,915. cited by applicant  
 Notice of Allowance, dated May 18, 2022, from corresponding U.S. Appl. No. 17/670,354. cited by applicant  
 Notice of Allowance, dated May 19, 2020, from corresponding U.S. Appl. No. 16/505,430. cited by applicant  
 Notice of Allowance, dated May 19, 2020, from corresponding U.S. Appl. No. 16/808,496. cited by applicant  
 Notice of Allowance, dated May 20, 2020, from corresponding U.S. Appl. No. 16/707,762. cited by applicant  
 Notice of Allowance, dated May 21, 2018, from corresponding U.S. Appl. No. 15/896,790. cited by applicant  
 Notice of Allowance, dated May 25, 2022, from corresponding U.S. Appl. No. 16/872,031. cited by applicant  
 Notice of Allowance, dated May 26, 2021, from corresponding U.S. Appl. No. 16/808,493. cited by applicant  
 Notice of Allowance, dated May 26, 2021, from corresponding U.S. Appl. No. 16/865,874. cited by applicant  
 Notice of Allowance, dated May 26, 2021, from corresponding U.S. Appl. No. 17/199,514. cited by applicant  
 Notice of Allowance, dated May 27, 2020, from corresponding U.S. Appl. No. 16/820,208. cited by applicant  
 Notice of Allowance, dated May 27, 2021, from corresponding U.S. Appl. No. 16/927,658. cited by applicant  
 Notice of Allowance, dated May 27, 2021, from corresponding U.S. Appl. No. 17/198,757. cited by applicant  
 Notice of Allowance, dated May 27, 2022, from corresponding U.S. Appl. No. 17/543,546. cited by applicant  
 Notice of Allowance, dated May 28, 2019, from corresponding U.S. Appl. No. 16/277,568. cited by applicant  
 Notice of Allowance, dated May 28, 2020, from corresponding U.S. Appl. No. 16/799,279. cited by applicant  
 Notice of Allowance, dated May 28, 2021, from corresponding U.S. Appl. No. 16/862,944. cited by applicant  
 Notice of Allowance, dated May 31, 2022, from corresponding U.S. Appl. No. 17/679,715. cited by applicant  
 Notice of Allowance, dated May 5, 2017, from corresponding U.S. Appl. No. 15/254,901. cited by applicant  
 Notice of Allowance, dated May 5, 2020, from corresponding U.S. Appl. No. 16/563,754. cited by applicant  
 Notice of Allowance, dated May 6, 2022, from corresponding U.S. Appl. No. 17/666,886. cited by applicant  
 Notice of Allowance, dated May 7, 2020, from corresponding U.S. Appl. No. 16/505,426. cited by applicant  
 Notice of Allowance, dated May 7, 2021, from corresponding U.S. Appl. No. 17/194,662. cited by applicant  
 Notice of Allowance, dated Nov. 14, 2019, from corresponding U.S. Appl. No. 16/436,616. cited by applicant  
 Notice of Allowance, dated Nov. 16, 2021, from corresponding U.S. Appl. No. 17/491,871. cited by applicant  
 Notice of Allowance, dated Nov. 16, 2022, from corresponding U.S. Appl. No. 17/860,255. cited by applicant  
 Notice of Allowance, dated Nov. 2, 2018, from corresponding U.S. Appl. No. 16/054,762. cited by applicant  
 Notice of Allowance, dated Nov. 21, 2022, from corresponding U.S. Appl. No. 17/831,713. cited by applicant  
 Notice of Allowance, dated Nov. 22, 2021, from corresponding U.S. Appl. No. 17/383,889. cited by applicant  
 Notice of Allowance, dated Nov. 22, 2022, from corresponding U.S. Appl. No. 17/828,953. cited by applicant  
 Notice of Allowance, dated Nov. 23, 2020, from corresponding U.S. Appl. No. 16/791,589. cited by applicant  
 Notice of Allowance, dated Nov. 24, 2020, from corresponding U.S. Appl. No. 17/027,019. cited by applicant  
 Notice of Allowance, dated Nov. 25, 2020, from corresponding U.S. Appl. No. 17/019,771. cited by applicant  
 Notice of Allowance, dated Nov. 26, 2019, from corresponding U.S. Appl. No. 16/563,735. cited by applicant  
 Notice of Allowance, dated Nov. 27, 2019, from corresponding U.S. Appl. No. 16/570,712. cited by applicant  
 Notice of Allowance, dated Nov. 27, 2019, from corresponding U.S. Appl. No. 16/577,634. cited by applicant  
 Notice of Allowance, dated Nov. 3, 2020, from corresponding U.S. Appl. No. 16/719,071. cited by applicant  
 Notice of Allowance, dated Nov. 5, 2019, from corresponding U.S. Appl. No. 16/560,965. cited by applicant  
 Notice of Allowance, dated Nov. 7, 2017, from corresponding U.S. Appl. No. 15/671,073. cited by applicant  
 Notice of Allowance, dated Nov. 8, 2018, from corresponding U.S. Appl. No. 16/042,642. cited by applicant  
 Notice of Allowance, dated Nov. 9, 2020, from corresponding U.S. Appl. No. 16/595,342. cited by applicant  
 Notice of Allowance, dated Nov. 9, 2022, from corresponding U.S. Appl. No. 17/187,329. cited by applicant  
 Notice of Allowance, dated Oct. 1, 2021, from corresponding U.S. Appl. No. 17/340,395. cited by applicant  
 Notice of Allowance, dated Oct. 10, 2019, from corresponding U.S. Appl. No. 16/277,539. cited by applicant

Notice of Allowance, dated Oct. 17, 2018, from corresponding U.S. Appl. No. 15/896,790. cited by applicant  
 Notice of Allowance, dated Oct. 17, 2018, from corresponding U.S. Appl. No. 16/054,672. cited by applicant  
 Notice of Allowance, dated Dec. 11, 2019, from corresponding U.S. Appl. No. 16/593,634. cited by applicant  
 Notice of Allowance, dated Dec. 12, 2017, from corresponding U.S. Appl. No. 15/169,643. cited by applicant  
 Notice of Allowance, dated Dec. 12, 2017, from corresponding U.S. Appl. No. 15/619,212. cited by applicant  
 Notice of Allowance, dated Dec. 12, 2017, from corresponding U.S. Appl. No. 15/619,382. cited by applicant  
 Notice of Allowance, dated Dec. 13, 2019, from corresponding U.S. Appl. No. 16/512,033. cited by applicant  
 Notice of Allowance, dated Dec. 13, 2021, from corresponding U.S. Appl. No. 16/908,081. cited by applicant  
 Notice of Allowance, dated Dec. 13, 2021, from corresponding U.S. Appl. No. 17/347,853. cited by applicant  
 Notice of Allowance, dated Dec. 15, 2020, from corresponding U.S. Appl. No. 16/989,086. cited by applicant  
 Notice of Allowance, dated Dec. 16, 2019, from corresponding U.S. Appl. No. 16/505,461. cited by applicant  
 Notice of Allowance, dated Dec. 17, 2020, from corresponding U.S. Appl. No. 17/034,772. cited by applicant  
 Notice of Allowance, dated Dec. 18, 2019, from corresponding U.S. Appl. No. 16/659,437. cited by applicant  
 Notice of Allowance, dated Dec. 2, 2021, from corresponding U.S. Appl. No. 16/901,654. cited by applicant  
 Notice of Allowance, dated Dec. 23, 2019, from corresponding U.S. Appl. No. 16/656,835. cited by applicant  
 Notice of Allowance, dated Dec. 23, 2020, from corresponding U.S. Appl. No. 17/068,557. cited by applicant  
 Notice of Allowance, dated Dec. 3, 2019, from corresponding U.S. Appl. No. 16/563,749. cited by applicant  
 Notice of Allowance, dated Dec. 30, 2021, from corresponding U.S. Appl. No. 16/938,520. cited by applicant  
 Notice of Allowance, dated Dec. 31, 2018, from corresponding U.S. Appl. No. 16/159,634. cited by applicant  
 Notice of Allowance, dated Dec. 31, 2019, from corresponding U.S. Appl. No. 16/404,399. cited by applicant  
 Notice of Allowance, dated Dec. 4, 2019, from corresponding U.S. Appl. No. 16/594,670. cited by applicant  
 Notice of Allowance, dated Dec. 5, 2017, from corresponding U.S. Appl. No. 15/633,703. cited by applicant  
 Notice of Allowance, dated Dec. 6, 2017, from corresponding U.S. Appl. No. 15/619,451. cited by applicant  
 Notice of Allowance, dated Dec. 6, 2017, from corresponding U.S. Appl. No. 15/619,459. cited by applicant  
 Notice of Allowance, dated Dec. 7, 2020, from corresponding U.S. Appl. No. 16/817,136. cited by applicant  
 Notice of Allowance, dated Dec. 8, 2021, from corresponding U.S. Appl. No. 17/397,472. cited by applicant  
 Notice of Allowance, dated Dec. 9, 2019, from corresponding U.S. Appl. No. 16/565,261. cited by applicant  
 Notice of Allowance, dated Dec. 9, 2020, from corresponding U.S. Appl. No. 16/404,491. cited by applicant  
 Notice of Allowance, dated Feb. 1, 2022, from corresponding U.S. Appl. No. 17/346,509. cited by applicant  
 Notice of Allowance, dated Feb. 10, 2020, from corresponding U.S. Appl. No. 16/552,765. cited by applicant  
 Notice of Allowance, dated Feb. 11, 2021, from corresponding U.S. Appl. No. 17/086,732. cited by applicant  
 Notice of Allowance, dated Feb. 12, 2020, from corresponding U.S. Appl. No. 16/572,182. cited by applicant  
 Notice of Allowance, dated Feb. 13, 2019, from corresponding U.S. Appl. No. 16/041,563. cited by applicant  
 Notice of Allowance, dated Feb. 14, 2019, from corresponding U.S. Appl. No. 16/226,272. cited by applicant  
 Notice of Allowance, dated Feb. 14, 2022, from corresponding U.S. Appl. No. 16/623,157. cited by applicant  
 Notice of Allowance, dated Feb. 19, 2019, from corresponding U.S. Appl. No. 16/159,632. cited by applicant  
 Notice of Allowance, dated Feb. 19, 2021, from corresponding U.S. Appl. No. 16/832,451. cited by applicant  
 Notice of Allowance, dated Feb. 22, 2022, from corresponding U.S. Appl. No. 17/535,065. cited by applicant  
 Notice of Allowance, dated Feb. 24, 2021, from corresponding U.S. Appl. No. 17/034,355. cited by applicant  
 Notice of Allowance, dated Feb. 24, 2021, from corresponding U.S. Appl. No. 17/068,198. cited by applicant  
 Notice of Allowance, dated Feb. 24, 2021, from corresponding U.S. Appl. No. 17/101,106. cited by applicant  
 Notice of Allowance, dated Feb. 24, 2021, from corresponding U.S. Appl. No. 17/101,253. cited by applicant  
 Notice of Allowance, dated Feb. 24, 2022, from corresponding U.S. Appl. No. 17/234,205. cited by applicant  
 Notice of Allowance, dated Feb. 24, 2022, from corresponding U.S. Appl. No. 17/549,170. cited by applicant  
 Notice of Allowance, dated Feb. 25, 2020, from corresponding U.S. Appl. No. 16/714,355. cited by applicant  
 Notice of Allowance, dated Feb. 25, 2021, from corresponding U.S. Appl. No. 17/106,469. cited by applicant  
 Notice of Allowance, dated Feb. 26, 2021, from corresponding U.S. Appl. No. 17/139,650. cited by applicant  
 Notice of Allowance, dated Feb. 27, 2019, from corresponding U.S. Appl. No. 16/041,468. cited by applicant  
 Notice of Allowance, dated Feb. 27, 2019, from corresponding U.S. Appl. No. 16/226,290. cited by applicant  
 Notice of Allowance, dated Feb. 3, 2021, from corresponding U.S. Appl. No. 16/827,039. cited by applicant  
 Notice of Allowance, dated Feb. 3, 2021, from corresponding U.S. Appl. No. 17/068,558. cited by applicant  
 Notice of Allowance, dated Feb. 4, 2022, from corresponding U.S. Appl. No. 17/520,272. cited by applicant  
 Tuomas Aura et al., Scanning Electronic Documents for Personally Identifiable Information, ACM, Oct. 30, 2006, retrieved online on Jun. 13, 2019, pp. 41-49  
 the Internet: URL: <http://delivery.acm.org/10.1145/1180000/1179608/p41-aura.pdf?> (Year: 2006). cited by applicant  
 Van Eijk et al., "The Impact of User Location on Cookie Notices (Inside and Outside of the European Union)," IEEE Security & Privacy Workshop on Technology  
 Consumer Protection (CONPRO '19), Jan. 1, 2019 (Year: 2019). cited by applicant  
 Vukovic et al., "Managing Enterprise IT Systems Using Online Communities," Jul. 9, 2011, IEEE, pp. 552-559. (Year: 2011). cited by applicant  
 Wang et al., "Revealing Key Non-Financial Factors for Online Credit-Scoring in E-Financing," 2013, IEEE, pp. 1-6 (Year: 2013). cited by applicant  
 Wang et al., "Secure and Efficient Access to Outsourced Data," ACM, pp. 55-65 (Year: 2009). cited by applicant  
 Weaver et al., "Understanding Information Preview in Mobile Email Processing", ACM, pp. 303-312, 2011 (Year: 2011). cited by applicant  
 Wong et al., "Ontology Mapping for the Interoperability Problem in Network Management," IEEE, pp. 2058-2068 (Year: 2005). cited by applicant  
 Wu et al., "Data Mining with Big Data," IEEE, Jan. 2014, pp. 97-107, vol. 26, No. 1, (Year: 2014). cited by applicant  
 www.truste.com (1), 200150207, Internet Archive Wayback Machine, [www.archive.org/2\\_7\\_2015](http://www.archive.org/2_7_2015). cited by applicant  
 Xu, et al., "GatorShare: A File System Framework for High-Throughput Data Management," ACM, pp. 776-786 (Year: 2010). cited by applicant  
 Yang et al., "DAC-MACS: Effective Data Access Control for Multiauthority Cloud Storage Systems," IEEE, pp. 1790-1801 (Year: 2013). cited by applicant  
 Yang et al., "Mining Web Access Sequence with Improved Apriori Algorithm," IEEE, 2017, pp. 780-784 (Year: 2017). cited by applicant  
 Ye et al., "An Evolution-Based Cache Scheme for Scalable Mobile Data Access," ACM, pp. 1-7 (Year: 2007). cited by applicant  
 Yin et al., "Multibank Memory Optimization for Parallel Data Access in Multiple Data Arrays", ACM, pp. 1-8 (Year: 2016). cited by applicant  
 Yiu et al., "Outsourced Similarity Search on Metric Data Assets", IEEE, pp. 338-352 (Year: 2012). cited by applicant  
 Yu, "Using Data from Social Media Websites to Inspire the Design of Assistive Technology", ACM, pp. 1-2 (Year: 2016). cited by applicant  
 Yu, et al., "Performance and Fairness Issues in Big Data Transfers," ACM, pp. 9-11 (Year: 2014). cited by applicant  
 Yue et al., "An Automatic HTTP Cookie Management System," Computer Networks, Elsevier, Amsterdam, NL, vol. 54, No. 13, Sep. 15, 2010, pp. 2182-2198  
 cited by applicant  
 Zannone, et al., "Maintaining Privacy on Derived Objects," ACM, pp. 10-19 (Year: 2005). cited by applicant  
 Zeldovich, Nickolai, et al, Making Information Flow Explicit in HiStar, OSDI '06: 7th USENIX Symposium on Operating Systems Design and Implementation



Association, p. 263-278. cited by applicant

Zhang et al., "Data Transfer Performance Issues for a Web Services Interface to Synchrotron Experiments", ACM, pp. 59-65 (Year: 2007). cited by applicant

Zhang et al., "Dynamic Topic Modeling for Monitoring Market Competition from Online Text and Image Data", ACM, pp. 1425-1434 (Year: 2015). cited by applicant

Zheng, et al., "Methodologies for Cross-Domain Data Fusion: An Overview," IEEE, pp. 16-34 (Year: 2015). cited by applicant

Zheng, et al., "Toward Assured Data Deletion in Cloud Storage," IEEE, vol. 34, No. 3, pp. 101-107 May/Jun. 2020 (Year: 2020). cited by applicant

Zhu, et al., "Dynamic Data Integration Using Web Services," IEEE, pp. 1-8 (Year: 2004). cited by applicant

Niu, et al., "Achieving Data Truthfulness and Privacy Preservation in Data Markets", IEEE Transactions on Knowledge and Data Engineering, IEEE Service Center, Alamos, CA, US, vol. 31, No. 1, Jan. 1, 2019, pp. 105-119 (Year 2019). cited by applicant

Notice of Filing Date for Petition for Post-Grant Review of related U.S. Pat. No. 9,691,090 dated Apr. 12, 2018. cited by applicant

Nouwens et al., "Dark Patterns after the GDPR: Scraping Consent Pop-ups and Demonstrating their Influence," ACM, pp. 1-13, Apr. 25, 2020 (Year: 2020). cited by applicant

O'Keefe et al., "Privacy-Preserving Data Linkage Protocols," Proceedings of the 2004 ACM Workshop on Privacy in the Electronic Society, 2004, pp. 94-102 (Year: 2004). cited by applicant

Olenski, Steve, For Consumers, Data Is A Matter Of Trust, CMO Network, Apr. 18, 2016, <https://www.forbes.com/sites/steveolenski/2016/04/18/for-consumers-data-is-a-matter-of-trust/#2e48496278b3>. cited by applicant

Ozdikis et al., "Tool Support for Transformation from an OWL Ontology to an HLA Object Model," ACM, pp. 1-6 (Year: 2010). cited by applicant

Paes, "Student Research Abstract: Automatic Detection of Cross-Browser Incompatibilities using Machine Learning and Screenshot Similarity," ACM, pp. 69-78 (Year: 2017). cited by applicant

Pearson, et al., "A Model-Based Privacy Compliance Checker," IJEER, vol. 5, No. 2, pp. 63-83, 2009, Nov. 21, 2008. [Online]. Available: <http://dx.doi.org/10.4018/ijeer.2009040104> (Year: 2008). cited by applicant

Pechenizkiy et al., "Process Mining Online Assessment Data," Educational Data Mining, pp. 279-288 (Year: 2009). cited by applicant

Petition for Post-Grant Review of related U.S. Pat. No. 9,691,090 dated Mar. 27, 2018. cited by applicant

Petrie et al., "The Relationship between Accessibility and Usability of Websites", ACM, pp. 397-406 (Year: 2007). cited by applicant

Ping et al., "Wide Area Placement of Data Replicas for Fast and Highly Available Data Access," ACM, pp. 1-8 (Year: 2011). cited by applicant

Popescu-Zeletin, "The Data Access and Transfer Support in a Local Heterogeneous Network (HMINET)", IEEE, pp. 147-152 (Year: 1979). cited by applicant

Porter, "De-Identified Data and Third Party Data Mining: The Risk of Re-Identification of Personal Information," Shidler JL Com. & Tech. 5, 2008, pp. 1-9 (Year: 2008). cited by applicant

Pretorius, et al., "Attributing Users Based on Web Browser History," 2017 IEEE Conference on Application, Information and Network Security (AINS), 2017, pp. 1-6 (Year: 2017). cited by applicant

Preuveneers et al., "Access Control with Delegated Authorization Policy Evaluation for Data-Driven Microservice Workflows," Future Internet 2017, MDPI, p. 1-11 (Year: 2017). cited by applicant

Qiu, et al., "Design and Application of Data Integration Platform Based on Web Services and XML," IEEE, pp. 253-256 (Year: 2016). cited by applicant

Qu et al., "Metadata Type System: Integrate Presentation, Data Models and Extraction to Enable Exploratory Browsing Interfaces," ACM, pp. 107-116 (Year: 2017). cited by applicant

Radu, et al., "Analyzing Risk Evaluation Frameworks and Risk Assessment Methods," IEEE, Dec. 12, 2020, pp. 1-6 (Year: 2020). cited by applicant

Rakers, "Managing Professional and Personal Sensitive Information," ACM, pp. 9-13, Oct. 24-27, 2010 (Year: 2010). cited by applicant

Reardon et al., User-Level Secure Deletion on Log-Structured File Systems, ACM, 2012, retrieved online on Apr. 22, 2021, pp. 1-11. Retrieved from the Internet: <http://citeseerx.ist.psu.edu/viewdoc/download;sessionid=450713515DC7F19F8ED09AE961D4B60E>. (Year: 2012). cited by applicant

Regulation (EU) 2016/679, "On the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation)," Official Journal of the European Union, May 4, 2016, pp. L 119/1-L 119/88 (Year: 2016). cited by applicant

Roesner et al., "Detecting and Defending Against Third-Party Tracking on the Web," 9th USENIX Symposium on Networked Systems Design and Implementation, 2013, pp. 1-14, ACM (Year: 2013). cited by applicant

Rozepz, "What is Google Privacy Checkup? Everything You Need to Know," Tom's Guide web post, Apr. 26, 2018, pp. 1-11 (Year: 2018). cited by applicant

Sachinopoulou et al., "Ontology-Based Approach for Managing Personal Health and Wellness Information," IEEE, pp. 1802-1805 (Year: 2007). cited by applicant

Salim et al., "Data Retrieval and Security using Lightweight Directory Access Protocol", IEEE, pp. 685-688 (Year: 2009). cited by applicant

Sanchez-Rola et al., "Can I Opt Out Yet?: GDPR and the Global Illusion of Cookie Control," Proceedings of the 2019 ACM Asia Conference on Computer and Communications Security, 2019, pp. 340-351 (Year: 2019). cited by applicant

Santhisree, et al., "Web Usage Data Clustering Using DbSCAN Algorithm and Set Similarities," IEEE, pp. 220-224 (Year: 2010). cited by applicant

Sanzo et al., "Analytical Modeling of Lock-Based Concurrency Control with Arbitrary Transaction Data Access Patterns," ACM, pp. 69-78 (Year: 2010). cited by applicant

Sarkar et al., "Towards Enforcement of the EU GDPR: Enabling Data Erasure," 2018 IEEE Confs on Internet of Things, Green Computing and Communications, 2018, pp. 1-6 (Year: 2018). cited by applicant

Physical and Social Computing, Smart Data, Blockchain, Computer and Information Technology, Congress on Cybermatics, 2018, pp. 222-229, IEEE (Year: 2018). cited by applicant

Schwartz, Edward J., et al, 2010 IEEE Symposium on Security and Privacy: All You Ever Wanted to Know About Dynamic Analysis and forward Symbolic Execution might have been afraid to ask), Carnegie Mellon University, IEEE Computer Society, 2010, p. 317-331. cited by applicant

Sedinic et al., "Security Risk Management in Complex Organization," May 29, 2015, IEEE, pp. 1331-1337 (Year: 2015). cited by applicant

Shahriar et al., "A Model-Based Detection of Vulnerable and Malicious Browser Extensions," IEEE, pp. 198-207 (Year: 2013). cited by applicant

Shankar et al., "Doppelganger: Better Browser Privacy Without the Bother," Proceedings of the 13th ACM Conference on Computer and Communications Security [Conference on Computer and Communications Security], New York, NY : ACM, US, Oct. 30, 2006, pp. 154-167 (Year: 2006). cited by applicant

Shulz et al., "Generative Data Models for Validation and Evaluation of Visualization Techniques," ACM, pp. 1-13 (Year: 2016). cited by applicant

Singh, et al., "A Metadata Catalog Service for Data Intensive Applications," ACM, pp. 1-17 (Year: 2003). cited by applicant

Sjosten et al., "Discovering Browser Extensions via Web Accessible Resources," ACM, pp. 329-336, Mar. 22, 2017 (Year: 2017). cited by applicant

Slezak, et al., "Brighthouse: An Analytic Data Warehouse for Ad-hoc Queries," ACM, pp. 1337-1345 (Year: 2008). cited by applicant

Socanu, et al., "Managing the Privacy and Security of eHealth Data," May 29, 2015, IEEE, pp. 1-8 (Year: 2015). cited by applicant

Srinivasan et al., "Descriptive Data Analysis of File Transfer Data," ACM, pp. 1-8 (Year: 2014). cited by applicant

Stack Overflow, "Is there a way to force a user to scroll to the bottom of a div?," Stack Overflow, pp. 1-11, Nov. 2013. [Online]. Available: <https://stackoverflow.com/questions/2745935/is-there-a-way-to-force-a-user-to-scroll-to-the-bottom-of-a-div> (Year: 2013). cited by applicant

Stern, Joanna, "iPhone Privacy Is Broken . . . and Apps Are to Blame", The Wall Street Journal, wsj.com, May 31, 2019. cited by applicant

Strodl, et al., "Personal & SOHO Archiving," Vienna University of Technology, Vienna, Austria, JCDL '08, Jun. 16-20, 2008, Pittsburgh, Pennsylvania, USA, 2008 (Year: 2008). cited by applicant

Sukumar et al., "Review on Modern Data Preprocessing Techniques in Web Usage Mining (WUM)," IEEE, 2016, pp. 64-69 (Year: 2016). cited by applicant

Symantec, Symantec Data Loss Prevention—Discover, monitor, and protect confidential data; 2008; Symantec Corporation; [http://www.mssuk.com/images/Symantec%2014552315\\_IRC\\_BR\\_DLP\\_03.09\\_sngl.pdf](http://www.mssuk.com/images/Symantec%2014552315_IRC_BR_DLP_03.09_sngl.pdf). cited by applicant

Tanasa et al., "Advanced Data Preprocessing for Intersites Web Usage Mining," IEEE, Mar. 2004, pp. 59-65 (Year: 2004). cited by applicant

The Cookie Collective, Optanon Cookie Policy Generator, The Cookie Collective, Year 2016, <http://web.archive.org/web/20160324062743/https://optanon.com/>. cited by applicant

Thomas et al., "MoOM—A Prototype Framework for Management of Ontology Mappings," IEEE, pp. 548-555 (Year: 2011). cited by applicant

TRUSTe Announces General Availability of Assessment Manager for Enterprises to Streamline Data Privacy Management with Automation, PRNewswire, TRUSTe, by applicant

Tsai et al, "Determinants of Intangible Assets Value: The Data Mining Approach," Knowledge Based System, pp. 67-77 <http://www.elsevier.com/locate/ksys> cited by applicant

Office Action, dated Oct. 10, 2018, from corresponding U.S. Appl. No. 16/055,944. cited by applicant

Office Action, dated Oct. 12, 2021, from corresponding U.S. Appl. No. 17/346,509. cited by applicant

Office Action, dated Oct. 14, 2020, from corresponding U.S. Appl. No. 16/927,658. cited by applicant

Office Action, dated Oct. 15, 2018, from corresponding U.S. Appl. No. 16/054,780. cited by applicant

Office Action, dated Oct. 15, 2021, from corresponding U.S. Appl. No. 16/908,081. cited by applicant

Office Action, dated Oct. 16, 2019, from corresponding U.S. Appl. No. 16/557,392. cited by applicant

Office Action, dated Oct. 16, 2020, from corresponding U.S. Appl. No. 16/808,489. cited by applicant

Office Action, dated Oct. 23, 2018, from corresponding U.S. Appl. No. 16/055,961. cited by applicant

Office Action, dated Oct. 25, 2022, from corresponding U.S. Appl. No. 17/836,865. cited by applicant

Office Action, dated Oct. 26, 2018, from corresponding U.S. Appl. No. 16/041,468. cited by applicant

Office Action, dated Oct. 27, 2022, from corresponding U.S. Appl. No. 17/161,159. cited by applicant

Office Action, dated Oct. 8, 2019, from corresponding U.S. Appl. No. 16/552,765. cited by applicant

Office Action, dated Sep. 1, 2017, from corresponding U.S. Appl. No. 15/619,459. cited by applicant

Office Action, dated Sep. 11, 2017, from corresponding U.S. Appl. No. 15/619,375. cited by applicant

Office Action, dated Sep. 11, 2017, from corresponding U.S. Appl. No. 15/619,478. cited by applicant

Office Action, dated Sep. 15, 2021, from corresponding U.S. Appl. No. 16/623,157. cited by applicant

Office Action, dated Sep. 16, 2019, from corresponding U.S. Appl. No. 16/277,715. cited by applicant

Office Action, dated Sep. 16, 2022, from corresponding U.S. Appl. No. 17/306,438. cited by applicant

Office Action, dated Sep. 19, 2017, from corresponding U.S. Appl. No. 15/671,073. cited by applicant

Office Action, dated Sep. 2, 2022, from corresponding U.S. Appl. No. 17/499,624. cited by applicant

Office Action, dated Sep. 22, 2017, from corresponding U.S. Appl. No. 15/619,278. cited by applicant

Office Action, dated Sep. 24, 2021, from corresponding U.S. Appl. No. 17/342,153. cited by applicant

Office Action, dated Sep. 4, 2020, from corresponding U.S. Appl. No. 16/989,086. cited by applicant

Office Action, dated Sep. 5, 2017, from corresponding U.S. Appl. No. 15/619,469. cited by applicant

Office Action, dated Sep. 6, 2017, from corresponding U.S. Appl. No. 15/619,479. cited by applicant

Office Action, dated Sep. 7, 2017, from corresponding U.S. Appl. No. 15/633,703. cited by applicant

Office Action, dated Sep. 8, 2017, from corresponding U.S. Appl. No. 15/619,251. cited by applicant

Office Action, dated Sep. 8, 2022, from corresponding U.S. Appl. No. 17/850,244. cited by applicant

Restriction Requirement, dated Apr. 10, 2019, from corresponding U.S. Appl. No. 16/277,715. cited by applicant

Restriction Requirement, dated Apr. 12, 2022, from corresponding U.S. Appl. No. 17/584,187. cited by applicant

Restriction Requirement, dated Apr. 13, 2020, from corresponding U.S. Appl. No. 16/817,136. cited by applicant

Restriction Requirement, dated Apr. 24, 2019, from corresponding U.S. Appl. No. 16/278,122. cited by applicant

Restriction Requirement, dated Aug. 7, 2019, from corresponding U.S. Appl. No. 16/410,866. cited by applicant

Restriction Requirement, dated Aug. 9, 2019, from corresponding U.S. Appl. No. 16/404,399. cited by applicant

Restriction Requirement, dated Dec. 17, 2021, from corresponding U.S. Appl. No. 17/475,244. cited by applicant

Restriction Requirement, dated Dec. 31, 2018, from corresponding U.S. Appl. No. 15/169,668. cited by applicant

Restriction Requirement, dated Dec. 9, 2019, from corresponding U.S. Appl. No. 16/565,395. cited by applicant

Restriction Requirement, dated Jan. 18, 2017, from corresponding U.S. Appl. No. 15/256,430. cited by applicant

Restriction Requirement, dated Jul. 28, 2017, from corresponding U.S. Appl. No. 15/169,658. cited by applicant

Restriction Requirement, dated Jun. 15, 2021, from corresponding U.S. Appl. No. 17/187,329. cited by applicant

Restriction Requirement, dated Jun. 15, 2021, from corresponding U.S. Appl. No. 17/222,556. cited by applicant

Restriction Requirement, dated Jun. 9, 2021, from corresponding U.S. Appl. No. 17/222,725. cited by applicant

Restriction Requirement, dated May 5, 2020, from corresponding U.S. Appl. No. 16/808,489. cited by applicant

Restriction Requirement, dated Nov. 10, 2021, from corresponding U.S. Appl. No. 17/366,754. cited by applicant

Restriction Requirement, dated Nov. 14, 2022, from corresponding U.S. Appl. No. 17/836,872. cited by applicant

Restriction Requirement, dated Nov. 15, 2019, from corresponding U.S. Appl. No. 16/586,202. cited by applicant

Restriction Requirement, dated Nov. 21, 2016, from corresponding U.S. Appl. No. 15/254,901. cited by applicant

Restriction Requirement, dated Nov. 5, 2019, from corresponding U.S. Appl. No. 16/563,744. cited by applicant

Restriction Requirement, dated Oct. 17, 2018, from corresponding U.S. Appl. No. 16/055,984. cited by applicant

Restriction Requirement, dated Oct. 6, 2021, from corresponding U.S. Appl. No. 17/340,699. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 11, 2018, from corresponding International Application No. PCT/US2018/043976. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 11, 2018, from corresponding International Application No. PCT/US2018/043977. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 11, 2018, from corresponding International Application No. PCT/US2018/044026. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 11, 2018, from corresponding International Application No. PCT/US2018/045240. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 12, 2017, from corresponding International Application No. PCT/US2017/036888. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 12, 2018, from corresponding International Application No. PCT/US2018/044046. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 16, 2018, from corresponding International Application No. PCT/US2018/045243. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 18, 2018, from corresponding International Application No. PCT/US2018/045249. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 20, 2017, from corresponding International Application No. PCT/US2017/036917. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 3, 2017, from corresponding International Application No. PCT/US2017/036912. cited by applicant

Written Opinion of the International Searching Authority, dated Sep. 1, 2017, from corresponding International Application No. PCT/US2017/036896. cited by applicant

Written Opinion of the International Searching Authority, dated Sep. 12, 2018, from corresponding International Application No. PCT/US2018/037504. cited by applicant

Written Opinion of the International Searching Authority, dated Sep. 15, 2021, from corresponding International Application No. PCT/US2021/033631. cited by applicant

Abdullah et al, "The Mapping Process of Unstructured Data to the Structured Data", ACM, pp. 151-155 (Year: 2013). cited by applicant

Acar, Gunes, et al, The Web Never Forgets, Computer and Communications Security, ACM, Nov. 3, 2014, pp. 674-689. cited by applicant

Aghasian, Erfan, et al, Scoring Users' Privacy Disclosure Across Multiple Online Social Networks, IEEE Access, Multidisciplinary Rapid Review Open Access Journal, 31, 2017, vol. 5, 2017. cited by applicant

Agosti et al, "Access and Exchange of Hierarchically Structured Resources on the Web with the NESTOR Framework", IEEE, pp. 659-662 (Year: 2009). cited by applicant

Ahmad et al, "Task-Oriented Access Model for Secure Data Sharing Over Cloud," ACM, pp. 1-7 (Year: 2015). cited by applicant

Ahmad, et al, "Performance of Resource Management Algorithms for Processable Bulk Data Transfer Tasks in Grid Environments," ACM, pp. 177-188 (Year: 2015). cited by applicant

Alaa et al, "Personalized Risk Scoring for Critical Care Prognosis Using Mixtures of Gaussian Processes," Apr. 27, 2017, IEEE, vol. 65, issue 1, pp. 207-217 (Year: 2017). cited by applicant

Ali et al, "Age Estimation from Facial Images Using Biometric Ratios and Wrinkle Analysis," IEEE, 2015, pp. 1-5 (Year: 2015). cited by applicant

Alkalha et al, "Investigating the Effects of Human Resource Policies on Organizational Performance: An Empirical Study on Commercial Banks Operating in the European Journal of Economics, Finance and Administrative Science, pp. 1-22 (Year: 2012). cited by applicant

Aman et al, "Detecting Data Tampering Attacks in Synchronphasor Networks using Time Hopping," IEEE, pp. 1-6 (Year: 2016). cited by applicant

Amar et al, "Privacy-Aware Infrastructure for Managing Personal Data," ACM, pp. 571-572, Aug. 22-26, 2016 (Year: 2016). cited by applicant

Antunes et al, "Preserving Digital Data in Heterogeneous Environments", ACM, pp. 345-348, 2009 (Year: 2009). cited by applicant

Ardagna, et al, "A Privacy-Aware Access Control System," Journal of Computer Security, 16:4, pp. 369-397 (Year: 2008). cited by applicant

Avepoint, Automating Privacy Impact Assessments, AvePoint, Inc. cited by applicant

Avepoint, AvePoint Privacy Impact Assessment 1: User Guide, Cumulative Update 2, Revision E, Feb. 2015, AvePoint, Inc. cited by applicant

Ball, et al, "Aspects of the Computer-Based Patient Record," Computers in Healthcare, Springer-Verlag New York Inc., pp. 1-23 (Year: 1992). cited by applicant

Banerjee et al, "Link Before You Share: Managing Privacy Policies through Blockchain," IEEE, pp. 4438-4447 (Year: 2017). cited by applicant

Bansal et al, "Integrating Big Data: A Semantic Extract-Transform-Load Framework," IEEE, pp. 42-50 (Year: 2015). cited by applicant

Bao et al, "Performance Modeling and Workflow Scheduling of Microservice-Based Applications in Clouds," IEEE Transactions on Parallel and Distributed Systems, No. 9, Sep. 2019, pp. 2101-2116 (Year: 2019). cited by applicant

Barker, "Personalizing Access Control by Generalizing Access Control," ACM, pp. 149-158 (Year: 2010). cited by applicant

Barr, "Amazon Rekognition Update—Estimated Age Range for Faces," AWS News Blog, Feb. 10, 2017, pp. 1-5 (Year: 2017). cited by applicant

Bayardo et al, "Technological Solutions for Protecting Privacy," Computer 36.9 (2003), pp. 115-118, (Year: 2003). cited by applicant

Berezovskiy et al, "A framework for dynamic data source identification and orchestration on the Web", ACM, pp. 1-8 Year: 2010). cited by applicant

Bertino et al, "On Specifying Security Policies for Web Documents with an XML-based Language," ACM, pp. 57-65 (Year: 2001). cited by applicant

Bertino et al, "Towards Mechanisms for Detection and Prevention of Data Exfiltration by Insiders," Mar. 22, 2011, ACM, pp. 10-19 (Year: 2011). cited by applicant

Bhargav-Spantzel et al., Receipt Management—Transaction History based Trust Establishment, 2007, ACM, p. 82-91. cited by applicant

Bhuvaneswaran et al, "Redundant Parallel Data Transfer Schemes for the Grid Environment", ACM, pp. 18 (Year: 2006). cited by applicant

Bieker, et al, "Privacy-Preserving Authentication Solutions—Best Practices for Implementation and EU Regulatory Perspectives," Oct. 29, 2014, IEEE, pp. 1-5 (Year: 2014). cited by applicant

Bin, et al, "Research on Data Mining Models for the Internet of Things," IEEE, pp. 1-6 (Year: 2010). cited by applicant

Bindschaedler et al, "Privacy Through Fake Yet Semantically Real Traces," arxiv.org, Cornell University Library, 201 Olin Library Cornell University Ithaca, NY, 27, 2015 (Year: 2015). cited by applicant

Bjorn Greif, "Cookie Pop-up Blocker: Cliqz Automatically Denies Consent Requests," Cliqz.com, pp. 1-9, Aug. 11, 2019 (Year: 2019). cited by applicant

Borgida, "Description Logics in Data Management," IEEE Transactions on Knowledge and Data Engineering, vol. 7, No. 5, Oct. 1995, pp. 671-682 (Year: 1995). cited by applicant

Brandt et al, "Efficient Metadata Management in Large Distributed Storage Systems," IEEE, pp. 1-9 (Year: 2003). cited by applicant

Bujlow et al, "Web Tracking: Mechanisms, Implications, and Defenses," Proceedings of the IEEE, Aug. 1, 2017, vol. 5, No. 8, pp. 1476-1510 (Year: 2017). cited by applicant

Byun, Ji-Won, Elisa Bertino, and Ninghui Li. "Purpose based access control of complex data for privacy protection." Proceedings of the tenth ACM symposium on access control models and technologies. ACM, 2005. (Year: 2005). cited by applicant

Carminati et al, "Enforcing Access Control Over Data Streams," ACM, pp. 21-30 (Year: 2007). cited by applicant

Carpineto et al, "Automatic Assessment of Website Compliance to the European Cookie Law with CoolCheck," Proceedings of the 2016 ACM on Workshop on Privacy in the Electronic Society, 2016, pp. 135-138 (Year: 2016). cited by applicant

Notice of Allowance, dated Jun. 16, 2022, from corresponding U.S. Appl. No. 17/119,080. cited by applicant

Notice of Allowance, dated Jun. 17, 2020, from corresponding U.S. Appl. No. 16/656,895. cited by applicant

Notice of Allowance, dated Jun. 18, 2019, from corresponding U.S. Appl. No. 16/410,566. cited by applicant

Notice of Allowance, dated Jun. 19, 2018, from corresponding U.S. Appl. No. 15/894,890. cited by applicant

Notice of Allowance, dated Jun. 19, 2019, from corresponding U.S. Appl. No. 16/042,673. cited by applicant

Notice of Allowance, dated Jun. 19, 2019, from corresponding U.S. Appl. No. 16/055,984. cited by applicant

Notice of Allowance, dated Jun. 2, 2021, from corresponding U.S. Appl. No. 17/198,581. cited by applicant

Notice of Allowance, dated Jun. 2, 2022, from corresponding U.S. Appl. No. 17/493,290. cited by applicant

Notice of Allowance, dated Jun. 21, 2019, from corresponding U.S. Appl. No. 16/404,439. cited by applicant

Notice of Allowance, dated Jun. 22, 2020, from corresponding U.S. Appl. No. 16/791,337. cited by applicant

Notice of Allowance, dated Jun. 23, 2022, from corresponding U.S. Appl. No. 17/588,645. cited by applicant

Notice of Allowance, dated Jun. 27, 2018, from corresponding U.S. Appl. No. 15/882,989. cited by applicant

Notice of Allowance, dated Jun. 29, 2022, from corresponding U.S. Appl. No. 17/675,118. cited by applicant

Notice of Allowance, dated Jun. 4, 2019, from corresponding U.S. Appl. No. 16/159,566. cited by applicant

Notice of Allowance, dated Jun. 5, 2019, from corresponding U.S. Appl. No. 16/220,899. cited by applicant

Notice of Allowance, dated Jun. 5, 2019, from corresponding U.S. Appl. No. 16/357,260. cited by applicant

Notice of Allowance, dated Jun. 6, 2018, from corresponding U.S. Appl. No. 15/875,570. cited by applicant

Notice of Allowance, dated Jun. 6, 2019, from corresponding U.S. Appl. No. 16/159,628. cited by applicant

Notice of Allowance, dated Jun. 7, 2021, from corresponding U.S. Appl. No. 17/099,270. cited by applicant

Notice of Allowance, dated Jun. 8, 2020, from corresponding U.S. Appl. No. 16/712,104. cited by applicant

Notice of Allowance, dated Jun. 8, 2022, from corresponding U.S. Appl. No. 17/722,551. cited by applicant

Notice of Allowance, dated Mar. 1, 2018, from corresponding U.S. Appl. No. 15/853,674. cited by applicant

Notice of Allowance, dated Mar. 1, 2019, from corresponding U.S. Appl. No. 16/059,911. cited by applicant

Notice of Allowance, dated Mar. 10, 2021, from corresponding U.S. Appl. No. 16/925,628. cited by applicant

Notice of Allowance, dated Mar. 10, 2021, from corresponding U.S. Appl. No. 17/128,666. cited by applicant

Notice of Allowance, dated Mar. 13, 2019, from corresponding U.S. Appl. No. 16/055,083. cited by applicant

Notice of Allowance, dated Mar. 14, 2019, from corresponding U.S. Appl. No. 16/055,944. cited by applicant

Notice of Allowance, dated Mar. 16, 2020, from corresponding U.S. Appl. No. 16/778,704. cited by applicant

Notice of Allowance, dated Mar. 16, 2021, from corresponding U.S. Appl. No. 17/149,380. cited by applicant

Notice of Allowance, dated Mar. 16, 2022, from corresponding U.S. Appl. No. 17/486,350. cited by applicant

Notice of Allowance, dated Mar. 17, 2020, from corresponding U.S. Appl. No. 16/560,885. cited by applicant

Notice of Allowance, dated Mar. 18, 2020, from corresponding U.S. Appl. No. 16/560,963. cited by applicant

Notice of Allowance, dated Mar. 19, 2021, from corresponding U.S. Appl. No. 17/013,757. cited by applicant

Notice of Allowance, dated Mar. 2, 2018, from corresponding U.S. Appl. No. 15/858,802. cited by applicant

Notice of Allowance, dated Mar. 2, 2022, from corresponding U.S. Appl. No. 16/872,130. cited by applicant

Notice of Allowance, dated Mar. 2, 2022, from corresponding U.S. Appl. No. 17/535,098. cited by applicant

Notice of Allowance, dated Mar. 21, 2022, from corresponding U.S. Appl. No. 17/366,754. cited by applicant

Notice of Allowance, dated Mar. 22, 2022, from corresponding U.S. Appl. No. 17/475,244. cited by applicant

Notice of Allowance, dated Mar. 22, 2022, from corresponding U.S. Appl. No. 17/504,102. cited by applicant

Notice of Allowance, dated Mar. 24, 2020, from corresponding U.S. Appl. No. 16/552,758. cited by applicant

Notice of Allowance, dated Mar. 25, 2019, from corresponding U.S. Appl. No. 16/054,780. cited by applicant

Notice of Allowance, dated Mar. 26, 2020, from corresponding U.S. Appl. No. 16/560,889. cited by applicant

Notice of Allowance, dated Mar. 26, 2020, from corresponding U.S. Appl. No. 16/578,712. cited by applicant

Notice of Allowance, dated Mar. 27, 2019, from corresponding U.S. Appl. No. 16/226,280. cited by applicant

Notice of Allowance, dated Mar. 28, 2022, from corresponding U.S. Appl. No. 17/499,609. cited by applicant

Notice of Allowance, dated Mar. 29, 2019, from corresponding U.S. Appl. No. 16/055,998. cited by applicant

Notice of Allowance, dated Mar. 31, 2020, from corresponding U.S. Appl. No. 16/563,744. cited by applicant

Notice of Allowance, dated Mar. 31, 2021, from corresponding U.S. Appl. No. 17/013,758. cited by applicant

Notice of Allowance, dated Mar. 31, 2021, from corresponding U.S. Appl. No. 17/162,205. cited by applicant

Notice of Allowance, dated Mar. 31, 2022, from corresponding U.S. Appl. No. 17/476,209. cited by applicant

Hodge, et al, "Managing Virtual Data Marts with Metapointer Tables," pp. 1-7 (Year: 2002). cited by applicant

Hjorral et al, "Evaluating Risk: IBM's Country Financial Risk and Treasury Risk Scorecards," Jul. 21, 2014, IBM, vol. 58, issue 4, pp. 2:1-2:9 (Year: 2014). cited by applicant

Hu, et al, "Attribute Considerations for Access Control Systems," NIST Special Publication 800-205, Jun. 2019, pp. 1-42 (Year: 2019). cited by applicant

Hu, et al, "Guide to Attribute Based Access Control (ABAC) Definition and Considerations (Draft)," NIST Special Publication 800-162, pp. 1-54 (Year: 2013). cited by applicant

Huang, et al, "A Study on Information Security Management with Personal Data Protection," IEEE, Dec. 9, 2011, pp. 624-630 (Year: 2011). cited by applicant

Huettner, "Digital Risk Management: Protecting Your Privacy, Improving Security, and Preparing for Emergencies," IEEE, pp. 136-138 (Year: 2006). cited by applicant

Huner et al, "Towards a Maturity Model for Corporate Data Quality Management", ACM, pp. 231-238, 2009 (Year: 2009). cited by applicant

Hunton & Williams LLP, The Role of Risk Management in Data Protection, Privacy Risk Framework and the Risk-based Approach to Privacy, Centre for Information Security, Workshop II, Nov. 23, 2014. cited by applicant

Huo et al, "A Cloud Storage Architecture Model for Data-Intensive Applications," IEEE, pp. 1-4 (Year: 2011). cited by applicant

Imran et al, "Searching in Cloud Object Storage by Using a Metadata Model", IEEE, 2014, retrieved online on Apr. 1, 2020, pp. 121-128. Retrieved from the Internet: <https://ieeexplore.ieee.org/stamp/stamp.jsp?> (Year: 2014). cited by applicant

Iordanou et al, "Tracing Cross Border Web Tracking," Oct. 31, 2018, pp. 329-342, ACM (Year: 2018). cited by applicant

Islam, et al, "Mixture Model Based Label Association Techniques for Web Accessibility," ACM, pp. 67-76 (Year: 2010). cited by applicant

Jayasinghe et al, "Matching Facial Images Using Age Related Morphing Changes," ISSRI, 2009, pp. 2901-2907 (Year: 2009). cited by applicant

Jensen, et al, "Temporal Data Management," IEEE Transactions on Knowledge and Data Engineering, vol. 11, No. 1, Jan./Feb. 1999, pp. 36-44 (Year: 1999). cited by applicant

Jiahao Chen et al. "Fairness Under Unawareness: Assessing Disparity when Protected Class is Unobserved," arxiv.org, Cornell University Library, 201 Olin Library, Ithaca, NY 14853, Nov. 27, 2018 (Nov. 27, 2018), Section 2, Figure 2. (Year 2018). cited by applicant

Joel Reardon et al., Secure Data Deletion from Persistent Media, ACM, Nov. 4, 2013, retrieved online on Jun. 13, 2019, pp. 271-283. Retrieved from the Internet: <http://delivery.acm.org/10.1145/2520000/2516699/p271-reardon.pdf?> (Year: 2013). cited by applicant

Jones et al, "AI and the Ethics of Automating Consent," IEEE, pp. 64-72, May 2018 (Year: 2018). cited by applicant

Joonbakhsh et al, "Mining and Extraction of Personal Software Process measures through IDE Interaction logs," ACM/IEEE, 2018, retrieved online on Dec. 20, 2018. Retrieved from the Internet: URL: <http://delivery.acm.org/10.1145/3200000/3196462/p78-joonbakhsh.pdf?> (Year: 2018). cited by applicant

Jun et al, "Scalable Multi-Access Flash Store for Big Data Analytics," ACM, pp. 55-64 (Year: 2014). cited by applicant

Khan et al, "Wrinkles Energy Based Age Estimation Using Discrete Cosine Transform," IEEE, 2015, pp. 1-4 (Year: 2015). cited by applicant

Kirkham, et al, "A Personal Data Store for an Internet of Subjects," IEEE, pp. 92-97 (Year: 2011). cited by applicant

Korba, Larry et al.; "Private Data Discovery for Privacy Compliance in Collaborative Environments"; Cooperative Design, Visualization, and Engineering; Springer, Heidelberg; Sep. 21, 2008; pp. 142-150. cited by applicant

Kristian et al, "Human Facial Age Classification Using Active Shape Module, Geometrical Feature, and Support Vector Machine on Early Growth Stage," ISBI, 2015, pp. 81-88 (Year: 2015). cited by applicant

Krol, Kat, et al, Control versus Effort in Privacy Warnings for Webforms, ACM, Oct. 24, 2016, pp. 13-23. cited by applicant

Lamb et al, "Role-Based Access Control for Data Service Integration", ACM, pp. 3-11 (Year: 2006). cited by applicant

Lasierra et al, "Data Management in Home Scenarios Using an Autonomic Ontology-Based Approach," IEEE, pp. 94-99 (Year: 2012). cited by applicant

Leadbetter, et al, "Where Big Data Meets Linked Data: Applying Standard Data Models to Environmental Data Streams," IEEE, pp. 2929-2937 (Year: 2016). cited by applicant

Lenzerini et al, "Ontology-based Data Management," ACM, pp. 5-6 (Year: 2011). cited by applicant

Lewis, James et al, "Microservices," Mar. 25, 2014 (Mar. 25, 2014), XP055907494, Retrieved from the Internet: <https://martinfowler.com/articles/microservices/> [retrieved on Mar. 31, 2022]. cited by applicant

Li, Ninghui, et al, t-Closeness: Privacy Beyond k-Anonymity and I-Diversity, IEEE, 2014, p. 106-115. cited by applicant

Liu et al, "A Novel Approach for Detecting Browser-based Silent Miner," IEEE, pp. 490-497 (Year: 2018). cited by applicant

Liu et al, "Cross-Geography Scientific Data Transferring Trends and Behavior," ACM, pp. 267-278 (Year: 2018). cited by applicant

Liu et al, "Overview on Ontology Mapping and Approach," IEEE, pp. 592-595 (Year: 2011). cited by applicant

Liu, Kun, et al, A Framework for Computing the Privacy Scores of Users in Online Social Networks, ACM Transactions on Knowledge Discovery from Data, vol. 14, No. 6, Dec. 2010, 30 pages. cited by applicant

Lu et al, "An HTTP Flooding Detection Method Based on Browser Behavior," IEEE, pp. 1151-1154 (Year: 2006). cited by applicant

Lu, "How Machine Learning Mitigates Racial Bias in the US Housing Market," Available as SSRN 3489519, pp. 1-73, Nov. 2019 (Year: 2019). cited by applicant

Luu, et al, "Combined Local and Holistic Facial Features for Age-Determination," 2010 11th Int. Conf. Control, Automation, Robotics and Vision, Singapore, 2010, pp. 900-904 (Year: 2010). cited by applicant

Maret et al, "Multimedia Information Interchange: Web Forms Meet Data Servers", IEEE, pp. 499-505 (Year: 1999). cited by applicant

Martin, et al, "Hidden Surveillance by Web Sites: Web Bugs in Contemporary Use," Communications of the ACM, vol. 46, No. 12, Dec. 2003, pp. 258-264. Retrieved from the Internet: <https://doi.org/10.1145/953460.953509>. (Year: 2003). cited by applicant

Matte et al, "Do Cookie Banners Respect my Choice?: Measuring Legal Compliance of Banners from IAB Europe's Transparency and Consent Framework," Proceedings of the Symposium on Security and Privacy (SP), 2020, pp. 791-809 (Year: 2020). cited by applicant

McGarth et al, "Digital Library Technology for Locating and Accessing Scientific Data", ACM, pp. 188-194 (Year: 1999). cited by applicant

Mesbah et al, "Crawling Ajax-Based Web Applications Through Dynamic Analysis of User Interface State Changes," ACM Transactions on the Web (TWEB), vol. 6, No. 3, Mar. 2012, pp. 1-30 (Year: 2012). cited by applicant

Milic et al, "Comparative Analysis of Metadata Models on e-Government Open Data Platforms," IEEE, pp. 119-130 (Year: 2021). cited by applicant

Moscato-Zea et al, "Datawarehouse Design for Educational Data Mining," IEEE, pp. 1-6 (Year: 2016). cited by applicant

Mudepalli et al, "An efficient data retrieval approach using blowfish encryption on cloud CipherText Retrieval in Cloud Computing" IEEE, pp. 267-271 (Year: 2018). cited by applicant

applicant  
Mundada et al, “Half-Baked Cookies: Hardening Cookie-Based Authentication for the Modern Web,” Proceedings of the 11th ACM on Asia Conference on Communications Security, 2016, pp. 675-685 (Year: 2016). cited by applicant  
Nemec et al, “Assessment of Query Execution Performance Using Selected Business Intelligence Tools and Experimental Agile Oriented Data Modeling Approach,” 2015, IEEE, pp. 1327-1333. (Year: 2015). cited by applicant  
Newman et al, “High Speed Scientific Data Transfers using Software Defined Networking,” ACM, pp. 1-9 (Year: 2015). cited by applicant  
Newman, “Email Archive Overviews using Subject Indexes”, ACM, pp. 652-653, 2002 (Year: 2002). cited by applicant  
Nishikawa, Taiji, English Translation of JP 2019154505, Aug. 27, 2019 (Year: 2019). cited by applicant  
Office Action, dated Mar. 25, 2020, from corresponding U.S. Appl. No. 16/791,006. cited by applicant  
Office Action, dated Mar. 27, 2019, from corresponding U.S. Appl. No. 16/278,120. cited by applicant  
Office Action, dated Mar. 30, 2018, from corresponding U.S. Appl. No. 15/894,890. cited by applicant  
Office Action, dated Mar. 30, 2018, from corresponding U.S. Appl. No. 15/896,790. cited by applicant  
Office Action, dated Mar. 30, 2021, from corresponding U.S. Appl. No. 17/151,399. cited by applicant  
Office Action, dated Mar. 4, 2019, from corresponding U.S. Appl. No. 16/237,083. cited by applicant  
Office Action, dated May 12, 2022, from corresponding U.S. Appl. No. 17/509,974. cited by applicant  
Office Action, dated May 14, 2020, from corresponding U.S. Appl. No. 16/808,497. cited by applicant  
Office Action, dated May 14, 2020, from corresponding U.S. Appl. No. 16/808,503. cited by applicant  
Office Action, dated May 15, 2020, from corresponding U.S. Appl. No. 16/808,493. cited by applicant  
Office Action, dated May 16, 2018, from corresponding U.S. Appl. No. 15/882,989. cited by applicant  
Office Action, dated May 16, 2022, from corresponding U.S. Appl. No. 17/679,750. cited by applicant  
Office Action, dated May 17, 2019, from corresponding U.S. Appl. No. 16/277,539. cited by applicant  
Office Action, dated May 18, 2021, from corresponding U.S. Appl. No. 17/196,570. cited by applicant  
Office Action, dated May 2, 2018, from corresponding U.S. Appl. No. 15/894,809. cited by applicant  
Office Action, dated May 2, 2019, from corresponding U.S. Appl. No. 16/104,628. cited by applicant  
Office Action, dated May 24, 2022, from corresponding U.S. Appl. No. 17/674,187. cited by applicant  
Office Action, dated May 29, 2020, from corresponding U.S. Appl. No. 16/862,944. cited by applicant  
Office Action, dated May 29, 2020, from corresponding U.S. Appl. No. 16/862,948. cited by applicant  
Office Action, dated May 29, 2020, from corresponding U.S. Appl. No. 16/863,226. cited by applicant  
Office Action, dated May 5, 2020, from corresponding U.S. Appl. No. 16/410,336. cited by applicant  
Office Action, dated May 9, 2022, from corresponding U.S. Appl. No. 16/840,943. cited by applicant  
Office Action, dated Nov. 1, 2017, from corresponding U.S. Appl. No. 15/169,658. cited by applicant  
Office Action, dated Nov. 10, 2021, from corresponding U.S. Appl. No. 17/380,485. cited by applicant  
Office Action, dated Nov. 10, 2021, from corresponding U.S. Appl. No. 17/409,999. cited by applicant  
Office Action, dated Nov. 10, 2022, from corresponding U.S. Appl. No. 17/670,341. cited by applicant  
Office Action, dated Nov. 12, 2020, from corresponding U.S. Appl. No. 17/034,355. cited by applicant  
Office Action, dated Nov. 12, 2020, from corresponding U.S. Appl. No. 17/034,772. cited by applicant  
Office Action, dated Nov. 12, 2021, from corresponding U.S. Appl. No. 17/346,586. cited by applicant  
Office Action, dated Nov. 12, 2021, from corresponding U.S. Appl. No. 17/373,444. cited by applicant  
Office Action, dated Nov. 15, 2018, from corresponding U.S. Appl. No. 16/059,911. cited by applicant  
Office Action, dated Nov. 15, 2019, from corresponding U.S. Appl. No. 16/552,758. cited by applicant  
Office Action, dated Nov. 15, 2022, from corresponding U.S. Appl. No. 17/200,698. cited by applicant  
Office Action, dated Nov. 16, 2021, from corresponding U.S. Appl. No. 17/370,650. cited by applicant  
Office Action, dated Nov. 16, 2021, from corresponding U.S. Appl. No. 17/486,350. cited by applicant  
Office Action, dated Nov. 18, 2019, from corresponding U.S. Appl. No. 16/560,885. cited by applicant  
Office Action, dated Nov. 18, 2019, from corresponding U.S. Appl. No. 16/560,889. cited by applicant  
Office Action, dated Nov. 18, 2019, from corresponding U.S. Appl. No. 16/572,347. cited by applicant  
Office Action, dated Nov. 18, 2022, from corresponding U.S. Appl. No. 17/836,454. cited by applicant  
Office Action, dated Nov. 19, 2019, from corresponding U.S. Appl. No. 16/595,342. cited by applicant  
Office Action, dated Nov. 20, 2019, from corresponding U.S. Appl. No. 16/595,327. cited by applicant  
Office Action, dated Nov. 23, 2018, from corresponding U.S. Appl. No. 16/042,673. cited by applicant  
Office Action, dated Nov. 23, 2021, from corresponding U.S. Appl. No. 17/013,756. cited by applicant  
Office Action, dated Nov. 24, 2020, from corresponding U.S. Appl. No. 16/925,628. cited by applicant  
Office Action, dated Nov. 26, 2021, from corresponding U.S. Appl. No. 16/925,550. cited by applicant  
Office Action, dated Nov. 29, 2022, from corresponding U.S. Appl. No. 17/838,939. cited by applicant  
Office Action, dated Nov. 4, 2021, from corresponding U.S. Appl. No. 17/491,906. cited by applicant  
Office Action, dated Nov. 8, 2021, from corresponding U.S. Appl. No. 16/872,130. cited by applicant  
Office Action, dated Oct. 10, 2018, from corresponding U.S. Appl. No. 16/041,563. cited by applicant  
Office Action, dated Oct. 10, 2018, from corresponding U.S. Appl. No. 16/055,083. cited by applicant  
Notice of Allowance, dated Feb. 8, 2022, from corresponding U.S. Appl. No. 17/342,153. cited by applicant  
Notice of Allowance, dated Jan. 1, 2021, from corresponding U.S. Appl. No. 17/026,727. cited by applicant  
Notice of Allowance, dated Jan. 11, 2022, from corresponding U.S. Appl. No. 17/371,350. cited by applicant  
Notice of Allowance, dated Jan. 12, 2022, from corresponding U.S. Appl. No. 17/334,948. cited by applicant  
Notice of Allowance, dated Jan. 12, 2022, from corresponding U.S. Appl. No. 17/463,775. cited by applicant  
Notice of Allowance, dated Jan. 14, 2020, from corresponding U.S. Appl. No. 16/277,715. cited by applicant  
Notice of Allowance, dated Jan. 15, 2021, from corresponding U.S. Appl. No. 17/030,714. cited by applicant  
Notice of Allowance, dated Jan. 18, 2018, from corresponding U.S. Appl. No. 15/619,478. cited by applicant  
Notice of Allowance, dated Jan. 18, 2019 from corresponding U.S. Appl. No. 16/159,635. cited by applicant  
Notice of Allowance, dated Jan. 2, 2020, from corresponding U.S. Appl. No. 16/410,296. cited by applicant  
Notice of Allowance, dated Jan. 23, 2018, from corresponding U.S. Appl. No. 15/619,251. cited by applicant  
Notice of Allowance, dated Jan. 24, 2022, from corresponding U.S. Appl. No. 17/340,699. cited by applicant  
Notice of Allowance, dated Jan. 25, 2021, from corresponding U.S. Appl. No. 16/410,336. cited by applicant  
Notice of Allowance, dated Jan. 26, 2018, from corresponding U.S. Appl. No. 15/619,469. cited by applicant  
Notice of Allowance, dated Jan. 26, 2022, from corresponding U.S. Appl. No. 17/491,906. cited by applicant  
Notice of Allowance, dated Jan. 29, 2020, from corresponding U.S. Appl. No. 16/278,119. cited by applicant  
Notice of Allowance, dated Jan. 31, 2022, from corresponding U.S. Appl. No. 17/472,948. cited by applicant

[illegible]

Office Action, dated Aug. 19, 2022, from corresponding U.S. Appl. No. 17/584,187. cited by applicant

Office Action, dated Aug. 2, 2022, from corresponding U.S. Appl. No. 17/670,354. cited by applicant

Office Action, dated Aug. 20, 2020, from corresponding U.S. Appl. No. 16/817,136. cited by applicant

Office Action, dated Aug. 23, 2017, from corresponding U.S. Appl. No. 15/626,052. cited by applicant

Office Action, dated Aug. 24, 2017, from corresponding U.S. Appl. No. 15/169,643. cited by applicant

Office Action, dated Aug. 24, 2017, from corresponding U.S. Appl. No. 15/619,451. cited by applicant

Office Action, dated Aug. 24, 2020, from corresponding U.S. Appl. No. 16/595,327. cited by applicant

Office Action, dated Aug. 27, 2019, from corresponding U.S. Appl. No. 16/410,296. cited by applicant

Notice of Allowance, dated Oct. 17, 2019, from corresponding U.S. Appl. No. 16/563,741. cited by applicant

Notice of Allowance, dated Oct. 18, 2022, from corresponding U.S. Appl. No. 16/840,943. cited by applicant

Notice of Allowance, dated Oct. 21, 2019, from corresponding U.S. Appl. No. 16/404,405. cited by applicant

Notice of Allowance, dated Oct. 21, 2020, from corresponding U.S. Appl. No. 16/834,812. cited by applicant

Notice of Allowance, dated Oct. 22, 2021, from corresponding U.S. Appl. No. 17/346,847. cited by applicant

Notice of Allowance, dated Oct. 25, 2022, from corresponding U.S. Appl. No. 17/711,331. cited by applicant

Notice of Allowance, dated Oct. 3, 2019, from corresponding U.S. Appl. No. 16/511,700. cited by applicant

Notice of Allowance, dated Sep. 1, 2021, from corresponding U.S. Appl. No. 17/196,570. cited by applicant

Notice of Allowance, dated Sep. 1, 2021, from corresponding U.S. Appl. No. 17/222,556. cited by applicant

Notice of Allowance, dated Sep. 1, 2022, from corresponding U.S. Appl. No. 17/480,377. cited by applicant

Notice of Allowance, dated Sep. 12, 2019, from corresponding U.S. Appl. No. 16/512,011. cited by applicant

Notice of Allowance, dated Sep. 12, 2022, from corresponding U.S. Appl. No. 17/674,187. cited by applicant

Notice of Allowance, dated Sep. 13, 2018, from corresponding U.S. Appl. No. 15/894,809. cited by applicant

Notice of Allowance, dated Sep. 13, 2018, from corresponding U.S. Appl. No. 15/894,890. cited by applicant

Notice of Allowance, dated Sep. 14, 2021, from corresponding U.S. Appl. No. 16/808,497. cited by applicant

Notice of Allowance, dated Sep. 16, 2020, from corresponding U.S. Appl. No. 16/915,097. cited by applicant

Notice of Allowance, dated Sep. 17, 2020, from corresponding U.S. Appl. No. 16/863,226. cited by applicant

Notice of Allowance, dated Sep. 18, 2018, from corresponding U.S. Appl. No. 15/894,819. cited by applicant

Notice of Allowance, dated Sep. 18, 2018, from corresponding U.S. Appl. No. 16/041,545. cited by applicant

Notice of Allowance, dated Sep. 18, 2020, from corresponding U.S. Appl. No. 16/812,795. cited by applicant

Notice of Allowance, dated Sep. 2, 2022, from corresponding U.S. Appl. No. 17/380,485. cited by applicant

Notice of Allowance, dated Sep. 23, 2020, from corresponding U.S. Appl. No. 16/811,793. cited by applicant

Notice of Allowance, dated Sep. 23, 2021, from corresponding U.S. Appl. No. 17/068,454. cited by applicant

Notice of Allowance, dated Sep. 24, 2021, from corresponding U.S. Appl. No. 17/334,939. cited by applicant

Notice of Allowance, dated Sep. 25, 2020, from corresponding U.S. Appl. No. 16/983,536. cited by applicant

Notice of Allowance, dated Sep. 27, 2017, from corresponding U.S. Appl. No. 15/626,052. cited by applicant

Notice of Allowance, dated Sep. 27, 2021, from corresponding U.S. Appl. No. 17/222,523. cited by applicant

Notice of Allowance, dated Sep. 28, 2018, from corresponding U.S. Appl. No. 16/041,520. cited by applicant

Notice of Allowance, dated Sep. 28, 2022, from corresponding U.S. Appl. No. 17/509,974. cited by applicant

Notice of Allowance, dated Sep. 28, 2022, from corresponding U.S. Appl. No. 17/689,683. cited by applicant

Notice of Allowance, dated Sep. 29, 2021, from corresponding U.S. Appl. No. 17/316,179. cited by applicant

Notice of Allowance, dated Sep. 30, 2022, from corresponding U.S. Appl. No. 17/867,068. cited by applicant

Notice of Allowance, dated Sep. 4, 2018, from corresponding U.S. Appl. No. 15/883,041. cited by applicant

Notice of Allowance, dated Sep. 4, 2020, from corresponding U.S. Appl. No. 16/808,500. cited by applicant

Notice of Allowance, dated Sep. 4, 2020, from corresponding U.S. Appl. No. 16/901,662. cited by applicant

Notice of Allowance, dated Sep. 9, 2021, from corresponding U.S. Appl. No. 17/334,909. cited by applicant

Agrawal et al, "Securing Electronic Health Records Without Impeding the Flow of Information," International Journal of Medical Informatics 76, 2007, pp. 4-10 (2007). cited by applicant

Avepoint, Installing and Configuring the APIA System, International Association of Privacy Professionals, AvePoint, Inc. cited by applicant

Bang et al, "Building an Effective and Efficient Continuous Web Application Security Program," 2016 International Conference on Cyber Security Situational Awareness and Data Analytics and Assessment (CyberSA), London, 2016, pp. 1-4 (Year: 2016). cited by applicant

Binns, et al, "Data Havens, or Privacy Sans Frontieres? A Study of International Personal Data Transfers," ACM, pp. 273-274 (Year: 2002). cited by applicant

Czeskis et al, "Lightweight Server Support for Browser-based CSRF Protection," Proceedings of the 22nd International Conference on World Wide Web, 2013, pp. 1-10 (Year: 2013). cited by applicant

Falahrastegar, Marjan, et al, Tracking Personal Identifiers Across the Web, Medical Image Computing and Computer-Assisted Intervention—Miccai 2015, 18th International Conference, Oct. 5, 2015, Munich, Germany. cited by applicant

Friedman et al, "Informed Consent in the Mozilla Browser: Implementing Value-Sensitive Design," Proceedings of the 35th Annual Hawaii International Conference on System Sciences, 2002, IEEE, pp. 1-10 (Year: 2002). cited by applicant

Ghiglieri, Marco et al.; Personal DLP for Facebook, 2014 IEEE International Conference on Pervasive Computing and Communication Workshops (Percom Workshops), 2014, IEEE; Mar. 24, 2014; pp. 629-634. cited by applicant

Gilda, et al, "Blockchain for Student Data Privacy and Consent," 2018 International Conference on Computer Communication and Informatics, Jan. 4-6, 2018, pp. 1-4 (Year: 2018). cited by applicant

Halevy, et al, "Schema Mediation in Peer Data Management Systems," IEEE, Proceedings of the 19th International Conference on Data Engineering, 2003, pp. 1-10 (2003). cited by applicant

App, Daily Dashboard, PIA Tool Stocked With New Templates for DPI, Infosec, International Association of Privacy Professionals, Apr. 22, 2014. cited by applicant

IAPP, ISO/IEC 27001 Information Security Management Template, Resource Center, International Association of Privacy Professionals. cited by applicant

International Search Report, dated Apr. 12, 2022, from corresponding International Application No. PCT/US2022/016735. cited by applicant

International Search Report, dated Aug. 15, 2017, from corresponding International Application No. PCT/US2017/036919. cited by applicant

Castro et al, "Creating Lightweight Ontologies for Dataset Description," IEEE, pp. 1-4 (Year: 2014). cited by applicant

Cerpzone, "How to Access Data on Data Archival Storage and Recovery System", <https://www.saj.usace.army.mil/Portals/44/docs/Environmental/Lake%20O%20Watershed/15February2017/How%20To%20Access%20Model%20Data%20Overview%2017-02-16-095535-633>, Feb. 16, 2017. cited by applicant

Cha et al, "A Data-Driven Security Risk Assessment Scheme for Personal Data Protection," IEEE, pp. 50510-50517 (Year: 2018). cited by applicant

Cha, et al, "Process-Oriented Approach for Validating Asset Value for Evaluating Information Security Risk," IEEE, Aug. 31, 2009, pp. 379-385 (Year: 2009). cited by applicant

Chang et al, "A Ranking Approach for Human Age Estimation Based on Face Images," IEEE, 2010, pp. 3396-3399 (Year: 2010). cited by applicant

Chapados et al, "Scoring Models for Insurance Risk Sharing Pool Optimization," 2008, IEEE, pp. 97-105 (Year: 2008). cited by applicant



Cheng, Raymond, et al, "Radiatus: A Shared-Nothing Server-Side Web Architecture," Proceedings of the Seventh ACM Symposium on Cloud Computing, Oct. 237-250 (Year: 2016). cited by applicant

Choi et al, "A Survey on Ontology Mapping," ACM, pp. 34-41 (Year: 2006). cited by applicant

Choi et al, "Retrieval Effectiveness of Table of Contents and Subject Headings," ACM, pp. 103-104 (Year: 2007). cited by applicant

Chowdhury et al, "A System Architecture for Subject-Centric Data Sharing," ACM, pp. 1-10 (Year: 2018). cited by applicant

Chowdhury et al, "Managing Data Transfers in Computer Clusters with Orchestra," ACM, pp. 98-109 (Year: 2011). cited by applicant

Civili et al, "Mastro Studio: Managing Ontology-Based Data Access Applications," ACM, pp. 1314-1317, Aug. 26-30, 2013 (Year: 2013). cited by applicant

Cruz et al, "Interactive User Feedback in Ontology Matching Using Signature Vectors," IEEE, pp. 1321-1324 (Year: 2012). cited by applicant

Cudre-Mauroux, "ESWC 2008 Ph.D. Symposium," The ESWC 2008 Ph.D. Symposium is sponsored by the Okkam project (<http://fp7.okkam.org/>), MIT, pp. 2008). cited by applicant

Cui et al, "Domain Ontology Management Environment," IEEE, pp. 1-9 (Year: 2000). cited by applicant

Decision Regarding Institution of Post-Grant Review in Case PGR2018-00056 for U.S. Pat. No. 9,691,090 B1, Oct. 11, 2018. cited by applicant

Degeling et al, "We Value Your Privacy . . . Now Take Some Cookies: Measuring the GDPRs Impact on Web Privacy," arxiv.org, Cornell University Library, 2018. cited by applicant

Cornell University, Ithaca, NY 14853, Aug. 15, 2018, pp. 1-15 (Year: 2019). cited by applicant

Dimou et al, "Machine-Interpretable Dataset and Service Descriptions for Heterogeneous Data Access and Retrieval", ACM, pp. 145-152 (Year: 2015). cited by applicant

Dokholyan et al, "Regulatory and Ethical Considerations for Linking Clinical and Administrative Databases," American Heart Journal 157.6 (2009), pp. 971-976 (Year: 2009). cited by applicant

Dowling, "Auditing Global HR Compliance," published May 23, 2014, retrieved from <https://www.shrm.org/resourcesandtools/hr-topics/global-hr/pages/audit-compliance.aspx> Jul. 2, 2022 (Year: 2014). cited by applicant

Dunkel et al, "Data Organization and Access for Efficient Data Mining", IEEE, pp. 522-529 (Year: 1999). cited by applicant

Dwork, Cynthia, Differential Privacy, Microsoft Research, p. 1-12. cited by applicant

Edinger et al, "Age and Gender Estimation of Unfiltered Faces," IEEE, 2014, pp. 2170-2179 (Year: 2014). cited by applicant

Emerson, et al, "A Data Mining Driven Risk Profiling Method for Road Asset Management," ACM, pp. 1267-1275 (Year: 2013). cited by applicant

Enck, William, et al, TaintDroid: An Information-Flow Tracking System for Realtime Privacy Monitoring on Smartphones, ACM Transactions on Computer Science and System Security, Vol. 2, Article 5, Jun. 2014, p. 5:1-5:29. cited by applicant

Everypixel Team, "A New Age Recognition API Detects the Age of People on Photos," May 20, 2019, pp. 1-5 (Year: 2019). cited by applicant

Falbo et al, "An Ontological Approach to Domain Engineering," ACM, pp. 351-358 (Year: 2002). cited by applicant

Fan et al, "Intrusion Investigations with Data-hiding for Computer Log-file Forensics," IEEE, pp. 1-6 (Year: 2010). cited by applicant

Final Written Decision Regarding Post-Grant Review in Case PGR2018-00056 for U.S. Pat. No. 9,691,090 B1, Oct. 10, 2019. cited by applicant

Francis, Andre, Business Mathematics and Statistics, South-Western Cengage Learning, 2008, Sixth Edition. cited by applicant

Friedman et al, "Data Mining with Differential Privacy," ACM, Jul. 2010, pp. 493-502 (Year: 2010). cited by applicant

Frikken, Keith B., et al, Yet Another Privacy Metric for Publishing Micro-data, Miami University, Oct. 27, 2008, p. 117-121. cited by applicant

Fung et al, "Discover Information and Knowledge from Websites using an Integrated Summarization and Visualization Framework", IEEE, pp. 232-235 (Year: 2008). cited by applicant

Gajare et al, "Improved Automatic Feature Selection Approach for Health Risk Prediction," Feb. 16, 2018, IEEE, pp. 816-819 (Year: 2018). cited by applicant

Geko et al, "An Ontology Capturing the Interdependence of the General Data Protection Regulation (GDPR) and Information Security," ACM, pp. 1-6, Nov. 1, 2018. cited by applicant

Golab, et al, "Issues in Data Stream Management," ACM, SIGMOD Record, vol. 32, No. 2, Jun. 2003, pp. 5-14 (Year: 2003). cited by applicant

Golfarelli et al, "Beyond Data Warehousing: What's Next in Business Intelligence?," ACM, pp. 1-6 (Year: 2004). cited by applicant

Gonçalves et al, "The XML Log Standard for Digital Libraries: Analysis, Evolution, and Deployment," IEEE, pp. 312-314 (Year: 2003). cited by applicant

Goni, Kyriaki, "Deletion Process\_Only you can see my history: Investigating Digital Privacy, Digital Oblivion, and Control on Personal Data Through an Intelligent Installation," ACM, 2016, retrieved online on Oct. 3, 2019, pp. 324-333. Retrieved from the Internet URL: <http://delivery.acm.org/10.1145/2920000/291.> cited by applicant

Gowadia et al, "RDF Metadata for XML Access Control," ACM, pp. 31-48 (Year: 2003). cited by applicant

Grolinger, et al, "Data Management in Cloud Environments: NoSQL and NewSQL Data Stores," Journal of Cloud Computing: Advances, Systems and Applications, Vol. 12, No. 1, (Year: 2013). cited by applicant

Guo, et al, "OPAL: A Passe-partout for Web Forms," ACM, pp. 353-356 (Year: 2012). cited by applicant

Gustarini, et al, "Evaluation of Challenges in Human Subject Studies "In-the-Wild" Using Subjects' Personal Smartphones," ACM, pp. 1447-1456 (Year: 2013). cited by applicant

Hacıgümüş, Hakan, et al., Executing SQL over Encrypted Data in the Database-Service-Provider Model, ACM, Jun. 4, 2002, pp. 216-227. cited by applicant

Han et al, "Demographic Estimation from Face Images: Human vs. Machine Performance," IEEE, 2015, pp. 1148-1161 (Year: 2015). cited by applicant

Hauch, et al, "Information Intelligence: Metadata for Information Discovery, Access, and Integration," ACM, pp. 793-798 (Year: 2005). cited by applicant

He et al, "A Crowdsourcing Framework for Detecting of Cross-Browser Issues in Web Application," ACM, pp. 1-4, Nov. 6, 2015 (Year: 2015). cited by applicant

Heil et al, "Downsizing and Rightsizing," <https://web.archive.org/web/20130523153311/https://www.referenceforbusiness.com/management/De-Ele/Downsizing/Downsizing.html> (Year: 2013). cited by applicant

Hernandez, et al, "Data Exchange with Data-Metadata Translations," ACM, pp. 260-273 (Year: 2008). cited by applicant

Hinde, "A Model to Assess Organisational Information Privacy Maturity Against the Protection of Personal Information Act" Dissertation University of Cape Town, 2014, pp. 1-121 (Year: 2014). cited by applicant

Advisory Action, dated Jan. 13, 2021, from corresponding U.S. Appl. No. 16/808,493. cited by applicant

Advisory Action, dated Jan. 13, 2021, from corresponding U.S. Appl. No. 16/862,944. cited by applicant

Advisory Action, dated Jan. 13, 2021, from corresponding U.S. Appl. No. 16/862,948. cited by applicant

Advisory Action, dated Jan. 13, 2021, from corresponding U.S. Appl. No. 16/862,952. cited by applicant

Advisory Action, dated Jan. 6, 2021, from corresponding U.S. Appl. No. 16/808,497. cited by applicant

Advisory Action, dated Jun. 19, 2020, from corresponding U.S. Appl. No. 16/595,342. cited by applicant

Advisory Action, dated Jun. 2, 2020, from corresponding U.S. Appl. No. 16/404,491. cited by applicant

Advisory Action, dated May 21, 2020, from corresponding U.S. Appl. No. 16/557,392. cited by applicant

Ex Parte Quayle Action, dated May 10, 2022, from corresponding U.S. Appl. No. 17/668,714. cited by applicant

Final Office Action, dated Apr. 1, 2022, from corresponding U.S. Appl. No. 17/370,650. cited by applicant

Final Office Action, dated Apr. 23, 2020, from corresponding U.S. Appl. No. 16/572,347. cited by applicant

Final Office Action, dated Apr. 25, 2022, from corresponding U.S. Appl. No. 17/149,421. cited by applicant

Final Office Action, dated Apr. 27, 2021, from corresponding U.S. Appl. No. 17/068,454. cited by applicant

Final Office Action, dated Apr. 28, 2022, from corresponding U.S. Appl. No. 16/925,550. cited by applicant

Final Office Action, dated Apr. 5, 2022, from corresponding U.S. Appl. No. 17/013,756. cited by applicant

Final Office Action, dated Apr. 7, 2020, from corresponding U.S. Appl. No. 16/595,327. cited by applicant

Final Office Action, dated Aug. 10, 2020, from corresponding U.S. Appl. No. 16/791,589. cited by applicant

Final Office Action, dated Aug. 27, 2021, from corresponding U.S. Appl. No. 17/161,159. cited by applicant



Final Office Action, dated Aug. 28, 2020, from corresponding U.S. Appl. No. 16/410,336. cited by applicant  
Final Office Action, dated Aug. 5, 2020, from corresponding U.S. Appl. No. 16/719,071. cited by applicant  
Final Office Action, dated Aug. 9, 2021, from corresponding U.S. Appl. No. 17/119,080. cited by applicant  
Final Office Action, dated Dec. 10, 2021, from corresponding U.S. Appl. No. 17/187,329. cited by applicant  
Final Office Action, dated Dec. 7, 2020, from corresponding U.S. Appl. No. 16/862,956. cited by applicant  
Final Office Action, dated Dec. 9, 2019, from corresponding U.S. Appl. No. 16/410,336. cited by applicant  
Final Office Action, dated Feb. 19, 2020, from corresponding U.S. Appl. No. 16/404,491. cited by applicant  
Final Office Action, dated Feb. 25, 2022, from corresponding U.S. Appl. No. 17/346,586. cited by applicant  
Final Office Action, dated Feb. 3, 2020, from corresponding U.S. Appl. No. 16/557,392. cited by applicant  
Final Office Action, dated Feb. 8, 2021, from corresponding U.S. Appl. No. 16/927,658. cited by applicant  
Final Office Action, dated Jan. 17, 2018, from corresponding U.S. Appl. No. 15/619,278. cited by applicant  
Final Office Action, dated Jan. 21, 2020, from corresponding U.S. Appl. No. 16/410,762. cited by applicant  
Final Office Action, dated Jan. 23, 2018, from corresponding U.S. Appl. No. 15/619,479. cited by applicant  
Final Office Action, dated Jan. 23, 2020, from corresponding U.S. Appl. No. 16/505,430. cited by applicant  
Final Office Action, dated Jul. 1, 2022, from corresponding U.S. Appl. No. 17/187,329. cited by applicant  
Final Office Action, dated Jul. 21, 2021, from corresponding U.S. Appl. No. 17/151,334. cited by applicant  
Final Office Action, dated Jul. 6, 2022, from corresponding U.S. Appl. No. 17/200,698. cited by applicant  
Final Office Action, dated Jul. 7, 2021, from corresponding U.S. Appl. No. 17/149,421. cited by applicant  
Final Office Action, dated Jun. 10, 2022, from corresponding U.S. Appl. No. 17/161,159. cited by applicant  
Final Office Action, dated Jun. 29, 2022, from corresponding U.S. Appl. No. 17/020,275. cited by applicant  
Final Office Action, dated Jun. 9, 2022, from corresponding U.S. Appl. No. 17/494,220. cited by applicant  
Final Office Action, dated Mar. 21, 2022, from corresponding U.S. Appl. No. 17/373,444. cited by applicant  
Final Office Action, dated Mar. 22, 2022, from corresponding U.S. Appl. No. 17/380,485. cited by applicant  
Final Office Action, dated Mar. 26, 2021, from corresponding U.S. Appl. No. 17/020,275. cited by applicant  
Final Office Action, dated Mar. 5, 2019, from corresponding U.S. Appl. No. 16/055,961. cited by applicant  
Final Office Action, dated Mar. 6, 2020, from corresponding U.S. Appl. No. 16/595,342. cited by applicant  
Final Office Action, dated May 12, 2022, from corresponding U.S. Appl. No. 17/499,624. cited by applicant  
Final Office Action, dated May 14, 2021, from corresponding U.S. Appl. No. 17/013,756. cited by applicant  
Final Office Action, dated May 16, 2022, from corresponding U.S. Appl. No. 17/480,377. cited by applicant  
Final Office Action, dated May 2, 2022, from corresponding U.S. Appl. No. 17/499,595. cited by applicant  
Final Office Action, dated May 24, 2022, from corresponding U.S. Appl. No. 17/499,582. cited by applicant  
Final Office Action, dated Nov. 29, 2017, from corresponding U.S. Appl. No. 15/619,237. cited by applicant  
Restriction Requirement, dated Sep. 15, 2020, from corresponding U.S. Appl. No. 16/925,628. cited by applicant  
Restriction Requirement, dated Sep. 9, 2019, from corresponding U.S. Appl. No. 16/505,426. cited by applicant  
Notice of Allowance, dated Apr. 12, 2017, from corresponding U.S. Appl. No. 15/256,419. cited by applicant  
Notice of Allowance, dated Apr. 12, 2022, from corresponding U.S. Appl. No. 17/479,807. cited by applicant  
Notice of Allowance, dated Apr. 14, 2022, from corresponding U.S. Appl. No. 17/572,276. cited by applicant  
Notice of Allowance, dated Apr. 17, 2020, from corresponding U.S. Appl. No. 16/593,639. cited by applicant  
Notice of Allowance, dated Apr. 19, 2021, from corresponding U.S. Appl. No. 17/164,029. cited by applicant  
Notice of Allowance, dated Apr. 2, 2019, from corresponding U.S. Appl. No. 16/160,577. cited by applicant  
Notice of Allowance, dated Apr. 2, 2021, from corresponding U.S. Appl. No. 17/162,006. cited by applicant  
Notice of Allowance, dated Apr. 20, 2022, from corresponding U.S. Appl. No. 17/573,808. cited by applicant  
Notice of Allowance, dated Apr. 22, 2021, from corresponding U.S. Appl. No. 17/163,701. cited by applicant  
Notice of Allowance, dated Apr. 25, 2018, from corresponding U.S. Appl. No. 15/883,041. cited by applicant  
Notice of Allowance, dated Apr. 27, 2022, from corresponding U.S. Appl. No. 17/573,999. cited by applicant  
Notice of Allowance, dated Apr. 28, 2021, from corresponding U.S. Appl. No. 17/135,445. cited by applicant  
Notice of Allowance, dated Apr. 28, 2021, from corresponding U.S. Appl. No. 17/181,828. cited by applicant  
Notice of Allowance, dated Apr. 28, 2022, from corresponding U.S. Appl. No. 17/592,922. cited by applicant  
Notice of Allowance, dated Apr. 28, 2022, from corresponding U.S. Appl. No. 17/670,352. cited by applicant  
Notice of Allowance, dated Apr. 29, 2020, from corresponding U.S. Appl. No. 16/700,049. cited by applicant  
Notice of Allowance, dated Apr. 29, 2022, from corresponding U.S. Appl. No. 17/387,421. cited by applicant  
Notice of Allowance, dated Apr. 30, 2020, from corresponding U.S. Appl. No. 16/565,265. cited by applicant  
Notice of Allowance, dated Apr. 30, 2020, from corresponding U.S. Appl. No. 16/820,346. cited by applicant  
Notice of Allowance, dated Apr. 30, 2021, from corresponding U.S. Appl. No. 16/410,762. cited by applicant  
Notice of Allowance, dated Apr. 4, 2022, from corresponding U.S. Appl. No. 17/493,332. cited by applicant  
Notice of Allowance, dated Apr. 4, 2022, from corresponding U.S. Appl. No. 17/572,298. cited by applicant  
Notice of Allowance, dated Apr. 8, 2019, from corresponding U.S. Appl. No. 16/228,250. cited by applicant  
Notice of Allowance, dated Apr. 8, 2020, from corresponding U.S. Appl. No. 16/791,348. cited by applicant  
Notice of Allowance, dated Apr. 9, 2020, from corresponding U.S. Appl. No. 16/791,075. cited by applicant  
Notice of Allowance, dated Aug. 10, 2020, from corresponding U.S. Appl. No. 16/671,444. cited by applicant  
Notice of Allowance, dated Aug. 10, 2020, from corresponding U.S. Appl. No. 16/788,633. cited by applicant  
Notice of Allowance, dated Aug. 12, 2020, from corresponding U.S. Appl. No. 16/719,488. cited by applicant  
Notice of Allowance, dated Aug. 12, 2021, from corresponding U.S. Appl. No. 16/881,832. cited by applicant  
Notice of Allowance, dated Aug. 14, 2018, from corresponding U.S. Appl. No. 15/989,416. cited by applicant  
Notice of Allowance, dated Aug. 18, 2017, from corresponding U.S. Appl. No. 15/619,455. cited by applicant  
Notice of Allowance, dated Aug. 20, 2019, from corresponding U.S. Appl. No. 16/241,710. cited by applicant  
Notice of Allowance, dated Aug. 22, 2022, from corresponding U.S. Appl. No. 17/499,595. cited by applicant  
Notice of Allowance, dated Aug. 24, 2018, from corresponding U.S. Appl. No. 15/619,479. cited by applicant  
Notice of Allowance, dated Aug. 26, 2019, from corresponding U.S. Appl. No. 16/443,374. cited by applicant  
Notice of Allowance, dated Aug. 26, 2020, from corresponding U.S. Appl. No. 16/808,503. cited by applicant  
Notice of Allowance, dated Aug. 28, 2019, from corresponding U.S. Appl. No. 16/278,120. cited by applicant  
Notice of Allowance, dated Aug. 3, 2022, from corresponding U.S. Appl. No. 17/668,714. cited by applicant  
Notice of Allowance, dated Aug. 30, 2018, from corresponding U.S. Appl. No. 15/996,208. cited by applicant  
Notice of Allowance, dated Aug. 31, 2021, from corresponding U.S. Appl. No. 17/326,901. cited by applicant  
Notice of Allowance, dated Aug. 4, 2021, from corresponding U.S. Appl. No. 16/895,278. cited by applicant

Notice of Allowance, dated Aug. 4, 2022, from corresponding U.S. Appl. No. 17/670,349. cited by applicant  
Notice of Allowance, dated Aug. 7, 2020, from corresponding U.S. Appl. No. 16/901,973. cited by applicant  
Notice of Allowance, dated Aug. 9, 2018, from corresponding U.S. Appl. No. 15/882,989. cited by applicant  
Notice of Allowance, dated Aug. 9, 2021, from corresponding U.S. Appl. No. 16/881,699. cited by applicant  
Notice of Allowance, dated Aug. 9, 2022, from corresponding U.S. Appl. No. 17/832,313. cited by applicant  
Notice of Allowance, dated Dec. 10, 2018, from corresponding U.S. Appl. No. 16/105,602. cited by applicant  
Notice of Allowance, dated Dec. 11, 2019, from corresponding U.S. Appl. No. 16/278,122. cited by applicant  
Office Action, dated Aug. 27, 2021, from corresponding U.S. Appl. No. 17/187,329. cited by applicant  
Office Action, dated Aug. 27, 2021, from corresponding U.S. Appl. No. 17/334,948. cited by applicant  
Office Action, dated Aug. 29, 2017, from corresponding U.S. Appl. No. 15/619,237. cited by applicant  
Office Action, dated Aug. 30, 2017, from corresponding U.S. Appl. No. 15/619,212. cited by applicant  
Office Action, dated Aug. 30, 2017, from corresponding U.S. Appl. No. 15/619,382. cited by applicant  
Office Action, dated Aug. 30, 2021, from corresponding U.S. Appl. No. 16/938,520. cited by applicant  
Office Action, dated Aug. 4, 2022, from corresponding U.S. Appl. No. 17/828,953. cited by applicant  
Office Action, dated Aug. 6, 2019, from corresponding U.S. Appl. No. 16/404,491. cited by applicant  
Office Action, dated Aug. 6, 2020, from corresponding U.S. Appl. No. 16/862,956. cited by applicant  
Office Action, dated Dec. 11, 2019, from corresponding U.S. Appl. No. 16/578,712. cited by applicant  
Office Action, dated Dec. 13, 2021, from corresponding U.S. Appl. No. 17/476,209. cited by applicant  
Office Action, dated Dec. 14, 2018, from corresponding U.S. Appl. No. 16/104,393. cited by applicant  
Office Action, dated Dec. 15, 2016, from corresponding U.S. Appl. No. 15/256,419. cited by applicant  
Office Action, dated Dec. 16, 2019, from corresponding U.S. Appl. No. 16/563,754. cited by applicant  
Office Action, dated Dec. 16, 2019, from corresponding U.S. Appl. No. 16/565,265. cited by applicant  
Office Action, dated Dec. 16, 2020, from corresponding U.S. Appl. No. 17/020,275. cited by applicant  
Office Action, dated Dec. 17, 2021, from corresponding U.S. Appl. No. 17/395,759. cited by applicant  
Office Action, dated Dec. 17, 2021, from corresponding U.S. Appl. No. 17/499,582. cited by applicant  
Office Action, dated Dec. 18, 2020, from corresponding U.S. Appl. No. 17/030,714. cited by applicant  
Office Action, dated Dec. 19, 2019, from corresponding U.S. Appl. No. 16/410,866. cited by applicant  
Office Action, dated Dec. 2, 2019, from corresponding U.S. Appl. No. 16/560,963. cited by applicant  
Office Action, dated Dec. 2, 2021, from corresponding U.S. Appl. No. 17/504,102. cited by applicant  
Office Action, dated Dec. 23, 2019, from corresponding U.S. Appl. No. 16/593,639. cited by applicant  
Office Action, dated Dec. 24, 2020, from corresponding U.S. Appl. No. 17/068,454. cited by applicant  
Office Action, dated Dec. 27, 2021, from corresponding U.S. Appl. No. 17/493,332. cited by applicant  
Office Action, dated Dec. 29, 2021, from corresponding U.S. Appl. No. 17/479,807. cited by applicant  
Office Action, dated Dec. 3, 2018, from corresponding U.S. Appl. No. 16/055,998. cited by applicant  
Office Action, dated Dec. 30, 2021, from corresponding U.S. Appl. No. 17/149,421. cited by applicant  
Office Action, dated Dec. 31, 2018, from corresponding U.S. Appl. No. 16/160,577. cited by applicant  
Office Action, dated Dec. 7, 2021, from corresponding U.S. Appl. No. 17/499,609. cited by applicant  
Office Action, dated Dec. 8, 2020, from corresponding U.S. Appl. No. 17/013,758. cited by applicant  
Office Action, dated Dec. 8, 2020, from corresponding U.S. Appl. No. 17/068,198. cited by applicant  
Office Action, dated Feb. 10, 2021, from corresponding U.S. Appl. No. 16/862,944. cited by applicant  
Office Action, dated Feb. 10, 2021, from corresponding U.S. Appl. No. 17/106,469. cited by applicant  
Office Action, dated Feb. 15, 2019, from corresponding U.S. Appl. No. 16/220,899. cited by applicant  
Office Action, dated Feb. 16, 2022, from corresponding U.S. Appl. No. 16/872,031. cited by applicant  
Office Action, dated Feb. 17, 2021, from corresponding U.S. Appl. No. 16/862,948. cited by applicant  
Office Action, dated Feb. 18, 2021, from corresponding U.S. Appl. No. 16/862,952. cited by applicant  
Office Action, dated Feb. 2, 2021, from corresponding U.S. Appl. No. 17/101,915. cited by applicant  
Office Action, dated Feb. 26, 2019, from corresponding U.S. Appl. No. 16/228,250. cited by applicant  
Office Action, dated Feb. 3, 2021, from corresponding U.S. Appl. No. 17/013,757. cited by applicant  
Office Action, dated Feb. 5, 2020, from corresponding U.S. Appl. No. 16/586,202. cited by applicant  
Office Action, dated Feb. 6, 2020, from corresponding U.S. Appl. No. 16/707,762. cited by applicant  
Office Action, dated Feb. 8, 2021, from corresponding U.S. Appl. No. 17/139,650. cited by applicant  
Office Action, dated Feb. 9, 2021, from corresponding U.S. Appl. No. 16/808,493. cited by applicant  
Office Action, dated Feb. 9, 2022, from corresponding U.S. Appl. No. 17/543,546. cited by applicant  
Office Action, dated Jan. 14, 2022, from corresponding U.S. Appl. No. 17/499,595. cited by applicant  
Office Action, dated Jan. 18, 2019, from corresponding U.S. Appl. No. 16/055,984. cited by applicant  
Office Action, dated Jan. 21, 2022, from corresponding U.S. Appl. No. 17/499,624. cited by applicant  
Office Action, dated Jan. 22, 2021, from corresponding U.S. Appl. No. 17/099,270. cited by applicant  
Invitation to Pay Additional Search Fees, dated Jan. 7, 2019, from corresponding International Application No. PCT/US2018/055773. cited by applicant  
Invitation to Pay Additional Search Fees, dated Jan. 8, 2019, from corresponding International Application No. PCT/US2018/055774. cited by applicant  
Invitation to Pay Additional Search Fees, dated Oct. 23, 2018, from corresponding International Application No. PCT/US2018/045296. cited by applicant  
Lebeau, Franck, et al, "Model-Based Vulnerability Testing for Web Applications," 2013 IEEE Sixth International Conference on Software Testing, Verification Workshops, pp. 445-452, IEEE, 2013 (Year: 2013). cited by applicant  
Liu, Yandong, et al, "Finding the Right Consumer: Optimizing for Conversion in Display Advertising Campaigns," Proceedings of the Fifth ACM International Web Search and Data Mining, Feb. 2, 2012, pp. 473-428 (Year: 2012). cited by applicant  
Lizar et al, "Usable Consents: Tracking and Managing Use of Personal Data with a Consent Transaction Receipt," Proceedings of the 2014 ACM International Conference on Pervasive and Ubiquitous Computing: Adjunct Publication, 2014, pp. 647-652 (Year: 2014). cited by applicant  
Ma Ziang, et al, "LibRadar: Fast and Accurate Detection of Third-Party Libraries in Android Apps," 2016 IEEE/ACM 38th IEEE International Conference on Engineering Companion (ICSE-C), ACM, May 14, 2016, pp. 653-656, DOI: <http://dx.doi.org/10.1145/2889160.2889178>, p. 653, r.col, par. 1-3; figure 3 (Year: 2016). cited by applicant  
Mandal, et al, "Automated Age Prediction Using Wrinkles Features of Facial Images and Neural Network," International Journal of Emerging Engineering Research Technology, vol. 5, Issue 2, Feb. 2017, pp. 12-20 (Year: 2017). cited by applicant  
Moiso et al, "Towards a User-Centric Personal Data Ecosystem The Role of the Bank of Individual's Data," 2012 16th International Conference on Intelligent Generation Networks, Berlin, 2012, pp. 202-209 (Year: 2012). cited by applicant  
Pfeifle, Sam, The Privacy Advisor, IAPP and AvePoint Launch New Free PIA Tool, International Association of Privacy Professionals, Mar. 5, 2014. cited by applicant  
Pfeifle, Sam, The Privacy Advisor, IAPP Heads to Singapore with APIA Template in Tow, International Association of Privacy Professionals, <https://iapp.org/>

heads-to-singapore-with-apia-template\_in\_tow/, Mar. 28, 2014, p. 1-3. cited by applicant

Qing-Jiang et al., “The (P, a, K) Anonymity Model for Privacy Protection of Personal Information in the Social Networks,” 2011 6th IEEE Joint International Technology and Artificial Intelligence Conference, vol. 2 IEEE, 2011, pp. 420-423 (Year: 2011). cited by applicant

Srivastava, Agrima, et al, Measuring Privacy Leaks in Online Social Networks, International Conference on Advances in Computing, Communications and Information (ICACCI), 2013. cited by applicant

Tanwar, et al., “Live Forensics Analysis: Violations of Business Security Policy,” 2014 International Conference on Contemporary Computing and Informatics, pp. 971-976 (Year: 2014). cited by applicant

Thuraisingham, “Security Issues for the Semantic Web,” Proceedings 27th Annual International Computer Software and Applications Conference, COMPSAC 2003, TX, USA, 2003, pp. 633-638 (Year: 2003). cited by applicant

Written Opinion of the International Searching Authority, dated Jun. 6, 2017, from corresponding International Application No. PCT/US2017/025611. cited by applicant

Written Opinion of the International Searching Authority, dated Apr. 12, 2022, from corresponding International Application No. PCT/US2022/016735. cited by applicant

Written Opinion of the International Searching Authority, dated Aug. 15, 2017, from corresponding International Application No. PCT/US2017/036919. cited by applicant

Written Opinion of the International Searching Authority, dated Aug. 21, 2017, from corresponding International Application No. PCT/US2017/036914. cited by applicant

Written Opinion of the International Searching Authority, dated Aug. 29, 2017, from corresponding International Application No. PCT/US2017/036898. cited by applicant

Written Opinion of the International Searching Authority, dated Aug. 8, 2017, from corresponding International Application No. PCT/US2017/036889. cited by applicant

Written Opinion of the International Searching Authority, dated Aug. 8, 2017, from corresponding International Application No. PCT/US2017/036890. cited by applicant

Written Opinion of the International Searching Authority, dated Aug. 8, 2017, from corresponding International Application No. PCT/US2017/036893. cited by applicant

Written Opinion of the International Searching Authority, dated Aug. 8, 2017, from corresponding International Application No. PCT/US2017/036901. cited by applicant

Written Opinion of the International Searching Authority, dated Aug. 8, 2017, from corresponding International Application No. PCT/US2017/036913. cited by applicant

Written Opinion of the International Searching Authority, dated Aug. 8, 2017, from corresponding International Application No. PCT/US2017/036920. cited by applicant

Written Opinion of the International Searching Authority, dated Dec. 14, 2018, from corresponding International Application No. PCT/US2018/045296. cited by applicant

Written Opinion of the International Searching Authority, dated Dec. 22, 2021, from corresponding International Application No. PCT/US2021/051217. cited by applicant

Written Opinion of the International Searching Authority, dated Feb. 11, 2022, from corresponding International Application No. PCT/US2021/053518. cited by applicant

Written Opinion of the International Searching Authority, dated Feb. 14, 2022, from corresponding International Application No. PCT/US2021/058274. cited by applicant

Written Opinion of the International Searching Authority, dated Jan. 14, 2019, from corresponding International Application No. PCT/US2018/046949. cited by applicant

Written Opinion of the International Searching Authority, dated Jan. 5, 2022, from corresponding International Application No. PCT/US2021/050497. cited by applicant

Written Opinion of the International Searching Authority, dated Jan. 7, 2019, from corresponding International Application No. PCT/US2018/055772. cited by applicant

Written Opinion of the International Searching Authority, dated Jun. 1, 2022, from corresponding International Application No. PCT/US2022/016930. cited by applicant

Written Opinion of the International Searching Authority, dated Jun. 21, 2017, from corresponding International Application No. PCT/US2017/025600. cited by applicant

Written Opinion of the International Searching Authority, dated Jun. 22, 2022, from corresponding International Application No. PCT/US2022/019358. cited by applicant

Written Opinion of the International Searching Authority, dated Jun. 24, 2022, from corresponding International Application No. PCT/US2022/019882. cited by applicant

Written Opinion of the International Searching Authority, dated Jun. 6, 2017, from corresponding International Application No. PCT/US2017/025605. cited by applicant

Written Opinion of the International Searching Authority, dated Mar. 14, 2019, from corresponding International Application No. PCT/US2018/055736. cited by applicant

Written Opinion of the International Searching Authority, dated Mar. 18, 2022, from corresponding International Application No. PCT/US2022/013733. cited by applicant

Written Opinion of the International Searching Authority, dated Mar. 4, 2019, from corresponding International Application No. PCT/US2018/055773. cited by applicant

Written Opinion of the International Searching Authority, dated Mar. 4, 2019, from corresponding International Application No. PCT/US2018/055774. cited by applicant

Written Opinion of the International Searching Authority, dated May 12, 2022, from corresponding International Application No. PCT/US2022/015929. cited by applicant

Written Opinion of the International Searching Authority, dated May 17, 2022, from corresponding International Application No. PCT/US2022/015241. cited by applicant

Written Opinion of the International Searching Authority, dated May 19, 2022, from corresponding International Application No. PCT/US2022/015637. cited by applicant

Written Opinion of the International Searching Authority, dated Nov. 12, 2021, from corresponding International Application No. PCT/US2021/043481. cited by applicant

Written Opinion of the International Searching Authority, dated Nov. 19, 2018, from corresponding International Application No. PCT/US2018/046939. cited by applicant

Written Opinion of the International Searching Authority, dated Nov. 3, 2021, from corresponding International Application No. PCT/US2021/040893. cited by applicant

Written Opinion of the International Searching Authority, dated Nov. 3, 2021, from corresponding International Application No. PCT/US2021/044910. cited by applicant

Written Opinion of the International Searching Authority, dated Oct. 11, 2018, from corresponding International Application No. PCT/US2018/043975. cited by applicant

Office Action, dated Jan. 24, 2020, from corresponding U.S. Appl. No. 16/505,426. cited by applicant

Office Action, dated Jan. 24, 2020, from corresponding U.S. Appl. No. 16/700,049. cited by applicant

Office Action, dated Jan. 25, 2022, from corresponding U.S. Appl. No. 17/494,220. cited by applicant

Office Action, dated Jan. 27, 2020, from corresponding U.S. Appl. No. 16/656,895. cited by applicant

Office Action, dated Jan. 28, 2020, from corresponding U.S. Appl. No. 16/712,104. cited by applicant

Office Action, dated Jan. 29, 2021, from corresponding U.S. Appl. No. 17/101,106. cited by applicant

Office Action, dated Jan. 31, 2022, from corresponding U.S. Appl. No. 17/493,290. cited by applicant

Office Action, dated Jan. 4, 2019, from corresponding U.S. Appl. No. 16/159,566. cited by applicant

Office Action, dated Jan. 4, 2019, from corresponding U.S. Appl. No. 16/159,628. cited by applicant

Office Action, dated Jan. 4, 2021, from corresponding U.S. Appl. No. 17/013,756. cited by applicant

Office Action, dated Jan. 4, 2022, from corresponding U.S. Appl. No. 17/480,377. cited by applicant

Office Action, dated Jan. 7, 2020, from corresponding U.S. Appl. No. 16/572,182. cited by applicant

Office Action, dated Jan. 7, 2022, from corresponding U.S. Appl. No. 17/387,421. cited by applicant

Office Action, dated Jul. 13, 2021, from corresponding U.S. Appl. No. 17/306,496. cited by applicant

Office Action, dated Jul. 15, 2021, from corresponding U.S. Appl. No. 17/020,275. cited by applicant

Office Action, dated Jul. 18, 2019, from corresponding U.S. Appl. No. 16/410,762. cited by applicant

Office Action, dated Jul. 19, 2021, from corresponding U.S. Appl. No. 17/316,179. cited by applicant

Office Action, dated Jul. 21, 2017, from corresponding U.S. Appl. No. 15/256,430. cited by applicant

Office Action, dated Jul. 21, 2021, from corresponding U.S. Appl. No. 16/901,654. cited by applicant

Office Action, dated Jul. 23, 2019, from corresponding U.S. Appl. No. 16/436,616. cited by applicant

Office Action, dated Jul. 24, 2020, from corresponding U.S. Appl. No. 16/404,491. cited by applicant

Office Action, dated Jul. 27, 2020, from corresponding U.S. Appl. No. 16/595,342. cited by applicant

Office Action, dated Jul. 27, 2022, from corresponding U.S. Appl. No. 17/831,713. cited by applicant

Office Action, dated Jul. 28, 2022, from corresponding U.S. Appl. No. 16/925,550. cited by applicant

Office Action, dated Jul. 7, 2022, from corresponding U.S. Appl. No. 17/370,650. cited by applicant

Office Action, dated Jun. 1, 2020, from corresponding U.S. Appl. No. 16/862,952. cited by applicant

Office Action, dated Jun. 1, 2022, from corresponding U.S. Appl. No. 17/306,496. cited by applicant

Office Action, dated Jun. 14, 2022, from corresponding U.S. Appl. No. 17/346,586. cited by applicant

Office Action, dated Jun. 16, 2022, from corresponding U.S. Appl. No. 17/689,683. cited by applicant

Office Action, dated Jun. 24, 2019, from corresponding U.S. Appl. No. 16/410,336. cited by applicant

Office Action, dated Jun. 24, 2021, from corresponding U.S. Appl. No. 17/234,205. cited by applicant



## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/573,999, filed Jan. 12, 2022, which is a continuation of U.S. patent application Ser. No. 17/334,909, filed May 31, 2021, now U.S. Pat. No. 11,227,247, issued Jan. 18, 2022, which is a continuation-in-part of U.S. patent application Ser. No. 17/086,732, filed Nov. 2, 2020, now U.S. Pat. No. 11,023,842, issued Jun. 1, 2021, which claims priority from U.S. Provisional Patent Application Ser. No. 62/929,583, filed Nov. 1, 2019, and is also a continuation-in-part of U.S. patent application Ser. No. 16/808,503, filed Mar. 4, 2020, now U.S. Pat. No. 10,885,485, issued Jan. 5, 2021, which claims priority from U.S. Provisional Patent Application Ser. No. 62/813,584, filed Mar. 4, 2019, and is also a continuation-in-part of U.S. patent application Ser. No. 16/714,355, filed Dec. 13, 2019, now U.S. Pat. No. 10,692,033, issued Jun. 23, 2020, which is a continuation of U.S. patent application Ser. No. 16/403,358, filed May 3, 2019, now U.S. Pat. No. 10,510,031, issued Dec. 17, 2019, which is a continuation of U.S. patent application Ser. No. 16/159,634, filed Oct. 13, 2018, now U.S. Pat. No. 10,282,692, issued May 7, 2019, which claims priority from U.S. Provisional Patent Application Ser. No. 62/572,096, filed Oct. 13, 2017 and U.S. Provisional Patent Application Ser. No. 62/728,435, filed Sep. 7, 2018, and is also a continuation-in-part of U.S. patent application Ser. No. 16/055,083, filed Aug. 4, 2018, now U.S. Pat. No. 10,289,870, issued May 14, 2019, which claims priority from U.S. Provisional Patent Application Ser. No. 62/547,530, filed Aug. 18, 2017, and is also a continuation-in-part of U.S. patent application Ser. No. 15/996,208, filed Jun. 1, 2018, now U.S. Pat. No. 10,181,051, issued Jan. 15, 2019, which claims priority from U.S. Provisional Patent Application Ser. No. 62/537,839, filed Jul. 27, 2017, and is also a continuation-in-part of U.S. patent application Ser. No. 15/853,674, filed Dec. 22, 2017, now U.S. Pat. No. 10,019,597, issued Jul. 10, 2018, which claims priority from U.S. Provisional Patent Application Ser. No. 62/541,613, filed Aug. 4, 2017, and is also a continuation-in-part of U.S. patent application Ser. No. 15/619,455, filed Jun. 10, 2017, now U.S. Pat. No. 9,851,966, issued Dec. 26, 2017, which is a continuation-in-part of U.S. patent application Ser. No. 15/254,901, filed Sep. 1, 2016, now U.S. Pat. No. 9,729,583, issued Aug. 8, 2017, which claims priority from: (1) U.S. Provisional Patent Application Ser. No. 62/360,123, filed Jul. 8, 2016; (2) U.S. Provisional Patent Application Ser. No. 62/353,802, filed Jun. 23, 2016; and (3) U.S. Provisional Patent Application Ser. No. 62/348,695, filed Jun. 10, 2016. The disclosures of all of the above patent applications are hereby incorporated herein by reference in their entirety.

## TECHNICAL FIELD

(1) This disclosure relates to a data processing system and methods for retrieving data regarding a plurality of privacy campaigns, and for using that data to assess a relative risk associated with the data privacy campaign, provide an audit schedule for each campaign, and electronically display campaign information.

## BACKGROUND

(2) Over the past years, privacy and security policies, and related operations have become increasingly important. Breaches in security, leading to the unauthorized access of personal data (which may include sensitive personal data) have become more frequent among companies and other organizations of all sizes. Such personal data may include, but is not limited to, personally identifiable information (PII), which may be information that directly (or indirectly) identifies an individual or entity. Examples of PII include names, addresses, dates of birth, social security numbers, and biometric identifiers such as a person's fingerprints or picture. Other personal data may include, for example, customers' Internet browsing habits, purchase history, or even their preferences (e.g., likes and dislikes, as provided or obtained through social media).

(3) Many organizations that obtain, use, and transfer personal data, including sensitive personal data, have begun to address these privacy and security issues. To manage personal data, many companies have attempted to implement operational policies and processes that comply with legal requirements, such as Canada's Personal Information Protection and Electronic Documents Act (PIPEDA) or the U.S.'s Health Insurance Portability and Accountability Act (HIPPA) protecting a patient's medical information. Many regulators recommend conducting privacy impact assessments, or data protection risk assessments along with data inventory mapping. For example, the GDPR requires data protection impact assessments. Additionally, the United Kingdom ICO's office provides guidance around privacy impact assessments. The OPC in Canada recommends certain personal information inventory practices, and the Singapore PDPA specifically mentions personal data inventory mapping.

(4) In implementing these privacy impact assessments, an individual may provide incomplete or incorrect information regarding personal data to be collected, for example, by new software, a new device, or a new business effort, for example, to avoid being prevented from collecting that personal data, or to avoid being subject to more frequent or more detailed privacy audits. In light of the above, there is currently a need for improved systems and methods for monitoring compliance with corporate privacy policies and applicable privacy laws in order to reduce a likelihood that an individual will successfully "game the system" by providing incomplete or incorrect information regarding current or future uses of personal data.

(5) Organizations that obtain, use, and transfer personal data often work with other organizations ("vendors") that provide services and/or products to the organizations. Organizations working with vendors may be responsible for ensuring that any personal data to which their vendors may have access is handled properly. However, organizations may have limited control over vendors and limited insight into their internal policies and procedures. Therefore, there is currently a need for improved systems and methods that help organizations ensure that their vendors handle personal data properly.

(6) Many organizations offer multiple services to customers and other users. Because each such service may use personal data (e.g., collect personal data, store personal data, retain personal data, etc.) in a different way than another such service, different privacy policies may apply to different services offered by an organization. Moreover, the geographical location of users of such services may vary, which may also affect the privacy policies that may apply to each such service. Therefore, there is currently a need for improved systems and methods of determining the applicable set of privacy policies for a particular combination of a user and a service.

## SUMMARY

(7) In accordance with various aspects, a method is provided that comprises: analyzing, by computing hardware, browser data associated with a browser application displaying a website to determine a user parameter identifying a geographical location of a user device executing the browser application; receiving, by the computing hardware, product or service information provided via a product or service parameter input displayed on the website; determining, by the computing hardware and based on the product or service information, a product or service parameter, wherein the product or service parameter comprises at least one of a particular product or service being provided or offered via the website, an entity offering or providing the particular product or service, a geographical location of the entity offering or providing the particular product or service, a type of the particular product or service, a subgroup of the entity offering or providing the particular product or service, or a geographical location of a computing system configured for at least one of hosting, providing, or facilitating access to the website; identifying, by the computing hardware and based on the website, a plurality of privacy policy rules; analyzing, by the computing hardware, the plurality of privacy policy rules using the user parameter and the product or service parameter to identify an applicable privacy policy rule from the plurality of privacy policy rules; identifying, by the computing hardware and based on the applicable privacy policy rule, an applicable privacy policy; configuring, by the computing hardware, a navigation element displayed on the website to navigate to a display element that presents the applicable privacy policy; receiving, by the computing hardware, an indication of a selection of the navigation element; and responsive to receiving the indication, transmitting, by the computing hardware, an instruction to the browser application causing the browser application to retrieve and present the display element.

(8) In some aspects, analyzing the plurality of privacy policy rules using the user parameter and the product or service parameter to identify the applicable privacy policy rule from the plurality of privacy policy rules is based on the applicable privacy policy rule having a priority higher than a second privacy policy rule of the plurality of privacy policy rules that is also applicable based on the user parameter and the product or service

parameter. In some aspects, analyzing the plurality of privacy policy rules is based on the applicable privacy policy rule having the priority higher than the second privacy policy rule is performed due to a conflict between the applicable privacy policy rule and the second privacy policy rule.

(9) In some aspects, the method further comprises assigning, by the computing hardware based on the user parameter, a first numeric value to the applicable privacy policy and a second numeric value to the second privacy policy rule, wherein the applicable privacy policy rule has the priority higher than the second privacy policy rule due to the first numeric value being higher than the second numeric value. In some aspects, receiving the product or service information comprises receiving a request sent by computing code included in a web page of the website and executing on the user device. In some aspects, the user parameter further comprises at least one of a language of a user of the user device, a territory of residence of the user, or a citizenship of the user. In some aspects, the user parameter further comprises a language of a user of the user device and the applicable privacy policy is provided in the language.

(10) In accordance with various aspects, a system comprising a non-transitory computer-readable medium storing instructions and a processing device communicatively coupled to the non-transitory computer-readable medium is provided. Accordingly, the processing device is configured to execute the instructions and thereby perform operations comprising: receiving a user parameter identifying a geographical location of a user device executing a user interface; determining product or service information based on user activity on the user interface; determining, based on the product or service information, a product or service parameter, wherein the product or service parameter comprises at least one of a particular product or service being provided or offered via the user interface, an entity offering or providing the particular product or service, a geographical location of the entity offering or providing the particular product or service, a type of the particular product or service, a subgroup of the entity offering or providing the particular product or service, or a geographical location of a computing system configured for at least one of hosting, providing, or facilitating access to the user interface; identifying a plurality of privacy policy rules; analyzing the plurality of privacy policy rules using the user parameter and the product or service parameter to identify an applicable privacy policy rule from the plurality of privacy policy rules; identifying, based on the applicable privacy policy rule, an applicable privacy policy; configuring a navigation element displayed on the user interface to navigate to a display element that presents the applicable privacy policy; receiving an indication of a selection of the navigation element; and responsive to receiving the indication, causing the user interface to present the display element.

(11) In some aspects, the user interface comprises at least one of a web page associated with a website or a display interface provided in a software application. In some aspects, identifying the plurality of privacy policy rules is based on at least one of the user parameter, the product or service parameter, the user interface, or the entity. In some aspects, the user parameter further comprises a language of a user of the user device and the applicable privacy policy is provided in the language.

(12) In some aspects, analyzing the plurality of privacy policy rules using the user parameter and the product or service parameter to identify the applicable privacy policy rule from the plurality of privacy policy rules is based on the applicable privacy policy rule having a priority higher than a second privacy policy rule of the plurality of privacy policy rules that is also applicable based on the user parameter and the product or service parameter. In some aspects, analyzing the plurality of privacy policy rules is based on the applicable privacy policy rule having the priority higher than the second privacy policy rule is performed due to a conflict between the applicable privacy policy rule and the second privacy policy rule. In some aspects, the operations further comprise assigning, based on the user parameter, a first numeric value to the applicable privacy policy and a second numeric value to the second privacy policy rule, wherein the applicable privacy policy rule has the priority higher than the second privacy policy rule due to the first numeric value being higher than the second numeric value.

(13) In accordance with various aspects, a non-transitory computer-readable medium storing computer-executable instructions is provided. Accordingly, the computer-executable instructions, when executed by one or more processing devices, configure the one or more processing devices to perform operations that comprise: receiving a user parameter identifying a geographical location of a user device executing a user interface; determining product or service information based on user activity on the user interface; determining, based on the product or service information, a product or service parameter, wherein the product or service parameter comprises at least one of a particular product or service being provided or offered via the user interface, an entity offering or providing the particular product or service, a geographical location of the entity offering or providing the particular product or service, a type of the particular product or service, a subgroup of the entity offering or providing the particular product or service, or a geographical location of a computing system configured for at least one of hosting, providing, or facilitating access to the user interface; identifying a plurality of policy rules; analyzing the plurality of policy rules using the user parameter and the product or service parameter to identify an applicable policy rule from the plurality of policy rules; identifying, based on the applicable policy rule, an applicable policy; configuring a navigation element displayed on the user interface to navigate to a display element that presents the applicable policy; receiving an indication of a selection of the navigation element; and responsive to receiving the indication, causing the user interface to present the display element.

(14) In some aspects, the user interface comprises at least one of a web page associated with a website or a display interface provided in a software application. In some aspects, identifying the plurality of policy rules is based on at least one of the user parameter, the product or service parameter, the user interface, or the entity. In some aspects, the operations further comprise assigning, based on the user parameter, a first numeric value to the applicable policy and a second numeric value to the second policy rule, wherein the applicable policy rule has the priority higher than the second policy rule due to the first numeric value being higher than the second numeric value.

(15) In some aspects, analyzing the plurality of policy rules using the user parameter and the product or service parameter to identify the applicable policy rule from the plurality of policy rules is based on the applicable policy rule having a priority higher than a second policy rule of the plurality of policy rules that is also applicable based on the user parameter and the product or service parameter. In some aspects, analyzing the plurality of policy rules is based on the applicable policy rule having the priority higher than the second policy rule is performed due to a conflict between the applicable policy rule and the second policy rule.

(16) The details of one or more embodiments of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter may become apparent from the description, the drawings, and the claims.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various embodiments of a system and method for operationalizing privacy compliance and assessing risk of privacy campaigns are described below. In the course of this description, reference will be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
- (2) FIG. 1 is a diagram illustrating an exemplary network environment in which the present systems and methods for operationalizing privacy compliance may operate.
- (3) FIG. 2 is a schematic diagram of a computer (such as the server 120; or user device 140, 150, 160, 170, 180, 190; and/or such as the vendor risk scanning server 1100, or one or more remote computing devices 1500) that is suitable for use in various embodiments;
- (4) FIG. 3 is a diagram illustrating an example of the elements (e.g., subjects, owner, etc.) that may be involved in privacy compliance.
- (5) FIG. 4 is a flow chart showing an example of a process performed by the Main Privacy Compliance Module.
- (6) FIG. 5 is a flow chart showing an example of a process performed by the Risk Assessment Module.
- (7) FIG. 6 is a flow chart showing an example of a process performed by the Privacy Audit Module.

(8) FIG. 7 is a flow chart showing an example of a process performed by the Data Flow Diagram Module.

(9) FIG. 8 is an example of a graphical user interface (GUI) showing a dialog that allows for the entry of description information related to a privacy campaign.

(10) FIG. 9 is an example of a notification, generated by the system, informing a business representative (e.g., owner) that they have been assigned to a particular privacy campaign.

(11) FIG. 10 is an example of a GUI showing a dialog allowing entry of the type of personal data that is being collected for a campaign.

(12) FIG. 11 is an example of a GUI that shows a dialog that allows collection of campaign data regarding the subject from which personal data was collected.

(13) FIG. 12 is an example of a GUI that shows a dialog for inputting information regarding where the personal data related to a campaign is stored.

(14) FIG. 13 is an example of a GUI that shows information regarding the access of personal data related to a campaign.

(15) FIG. 14 is an example of an instant messaging session overlaid on top of a GUI, wherein the GUI contains prompts for the entry or selection of campaign data.

(16) FIG. 15 is an example of a GUI showing an inventory page.

(17) FIG. 16 is an example of a GUI showing campaign data, including a data flow diagram.

(18) FIG. 17 is an example of a GUI showing a web page that allows editing of campaign data.

(19) FIGS. 18A-18B depict a flow chart showing an example of a process performed by the Data Privacy Compliance Module.

(20) FIGS. 19A-19B depict a flow chart showing an example of a process performed by the Privacy Assessment Report Module.

(21) FIG. 20 is a flow chart showing an example of a process performed by the Privacy Assessment Monitoring Module according to particular embodiments.

(22) FIG. 21 is a flow chart showing an example of a process performed by the Privacy Assessment Modification Module.

(23) FIG. 22 depicts an exemplary vendor risk scanning system according to particular embodiments.

(24) FIG. 23 is a flow chart showing an example of a process performed by the Vendor Incident Notification Module according to particular embodiments.

(25) FIG. 24 is a flow chart showing an example of a process performed by the Vendor Compliance Demonstration Module according to particular embodiments.

(26) FIG. 25 is a flow chart showing an example of a process performed by the Vendor Information Update Module according to particular embodiments.

(27) FIG. 26 is a flow chart showing an example of a process performed by the Vendor Privacy Risk Score Calculation Module according to particular embodiments.

(28) FIG. 27 is a flow chart showing an example of a process performed by the Vendor Privacy Risk Determination Module according to particular embodiments.

(29) FIG. 28 is a flow chart showing an example of a process performed by the Dynamic Vendor Privacy Training Material Generation Module according to particular embodiments.

(30) FIG. 29 is a flow chart showing an example of a process performed by the Dynamic Vendor Privacy Training Material Update Module according to particular embodiments.

(31) FIG. 30 is an example of a GUI showing a listing of vendors.

(32) FIG. 31 is an example of a GUI showing incident details.

(33) FIG. 32 is another example of a GUI showing incident details.

(34) FIG. 33 is an example of a GUI showing a vendor-related task.

(35) FIG. 34 is an example of a GUI showing a listing of vendor-related tasks.

(36) FIG. 35 is another example of a GUI showing a listing of vendors.

(37) FIG. 36 is another example of a GUI showing a listing of vendors.

(38) FIG. 37 is an example of a GUI allowing entry of vendor information.

(39) FIG. 38 is an example of a GUI showing a listing of vendor-related documents and allowing the addition of vendor-related documents.

(40) FIG. 39 is an example of a GUI showing details of vendor-related documents.

(41) FIG. 40 is an example of a GUI showing the analysis of vendor information.

(42) FIG. 41 is an example of a GUI showing an overview of vendor information.

(43) FIG. 42 is an example of a GUI showing vendor information details.

(44) FIG. 43 is an example of a GUI for requesting a vendor assessment.

(45) FIG. 44 is an example of a GUI indicating the detection of a vendor assessment.

(46) FIG. 45 is an example of a GUI allowing entry of vendor assessment information.

(47) FIG. 46 is another example of a GUI allowing entry of vendor assessment information.

(48) FIG. 47 is an example of a GUI showing a listing of vendors and an indication of a change in vendor information.

(49) FIG. 48 is another example of a GUI showing a listing of vendors.

(50) FIG. 49 is another example of a GUI showing an overview of vendor information.

(51) FIG. 50 is another example of a GUI showing vendor information details.

(52) FIG. 51 is another example of a GUI showing a listing of vendors.

(53) FIG. 52 is another example of a GUI showing an overview of vendor information.

(54) FIG. 53 is another example of a GUI showing a listing of vendors and an indication of a change in vendor information.

(55) FIG. 54 illustrates an exemplary data structure representing an aspect of an ontology that may be used to determine disclosure requirements for various territories according to various embodiments.

(56) FIG. 55 is a flow chart showing an example of a process performed by the Disclosure Compliance Module according to particular embodiments.

(57) FIG. 56 is an example of a GUI indicating territories that require notification of a data breach.

(58) FIG. 57 is an example of a GUI indicating data breach notification details for a particular territory.

(59) FIG. 58 illustrates an exemplary data structure representing an aspect of an ontology that may be used to determine compliance with various privacy standards and regulations according to various embodiments.

(60) FIG. 59 is a flow chart showing an example of a process performed by the Privacy Standard Compliance Module according to particular embodiments.

(61) FIG. 60 illustrates an exemplary data structure representing an aspect of an ontology that may be used to determine an entity's compliance readiness for various and regions territories according to various embodiments.

(62) FIG. 61 is a flow chart showing an example of a process performed by the Global Readiness Assessment Module according to particular embodiments.

(63) FIG. 62 is an example of a GUI allowing user selection of territories and regions for compliance readiness assessment.

(64) FIG. 63 is an example of a GUI showing user selection of territories and regions for compliance readiness assessment.

(65) FIG. 64 is an example of a GUI showing compliance details for regulations associated with a territory or region selected for compliance



readiness assessment.

(66) FIG. 65 is an example of a GUI showing the results of a compliance readiness assessment.

(67) FIG. 66 is a flow chart showing an example of a process performed by the Disclosure Prioritization Module according to particular embodiments.

(68) FIG. 67 is a flow chart showing an example of a process performed by the Data Breach Reporting Module according to particular embodiments.

(69) FIG. 68 is a flow chart showing an example of a process performed by the Regulatory Conflict Resolution Module according to particular embodiments.

(70) FIG. 69 is an example of a GUI allowing user entry of data breach information for disclosure requirement analysis and data breach reporting.

(71) FIG. 70 is an example of another GUI allowing user entry of data breach information for disclosure requirement analysis and data breach reporting.

(72) FIG. 71 is an example of a GUI showing a heat map of jurisdictions in which reporting of a data breach may be required and associated reporting tasks.

(73) FIG. 72 is an example of a GUI showing a map of jurisdictions in which reporting of a data breach may be required and associated reporting tasks.

(74) FIG. 73 is an example of a GUI showing a listing of data breach reporting tasks.

(75) FIG. 74 is an example of a GUI allowing user entry of information as response to questions in a master questionnaire.

(76) FIG. 75 is a flow chart showing an example of a process performed by the Privacy Policy Bundle Determination Module according to particular embodiments.

(77) FIG. 76 is an example of a GUI allowing a user to define parameters that may be associated with a particular privacy policy rule.

(78) FIG. 77 is an example of a GUI allowing a user to define a particular privacy policy rule.

(79) FIG. 78 is an example of a GUI allowing a user to define a particular privacy policy rule group.

(80) FIG. 79 is an example of a GUI illustrating details of a particular privacy policy rule group.

(81) FIG. 80 is an example of a GUI allowing a user to refine a request for privacy policy information by providing criteria that can be used to determine the applicable privacy policy.

#### DETAILED DESCRIPTION

(82) Various embodiments now will be described more fully hereinafter with reference to the accompanying drawings. It should be understood that the invention may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. Like numbers refer to like elements throughout.

#### Overview

(83) According to exemplary embodiments, a system for operationalizing privacy compliance is described herein. The system may be comprised of one or more servers and client computing devices that execute software modules that facilitate various functions.

(84) A Main Privacy Compliance Module is operable to allow a user to initiate the creation of a privacy campaign (i.e., a business function, system, product, technology, process, project, engagement, initiative, campaign, etc.), that may utilize personal data collected from one or more persons or entities). The personal data may contain PII that may be sensitive personal data. The user can input information such as the name and description of the campaign. The user may also select whether he/she will take ownership of the campaign (i.e., be responsible for providing the information needed to create the campaign and oversee the conducting of privacy audits related to the campaign), or assign the campaign to one or more other persons. The Main Privacy Compliance Module can generate a sequence or series of GUI windows that facilitate the entry of campaign data representative of attributes related to the privacy campaign (e.g., attributes that might relate to the description of the personal data, what personal data is collected, whom the data is collected from, the storage of the data, and access to that data).

(85) Based on the information input, a Risk Assessment Module may be operable to take into account Weighting Factors and Relative Risk Ratings associated with the campaign in order to calculate a numerical Risk Level associated with the campaign, as well as an Overall Risk Assessment for the campaign (i.e., low-risk, medium risk, or high risk). The Risk Level may be indicative of the likelihood of a breach involving personal data related to the campaign being compromised (i.e., lost, stolen, accessed without authorization, inadvertently disclosed, maliciously disclosed, etc.). An inventory page can visually depict the Risk Level for one or more privacy campaigns.

(86) After the Risk Assessment Module has determined a Risk Level for a campaign, a Privacy Audit Module may be operable to use the Risk Level to determine an audit schedule for the campaign. The audit schedule may be editable, and the Privacy Audit Module also facilitates the privacy audit process by sending alerts when a privacy audit is impending, or sending alerts when a privacy audit is overdue.

(87) The system may also include a Data Flow Diagram Module for generating a data flow diagram associated with a campaign. An exemplary data flow diagram displays one or more shapes representing the source from which data associated with the campaign is derived, the destination (or location) of that data, and which departments or software systems may have access to the data. The Data Flow Diagram Module may also generate one or more security indicators for display. The indicators may include, for example, an “eye” icon to indicate that the data is confidential, a “lock” icon to indicate that the data, and/or a particular flow of data, is encrypted, or an “unlocked lock” icon to indicate that the data, and/or a particular flow of data, is not encrypted. Data flow lines may be colored differently to indicate whether the data flow is encrypted or unencrypted.

(88) The system also provides for a Communications Module that facilitates the creation and transmission of notifications and alerts (e.g., via email). The Communications Module may also instantiate an instant messaging session and overlay the instant messaging session over one or more portions of a GUI in which a user is presented with prompts to enter or select information.

(89) In particular embodiments, a vendor risk scanning system is configured to scan one or more webpages associated with a particular vendor (e.g., provider of particular software, particular entity, etc.) in order to identify one or more vendor attributes. In particular embodiments, the system may be configured to scan the one or more web pages to identify one or more vendor attributes such as, for example: (1) one or more security certifications that the vendor does or does not have (e.g., ISO 27001, SOC II Type 2, etc.); (2) one or more awards and/or recognitions that the vendor has received (e.g., one or more security awards); (3) one or more security policies and/or 3.sup.rd party vendor parties; (4) one or more privacy policies and/or cookie policies for the one or more webpages; (5) one or more key partners or potential sub processors of one or more services associated with the vendor; and/or (6) any other suitable vendor attribute. Other suitable vendor attributes may include, for example, membership in a Privacy Shield, use of Standardized Information Gathering (SIG), etc.

(90) In various embodiments, the system is configured to scan the one or more webpages by: (1) scanning one or more pieces of computer code associated with the one or more webpages (e.g., HTML, Java, etc.); (2) scanning one or more contents of the one or more webpages (e.g., using one or more natural language processing techniques); (3) scanning for one or more particular images on the one or more webpages (e.g., one or more images that indicate membership in a particular organization, receipt of a particular award etc.; and/or (4) using any other suitable scanning technique. The system may, for example, identify one or more image hosts of one or more images identified on the website, analyze the contents of a particular identified privacy or cookie policy that is displayed on the one or more webpages, etc. The system may, for example, be configured to automatically detect the one or more vendor attributes described above.

(91) In various embodiments, the system may, for example: (1) analyze the one or more vendor attributes; and (2) calculate a risk rating for the vendor based at least in part on the one or more vendor attributes. In particular embodiments, the system is configured to automatically assign a suitable weighting factor to each of the one or more vendor attributes when calculating the risk rating. In particular embodiments, the system is



configured to analyze one or more pieces of vendor's published applications available to one or more customers for download via the one or more webpages to detect one or more privacy disclaimers associated with the published applications. The system may then, for example, be configured to use one or more text matching techniques to determine whether the one or more privacy disclaimers contain one or more pieces of language required by one or more prevailing industry or legal requirements related to data privacy. The system may, for example, be configured to assign a relatively low risk score to a vendor whose software (e.g., and/or webpages) includes required privacy disclaimers, and configured to assign a relatively high risk score to a vendor whose one or more webpages do not include such disclaimers.

(92) In another example, the system may be configured to analyze one or more websites associated with a particular vendor for one or more privacy notices, one or more blog posts, one or more preference centers, and/or one or more control centers. The system may, for example, calculate the vendor risk score based at least in part on a presence of one or more suitable privacy notices, one or more contents of one or more blog posts on the vendor site (e.g., whether the vendor site has one or more blog posts directed toward user privacy), a presence of one or more preference or control centers that enable visitors to the site to opt in or out of certain data collection policies (e.g., cookie policies, etc.), etc.

(93) In particular other embodiments, the system may be configured to determine whether the particular vendor holds one or more security certifications. The one or more security certifications may include, for example: (1) system and organization control (SOC); (2) International Organization for Standardization (ISO); (3) Health Insurance Portability and Accountability ACT (HIPPA); (4) etc. In various embodiments, the system is configured to access one or more public databases of security certifications to determine whether the particular vendor holds any particular certification. The system may then determine the privacy awareness score based on whether the vendor holds one or more security certifications (e.g., the system may calculate a relatively higher score depending on one or more particular security certifications held by the vendor). The system may be further configured to scan a vendor website for an indication of the one or more security certifications. The system may, for example, be configured to identify one or more images indicated receipt of the one or more security certifications, etc.

(94) In still other embodiments, the system is configured to analyze one or more social networking sites (e.g., LinkedIn, Facebook, etc.) and/or one or more business related job sites (e.g., one or more job-posting sites, one or more corporate websites, etc.) or other third-party websites that are associated with the vendor (e.g., but not maintained by the vendor). The system may, for example, use social networking and other data to identify one or more employee titles of the vendor, one or more job roles for one or more employees of the vendor, one or more job postings for the vendor, etc. The system may then analyze the one or more job titles, postings, listings, roles, etc. to determine whether the vendor has or is seeking one or more employees that have a role associated with data privacy or other privacy concerns. In this way, the system may determine whether the vendor is particularly focused on privacy or other related activities. The system may then calculate a privacy awareness score and/or risk rating based on such a determination (e.g., a vendor that has one or more employees whose roles or titles are related to privacy may receive a relatively higher privacy awareness score).

(95) In particular embodiments, the system may be configured to calculate the privacy awareness score using one or more additional factors such as, for example: (1) public information associated with one or more events that the vendor is attending; (2) public information associated with one or more conferences that the vendor has participated in or is planning to participate in; (3) etc. In some embodiments, the system may calculate a privacy awareness score based at least in part on one or more government relationships with the vendor. For example, the system may be configured to calculate a relatively high privacy awareness score for a vendor that has one or more contracts with one or more government entities (e.g., because an existence of such a contract may indicate that the vendor has passed one or more vetting requirements imposed by the one or more government entities).

(96) In any embodiment described herein, the system may be configured to assign, identify, and/or determine a weighting factor for each of a plurality of factors used to determine a risk rating score for a particular vendor. For example, when calculating the rating, the system may assign a first weighting factor to whether the vendor has one or more suitable privacy notices posted on the vendor website, a second weighting factor to whether the vendor has one or more particular security certifications, etc. The system may, for example, assign one or more weighting factors using any suitable technique described herein with relation to risk rating determination. In some embodiments, the system may be configured to receive the one or more weighting factors (e.g., from a user). In other embodiments, the system may be configured to determine the one or more weighting factors based at least in part on a type of the factor.

(97) In any embodiment described herein, the system may be configured to determine an overall risk rating for a particular vendor (e.g., particular piece of vendor software) based in part on the privacy awareness score. In other embodiments, the system may be configured to determine an overall risk rating for a particular vendor based on the privacy awareness rating in combination with one or more additional factors (e.g., one or more additional risk factors described herein). In any such embodiment, the system may assign one or more weighting factors or relative risk ratings to each of the privacy awareness score and other risk factors when calculating an overall risk rating. The system may then be configured to provide the risk score for the vendor, software, and/or service for use in calculating a risk of undertaking a particular processing activity that utilizes the vendor, software, and/or service (e.g., in any suitable manner described herein).

(98) In a particular example, the system may be configured to identify whether the vendor is part of a Privacy Shield arrangement. In particular, a privacy shield arrangement may facilitate monitoring of an entity's compliance with one or more commitments and enforcement of those commitments under the privacy shield. In particular, an entity entering a privacy shield arrangement may, for example: (1) be obligated to publicly commit to robust protection of any personal data that it handles; (2) be required to establish a clear set of safeguards and transparency mechanisms on who can access the personal data it handles; and/or (3) be required to establish a redress right to address complaints about improper access to the personal data.

(99) In a particular example of a privacy shield, a privacy shield between the United States and Europe may involve, for example: (1) establishment of responsibility by the U.S. Department of Commerce to monitor an entity's compliance (e.g., a company's compliance) with its commitments under the privacy shield; and (2) establishment of responsibility of the Federal Trade Commission having enforcement authority over the commitments. In a further example, the U.S. Department of Commerce may designate an ombudsman to hear complaints from Europeans regarding U.S. surveillance that affects personal data of Europeans.

(100) In some embodiments, the one or more regulations may include a regulation that allows data transfer to a country or entity that participates in a safe harbor and/or privacy shield as discussed herein. The system may, for example, be configured to automatically identify a transfer that is subject to a privacy shield and/or safe harbor as 'low risk.' In this example, U.S. Privacy Shield members may be maintained in a database of privacy shield members (e.g., on one or more particular webpages such as at [www.privacyshield.gov](http://www.privacyshield.gov)). The system may be configured to scan such webpages to identify whether the vendor is part of the privacy shield.

(101) In particular embodiments, the system may be configured to monitor the one or more websites (e.g., one or more webpages) to identify one or more changes to the one or more vendor attributes. For example, a vendor may update a privacy policy for the website (e.g., to comply with one or more legal or policy changes). In some embodiments, a change in a privacy policy may modify a relationship between a website and its users. In such embodiments, the system may be configured to: (1) determine that a particular website has changed its privacy policy; and (2) perform a new scan of the website in response to determining the change. The system may, for example, scan a website's privacy policy at a first time and a second time to determine whether a change has occurred. The system may be configured to analyze the change in privacy policy to determine whether to modify the calculated risk rating for the vendor (e.g., based on the change).

(102) The system may, for example, be configured to continuously monitor for one or more changes. In other embodiments, the system may be configured to scan for one or more changes according to a particular schedule (e.g., hourly, daily, weekly, or any other suitable schedule.). For example, the system may be configured to scan the one or more webpages on an ongoing basis to determine whether the one or more vendor

attributes have changed (e.g., if the vendor did not renew its Privacy Shield membership, lost its ISO certification, etc.).

#### Technical Contributions of Various Embodiments

(103) An entity that handles (e.g., collects, receives, transmits, stores, processes, shares, and/or the like) sensitive and/or personal information associated with particular individuals (e.g., personally identifiable information (PII) data, sensitive data, personal data, etc.) may be subject to various laws, regulations, and/or requirements regarding the handling of such personal data. As part of such laws, regulations, and/or requirements, an entity may be required to create, enforce, and make available to users one or more privacy policies. Because the applicable laws, regulations, and/or requirements may vary based on the jurisdiction of the entity, the jurisdiction of a user (“data subject”), the products or services involved, and/or the jurisdiction in which user data is handled, the applicable privacy policies may differ from situation to situation. Due to the vast number of possible situations in which a privacy policy may apply and the varying requirements of each, there may be large numbers of potentially applicable privacy policies for any one particular situation. Therefore, an entity may need to ensure that it selects the most appropriate privacy policy to enforce and provide to users in any particular situation so that the entity can remain in compliance with applicable laws, regulations, and/or requirements.

(104) Accordingly, various embodiments of present disclosure overcome many of the technical challenges associated with configuring and generating a graphical user interface that presents the one or more applicable privacy policies that may be applicable to particular situation from among multiple potentially applicable privacy policies. Specifically, various embodiments of the disclosure are directed to a computational framework configured for using data detected at a user device, such as browser state data, to determine the appropriate navigational elements to present to the user in a graphical user interface in order to display one or more elements of one or more privacy policy datasets. User device data (e.g., state data) may include or indicate one or more parameters that may be used to determine one or more privacy policy datasets that apply to the user and/or user device. The system may identify one or more navigational elements that correspond to the determined one or more privacy policy datasets to configure on a graphical user interface. The system may also, or instead, identify one or more navigational elements that do not correspond to the determined one or more privacy policy datasets and may, in response, determine to exclude those one or more navigational elements from a graphical user interface presented to the user. These navigational elements included in the graphical user interface, when activated, may cause the user device to retrieve and present the corresponding one or more privacy policy datasets to the user. For example, in response to detecting the selection of a particular navigational element, the system may transmit an instruction causing a browser application executing on a user device to display a privacy policy dataset associated with that particular navigational element (see, e.g., FIG. 80 and related text). The system may determine and/or use, from user device data, one or more (e.g., any combination) of various user criteria, entity criteria, website/webpage criteria, associated product and/or service criteria, etc. as parameters to determine navigational elements and/or privacy policy datasets associated with navigational elements.

(105) In various embodiments described herein (see, e.g., FIGS. 75-79 and related text), the system (e.g., a rules engine executed by the system) may use parameters to identify a particular privacy policy rule group that may be most applicable to the situation represented by the parameters. The system can then evaluate the rules within the rule group to identify the most applicable rule. Once the most applicable rule is determined, the system may determine the appropriate navigational element to configure on a graphical user interface that, when activated by a user, causes the user device to retrieve and presents the privacy policy dataset associated with that rule to the user and/or take other actions based on the identified most applicable rule.

(106) For example, the system may organize particular privacy policy rules into groups that each associated with a particular division of an entity. Each rule may represent privacy aspects that are associated with one or more particular products associated with that division and/or types of users of such products. In various embodiments, the system may receive a user-initiated request for privacy policy information. The request may include parameters automatically derived from available data (e.g., browser data, website data, etc.) and/or parameters based on user-provided information (e.g., solicited from the user or otherwise provided by the user). The system may analyze this request to determine a most applicable privacy policy rule group based on one or more of the parameters included in the request. The system may then determine one or more applicable rules within the rule group based on one or more of the parameters included in the request. If there is more than one applicable rule, the system may use a prioritization level or score to select the most applicable rule. If there are no applicable rules, the system may select a default rule as the most applicable rule. The system may determine the appropriate navigational element to configure on a graphical user interface that, when activated by a user, causes the user device to retrieve and present, to the user, a privacy policy dataset associated with the most applicable rule.

(107) Accordingly, various embodiments of the disclosure provided herein are more effective, efficient, timely, accurate, and faster in determining the appropriate graphical user interface elements for presentation on a graphical user interface to a user to in a specific situation so that the user may obtain applicable privacy policy information. In addition, various embodiments provided herein may facilitate the identification of appropriate graphical user interface elements by using the privacy policy rule and privacy policy rule group data structures described herein. By properly identifying the most applicable graphical user interface elements for a specific situation, the various embodiments help ensure that a user can be provided with accurate privacy policy information and that the entity can remain in compliance with the applicable requirements and regulations. This is especially advantageous when an entity has many webpages and/or websites accessible from many different locations and by many different types of users. In facilitating the identification of the applicable graphical user interface elements in many various types of situations, the various embodiments of the present disclosure make major technical contributions to improving the computational efficiency and reliability of various privacy management systems and procedures for determining and providing graphical user interface elements that correspond to privacy policy datasets that are applicable to a particular user in a particular situation. This in turn translates to more computationally efficient software systems. Further detail is now provided for different aspects of various embodiments of the disclosure.

#### (108) Exemplary Technical Platforms

(109) As will be appreciated by one skilled in the relevant field, a system for operationalizing privacy compliance and assessing risk of privacy campaigns may be, for example, embodied as a computer system, a method, or a computer program product. Accordingly, various embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment, or an embodiment combining software and hardware aspects. Furthermore, particular embodiments may take the form of a computer program product stored on a computer-readable storage medium having computer-readable instructions (e.g., software) embodied in the storage medium. Various embodiments may take the form of web, mobile, wearable computer-implemented, computer software. Any suitable computer-readable storage medium may be utilized including, for example, hard disks, compact disks, DVDs, optical storage devices, and/or magnetic storage devices.

(110) Various embodiments are described below with reference to block diagrams and flowchart illustrations of methods, apparatuses (e.g., systems) and computer program products. It should be understood that each step of the block diagrams and flowchart illustrations, and combinations of steps in the block diagrams and flowchart illustrations, respectively, may be implemented by a computer executing computer program instructions. These computer program instructions may be loaded onto a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions which execute on the computer or other programmable data processing apparatus to create means for implementing the functions specified in the flowchart step or steps

(111) These computer program instructions may also be stored in a computer-readable memory that may direct a computer or other programmable data processing apparatus to function in a particular manner such that the instructions stored in the computer-readable memory produce an article of manufacture that is configured for implementing the function specified in the flowchart step or steps. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer implemented process such that the instructions that execute on the computer or other programmable apparatus provide steps for implementing the functions specified in the flowchart step or steps.

(112) According to steps of the block diagrams and flowchart illustrations support combinations of mechanisms for performing the specified functions, combinations of steps for performing the specified functions, and program instructions for performing the specified functions. It should also be understood that each step of the block diagrams and flowchart illustrations, and combinations of steps in the block diagrams and flowchart illustrations, may be implemented by special purpose hardware-based computer systems that perform the specified functions or steps, or combinations of special purpose hardware and other hardware executing appropriate computer instructions.

(113) Example System Architecture

(114) FIG. 1 is a block diagram of a System **100** according to a particular embodiment. As may be understood from this figure, the System **100** includes one or more computer networks **110**, a Server **120**, a Storage Device **130** (which may contain one or more databases of information), one or more remote client computing devices such as a tablet computer **140**, a desktop or laptop computer **150**, or a handheld computing device **160**, such as a cellular phone, browser and Internet capable set-top boxes **170** connected with a TV **180**, or even smart TVs **180** having browser and Internet capability. The client computing devices attached to the network may also include copiers/printers **190** having hard drives (a security risk since copies/prints may be stored on these hard drives). The Server **120**, client computing devices, and Storage Device **130** may be physically located in a central location, such as the headquarters of the organization, for example, or in separate facilities. The devices may be owned or maintained by employees, contractors, or other third parties (e.g., a cloud service provider). In particular embodiments, the one or more computer networks **110** facilitate communication between the Server **120**, one or more client computing devices **140**, **150**, **160**, **170**, **180**, **190**, and Storage Device **130**.

(115) The one or more computer networks **110** may include any of a variety of types of wired or wireless computer networks such as the Internet, a private intranet, a public switched telephone network (PSTN), or any other type of network. The communication link between the Server **120**, one or more client computing devices **140**, **150**, **160**, **170**, **180**, **190**, and Storage Device **130** may be, for example, implemented via a Local Area Network (LAN) or via the Internet.

(116) Example Computer Architecture Used Within the System

(117) FIG. 2 illustrates a diagrammatic representation of the architecture of a computer **200** that may be used within the System **100**, for example, as a client computer (e.g., one of computing devices **140**, **150**, **160**, **170**, **180**, **190**, shown in FIG. 1), or as a server computer (e.g., Server **120** shown in FIG. 1). In exemplary embodiments, the computer **200** may be suitable for use as a computer within the context of the System **100** that is configured to operationalize privacy compliance and assess risk of privacy campaigns. In particular embodiments, the computer **200** may be connected (e.g., networked) to other computers in a LAN, an intranet, an extranet, and/or the Internet. As noted above, the computer **200** may operate in the capacity of a server or a client computer in a client-server network environment, or as a peer computer in a peer-to-peer (or distributed) network environment. The computer **200** may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any other computer capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that computer. Further, while only a single computer is illustrated, the term “computer” shall also be taken to include any collection of computers that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

(118) An exemplary computer **200** includes a processing device **202**, a main memory **204** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **206** (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage device **218**, which communicate with each other via a bus **232**.

(119) The processing device **202** represents one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device **202** may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or processor implementing other instruction sets, or processors implementing a combination of instruction sets. The processing device **202** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **202** may be configured to execute processing logic **226** for performing various operations and steps discussed herein.

(120) The computer **200** may further include a network interface device **208**. The computer **200** also may include a video display unit **210** (e.g., a liquid crystal display (LCD) or a cathode ray tube (CRT)), an alphanumeric input device **212** (e.g., a keyboard), a cursor control device **214** (e.g., a mouse), and a signal generation device **216** (e.g., a speaker). The data storage device **218** may include a non-transitory computer-readable storage medium **230** (also known as a non-transitory computer-readable storage medium or a non-transitory computer-readable medium) on which is stored one or more sets of instructions **222** (e.g., software, software modules) embodying any one or more of the methodologies or functions described herein. The instructions **222** may also reside, completely or at least partially, within main memory **204** and/or within processing device **202** during execution thereof by computer **200**—main memory **204** and processing device **202** also constituting computer-accessible storage media. The instructions **222** may further be transmitted or received over a network **110** via network interface device **208**.

(121) While the computer-readable storage medium **230** is shown in an exemplary embodiment to be a single medium, the terms “computer-readable storage medium” and “machine-accessible storage medium” should be understood to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “computer-readable storage medium” should also be understood to include any medium that is capable of storing, encoding or carrying a set of instructions for execution by the computer and that cause the computer to perform any one or more of the methodologies of the present invention. The term “computer-readable storage medium” should accordingly be understood to include, but not be limited to, solid-state memories, optical and magnetic media, etc.

(122) Exemplary System Platform

(123) According to various embodiments, the processes and logic flows described in this specification may be performed by a system (e.g., System **100**) that includes, but is not limited to, one or more programmable processors (e.g., processing device **202**) executing one or more computer program modules to perform functions by operating on input data and generating output, thereby tying the process to a particular machine (e.g., a machine programmed to perform the processes described herein). This includes processors located in one or more of client computers (e.g., client computing devices **140**, **150**, **160**, **170**, **180**, **190** of FIG. 1). These devices connected to network **110** may access and execute one or more Internet browser-based program modules that are “served up” through the network **110** by one or more servers (e.g., server **120** of FIG. 1), and the data associated with the program may be stored on a one or more storage devices, which may reside within a server or computing device (e.g., Main Memory **204**, Static Memory **206**), be attached as a peripheral storage device to the one or more servers or computing devices, or attached to the network (e.g., Storage Device **130**).

(124) The System **100** facilitates the acquisition, storage, maintenance, use, and retention of campaign data associated with a plurality of privacy campaigns within an organization. In doing so, various aspects of the System **100** initiates and creates a plurality of individual data privacy campaign records that are associated with a variety of privacy-related attributes and assessment related meta-data for each campaign. These data elements may include: the subjects of the sensitive information, the respective person or entity responsible for each campaign (e.g., the campaign's “owner”), the location where the personal data will be stored, the entity or entities that will access the data, the parameters according to which the personal data will be used and retained, the Risk Level associated with a particular campaign (as well as assessments from which the Risk Level is calculated), an audit schedule, and other attributes and meta-data. The System **100** may also be adapted to facilitate the setup and auditing of each privacy campaign. These modules may include, for example, a Main Privacy Compliance Module, a Risk Assessment Module, a Privacy Audit Module, a Data Flow Diagram Module, a Communications Module (examples of which are described below), a Privacy Assessment Monitoring Module, and a Privacy Assessment Modification Module. It is to be understood that these are examples of modules of various embodiments, but the functionalities

performed by each module as described may be performed by more (or less) modules. Further, the functionalities described as being performed by one module may be performed by one or more other modules.

(125) A. Example Elements Related to Privacy Campaigns

(126) FIG. 3 provides a high-level visual overview of example “subjects” for particular data privacy campaigns, exemplary campaign “owners,” various elements related to the storage and access of personal data, and elements related to the use and retention of the personal data. Each of these elements may, in various embodiments, be accounted for by the System 100 as it facilitates the implementation of an organization's privacy compliance policy.

(127) As may be understood from FIG. 3, sensitive information may be collected by an organization from one or more subjects 300. Subjects may include customers whose information has been obtained by the organization. For example, if the organization is selling goods to a customer, the organization may have been provided with a customer's credit card or banking information (e.g., account number, bank routing number), social security number, or other sensitive information.

(128) An organization may also possess personal data originating from one or more of its business partners. Examples of business partners are vendors that may be data controllers or data processors (which have different legal obligations under EU data protection laws). Vendors may supply a component or raw material to the organization, or an outside contractor responsible for the marketing or legal work of the organization. The personal data acquired from the partner may be that of the partners, or even that of other entities collected by the partners. For example, a marketing agency may collect personal data on behalf of the organization, and transfer that information to the organization. Moreover, the organization may share personal data with one of its partners. For example, the organization may provide a marketing agency with the personal data of its customers so that it may conduct further research.

(129) Other subjects 300 include the organization's own employees. Organizations with employees often collect personal data from their employees, including address and social security information, usually for payroll purposes, or even prior to employment, for conducting credit checks. The subjects 300 may also include minors. It is noted that various corporate privacy policies or privacy laws may require that organizations take additional steps to protect the sensitive privacy of minors.

(130) Still referring to FIG. 3, within an organization, a particular individual (or groups of individuals) may be designated to be an “owner” of a particular campaign to obtain and manage personal data. These owners 310 may have any suitable role within the organization. In various embodiments, an owner of a particular campaign will have primary responsibility for the campaign, and will serve as a resident expert regarding the personal data obtained through the campaign, and the way that the data is obtained, stored, and accessed. As shown in FIG. 3, an owner may be a member of any suitable department, including the organization's marketing, HR, R&D, or IT department. As will be described below, in exemplary embodiments, the owner can always be changed, and owners can sub-assign other owners (and other collaborators) to individual sections of campaign data input and operations.

(131) Referring still to FIG. 3, the system may be configured to account for the use and retention 315 of personal data obtained in each particular campaign. The use and retention of personal data may include how the data is analyzed and used within the organization's operations, whether the data is backed up, and which parties within the organization are supporting the campaign.

(132) The system may also be configured to help manage the storage and access 320 of personal data. As shown in FIG. 3, a variety of different parties may access the data, and the data may be stored in any of a variety of different locations, including on-site, or in “the cloud”, i.e., on remote servers that are accessed via the Internet or other suitable network.

(133) B. Main Compliance Module

(134) FIG. 4 illustrates an exemplary process for operationalizing privacy compliance. Main Privacy Compliance Module 400, which may be executed by one or more computing devices of System 100, may perform this process. In exemplary embodiments, a server (e.g., server 140) in conjunction with a client computing device having a browser, execute the Main Privacy Compliance Module (e.g., computing devices 140, 150, 160, 170, 180, 190) through a network (network 110). In various exemplary embodiments, the Main Privacy Compliance Module 400 may call upon other modules to perform certain functions. In exemplary embodiments, the software may also be organized as a single module to perform various computer executable routines.

(135) I. Adding a Campaign

(136) The process may begin at step 405, wherein the Main Privacy Compliance Module 400 of the System 100 receives a command to add a privacy campaign. In exemplary embodiments, the user selects an on-screen button (e.g., the Add Data Flow button 1555 of FIG. 15) that the Main Privacy Compliance Module 400 displays on a landing page, which may be displayed in a graphical user interface (GUI), such as a window, dialog box, or the like. The landing page may be, for example, the inventory page 1500 below. The inventory page 1500 may display a list of one or more privacy campaigns that have already been input into the System 100. As mentioned above, a privacy campaign may represent, for example, a business operation that the organization is engaged in, or some business record, that may require the use of personal data, which may include the personal data of a customer or some other entity. Examples of campaigns might include, for example, Internet Usage History, Customer Payment Information, Call History Log, Cellular Roaming Records, etc. For the campaign “Internet Usage History,” a marketing department may need customers' on-line browsing patterns to run analytics. This might entail retrieving and storing customers' IP addresses, MAC address, URL history, subscriber ID, and other information that may be considered personal data (and even sensitive personal data). As will be described herein, the System 100, through the use of one or more modules, including the Main Privacy Compliance Module 400, creates a record for each campaign. Data elements of campaign data may be associated with each campaign record that represents attributes such as: the type of personal data associated with the campaign; the subjects having access to the personal data; the person or persons within the company that take ownership (e.g., business owner) for ensuring privacy compliance for the personal data associated with each campaign; the location of the personal data; the entities having access to the data; the various computer systems and software applications that use the personal data; and the Risk Level (see below) associated with the campaign.

(137) II. Entry of Privacy Campaign Related Information, Including Owner

(138) At step 410, in response to the receipt of the user's command to add a privacy campaign record, the Main Privacy Compliance Module 400 initiates a routine to create an electronic record for a privacy campaign, and a routine for the entry data inputs of information related to the privacy campaign. The Main Privacy Compliance Module 400 may generate one or more graphical user interfaces (e.g., windows, dialog pages, etc.), which may be presented one GUI at a time. Each GUI may show prompts, editable entry fields, check boxes, radial selectors, etc., where a user may enter or select privacy campaign data. In exemplary embodiments, the Main Privacy Compliance Module 400 displays on the graphical user interface a prompt to create an electronic record for the privacy campaign. A user may choose to add a campaign, in which case the Main Privacy Compliance Module 400 receives a command to create the electronic record for the privacy campaign, and in response to the command, creates a record for the campaign and digitally stores the record for the campaign. The record for the campaign may be stored in, for example, a Storage Device 130, or a storage device associated with the Main Privacy Compliance Module (e.g., a hard drive residing on Server 120, or a peripheral hard drive attached to Server 120).

(139) The user may be a person who works in the Chief Privacy Officer's organization (e.g., a privacy office rep, or privacy officer). The privacy officer may be the user that creates the campaign record, and enters initial portions of campaign data (e.g., “high level” data related to the campaign), for example, a name for the privacy campaign, a description of the campaign, and a business group responsible for administering the privacy operations related to that campaign (for example, though the GUI shown in FIG. 6). The Main Privacy Compliance Module 400 may also prompt the user to enter a person or entity responsible for each campaign (e.g., the campaign's “owner”). The owner may be tasked with the responsibility for ensuring or attempting to ensure that the privacy policies or privacy laws associated with personal data related to a particular privacy campaign are

being completed with. In exemplary embodiments, the default owner may be the person who initiated the creation of the privacy campaign. That owner may be a person who works in the Chief Privacy Officer's organization (e.g., a privacy office rep, or privacy officer). The initial owner of the campaign may designate someone else to be the owner of the campaign. The designee may be, for example, a representative of some business unit within the organization (a business rep). Additionally, more than one owner may be assigned. For example, the user may assign a primary business rep, and may also assign a privacy office rep as owners of the campaign.

(140) In many instances, some or most of the required information related to the privacy campaign record might not be within the knowledge of the default owner (i.e., the privacy office rep). The Main Privacy Compliance Module **400** can be operable to allow the creator of the campaign record (e.g., a privacy officer rep) to designate one or more other collaborators to provide at least one of the data inputs for the campaign data. Different collaborators, which may include the one or more owners, may be assigned to different questions, or to specific questions within the context of the privacy campaign. Additionally, different collaborators may be designated to respond to pats of questions. Thus, portions of campaign data may be assigned to different individuals.

(141) Still referring to FIG. **4**, if at step **415** the Main Privacy Compliance Module **400** has received an input from a user to designate a new owner for the privacy campaign that was created, then at step **420**, the Main Privacy Compliance Module **400** may notify that individual via a suitable notification that the privacy campaign has been assigned to him or her. Prior to notification, the Main Privacy Compliance Module **400** may display a field that allows the creator of the campaign to add a personalized message to the newly assigned owner of the campaign to be included with that notification. In exemplary embodiments, the notification may be in the form of an email message. The email may include the personalized message from the assignor, a standard message that the campaign has been assigned to him/her, the deadline for completing the campaign entry, and instructions to log in to the system to complete the privacy campaign entry (along with a hyperlink that takes the user to a GUI providing access to the Main Privacy Compliance Module **400**). Also included may be an option to reply to the email if an assigned owner has any questions, or a button that when clicked on, opens up a chat window (i.e., instant messenger window) to allow the newly assigned owner and the assignor a GUI in which they are able to communicate in real-time. An example of such a notification appears in FIG. **16** below. In addition to owners, collaborators that are assigned to input portions of campaign data may also be notified through similar processes. In exemplary embodiments, The Main Privacy Compliance Module **400** may, for example through a Communications Module, be operable to send collaborators emails regarding their assignment of one or more portions of inputs to campaign data. Or through the Communications Module, selecting the commentators button brings up one or more collaborators that are on-line (with the off-line users still able to see the messages when they are back on-line. Alerts indicate that one or more emails or instant messages await a collaborator.

(142) At step **425**, regardless of whether the owner is the user (i.e., the creator of the campaign), "someone else" assigned by the user, or other collaborators that may be designated with the task of providing one or more items of campaign data, the Main Privacy Compliance Module **400** may be operable to electronically receive campaign data inputs from one or more users related to the personal data related to a privacy campaign through a series of displayed computer-generated graphical user interfaces displaying a plurality of prompts for the data inputs. In exemplary embodiments, through a step-by-step process, the Main Privacy Campaign Module may receive from one or more users' data inputs that include campaign data like: (1) a description of the campaign; (2) one or more types of personal data to be collected and stored as part of the campaign; (3) individuals from which the personal data is to be collected; (4) the storage location of the personal data, and (5) information regarding who will have access to the personal data. These inputs may be obtained, for example, through the graphical user interfaces shown in FIGS. **8** through **13**, wherein the Main Privacy Compliance Module **400** presents on sequentially appearing GUIs the prompts for the entry of each of the enumerated campaign data above. The Main Privacy Compliance Module **400** may process the campaign data by electronically associating the campaign data with the record for the campaign and digitally storing the campaign data with the record for the campaign. The campaign data may be digitally stored as data elements in a database residing in a memory location in the server **120**, a peripheral storage device attached to the server, or one or more storage devices connected to the network (e.g., Storage Device **130**). If campaign data inputs have been assigned to one or more collaborators, but those collaborators have not input the data yet, the Main Privacy Compliance Module **400** may, for example through the Communications Module, sent an electronic message (such as an email) alerting the collaborators and owners that they have not yet supplied their designated portion of campaign data.

(143) III. Privacy Campaign Information Display

(144) At step **430**, Main Privacy Compliance Module **400** may, in exemplary embodiments, call upon a Risk Assessment Module **430** that may determine and assign a Risk Level for the privacy campaign, based wholly or in part on the information that the owner(s) have input. The Risk Assessment Module **430** will be discussed in more detail below.

(145) At step **432**, Main Privacy Compliance Module **400** may in exemplary embodiments, call upon a Privacy Audit Module **432** that may determine an audit schedule for each privacy campaign, based, for example, wholly or in part on the campaign data that the owner(s) have input, the Risk Level assigned to a campaign, and/or any other suitable factors. The Privacy Audit Module **432** may also be operable to display the status of an audit for each privacy campaign. The Privacy Audit Module **432** will be discussed in more detail below.

(146) At step **435**, the Main Privacy Compliance Module **400** may generate and display a GUI showing an inventory page (e.g., inventory page **1500**) that includes information associated with each campaign. That information may include information input by a user (e.g., one or more owners), or information calculated by the Main Privacy Compliance Module **400** or other modules. Such information may include for example, the name of the campaign, the status of the campaign, the source of the campaign, the storage location of the personal data related to the campaign, etc. The inventory page **1500** may also display an indicator representing the Risk Level (as mentioned, determined for each campaign by the Risk Assessment Module **430**), and audit information related to the campaign that was determined by the Privacy Audit Module (see below). The inventory page **1500** may be the landing page displayed to users that access the system. Based on the login information received from the user, the Main Privacy Compliance Module may determine which campaigns and campaign data the user is authorized to view, and display only the information that the user is authorized to view. Also from the inventory page **1500**, a user may add a campaign (discussed above in step **405**), view more information for a campaign, or edit information related to a campaign (see, e.g., FIGS. **15**, **16**, **17**).

(147) If other commands from the inventory page are received (e.g., add a campaign, view more information, edit information related to the campaign), then step **440**, **445**, and/or **450** may be executed.

(148) At step **440**, if a command to view more information has been received or detected, then at step **445**, the Main Privacy Compliance Module **400** may present more information about the campaign, for example, on a suitable inventory page **1500**. At this step, the Main Privacy Compliance Module **400** may invoke a Data Flow Diagram Module (described in more detail below). The Data Flow Diagram Module may generate a flow diagram that shows, for example, visual indicators indicating whether data is confidential and/or encrypted (see, e.g., FIG. **1600** below).

(149) At step **450**, if the system has received a request to edit a campaign, then, at step **455**, the system may display a dialog page that allows a user to edit information regarding the campaign (e.g., edit campaign dialog **1700**).

(150) At step **460**, if the system has received a request to add a campaign, the process may proceed back to step **405**.

(151) C. Risk Assessment Module

(152) FIG. **5** illustrates an exemplary process for determining a Risk Level and Overall Risk Assessment for a particular privacy campaign performed by Risk Assessment Module **430**.

(153) I. Determining Risk Level

(154) In exemplary embodiments, the Risk Assessment Module **430** may be operable to calculate a Risk Level for a campaign based on the campaign data related to the personal data associated with the campaign. The Risk Assessment Module may associate the Risk Level with the record for the campaign and digitally store the Risk Level with the record for the campaign.

(155) The Risk Assessment Module **430** may calculate this Risk Level based on any of various factors associated with the campaign. The Risk Assessment Module **430** may determine a plurality of weighting factors based upon, for example: (1) the nature of the sensitive information collected as part of the campaign (e.g., campaigns in which medical information, financial information or non-public personal identifying information is collected may be indicated to be of higher risk than those in which only public information is collected, and thus may be assigned a higher numerical weighting factor); (2) the location in which the information is stored (e.g., campaigns in which data is stored in the cloud may be deemed higher risk than campaigns in which the information is stored locally); (3) the number of individuals who have access to the information (e.g., campaigns that permit relatively large numbers of individuals to access the personal data may be deemed more risky than those that allow only small numbers of individuals to access the data); (4) the length of time that the data will be stored within the system (e.g., campaigns that plan to store and use the personal data over a long period of time may be deemed more risky than those that may only hold and use the personal data for a short period of time); (5) the individuals whose sensitive information will be stored (e.g., campaigns that involve storing and using information of minors may be deemed of greater risk than campaigns that involve storing and using the information of adults); (6) the country of residence of the individuals whose sensitive information will be stored (e.g., campaigns that involve collecting data from individuals that live in countries that have relatively strict privacy laws may be deemed more risky than those that involve collecting data from individuals that live in countries that have relative lax privacy laws). It should be understood that any other suitable factors may be used to assess the Risk Level of a particular campaign, including any new inputs that may need to be added to the risk calculation.

(156) In particular embodiments, one or more of the individual factors may be weighted (e.g., numerically weighted) according to the deemed relative importance of the factor relative to other factors (i.e., Relative Risk Rating).

(157) These weightings may be customized from organization to organization, and/or according to different applicable laws. In particular embodiments, the nature of the sensitive information will be weighted higher than the storage location of the data, or the length of time that the data will be stored.

(158) In various embodiments, the system uses a numerical formula to calculate the Risk Level of a particular campaign. This formula may be, for example:  $\text{Risk Level for campaign} = (\text{Weighting Factor of Factor 1}) * (\text{Relative Risk Rating of Factor 1}) + (\text{Weighting Factor of Factor 2}) * (\text{Relative Risk Rating of Factor 2}) + \dots + (\text{Weighting Factor of Factor N}) * (\text{Relative Risk Rating of Factor N})$ . As a simple example, the Risk Level for a campaign that only collects publicly available information for adults and that stores the information locally for a short period of several weeks might be determined as  $\text{Risk Level} = (\text{Weighting Factor of Nature of Sensitive Information}) * (\text{Relative Risk Rating of Particular Sensitive Information to be Collected}) + (\text{Weighting Factor of Individuals from which Information is to be Collected}) * (\text{Relative Risk Rating of Individuals from which Information is to be Collected}) + (\text{Weighting Factor of Duration of Data Retention}) * (\text{Relative Risk Rating of Duration of Data Retention}) + (\text{Weighting Factor of Individuals from which Data is to be Collected}) * (\text{Relative Risk Rating of Individuals from which Data is to be Collected})$ . In this example, the Weighting Factors may range, for example from 1-5, and the various Relative Risk Ratings of a factor may range from 1-10. However, the system may use any other suitable ranges.

(159) In particular embodiments, the Risk Assessment Module **430** may have default settings for assigning Overall Risk Assessments to respective campaigns based on the numerical Risk Level value determined for the campaign, for example, as described above. The organization may also modify these settings in the Risk Assessment Module **430** by assigning its own Overall Risk Assessments based on the numerical Risk Level. For example, the Risk Assessment Module **430** may, based on default or user assigned settings, designate: (1) campaigns with a Risk Level of 1-7 as “low risk” campaigns, (2) campaigns with a Risk Level of 8-15 as “medium risk” campaigns; (3) campaigns with a Risk Level of over 16 as “high risk” campaigns. As show below, in an example inventory page **1500**, the Overall Risk Assessment for each campaign can be indicated by up/down arrow indicators, and further, the arrows may have different shading (or color, or portions shaded) based upon this Overall Risk Assessment. The selected colors may be conducive for viewing by those who suffer from color blindness.

(160) Thus, the Risk Assessment Module **430** may be configured to automatically calculate the numerical Risk Level for each campaign within the system, and then use the numerical Risk Level to assign an appropriate Overall Risk Assessment to the respective campaign. For example, a campaign with a Risk Level of 5 may be labeled with an Overall Risk Assessment as “Low Risk”. The system may associate both the Risk Level and the Overall Risk Assessment with the campaign and digitally store them as part of the campaign record.

(161) II. Exemplary Process for Assessing Risk

(162) Accordingly, as shown in FIG. 5, in exemplary embodiments, the Risk Assessment Module **430** electronically retrieves from a database (e.g., storage device **130**) the campaign data associated with the record for the privacy campaign. It may retrieve this information serially, or in parallel. At step **505**, the Risk Assessment Module **430** retrieves information regarding (1) the nature of the sensitive information collected as part of the campaign. At step **510**, the Risk Assessment Module **430** retrieves information regarding the (2) the location in which the information related to the privacy campaign is stored. At step **515**, the Risk Assessment Module **430** retrieves information regarding (3) the number of individuals who have access to the information. At step **520**, the Risk Assessment Module retrieves information regarding (4) the length of time that the data associated with a campaign will be stored within the System **100**. At step **525**, the Risk Assessment Module retrieves information regarding (5) the individuals whose sensitive information will be stored. At step **530**, the Risk Assessment Module retrieves information regarding (6) the country of residence of the individuals whose sensitive information will be stored.

(163) At step **535**, the Risk Assessment Module takes into account any user customizations to the weighting factors related to each of the retrieved factors from steps **505**, **510**, **515**, **520**, **525**, and **530**. At steps **540** and **545**, the Risk Assessment Module applies either default settings to the weighting factors (which may be based on privacy laws), or customizations to the weighting factors. At step **550**, the Risk Assessment Module determines a plurality of weighting factors for the campaign. For example, for the factor related to the nature of the sensitive information collected as part of the campaign, a weighting factor of 1-5 may be assigned based on whether non-public personal identifying information is collected.

(164) At step **555**, the Risk Assessment Module takes into account any user customizations to the Relative Risk assigned to each factor, and at step **560** and **565**, will either apply default values (which can be based on privacy laws) or the customized values for the Relative Risk. At step **570**, the Risk Assessment Module assigns a relative risk rating for each of the plurality of weighting factors. For example, the relative risk rating for the location of the information of the campaign may be assigned a numerical number (e.g., from 1-10) that is lower than the numerical number assigned to the Relative Risk Rating for the length of time that the sensitive information for that campaign is retained.

(165) At step **575**, the Risk Assessment Module **430** calculates the relative risk assigned to the campaign based upon the plurality of Weighting Factors and the Relative Risk Rating for each of the plurality of factors. As an example, the Risk Assessment Module **430** may make this calculation using the formula of  $\text{Risk Level} = (\text{Weighting Factor of Factor 1}) * (\text{Relative Risk Rating of Factor 1}) + (\text{Weighting Factor of Factor 2}) * (\text{Relative Risk Rating of Factor 2}) + \dots + (\text{Weighting Factor of Factor N}) * (\text{Relative Risk Rating of Factor N})$ .

(166) At step **580**, based upon the numerical value derived from step **575**, the Risk Assessment Module **430** may determine an Overall Risk Assessment for the campaign. The Overall Risk Assessment determination may be made for the privacy campaign may be assigned based on the following criteria, which may be either a default or customized setting: (1) campaigns with a Risk Level of 1-7 as “low risk” campaigns, (2) campaigns with a Risk Level of 8-15 as “medium risk” campaigns; (3) campaigns with a Risk Level of over 16 as “high risk” campaigns. The Overall Risk Assessment is then associated and stored with the campaign record.

(167) D. Privacy Audit Module

(168) The System **100** may determine an audit schedule for each campaign, and indicate, in a particular graphical user interface (e.g., inventory page **1500**), whether a privacy audit is coming due (or is past due) for each particular campaign and, if so, when the audit is/was due. The System **100** may also be operable to provide an audit status for each campaign, and alert personnel of upcoming or past due privacy audits. To further the retention of

evidence of compliance, the System **100** may also receive and store evidence of compliance. A Privacy Audit Module **432** may facilitate these functions.

(169) I. Determining a Privacy Audit Schedule and Monitoring Compliance

(170) In exemplary embodiments, the Privacy Audit Module **432** is adapted to automatically schedule audits and manage compliance with the audit schedule. In particular embodiments, the system may allow a user to manually specify an audit schedule for each respective campaign. The Privacy Audit Module **432** may also automatically determine, and save to memory, an appropriate audit schedule for each respective campaign, which in some circumstances, may be editable by the user.

(171) The Privacy Audit Module **432** may automatically determine the audit schedule based on the determined Risk Level of the campaign. For example, all campaigns with a Risk Level less than 10 may have a first audit schedule and all campaigns with a Risk Level of 10 or more may have a second audit schedule. The Privacy Audit Module may also be operable determine the audit schedule based on the Overall Risk Assessment for the campaign (e.g., “low risk” campaigns may have a first predetermined audit schedule, “medium risk” campaigns may have a second predetermined audit schedule, “high risk” campaigns may have a third predetermined audit schedule, etc.).

(172) In particular embodiments, the Privacy Audit Module **432** may automatically facilitate and monitor compliance with the determined audit schedules for each respective campaign. For example, the system may automatically generate one or more reminder emails to the respective owners of campaigns as the due date approaches. The system may also be adapted to allow owners of campaigns, or other users, to submit evidence of completion of an audit (e.g., by for example, submitting screen shots that demonstrate that the specified parameters of each campaign are being followed). In particular embodiments, the system is configured for, in response to receiving sufficient electronic information documenting completion of an audit, resetting the audit schedule (e.g., scheduling the next audit for the campaign according to a determined audit schedule, as determined above).

(173) II. Exemplary Privacy Audit Process

(174) FIG. **6** illustrates an exemplary process performed by a Privacy Audit Module **432** for assigning a privacy audit schedule and facilitating and managing compliance for a particular privacy campaign. At step **605**, the Privacy Audit Module **432** retrieves the Risk Level associated with the privacy campaign. In exemplary embodiments, the Risk Level may be a numerical number, as determined above by the Risk Assessment Module **430**. If the organization chooses, the Privacy Audit Module **432** may use the Overall Risk Assessment to determine which audit schedule for the campaign to assign.

(175) At step **610**, based on the Risk Level of the campaign (or the Overall Risk Assessment), or based on any other suitable factor, the Privacy Audit Module **432** can assign an audit schedule for the campaign. The audit schedule may be, for example, a timeframe (i.e., a certain amount of time, such as number of days) until the next privacy audit on the campaign to be performed by the one or more owners of the campaign. The audit schedule may be a default schedule. For example, the Privacy Audit Module can automatically apply an audit schedule of 120 days for any campaign having Risk Level of 10 and above. These default schedules may be modifiable. For example, the default audit schedule for campaigns having a Risk Level of 10 and above can be changed from 120 days to 150 days, such that any campaign having a Risk Level of 10 and above is assigned the customized default audit schedule (i.e., 150 days). Depending on privacy laws, default policies, authority overrides, or the permission level of the user attempting to modify this default, the default might not be modifiable.

(176) At step **615**, after the audit schedule for a particular campaign has already been assigned, the Privacy Audit Module **432** determines if a user input to modify the audit schedule has been received. If a user input to modify the audit schedule has been received, then at step **620**, the Privacy Audit Module **432** determines whether the audit schedule for the campaign is editable (i.e., can be modified). Depending on privacy laws, default policies, authority overrides, or the permission level of the user attempting to modify the audit schedule, the campaign's audit schedule might not be modifiable.

(177) At step **625**, if the audit schedule is modifiable, then the Privacy Audit Module will allow the edit and modify the audit schedule for the campaign. If at step **620** the Privacy Audit Module determines that the audit schedule is not modifiable, in some exemplary embodiments, the user may still request permission to modify the audit schedule. For example, the Privacy Audit Module **432** can at step **630** provide an indication that the audit schedule is not editable, but also provide an indication to the user that the user may contact through the system one or more persons having the authority to grant or deny permission to modify the audit schedule for the campaign (i.e., administrators) to gain permission to edit the field. The Privacy Audit Module **432** may display an on-screen button that, when selected by the user, sends a notification (e.g., an email) to an administrator. The user can thus make a request to modify the audit schedule for the campaign in this manner.

(178) At step **635**, the Privacy Audit Module may determine whether permission has been granted by an administrator to allow a modification to the audit schedule. It may make this determination based on whether it has received input from an administrator to allow modification of the audit schedule for the campaign. If the administrator has granted permission, the Privacy Audit Module **432** at step **635** may allow the edit of the audit schedule. If at step **640**, a denial of permission is received from the administrator, or if a certain amount of time has passed (which may be customized or based on a default setting), the Privacy Audit Module **432** retains the audit schedule for the campaign by not allowing any modifications to the schedule, and the process may proceed to step **645**. The Privacy Audit Module may also send a reminder to the administrator that a request to modify the audit schedule for a campaign is pending.

(179) At step **645**, the Privacy Audit Module **432** determines whether a threshold amount of time (e.g., number of days) until the audit has been reached. This threshold may be a default value, or a customized value. If the threshold amount of time until an audit has been reached, the Privacy Audit Module **432** may at step **650** generate an electronic alert. The alert can be a message displayed to the collaborator the next time the collaborator logs into the system, or the alert can be an electronic message sent to one or more collaborators, including the campaign owners. The alert can be, for example, an email, an instant message, a text message, or one or more of these communication modalities. For example, the message may state, “This is a notification that a privacy audit for Campaign Internet Browsing History is scheduled to occur in 90 days.” More than one threshold may be assigned, so that the owner of the campaign receives more than one alert as the scheduled privacy audit deadline approaches. If the threshold number of days has not been reached, the Privacy Audit Module **432** will continue to evaluate whether the threshold has been reached (i.e., back to step **645**).

(180) In exemplary embodiments, after notifying the owner of the campaign of an impending privacy audit, the Privacy Audit Module may determine at step **655** whether it has received any indication or confirmation that the privacy audit has been completed. In example embodiments, the Privacy Audit Module allows for evidence of completion to be submitted, and if sufficient, the Privacy Audit Module **432** at step **660** resets the counter for the audit schedule for the campaign. For example, a privacy audit may be confirmed upon completion of required electronic forms in which one or more collaborators verify that their respective portions of the audit process have been completed. Additionally, users can submit photos, screen shots, or other documentation that show that the organization is complying with that user's assigned portion of the privacy campaign. For example, a database administrator may take a screen shot showing that all personal data from the privacy campaign is being stored in the proper database and submit that to the system to document compliance with the terms of the campaign.

(181) If at step **655**, no indication of completion of the audit has been received, the Privacy Audit Module **432** can determine at step **665** whether an audit for a campaign is overdue (i.e., expired). If it is not overdue, the Privacy Audit Module **432** will continue to wait for evidence of completion (e.g., step **655**). If the audit is overdue, the Privacy Audit Module **432** at step **670** generates an electronic alert (e.g., an email, instant message, or text message) to the campaign owner(s) or other administrators indicating that the privacy audit is overdue, so that the organization can take responsive or remedial measures.

(182) In exemplary embodiments, the Privacy Audit Module **432** may also receive an indication that a privacy audit has begun (not shown), so that



the status of the audit as pending, the Privacy Audit Module **432** may be operable to generate reminders to be sent to the campaign owner(s), for example, to remind the owner of the deadline for completing the audit.

(183) E. Data Flow Diagram Module

(184) The system **100** may be operable to generate a data flow diagram based on the campaign data entered and stored, for example in the manner described above.

(185) I. Display of Security Indicators and Other Information

(186) In various embodiments, a Data Flow Diagram Module is operable to generate a flow diagram for display containing visual representations (e.g., shapes) representative of one or more parts of campaign data associated with a privacy campaign, and the flow of that information from a source (e.g., customer), to a destination (e.g., an internet usage database), to which entities and computer systems have access (e.g., customer support, billing systems). Data Flow Diagram Module may also generate one or more security indicators for display. The indicators may include, for example, an “eye” icon to indicate that the data is confidential, a “lock” icon to indicate that the data, and/or a particular flow of data, is encrypted, or an “unlocked lock” icon to indicate that the data, and/or a particular flow of data, is not encrypted. In the example shown in FIG. **16**, the dotted arrow lines generally depict respective flows of data and the locked or unlocked lock symbols indicate whether those data flows are encrypted or unencrypted. The color of dotted lines representing data flows may also be colored differently based on whether the data flow is encrypted or non-encrypted, with colors conducive for viewing by those who suffer from color blindness.

(187) II. Exemplary Process Performed by Data Flow Diagram Module

(188) FIG. **7** shows an example process performed by the Data Flow Diagram Module **700**. At step **705**, the Data Flow Diagram retrieves campaign data related to a privacy campaign record. The campaign data may indicate, for example, that the sensitive information related to the privacy campaign contains confidential information, such as the social security numbers of a customer.

(189) At step **710**, the Data Flow Diagram Module **700** is operable to display on-screen objects (e.g., shapes) representative of the Source, Destination, and Access, which indicate that information below the heading relates to the source of the personal data, the storage destination of the personal data, and access related to the personal data. In addition to campaign data regarding Source, Destination, and Access, the Data Flow Diagram Module **700** may also account for user defined attributes related to personal data, which may also be displayed as on-screen objects. The shape may be, for example, a rectangular box (see, e.g., FIG. **16**). At step **715**, the Data Flow Diagram Module **700** may display a hyperlink label within the on-screen object (e.g., as shown in FIG. **16**, the word “Customer” may be a hyperlink displayed within the rectangular box) indicative of the source of the personal data, the storage destination of the personal data, and access related to the personal data, under each of the respective headings. When a user hovers over the hyperlinked word, the Data Flow Diagram is operable to display additional campaign data relating to the campaign data associated with the hyperlinked word. The additional information may also be displayed in a pop up, or a new page. For example, FIG. **16** shows that if a user hovers over the words “Customer,” the Data Flow Diagram Module **700** displays what customer information is associated with the campaign (e.g., the Subscriber ID, the IP and Mac Addresses associated with the Customer, and the customer's browsing and usage history). The Data Flow Diagram Module **700** may also generate for display information relating to whether the source of the data includes minors, and whether consent was given by the source to use the sensitive information, as well as the manner of the consent (for example, through an End User License Agreement (EULA)).

(190) At step **720**, the Data Flow Diagram Module **700** may display one or more parameters related to backup and retention of personal data related to the campaign, including in association with the storage destination of the personal data. As an example, Data Flow Diagram **1615** of FIG. **16** displays that the information in the Internet Usage database is backed up, and the retention related to that data is Unknown.

(191) At step **725**, the Data Flow Diagram Module **700** determines, based on the campaign data associated with the campaign, whether the personal data related to each of the hyperlink labels is confidential. At step **730**, if the personal data related to each hyperlink label is confidential, the Data Flow Diagram Module **700** generates visual indicator indicating confidentiality of that data (e.g., an “eye” icon, as show in Data Flow Diagram **1615**). If there is no confidential information for that box, then at step **735**, no indicators are displayed. While this is an example of the generation of indicators for this particular hyperlink, in exemplary embodiments, any user defined campaign data may visual indicators that may be generated for it.

(192) At step **740**, the Data Flow Diagram Module **700** determined whether any of the data associated with the source, stored in a storage destination, being used by an entity or application, or flowing to one or more entities or systems (i.e., data flow) associated with the campaign is designated as encrypted. If the data is encrypted, then at step **745** the Data Flow Diagram Module **700** may generate an indicator that the personal data is encrypted (e.g., a “lock” icon). If the data is non-encrypted, then at step **750**, the Data Flow Diagram Module **700** displays an indicator to indicate that the data or particular flow of data is not encrypted. (e.g., an “unlocked lock” icon). An example of a data flow diagram is depicted in FIG. **9**. Additionally, the data flow diagram lines may be colored differently to indicate whether the data flow is encrypted or unencrypted, wherein the colors can still be distinguished by a color-blind person.

(193) F. Communications Module

(194) In exemplary embodiments, a Communications Module of the System **100** may facilitate the communications between various owners and personnel related to a privacy campaign. The Communications Module may retain contact information (e.g., emails or instant messaging contact information) input by campaign owners and other collaborators. The Communications Module can be operable to take a generated notification or alert (e.g., alert in step **670** generated by Privacy Audit Module **432**) and instantiate an email containing the relevant information. As mentioned above, the Main Privacy Compliance Module **400** may, for example through a communications module, be operable to send collaborators emails regarding their assignment of one or more portions of inputs to campaign data. Or through the communications module, selecting the commentators button brings up one or more collaborators that are on-line

(195) In exemplary embodiments, the Communications Module can also, in response to a user request (e.g., depressing the “comment” button show in FIG. **9**, FIG. **10**, FIG. **11**, FIG. **12**, FIG. **13**, FIG. **16**), instantiate an instant messaging session and overlay the instant messaging session over one or more portions of a GUI, including a GUI in which a user is presented with prompts to enter or select information. An example of this instant messaging overlay feature orchestrated by the Communications Module is shown in FIG. **14**. While a real-time message session may be generated, off-line users may still be able to see the messages when they are back on-line.

(196) The Communications Module may facilitate the generation of alerts that indicate that one or more emails or instant messages await a collaborator.

(197) If campaign data inputs have been assigned to one or more collaborators, but those collaborators have not input the data yet, the Communications Module, may facilitate the sending of an electronic message (such as an email) alerting the collaborators and owners that they have not yet supplied their designated portion of campaign data.

(198) Exemplary User Experience

(199) In the exemplary embodiments of the system for operationalizing privacy compliance, adding a campaign (i.e., data flow) comprises gathering information that includes several phases: (1) a description of the campaign; (2) the personal data to be collected as part of the campaign; (3) who the personal data relates to; (4) where the personal data be stored; and (5) who will have access to the indicated personal data.

(200) A. FIG. **8**: Campaign Record Creation and Collaborator Assignment

(201) FIG. **8** illustrates an example of the first phase of information gathering to add a campaign. In FIG. **8**, a description entry dialog **800** may have several fillable/editable fields and drop-down selectors. In this example, the user may fill out the name of the campaign in the Short Summary (name)



field **805**, and a description of the campaign in the Description field **810**. The user may enter or select the name of the business group (or groups) that will be accessing personal data for the campaign in the Business Group field **815**. The user may select the primary business representative responsible for the campaign (i.e., the campaign's owner), and designate him/herself, or designate someone else to be that owner by entering that selection through the Someone Else field **820**. Similarly, the user may designate him/herself as the privacy office representative owner for the campaign, or select someone else from the second Someone Else field **825**. At any point, a user assigned as the owner may also assign others the task of selecting or answering any question related to the campaign. The user may also enter one or more tag words associated with the campaign in the Tags field **830**. After entry, the tag words may be used to search for campaigns, or used to filter for campaigns (for example, under Filters). The user may assign a due date **835** for completing the campaign entry, and turn reminders for the campaign on or off. The user may save and continue, or assign and close.

(202) In example embodiments, some of the fields may be filled in by a user, with suggest-as-you-type display of possible field entries (e.g., Business Group field **815**), and/or may include the ability for the user to select items from a drop-down selector (e.g., drop-down selectors **840a**, **840b**, **840c**). The system may also allow some fields to stay hidden or unmodifiable to certain designated viewers or categories of users. For example, the purpose behind a campaign may be hidden from anyone who is not the chief privacy officer of the company, or the retention schedule may be configured so that it cannot be modified by anyone outside of the organization's legal department.

(203) B. FIG. 9: Collaborator Assignment Notification and Description Entry

(204) Moving to FIG. 9, in example embodiments, if another business representative (owner), or another privacy office representative has been assigned to the campaign (e.g., John Doe in FIG. 8), the system may send a notification (e.g., an electronic notification) to the assigned individual, letting them know that the campaign has been assigned to him/her. FIG. 9 shows an example notification **900** sent to John Doe that is in the form of an email message. The email informs him that the campaign "Internet Usage Tracking" has been assigned to him, and provides other relevant information, including the deadline for completing the campaign entry and instructions to log in to the system to complete the campaign (data flow) entry (which may be done, for example, using a suitable "wizard" program). The user that assigned John ownership of the campaign may also include additional comments **905** to be included with the notification **900**. Also included may be an option to reply to the email if an assigned owner has any questions.

(205) In this example, if John selects the hyperlink Privacy Portal **910**, he is able to access the system, which displays a landing page **915**. The landing page **915** displays a Getting Started section **920** to familiarize new owners with the system, and also display an "About This Data Flow" section **930** showing overview information for the campaign.

(206) C. FIG. 10: What Personal Data is Collected

(207) Moving to FIG. 10, after the first phase of campaign addition (i.e., description entry phase), the system may present the user (who may be a subsequently assigned business representative or privacy officer) with a dialog **1000** from which the user may enter in the type of personal data being collected.

(208) In addition, questions are described generally as transitional questions, but the questions may also include one or more smart questions in which the system is configured to: (1) pose an initial question to a user and, (2) in response to the user's answer satisfying certain criteria, presenting the user with one or more follow-up questions. For example, in FIG. 10, if the user responds with a choice to add personal data, the user may be additionally presented follow-up prompts, for example, the select personal data window overlaying screen **1005** that includes commonly used selections may include, for example, particular elements of an individual's contact information (e.g., name, address, email address), Financial/Billing Information (e.g., credit card number, billing address, bank account number), Online Identifiers (e.g., IP Address, device type, MAC Address), Personal Details (Birthdate, Credit Score, Location), or Telecommunication Data (e.g., Call History, SMS History, Roaming Status). The System **100** is also operable to pre-select or automatically populate choices—for example, with commonly-used selections, some of the boxes may already be checked. The user may also use a search/add tool **1010** to search for other selections that are not commonly used and add another selection. Based on the selections made, the user may be presented with more options and fields. For example, if the user selected "Subscriber ID" as personal data associated with the campaign, the user may be prompted to add a collection purpose under the heading Collection Purpose **1015**, and the user may be prompted to provide the business reason why a Subscriber ID is being collected under the "Describe Business Need" heading **1020**.

(209) D. FIG. 11: Who Personal Data is Collected from

(210) As displayed in the example of FIG. 11, the third phase of adding a campaign may relate to entering and selecting information regarding who the personal data is gathered from. As noted above, the personal data may be gathered from, for example, one or more Subjects **300**. In the exemplary "Collected From" dialog **1100**, a user may be presented with several selections in the "Who Is It Collected From" section **1105**. These selections may include whether the personal data was to be collected from an employee, customer, or other entity. Any entities that are not stored in the system may be added. The selections may also include, for example, whether the data was collected from a current or prospective subject (e.g., a prospective employee may have filled out an employment application with his/her social security number on it). Additionally, the selections may include how consent was given, for example through an end user license agreement (EULA), on-line Opt-in prompt, Implied consent, or an indication that the user is not sure. Additional selections may include whether the personal data was collected from a minor, and where the subject is located.

(211) E. FIG. 12: Where is the Personal Data Stored

(212) FIG. 12 shows an example "Storage Entry" dialog screen **1200**, which is a graphical user interface that a user may use to indicate where particular sensitive information is to be stored within the system. From this section **1210**, a user may specify, in this case for the Internet Usage History campaign, the primary destination of the personal data **1220** and how long the personal data is to be kept **1230**. The personal data may be housed by the organization (in this example, an entity called "Acme") or a third party. The user may specify an application associated with the personal data's storage (in this example, ISP Analytics), and may also specify the location of computing systems (e.g., servers) that will be storing the personal data (e.g., a Toronto data center). Other selections indicate whether the data will be encrypted and/or backed up.

(213) The system also allows the user to select whether the destination settings are applicable to all the personal data of the campaign, or just select data (and if so, which data). In FIG. 12, the user may also select and input options related to the retention of the personal data collected for the campaign (e.g., How Long Is It Kept **1230**). The retention options may indicate, for example, that the campaign's personal data should be deleted after a per-determined period of time has passed (e.g., on a particular date), or that the campaign's personal data should be deleted in accordance with the occurrence of one or more specified events (e.g., in response to the occurrence of a particular event, or after a specified period of time passes after the occurrence of a particular event), and the user may also select whether backups should be accounted for in any retention schedule. For example, the user may specify that any backups of the personal data should be deleted (or, alternatively, retained) when the primary copy of the personal data is deleted.

(214) F. FIG. 13: Who and What Systems have Access to Personal Data

(215) FIG. 13 describes an example Access entry dialog screen **1300**. As part of the process of adding a campaign or data flow, the user may specify in the "Who Has Access" section **1305** of the dialog screen **1300**. In the example shown, the Customer Support, Billing, and Government groups within the organization are able to access the Internet Usage History personal data collected by the organization. Within each of these access groups, the user may select the type of each group, the format in which the personal data was provided, and whether the personal data is encrypted. The access level of each group may also be entered. The user may add additional access groups via the Add Group button **1310**.

(216) G. Facilitating Entry of Campaign Data, Including Chat Shown in FIG. 14

(217) As mentioned above, to facilitate the entry of data collected through the example GUIs shown in FIGS. 8 through 12, in exemplary embodiments, the system is adapted to allow the owner of a particular campaign (or other user) to assign certain sections of questions, or individual

questions, related to the campaign to contributors other than the owner. This may eliminate the need for the owner to contact other users to determine information that they don't know and then enter the information into the system themselves. Rather, in various embodiments, the system facilitates the entry of the requested information directly into the system by the assigned users.

(218) In exemplary embodiments, after the owner assigns a respective responsible party to each question or section of questions that need to be answered in order to fully populate the data flow, the system may automatically contact each user (e.g., via an appropriate electronic message) to inform the user that they have been assigned to complete the specified questions and/or sections of questions, and provide those users with instructions as to how to log into the system to enter the data. The system may also be adapted to periodically follow up with each user with reminders until the user completes the designated tasks. As discussed elsewhere herein, the system may also be adapted to facilitate real-time text or voice communications between multiple collaborators as they work together to complete the questions necessary to define the data flow. Together, these features may reduce the amount of time and effort needed to complete each data flow.

(219) To further facilitate collaboration, as shown FIG. 14, in exemplary embodiments, the System 100 is operable to overlay an instant messaging session over a GUI in which a user is presented with prompts to enter or select information. In FIG. 14, a communications module is operable to create an instant messaging session window 1405 that overlays the Access entry dialog screen 1300. In exemplary embodiments, the Communications Module, in response to a user request (e.g., depressing the "comment" button shown in FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 16), instantiates an instant messaging session and overlays the instant messaging session over one or more portions of the GUI.

(220) H: FIG. 15: Campaign Inventory Page

(221) After new campaigns have been added, for example using the exemplary processes explained in regard to FIGS. 8-13, the users of the system may view their respective campaign or campaigns, depending on whether they have access to the campaign. The chief privacy officer, or another privacy office representative, for example, may be the only user that may view all campaigns. A listing of all of the campaigns within the system may be viewed on, for example, inventory page 1500 (see below). Further details regarding each campaign may be viewed via, for example, campaign information page 1600, which may be accessed by selecting a particular campaign on the inventory page 1500. And any information related to the campaign may be edited or added through, for example, the edit campaign dialog 1700 screen (see FIG. 17). Certain fields or information may not be editable, depending on the particular user's level of access. A user may also add a new campaign using a suitable user interface, such as the graphical user interface shown in FIG. 15 or FIG. 16.

(222) In example embodiments, the System 100 (and more particularly, the Main Privacy Compliance Module 400) may use the history of past entries to suggest selections for users during campaign creation and entry of associated data. As an example, in FIG. 10, if most entries that contain the term "Internet" and have John Doe as the business rep assigned to the campaign have the items Subscriber ID, IP Address, and MAC Address selected, then the items that are commonly used may display as pre-selected items the Subscriber ID, IP address, and MAC Address each time a campaign is created having Internet in its description and John Doe as its business rep.

(223) FIG. 15 describes an example embodiment of an inventory page 1500 that may be generated by the Main Privacy Compliance Module 400. The inventory page 1500 may be represented in a graphical user interface. Each of the graphical user interfaces (e.g., webpages, dialog boxes, etc.) presented in this application may be, in various embodiments, an HTML-based page capable of being displayed on a web browser (e.g., Firefox, Internet Explorer, Google Chrome, Opera, etc.), or any other computer-generated graphical user interface operable to display information, including information having interactive elements (e.g., an iOS, Mac OS, Android, Linux, or Microsoft Windows application). The webpage displaying the inventory page 1500 may include typical features such as a scroll-bar, menu items, as well as buttons for minimizing, maximizing, and closing the webpage. The inventory page 1500 may be accessible to the organization's chief privacy officer, or any other of the organization's personnel having the need, and/or permission, to view personal data.

(224) Still referring to FIG. 15, inventory page 1500 may display one or more campaigns listed in the column heading Data Flow Summary 1505, as well as other information associated with each campaign, as described herein. Some of the exemplary listed campaigns include Internet Usage History 1510, Customer Payment Information, Call History Log, Cellular Roaming Records, etc. A campaign may represent, for example, a business operation that the organization is engaged in may require the use of personal data, which may include the personal data of a customer. In the campaign Internet Usage History 1510, for example, a marketing department may need customers' on-line browsing patterns to run analytics. Examples of more information that may be associated with the Internet Usage History 1510 campaign will be presented in FIG. 4 and FIG. 5. In example embodiments, clicking on (i.e., selecting) the column heading Data Flow Summary 1505 may result in the campaigns being sorted either alphabetically, or reverse alphabetically.

(225) The inventory page 1500 may also display the status of each campaign, as indicated in column heading Status 1515. Exemplary statuses may include "Pending Review", which means the campaign has not been approved yet, "Approved," meaning the data flow associated with that campaign has been approved, "Audit Needed," which may indicate that a privacy audit of the personal data associated with the campaign is needed, and "Action Required," meaning that one or more individuals associated with the campaign must take some kind of action related to the campaign (e.g., completing missing information, responding to an outstanding message, etc.). In certain embodiments, clicking on (i.e., selecting) the column heading Status 1515 may result in the campaigns being sorted by status.

(226) The inventory page 1500 of FIG. 15 may list the "source" from which the personal data associated with a campaign originated, under the column heading "Source" 1520. The sources may include one or more of the subjects 300 in example FIG. 3. As an example, the campaign "Internet Usage History" 1510 may include a customer's IP address or MAC address. For the example campaign "Employee Reference Checks", the source may be a particular employee. In example embodiments, clicking on (i.e., selecting) the column heading Source 1520 may result in the campaigns being sorted by source.

(227) The inventory page 1500 of FIG. 15 may also list the "destination" of the personal data associated with a particular campaign under the column heading Destination 1525. Personal data may be stored in any of a variety of places, for example on one or more storage devices 280 that are maintained by a particular entity at a particular location. Different custodians may maintain one or more of the different storage devices. By way of example, referring to FIG. 15, the personal data associated with the campaign Internet Usage History 1510 may be stored in a repository located at the Toronto data center, and the repository may be controlled by the organization (e.g., Acme corporation) or another entity, such as a vendor of the organization that has been hired by the organization to analyze the customer's internet usage history. Alternatively, storage may be with a department within the organization (e.g., its marketing department). In example embodiments, clicking on (i.e., selecting) the column heading Destination 1525 may result in the campaigns being sorted by destination.

(228) On the inventory page 1500, the column heading "Access" 1530 may show the number of transfers that the personal data associated with a campaign has undergone. In example embodiments, clicking on (i.e., selecting) the column heading "Access" 1530 may result in the campaigns being sorted by Access.

(229) The column with the heading Audit 1535 shows the status of any privacy audits associated with the campaign. Privacy audits may be pending, in which an audit has been initiated but yet to be completed. The audit column may also show for the associated campaign how many days have passed since a privacy audit was last conducted for that campaign. (e.g., 140 days, 360 days). If no audit for a campaign is currently required, an "OK" or some other type of indication of compliance (e.g., a "thumbs up" indicia) may be displayed for that campaign's audit status. Campaigns may also be sorted based on their privacy audit status by selecting or clicking on the heading Audit 1535.

(230) In example inventory page 1500, an indicator under the heading Risk 1540 may also display an indicator as to the Risk Level associated with the personal data for a particular campaign. As described earlier, a risk assessment may be made for each campaign based on one or more factors that may be obtained by the system. The indicator may, for example, be a numerical score (e.g., Risk Level of the campaign), or, as in the example shown

in FIG. 15, it may be arrows that indicate the Overall Risk Assessment for the campaign. The arrows may be of different shades or different colors (e.g., red arrows indicating “high risk” campaigns, yellow arrows indicating “medium risk” campaigns, and green arrows indicating “low risk” campaigns). The direction of the arrows—for example, pointing upward or downward, may also provide a quick indication of Overall Risk Assessment for users viewing the inventory page **1500**. Each campaign may be sorted based on the Risk Level associated with the campaign. (231) The example inventory page **1500** may comprise a filter tool, indicated by Filters **1545**, to display only the campaigns having certain information associated with them. For example, as shown in FIG. 15, under Collection Purpose **1550**, checking the boxes “Commercial Relations,” “Provide Products/Services,” “Understand Needs,” “Develop Business & Ops,” and “Legal Requirement” will result the display under the column heading Data Flow Summary **1505** of only the campaigns that meet those selected collection purpose requirements.

(232) From example inventory page **1500**, a user may also add a campaign by selecting (i.e., clicking on) the Add Data Flow button **1555**. Once this selection has been made, the system initiates a routine to guide the user in a phase-by-phase manner through the process of creating a new campaign (further details herein). An example of the multi-phase GUIs in which campaign data associated with the added privacy campaign may be input and associated with the privacy campaign record is described in FIG. 8-13 above.

(233) From the example inventory page **1500**, a user may view the information associated with each campaign in more depth, or edit the information associated with each campaign. To do this, the user may, for example, click on or select the name of the campaign (i.e., click on Internet Usage History **1510**). As another example, the user may select a button displayed on screen indicating that the campaign data is editable (e.g., edit button **1560**).

(234) I. FIG. 16: Campaign Information Page and Data Flow Diagram

(235) FIG. 16 shows an example of information associated with each campaign being displayed in a campaign information page **1600**. Campaign information page **1600** may be accessed by selecting (i.e., clicking on), for example, the edit button **1560**. In this example, Personal Data Collected section **1605** displays the type of personal data collected from the customer for the campaign Internet Usage History. The type of personal data, which may be stored as data elements associated with the Internet Usage History campaign digital record entry. The type of information may include, for example, the customer's Subscriber ID, which may be assigned by the organization (e.g., a customer identification number, customer account number). The type of information may also include data associated with a customer's premises equipment, such as an IP Address, MAC Address, URL History (i.e., websites visited), and Data Consumption (i.e., the number of megabytes or gigabytes that the user has download).

(236) Still referring to FIG. 16, the “About this Data Flow” section **1610** displays relevant information concerning the campaign, such as the purpose of the campaign. In this example, a user may see that the Internet Usage History campaign is involved with the tracking of internet usage from customers in order to bill appropriately, manage against quotas, and run analytics. The user may also see that the business group that is using the sensitive information associated with this campaign is the Internet group. A user may further see that the next privacy audit is scheduled for Jun. 10, 2016, and that the last update of the campaign entry was Jan. 2, 2015. The user may also select the “view history” hyperlink to display the history of the campaign.

(237) FIG. 16 also depicts an example of a Data Flow Diagram **1615** generated by the system, based on information provided for the campaign. The Data Flow Diagram **1615** may provide the user with a large amount of information regarding a particular campaign in a single compact visual. In this example, for the campaign Internet Usage History, the user may see that the source of the personal data is the organization's customers. In example embodiments, as illustrated, hovering the cursor (e.g., using a touchpad, or a mouse) over the term “Customers” may cause the system to display the type of sensitive information obtained from the respective consumers, which may correspond with the information displayed in the “Personal Data Collected” section **1605**.

(238) In various embodiments, the Data Flow Diagram **1615** also displays the destination of the data collected from the User (in this example, an Internet Usage Database), along with associated parameters related to backup and deletion. The Data Flow Diagram **1615** may also display to the user which department(s) and what system(s) have access to the personal data associated with the campaign. In this example, the Customer Support Department has access to the data, and the Billing System may retrieve data from the Internet Usage Database to carry out that system's operations. In the Data Flow Diagram **1615**, one or more security indicators may also be displayed. The one or more security indicators may include, for example, an “eye” icon to indicate that the data is confidential, a “lock” icon to indicate that the data, and/or a particular flow of data, is encrypted, or an “unlocked lock” icon to indicate that the data, and/or a particular flow of data, is not encrypted. In the example shown in FIG. 16, the dotted arrow lines generally depict respective flows of data and the locked or unlocked lock symbols indicate whether those data flows are encrypted or unencrypted.

(239) Campaign information page **1600** may also facilitate communications among the various personnel administrating the campaign and the personal data associated with it. Collaborators may be added through the Collaborators button **1625**. The system may draw information from, for example, an active directory system, to access the contact information of collaborators.

(240) If Comment button **1630** is selected, a real-time communication session (e.g., an instant messaging session) among all (or some) of the collaborators may be instantiated and overlaid on top of the page **1600**. This may be helpful, for example, in facilitating population of a particular page of data by multiple users. In example embodiments, the Collaborators button **1625** and Comment button **1630** may be included on any graphical user interface described herein, including dialog boxes in which information is entered or selected. Likewise, any instant messaging session may be overlaid on top of a webpage or dialog box. The system may also use the contact information to send one or more users associated with the campaign periodic updates, or reminders. For example, if the deadline to finish entering the campaign data associated with a campaign is upcoming in three days, the business representative of that assigned campaign may be sent a message reminding him or her that the deadline is in three days.

(241) Like inventory page **1500**, campaign information page **1600** also allows for campaigns to be sorted based on risk (e.g., Sort by Risk **1635**). Thus, for example, a user is able to look at the information for campaigns with the highest risk assessment.

(242) J. FIG. 17: Edit Campaign Dialog

(243) FIG. 17 depicts an example of a dialog box—the edit campaign dialog **1700**. The edit campaign dialog **1700** may have editable fields associated with a campaign. In this example, the information associated with the Internet Usage History campaign may be edited via this dialog. This includes the ability for the user to change the name of the campaign, the campaign's description, the business group, the current owner of the campaign, and the particular personal data that is associated with the campaign (e.g., IP address, billing address, credit score, etc.). In example embodiments, the edit campaign dialog **1700** may also allow for the addition of more factors, checkboxes, users, etc.

(244) The system **100** also includes a Historical Record Keeping Module, wherein every answer, change to answer, as well as assignment/re-assignment of owners and collaborators is logged for historical record keeping.

(245) Automated Approach to Demonstrating Privacy by Design, and Integration with Software Development and Agile Tools for Privacy Design

(246) In particular embodiments, privacy by design can be used in the design phase of a product (e.g., hardware or software), which is a documented approach to managing privacy risks. One of the primary concepts is evaluating privacy impacts, and making appropriate privacy-protecting changes during the design of a project, before the project go-live.

(247) In various embodiments, the system is adapted to automate this process with the following capabilities: (1) initial assessment; (2) gap analysis/recommended steps; and/or (3) final/updated assessment. These capabilities are discussed in greater detail below.

(248) Initial Assessment

(249) In various embodiments, when a business team within a particular organization is planning to begin a privacy campaign, the system presents the business team with a set of assessment questions that are designed to help one or more members of the organization's privacy team to understand what the business team's plans are, and to understand whether the privacy campaign may have a privacy impact on the organization. The questions

may also include a request for the business team to provide the “go-live” date, for the privacy campaign. In response to receiving the answers to these questions, the system stores the answers to the system's memory and makes the answers available to the organization's privacy team. The system may also add the “go-live” date to one or more electronic calendars (e.g., the system's electronic docket).

(250) In some implementations, the initial assessment can include an initial privacy impact assessment that evaluates one or more privacy impact features of the proposed design of the product. The initial privacy impact assessment incorporates the respective answers for the plurality of question/answer pairings in the evaluation of the one or more privacy impact features. The privacy impact features may, for example, be related to how the proposed design of the new product will collect, use, store, and/or manage personal data. One or more of these privacy impact features can be evaluated, and the initial privacy assessment can be provided to identify results of the evaluation.

(251) Gap Analysis Recommended Steps

(252) After the system receives the answers to the questions, one or more members of the privacy team may review the answers to the questions. The privacy team may then enter, into the system, guidance and/or recommendations regarding the privacy campaign. In some implementations, the privacy team may input their recommendations into the privacy compliance software. In particular embodiments, the system automatically communicates the privacy team's recommendations to the business team and, if necessary, reminds one or more members of the business team to implement the privacy team's recommendations before the go-live date. The system may also implement one or more audits (e.g., as described above) to make sure that the business team incorporates the privacy team's recommendations before the “go-live” date.

(253) The recommendations may include one or more recommended steps that can be related to modifying one or more aspects of how the product will collect, use, store, and/or manage personal data. The recommended steps may include, for example: (1) limiting the time period that personal data is held by the system (e.g., seven days); (2) requiring the personal data to be encrypted when communicated or stored; (3) anonymizing personal data; or (4) restricting access to personal data to a particular, limited group of individuals. The one or more recommended steps may be provided to address a privacy concern with one or more of the privacy impact features that were evaluated in the initial privacy impact assessment.

(254) In response to a recommended one or more steps being provided (e.g., by the privacy compliance officers), the system may generate one or more tasks in suitable project management software that is used in managing the proposed design of the product at issue. In various embodiments, the one or more tasks may be tasks that, if recommended, would individually or collectively complete one or more (e.g., all of) the recommended steps. For example, if the one or more recommended steps include requiring personal data collected by the product to be encrypted, then the one or more tasks may include revising the product so that it encrypts any personal data that it collects.

(255) The one or more tasks may include, for example, different steps to be performed at different points in the development of the product. In particular embodiments, the computer software application may also monitor, either automatically or through suitable data inputs, the development of the product to determine whether the one or more tasks have been completed.

(256) Upon completion of each respective task in the one or more tasks, the system may provide a notification that the task has been completed. For example, the project management software may provide a suitable notification to the privacy compliance software that the respective task has been completed.

(257) Final/Updated Assessment

(258) Once the mitigation steps and recommendations are complete, the system may (e.g., automatically) conduct an updated review to assess any privacy risks associated with the revised product.

(259) In particular embodiments, the system includes unique reporting and historical logging capabilities to automate Privacy-by-Design reporting and/or privacy assessment reporting. In various embodiments, the system is adapted to: (1) measure/analyze the initial assessment answers from the business team; (2) measure recommendations for the privacy campaign; (3) measure any changes that were implemented prior to the go-live date; (4) automatically differentiate between: (a) substantive privacy protecting changes, such as the addition of encryption, anonymization, or minimizations; and (b) non-substantive changes, such as spelling correction.

(260) The system may also be adapted to generate a privacy assessment report showing that, in the course of a business's normal operations: (1) the business evaluates projects prior to go-live for compliance with one or more privacy-related regulations or policies; and (2) related substantive recommendations are made and implemented prior to go-live. This may be useful in documenting that privacy-by-design is being effectively implemented for a particular privacy campaign.

(261) The privacy assessment report may, in various embodiments, include an updated privacy impact assessment that evaluates the one or more privacy impact features after the one or more recommended steps discussed above are implemented. The system may generate this updated privacy impact assessment automatically by, for example, automatically modifying any answers from within the question/answer pairings of the initial impact privacy assessment to reflect any modifications to the product that have been made in the course of completing the one or more tasks that implement the one or more substantive recommendations. For example, if a particular question from the initial privacy impact assessment indicated that certain personal data was personally identifiable data, and a recommendation was made to anonymize the data, the question/answer pairing for the particular question could be revised so the answer to the question indicates that the data has been anonymized. Any revised question/answer pairings may then be used to complete an updated privacy assessment report.

(262) FIGS. 18A and 18B show an example process performed by a Data Privacy Compliance Module **1800**. In executing the Data Privacy Compliance Module **1800**, the system begins at Step **1802**, where it presents a series of questions to a user (e.g., via a suitable computer display screen or other user-interface, such as a voice-interface) regarding the design and/or anticipated operation of the product. This may be done, for example, by having a first software application (e.g., a data privacy software application or other suitable application) present the user with a template of questions regarding the product (e.g., for use in conducting an initial privacy impact assessment for the product). Such questions may include, for example, data mapping questions and other questions relevant to the product's design and/or anticipated operation.

(263) Next, at Step **1804**, the system receives, via a first computer software application, from a first set of one or more users (e.g., product designers, such as software designers, or other individuals who are knowledgeable about the product), respective answers to the questions regarding the product and associates the respective answers with their corresponding respective questions within memory to create a plurality of question/answer pairings regarding the proposed design of the product (e.g., software, a computerized electro-mechanical product, or other product).

(264) Next, at Step **1806**, the system presents a question to one or more users requesting the scheduled implantation date for the product. At Step **1808**, the system receives this response and saves the scheduled implementation date to memory.

(265) Next, after receiving the respective answers at Step **1804**, the system displays, at Step **1810**, the respective answers (e.g., along with their respective questions and/or a summary of the respective questions) to a second set of one or more users (e.g., one or more privacy officers from the organization that is designing the product), for example, in the form a plurality of suitable question/answer pairings. As an aside, within the context of this specification, pairings of an answer and either its respective question or a summary of the question may be referred to as a “question/answer” pairing. As an example, the question “Is the data encrypted?” and respective answer “Yes” may be represented, for example, in either of the following question/answer pairings: (1) “The data is encrypted”; and (2) “Data encrypted? Yes”. Alternatively, the question/answer pairing may be represented as a value in a particular field in a data structure that would convey that the data at issue is encrypted.

(266) The system then advances to Step **1812**, where it receives, from the second set of users, one or more recommended steps to be implemented as part of the proposed design of the product and before the implementation date, the one or more recommended steps comprising one or more steps that facilitate the compliance of the product with the one or more privacy standards and/or policies. In particular embodiments in which the product is a software application or an electro-mechanical device that runs device software, the one or more recommended steps may comprise modifying the software application or device software to comply with one or more privacy standards and/or policies.

(267) Next, at Step **1814**, in response to receiving the one or more recommended steps, the system automatically initiates the generation of one or more tasks in a second computer software application (e.g., project management software) that is to be used in managing the design of the product. In particular embodiments, the one or more tasks comprise one or more tasks that, if completed, individually and/or collectively would result in the completion of the one or more recommended steps. The system may do this, for example, by facilitating communication between the first and second computer software applications via a suitable application programming interface (API).

(268) The system then initiates a monitoring process for determining whether the one or more tasks have been completed. This step may, for example, be implemented by automatically monitoring which changes (e.g., edits to software code) have been made to the product, or by receiving manual input confirming that various tasks have been completed.

(269) At Step **1818**, the system receives a notification that the at least one task has been completed. Finally, at Step **1816**, at least partially in response to the first computer software application being provided with the notification that the task has been completed, the system generates an updated privacy assessment for the product that reflects the fact that the task has been completed. The system may generate this updated privacy impact assessment automatically by, for example, automatically modifying any answers from within the question/answer pairings of the initial impact privacy assessment to reflect any modifications to the product that have been made in the course of completing the one or more tasks that implement the one or more substantive recommendations. For example, if a particular question from the initial privacy impact assessment indicated that certain personal data was personally-identifiable data, and a recommendation was made to anonymize the data, the question/answer pairing for the particular question could be revised so that the answer to the question indicates that the data has been anonymized. Any revised question/answer pairings may then be used to complete an updated privacy assessment report.

(270) FIGS. **19A-19B** depict the operation of a Privacy Assessment Reporting Module **1900**. In various embodiments, when the system executes the Privacy Assessment Reporting Module **1900**, the system begins, at Step **1902**, where it presents a series of questions to a user (e.g., via a suitable computer display screen or other user-interface, such as a voice-interface) regarding the design and/or anticipated operation of the product. This may be done, for example, by having a first software application (e.g., a data privacy software application or other suitable application) present the user with a template of questions regarding the product (e.g., for use in conducting an initial privacy impact assessment for the product). Such questions may include, for example, data mapping questions and other questions relevant to the product's design and/or anticipated operation.

(271) Next, at Step **1904**, the system receives, e.g., via a first computer software application, from a first set of one or more users (e.g., product designers, such as software designers, or other individuals who are knowledgeable about the product), respective answers to the questions regarding the product and associates the respective answers with their corresponding respective questions within memory to create a plurality of question/answer pairings regarding the proposed design of the product (e.g., software, a computerized electro-mechanical product, or other product).

(272) Next, at Step **1906**, the system presents a question to one or more users requesting the scheduled implantation date for the product. At Step **1908**, the system receives this response and saves the scheduled implementation date to memory.

(273) Next, after receiving the respective answers at Step **1904**, the system displays, at Step **1910**, the respective answers (e.g., along with their respective questions and/or a summary of the respective questions) to a second set of one or more users (e.g., one or more privacy officers from the organization that is designing the product), for example, in the form a plurality of suitable question/answer pairings. As an aside, within the context of this specification, pairings of an answer and either its respective question or a summary of the question may be referred to as a "question/answer" pairing. As an example, the question "Is the data encrypted?" and respective answer "Yes" may be represented, for example, in either of the following question/answer pairings: (1) "The data is encrypted"; and (2) "Data encrypted? Yes". Alternatively, the question/answer pairing may be represented as a value in a particular field in a data structure that would convey that the data at issue is encrypted.

(274) The system then advances to Step **1912**, where it receives, from the second set of users, one or more recommended steps to be implemented as part of the proposed design of the product and before the implementation date, the one or more recommended steps comprising one or more steps that facilitate the compliance of the product with the one or more privacy standards and/or policies. In particular embodiments in which the product is a software application or an electro-mechanical device that runs device software, the one or more recommended steps may comprise modifying the software application or device software to comply with one or more privacy standards and/or policies.

(275) Next, at Step **1914**, in response to receiving the one or more recommended steps, the system automatically initiates the generation of one or more tasks in a second computer software application (e.g., project management software) that is to be used in managing the design of the product. In particular embodiments, the one or more tasks comprise one or more tasks that, if completed, individually and/or collectively would result in the completion of the one or more recommended steps.

(276) The system then initiates a monitoring process for determining whether the one or more tasks have been completed. This step may, for example, be implemented by automatically monitoring which changes (e.g., edits to software code) have been made to the product, or by receiving manual input confirming that various tasks have been completed.

(277) The system then advances to Step **1916**, where it receives a notification that the at least one task has been completed. Next, at Step **1918**, at least partially in response to the first computer software application being provided with the notification that the task has been completed, the system generates an updated privacy assessment for the product that reflects the fact that the task has been completed. The system may generate this updated privacy impact assessment automatically by, for example, automatically modifying any answers from within the question/answer pairings of the initial impact privacy assessment to reflect any modifications to the product that have been made in the course of completing the one or more tasks that implement the one or more substantive recommendations. For example, if a particular question from the initial privacy impact assessment indicated that certain personal data was personally-identifiable data, and a recommendation was made to anonymize the data, the question/answer pairing for the particular question could be revised so that the answer to the question indicates that the data has been anonymized. Any revised question/answer pairings may then be used to complete an updated privacy assessment report.

(278) As discussed above, at Step **1920**, the system may then analyze the one or more revisions that have made to the product to determine whether the one or more revisions substantively impact the product's compliance with one or more privacy standards. Finally, at Step **1922**, the system generates a privacy-by-design report that may, for example, include a listing of any of the one or more revisions that have been made and that substantively impact the product's compliance with one or more privacy standards.

(279) In various embodiments, the privacy-by-design report may also comprise, for example, a log of data demonstrating that the business, in the normal course of its operations: (1) conducts privacy impact assessments on new products before releasing them; and (2) implements any changes needed to comply with one or more privacy policies before releasing the new products. Such logs may include data documenting the results of any privacy impact assessments conducted by the business (and/or any particular sub-part of the business) on new products before each respective new product's launch date, any revisions that the business (and/or any particular sub-part of the business) make to new products before the launch of the product. The report may also optionally include the results of any updated privacy impact assessments conducted on products after the products have been revised to comply with one or more privacy regulations and/or policies. The report may further include a listing of any changes that the business has made to particular products in response to initial impact privacy assessment results for the products. The system may also list which of the listed changes were determined, by the system, to be substantial changes (e.g., that the changes resulted in advancing the product's compliance with one or more privacy regulations).

(280) Additional Aspects of System

(281) 1. Standardized and Customized Assessment of Vendors' Compliance with Privacy and/or Security Policies

(282) In particular embodiments, the system may be adapted to: (1) facilitate the assessment of one or more vendors' compliance with one or more privacy and/or security policies; and (2) allow organizations (e.g., companies or other organizations) who do business with the vendors to create,

view and/or apply customized criteria to information periodically collected by the system to evaluate each vendor's compliance with one or more of the company's specific privacy and/or security policies. In various embodiments, the system may also flag any assessments, projects, campaigns, and/or data flows that the organization has documented and maintained within the system if those data flows are associated with a vendor that has its rating changed so that the rating meets certain criteria (e.g., if the vendor's rating falls below a predetermined threshold).

(283) In particular embodiments: The system may include an online portal and community that includes a listing of all supported vendors. An appropriate party (e.g., the participating vendor or a member of the on-line community) may use the system to submit an assessment template that is specific to a particular vendor. If the template is submitted by the vendor itself, the template may be tagged in any appropriate way as "official" An instance for each organization using the system (i.e., customer) is integrated with this online community/portal so that the various assessment templates can be directly fed into that organization's instance of the system if the organization wishes to use it. Vendors may subscribe to a predetermined standardized assessment format. Assessment results may also be stored in the central community/portal. A third-party privacy and/or security policy compliance assessor, on a schedule, may (e.g., periodically) complete the assessment of the vendor. Each organization using the system can subscribe to the results (e.g., once they are available). Companies can have one or more customized rules set up within the system for interpreting the results of assessments in their own unique way. For example: Each customer can weight each question within an assessment as desired and set up addition/multiplication logic to determine an aggregated risk score that takes into account the customized weightings given to each question within the assessment. Based on new assessment results—the system may notify each customer if the vendor's rating falls, improves, or passes a certain threshold. The system can flag any assessments, projects, campaigns, and/or data flows that the customer has documented and maintained within the system if those data flows are associated with a vendor that has its rating changed.

#### 2. Privacy Policy Compliance System that Facilitates Communications with Regulators (Including Translation Aspect)

(284) In particular embodiments, the system is adapted to interface with the computer systems of regulators (e.g., government regulatory agencies) that are responsible for approving privacy campaigns. This may, for example, allow the regulators to review privacy campaign information directly within particular instances of the system and, in some embodiments, approve the privacy campaigns electronically.

(285) In various embodiments, the system may implement this concept by: Exporting relevant data regarding the privacy campaign, from an organization's instance of the system (e.g., customized version of the system) in standardized format (e.g., PDF or Word) and sending the extracted data to an appropriate regulator for review (e.g., in electronic or paper format). Either regular provides the format that the system codes to, or the organization associated with the system provides a format that the regulators are comfortable with. Send secure link to regulator that gives them access to comment and leave feedback Gives the regulator direct access to the organization's instance of the system with a limited and restricted view of just the projects and associated audit and commenting logs the organization needs reviewed. Regulator actions are logged historically and the regulator can leave guidance, comments, and questions, etc. Have portal for regulator that securely links to the systems of their constituents.

(286) Details: When submitted—the PIAs are submitted with requested priority—standard or expedited. DPA specifies how many expedited requests individuals are allowed to receive. Either the customer or DPA can flag a PIA or associated comments/guidance on the PIA with "needs translation" and that can trigger an automated or manual language translation. Regulator could be a DPA "data protection authority" in any EU country, or other country with similar concept like FTC in US, or OPC in Canada.

#### 3. Systems/Methods for Measuring the Privacy Maturity of a Business Group within an Organization.

(287) In particular embodiments, the system is adapted for automatically measuring the privacy of a business group, or other group, within a particular organization that is using the system. This may provide an automated way of measuring the privacy maturity, and one or more trends of change in privacy maturity of the organization, or a selected sub-group of the organization.

(288) In various embodiments, the organization using the system can customize one or more algorithms used by the system to measure the privacy maturity of a business group (e.g., by specifying one or more variables and/or relative weights for each variable in calculating a privacy maturity score for the group). The following are examples of variables that may be used in this process: Issues/Risks found in submitted assessments that are unmitigated or uncaught prior to the assessment being submitted to the privacy office % of privacy assessments with high issues/total assessments % with medium % with low Size and type of personal data used by the group Total assessments done Number of projects/campaigns with personal data Amount of personal data Volume of data transfers to internal and external parties Training of the people in the group Number or % of individuals who have watched training, readings, or videos Number or % of individuals who have completed quizzes or games for privacy training Number or % of individuals who have attended privacy events either internally or externally Number or % of individuals who are members of IAPP Number or % of individuals who have been specifically trained in privacy either internally or externally, formally (IAPP certification) or informally Usage of an online version of the system, or mobile training or communication portal that customer has implemented Other factors

#### 4. Automated Assessment of Compliance (Scan App or Website to Determine Behavior/Compliance with Privacy Policies)

(289) In various embodiments, instead of determining whether an organization complies with the defined parameters of a privacy campaign by, for example, conducting an audit as described above (e.g., by asking users to answer questions regarding the privacy campaign, such as "What is collected?" "what cookies are on your website", etc.), the system may be configured to automatically determine whether the organization is complying with one or more aspects of the privacy policy.

(290) For example, during the audit process, the system may obtain a copy of a software application (e.g., an "app") that is collecting and/or using sensitive user information, and then automatically analyze the app to determine whether the operation of the app is complying with the terms of the privacy campaign that govern use of the app.

(291) Similarly, the system may automatically analyze a website that is collecting and/or using sensitive user information to determine whether the operation of the web site is complying with the terms of the privacy campaign that govern use of the web site.

(292) In regard to various embodiments of the automatic application-analyzing embodiment referenced above: The typical initial questions asked during an audit may be replaced by a request to "Upload your app here". After the app is uploaded to the system, the system detects what privacy permissions and data the app is collecting from users. This is done by having the system use static or behavioral analysis of the application, or by having the system integrate with a third-party system or software (e.g., Veracode), which executes the analysis. During the analysis of the app, the system may detect, for example, whether the app is using location services to detect the location of the user's mobile device. In response to determining that the app is collecting one or more specified types of sensitive information (e.g., the location of the user's mobile device), the system may automatically request follow up information from the user by posing one or more questions to the user, such as: For what business reason is the data being collected? How is the user's consent given to obtain the data? Would users be surprised that the data is being collected? Is the data encrypted at rest and/or in motion? What would happen if the system did not collect this data? What business impact would it have? In various embodiments, the system is adapted to allow each organization to define these follow-up questions, but the system asks the questions (e.g., the same questions, or a customized list of questions) for each privacy issue that is found in the app. In various embodiments, after a particular app is scanned a first time, when the app is scanned, the system may only detect and analyze any changes that have been made to the app since the previous scan of the app. In various embodiments, the system is adapted to (optionally) automatically monitor (e.g., continuously monitor) one or more online software application marketplaces (such as Microsoft, Google, or Apple's App Store) to determine whether the application has changed. If so, the system may, for example: (1) automatically scan the application as discussed above; and (2) automatically notify one or more designated individuals (e.g., privacy office representatives) that an app was detected that the business failed to perform a privacy assessment on prior to launching the application.

(293) In regard to various embodiments of the automatic application-analyzing embodiment referenced above: The system prompts the user to enter the URL of the website to be analyzed, and, optionally, the URL to the privacy policy that applies to the web site. The system then scans the website

for cookies, and/or other tracking mechanisms, such as fingerprinting technologies and/or 3rd party SDKs. The system may then optionally ask the user to complete a series of one or more follow-up questions for each of these items found during the scan of the website. This may help the applicable privacy office craft a privacy policy to be put on the website to disclose the use of the tracking technologies and SDK's used on the website. The system may then start a continuous monitoring of the website site to detect whether any new cookies, SDKs, or tracking technologies are used. In various embodiments, the system is configured to, for example, generate an alert to an appropriate individual (e.g., a designated privacy officer) to inform them of the change to the website. The privacy officer may use this information, for example, to determine whether to modify the privacy policy for the website or to coordinate discontinuing use of the new tracking technologies and/or SDK's. In various embodiments, the system may also auto-detect whether any changes have been made to the policy or the location of the privacy policy link on the page and, in response to auto-detecting such changes, trigger an audit of the project. It should be understood that the above methods of automatically assessing behavior and/or compliance with one or more privacy policies may be done in any suitable way (e.g., ways other than website scanning and app scanning). For example, the system may alternatively, or in addition, automatically detect, scan and/or monitor any appropriate technical system(s) (e.g., computer system and/or system component or software), cloud services, apps, websites and/or data structures, etc.

#### 5. System Integration with DLP Tools.

(294) DLP tools are traditionally used by information security professionals. Various DLP tools discover where confidential, sensitive, and/or personal information is stored and use various techniques to automatically discover sensitive data within a particular computer system—for example, in emails, on a particular network, in databases, etc. DLP tools can detect the data, what type of data, the amount of data, and whether the data is encrypted. This may be valuable for security professionals, but these tools are typically not useful for privacy professionals because the tools typically cannot detect certain privacy attributes that are required to be known to determine whether an organization is in compliance with particular privacy policies.

(295) For example, traditional DLP tools cannot typically answer the following questions: Who was the data collected from (data subject)? Where are those subjects located? Are they minors? How was consent to use the data received? What is the use of the data? Is the use consistent with the use specified at the time of consent? What country is the data stored in and/or transferred to? Etc. In various embodiments, the system is adapted to integrate with appropriate DLP and/or data discovery tools (e.g., INFORMATICA) and, in response to data being discovered by those tools, to show each area of data that is discovered as a line-item in a system screen via integration. The system may do this, for example, in a manner that is similar to pending transactions in a checking account that have not yet been reconciled. A designated privacy officer may then select one of those—and either match it up (e.g., reconcile it) with an existing data flow or campaign in the system OR trigger a new assessment to be done on that data to capture the privacy attributes and data flow.

#### 6. System for Generating an Organization's Data Map by Campaign, by System, or by Individual Data Attributes.

(296) In particular embodiments, the system may be adapted to allow users to specify various criteria, and then to display, to the user, any data maps that satisfy the specified criteria. For example, the system may be adapted to display, in response to an appropriate request: (1) all of a particular customer's data flows that are stored within the system; (2) all of the customer's data flows that are associated with a particular campaign; and/or (3) all of the customer's data flows that involve a particular address.

(297) Similarly, the system may be adapted to allow privacy officers to document and input the data flows into the system in any of a variety of different ways, including: Document by process The user initiates an assessment for a certain business project and captures the associated data flows (including the data elements related to the data flows and the systems they are stored in). Document by element The user initiates an audit of a data element—such as SSN—and tries to identify all data structures associated with the organization that include the SSN. The system may then document this information (e.g., all of the organization's systems and business processes that involve the business processes.) Document by system The user initiates an audit of a database, and the system records, in memory, the results of the audit.

#### 7. Privacy Policy Compliance System that Allows Users to Attach Emails to Individual Campaigns.

(298) Privacy officers frequently receive emails (or other electronic messages) that are associated with an existing privacy assessment or campaign, or a potential future privacy assessment. For record keeping and auditing purposes, the privacy officer may wish to maintain those emails in a central storage location, and not in email. In various embodiments, the system is adapted to allow users to automatically attach the email to an existing privacy assessment, data flow, and/or privacy campaign. Alternatively, or additionally, the system may allow a user to automatically store emails within a data store associated with the system, and to store the emails as “unassigned”, so that they may later be assigned to an existing privacy assessment, data flow, and/or privacy campaign. In various embodiments, the system is adapted to allow a user to store an email using: a browser plugin-extension that captures webmail; a Plug-in directly with office 365 or google webmail (or other suitable email application); a Plug-in with email clients on computers such as Outlook; via an integrated email alias that the email is forwarded to; or any other suitable configuration

#### 8. Various Aspects of Related Mobile Applications

(299) In particular embodiments, the system may use a mobile app (e.g., that runs on a particular mobile device associated by a user) to collect data from a user. The mobile app may be used, for example, to collect answers to screening questions. The app may also be adapted to allow users to easily input data documenting and/or reporting a privacy incident. For example, the app may be adapted to assist a user in using their mobile device to capture an image of a privacy incident (e.g., a screen shot documenting that data has been stored in an improper location, or that a printout of sensitive information has been left in a public workspace within an organization.)

(300) The mobile app may also be adapted to provide incremental training to individuals. For example, the system may be adapted to provide incremental training to a user (e.g., in the form of the presentation of short lessons on privacy). Training sessions may be followed by short quizzes that are used to allow the user to assess their understanding of the information and to confirm that they have completed the training.

#### (301) 9. Automatic Generation of Personal Data Inventory for Organization

(302) In particular embodiments, the system is adapted to generate and display an inventory of the personal data that an organization collects and stores within its systems (or other systems). As discussed above, in various embodiments, the system is adapted to conduct privacy impact assessments for new and existing privacy campaigns. During a privacy impact assessment for a particular privacy campaign, the system may ask one or more users a series of privacy impact assessment questions regarding the particular privacy campaign and then store the answers to these questions in the system's memory, or in memory of another system, such a third-party computer server.

(303) Such privacy impact assessment questions may include questions regarding: (1) what type of data is to be collected as part of the campaign; (2) who the data is to be collected from; (3) where the data is to be stored; (4) who will have access to the data; (5) how long the data will be kept before being deleted from the system's memory or archived; and/or (6) any other relevant information regarding the campaign.

(304) The system may store the above information, for example, in any suitable data structure, such as a database. In particular embodiments, the system may be configured to selectively (e.g., upon request by an authorized user) generate and display a personal data inventory for the organization that includes, for example, all of the organization's current active campaigns, all of the organization's current and past campaigns, or any other listing of privacy campaigns that, for example, satisfy criteria specified by a user. The system may be adapted to display and/or export the data inventory in any suitable format (e.g., in a table, a spreadsheet, or any other suitable format).

#### (305) 10. Integrated/Automated Solution for Privacy Risk Assessments

(306) Continuing with Concept 9, above, in various embodiments, the system may execute multiple integrated steps to generate a personal data inventory for a particular organization. For example, in a particular embodiment, the system first conducts a Privacy Threshold Assessment (PTA) by asking a user a relatively short set of questions (e.g., between 1 and 15 questions) to quickly determine whether the risk associated with the campaign may potentially exceed a pre-determined risk threshold (e.g., whether the campaign is a potentially high-risk campaign). The system may do this, for



example, by using any of the above techniques to assign a collective risk score to the user's answers and determining whether the collective risk score exceeds a particular risk threshold value. Alternatively, the system may be configured to determine that the risk associated with the campaign exceeds the risk threshold value if the user answers a particular one or more of the questions in a certain way.

(307) The system may be configured for, in response to the user's answers to one or more of the questions within the Privacy Threshold Assessment indicating that the campaign exceeds, or may potentially exceed, a pre-determined risk threshold, presenting the user with a longer set of detailed questions regarding the campaign (e.g., a Privacy Impact Assessment). The system may then use the user's answers to this longer list of questions to assess the overall risk of the campaign, for example, as described above.

(308) In particular embodiments, the system may be configured for, in response to the user's answers to one or more of the questions within the Privacy Threshold Assessment indicating that the campaign does not exceed, or does not potentially exceed, a pre-determined risk threshold, not presenting the user with a longer set of detailed questions regarding the campaign (e.g., a Privacy Impact Assessment). In such a case, the system may simply save an indication to memory that the campaign is a relatively low risk campaign.

(309) Accordingly, in particular embodiments, the system may be adapted to automatically initiate a Privacy Impact Assessment if the results of a shorter Privacy Threshold Assessment satisfy certain criteria. Additionally, or alternatively, in particular embodiments, the system may be adapted to allow a privacy officer to manually initiate a Privacy Impact Assessment for a particular campaign.

(310) In particular embodiments, built into the Privacy Threshold Assessment and the Privacy Impact Assessment are the data mapping questions and/or sub-questions of how the personal data obtained through the campaign will be collected, used, stored, accessed, retained, and/or transferred, etc. In particular embodiments: (1) one or more of these questions are asked in the Privacy Threshold Assessment; and (2) one or more of the questions are asked in the Privacy Impact Assessment. In such embodiments, the system may obtain the answers to each of these questions, as captured during the Privacy Threshold Assessment and the Privacy Impact Assessment, and then use the respective answers to generate the end-to-end data flow for the relevant privacy campaign.

(311) The system may then link all of the data flows across all of the organization's privacy campaigns together in order to show a complete evergreen version of the personal data inventory of the organization. Thus, the system may efficiently generate the personal data inventory of an organization (e.g., through the use of reduced computer processing power) by automatically gathering the data needed to prepare the personal data inventory while conducting Privacy Threshold Assessments and Privacy Impact Assessments.

(312) System for Preventing Individuals from Trying to Game the System

(313) As discussed above, in particular embodiments, the system is adapted to display a series of threshold questions for particular privacy campaigns and to use conditional logic to assess whether to present additional, follow-up questions to the user. There may, for example, be situations in which a user may answer, or attempt to answer, one or more of the threshold questions incorrectly (e.g., dishonestly) in an attempt to avoid needing to answer additional questions. This type of behavior can present serious potential problems for the organization because the behavior may result in privacy risks associated with a particular privacy campaign being hidden due to the incorrect answer or answers.

(314) To address this issue, in various embodiments, the system maintains a historical record of every button press (e.g., un-submitted system input) that an individual makes when a question is presented to them. In particular embodiments, actively monitoring the user's system inputs may include, for example, monitoring, recording, tracking, and/or otherwise taking account of the user's system inputs. These system inputs may include, for example: (1) one or more mouse inputs; (2) one or more keyboard (e.g., text) inputs; (3) one or more touch inputs; and/or (4) any other suitable inputs (e.g., such as one or more vocal inputs, etc.). In various embodiments, the system is configured to actively monitor the user's system inputs, for example: (1) while the user is viewing one or more graphical user interfaces for providing information regarding or responses to questions regarding one or more privacy campaigns; (2) while the user is logged into a privacy portal; and/or (3) in any other suitable situation related to the user providing information related to the collection or storage of personal data (e.g., in the context of a privacy campaign). Additionally, the system tracks, and saves to memory, each incidence of the individual changing their answer to a question (e.g., (a) before formally submitting the answer by pressing an "enter" key, or other "submit" key on a user interface, such as a keyboard or graphical user interface on a touch-sensitive display screen; or (b) after initially submitting the answer).

(315) The system may also be adapted to automatically determine whether a particular question (e.g., threshold question) is a "critical" question that, if answered in a certain way, would cause the conditional logic trigger to present the user with one or more follow-up questions. For example, the system may, in response to receiving the user's full set of answers to the threshold questions, automatically identify any individual question within the series of threshold questions that, if answered in a particular way (e.g., differently than the user answered the question) would have caused the system to display one or more follow up questions. The system may then flag those identified questions, in the system's memory, as "critical" questions.

(316) Alternatively, the system may be adapted to allow a user (e.g., a privacy officer of an organization) who is drafting a particular threshold question that, when answered in a particular way, will automatically trigger the system to display one or more follow up questions to the user, to indicate that is a "critical" threshold question. The system may then save this "critical" designation of the question to the system's computer memory.

(317) In various embodiments, the system is configured, for any questions that are deemed "critical" (e.g., either by the system, or manually, as discussed above), to determine whether the user exhibited any abnormal behavior when answering the question. For example, the system may check to see whether the user changed their answer once, or multiple times, before submitting their answer to the question (e.g., by tracking the user's keystrokes while they are answering the threshold question, as described above). As another example, the system may determine whether it took the user longer than a pre-determined threshold amount of time (e.g., 5 minutes, 3 minutes, etc. . . .) to answer the critical threshold question.

(318) In particular embodiments, the system may be adapted, in response to determining that the user exhibited abnormal behavior when answering the critical threshold question, to automatically flag the threshold question and the user's answer to that question for later follow up by a designated individual or team (e.g., a member of the organization's privacy team). In particular embodiments, the system may also, or alternatively, be adapted to automatically generate and transmit a message to one or more individuals (e.g., the organization's chief privacy officer) indicating that the threshold question may have been answered incorrectly and that follow-up regarding the question may be advisable. After receiving the message, the individual may, in particular embodiments, follow up with the individual who answered the question, or conduct other additional research, to determine whether the question was answered accurately.

(319) In particular embodiments, the system is configured to monitor a user's context as the user provides responses for a computerized privacy questionnaire. The user context may take in to account a multitude of different user factors to incorporate information about the user's surroundings and circumstances. One user factor may be the amount of time a user takes to respond to one or more particular questions or the complete computerized privacy questionnaire. For example, if the user rushed through the computerized privacy questionnaire, the system may indicate that user abnormal behavior occurred in providing the one or more responses. In some implementations, the system may include a threshold response time for each question of the computerized privacy questionnaire (e.g., this may be a different threshold response time for each question) or the complete computerized privacy questionnaire. The system may compare the response time for each of the one or more responses to its associated threshold response time, and/or the system may compare the response time for completion of the computerized privacy questionnaire to the associated threshold response time for completion of the full computerized privacy questionnaire. The system may be configured to indicate that user abnormal behavior occurred in providing the one or more responses when either the response time is a longer period of time (e.g., perhaps indicating that the user is being dishonest) or shorter period of time (e.g., perhaps indicating that the user is rushing through the computerized privacy questionnaire and the responses may be inaccurate) than the threshold response time.

(320) Another user factor may be a deadline for initiation or completion of the computerized privacy questionnaire. For example, if the user initiated



or completed the computerized privacy questionnaire after a particular period of time (e.g., an initiation time or a completion time), the system may indicate that user abnormal behavior occurred in providing the one or more responses. The certain period of time may be preset, user-defined, and/or adjusted by the user, and may be a threshold time period. Additionally, in some implementations, the user factors may be adjusted based on one another. For example, if the user initiated the computerized privacy questionnaire close to a deadline for the computerized privacy questionnaire, then the threshold response time for each question of the computerized privacy questionnaire or the complete computerized privacy questionnaire may be modified (e.g., the threshold response time may be increased to ensure that the user does not rush through the privacy questionnaire close to the deadline).

(321) Additionally, another user factor may incorporate a location in which the user conducted the privacy questionnaire. For example, if the user conducted the privacy questionnaire in a distracting location (e.g., at the movies or airport), the system may indicate that user abnormal behavior occurred. The system may use GPS tracking data associated with the electronic device (e.g., laptop, smart phone) on which the user conducted the privacy questionnaire to determine the location of the user. The system may include one or more particular locations or types of locations that are designated as locations in which the user may be distracted, or otherwise provide less accurate results. The locations may be specific to each user or the same locations for all users, and the locations may be adjusted (e.g., added, removed, or otherwise modified). The types of locations may be locations such as restaurants, entertainment locations, mass transportation points (e.g., airports, train stations), etc.

(322) In particular embodiments, the system is configured to determine a type of connection via which the user is accessing the questionnaire. For example, the system may determine that the user is accessing the questionnaire while connect to a public wireless network (e.g., at an airport, coffee shop, etc.). The system may further determine that the user is connect to a wireless or other network such as a home network (e.g., at the user's house). In such examples, the system may determine that the user may be distracted based on a location inferred based on one or more connections identified for the computing device via which the user is accessing the questionnaire. In other embodiments, the system may determine that the user is connect via a company network (e.g., a network associated with the entity providing the questionnaire for completion). In such embodiments, the system may be configured to determine that the user is focused on the questionnaire (e.g., by virtue of the user being at work while completing it).

(323) Moreover, another user factor may involve determining the electronic activities the user is performing on the user's electronic device while they are completing the privacy questionnaire. This factor may also be related to determining if the user is distracted when completing the privacy questionnaire. For example, the system may determine whether the user interacted, on the electronic device, with one or more web browsers or software applications that are unrelated to conducting the computerized privacy questionnaire (e.g., by determining whether the user accessed one or more other active browsing windows, or whether a browsing window in which the user is completing the questionnaire becomes inactive while the user is completing it). If the system determines that such unrelated electronic activities were interacted with, the system may indicate that user abnormal behavior occurred in completing the privacy questionnaire. Further, the electronic activities may be preset, user-specific, and/or modified. The user factors above are provided by way of example, and more, fewer, or different user factors may be included as part of the system. In some embodiments, the system may incorporate the user's electronic device camera to determine if the user is exhibiting abnormal behavior (e.g., pupils dilated/blinking a lot could indicate deception in responding to the privacy questionnaire).

(324) In some implementations, the system may use one or more of the user factors to calculate a user context score. Each of the user factors may include a user factor rating to indicate a likelihood that user abnormal behavior occurred with respect to that particular user factor. The user context score may be calculated based on each of the user factor ratings. In some embodiments, a weighting factor may be applied to each user factor (e.g., this may be specific for each organization) for the calculation of the user context score. Additionally, in some embodiments, if one or more user factor ratings is above a certain rating (i.e., indicating a very likelihood of user abnormal behavior for that particular user factor), then the user context score may automatically indicate that user abnormal behavior occurred in completing the privacy questionnaire. The user context score may be compared to a threshold user context score that may be preset, user or organization defined, and/or modified. If the system determines that the user context score is greater than the threshold user context score (i.e., indicates a higher likelihood of user abnormal behavior than the likelihood defined by threshold), then the system may indicate that user abnormal behavior occurred in conducting the privacy questionnaire.

(325) In some implementations, the submitted input of the user to one or more responses may include a particular type of input that may cause the system to provide one or more follow up questions. The follow up questions may be provided for the user justify the particular type of input response that was provided. The particular type of input may be responses that are indefinite, indicate the user is unsure of the appropriate response (e.g., "I do not know"), or intimate that the user is potentially being untruthful in the response. For example, if the user provides a response of "I do not know" (e.g., by selecting in a list or inputting in a text box), the system may be configured to provided one or more follow up questions to further determine why the user "does not know" the answer to the specific inquiry or if the user is being truthful in saying they "do not know."

(326) In some implementations, the system may, for each of the one or more responses to one or more questions in the computerized privacy questionnaire, determine a confidence factor score. The confidence factor score may be based on the user context of the user as the user provides the one or more responses and/or the one or more system inputs from the user the comprise the one or more responses. For example, if the user was in a distracting environment when the user provided a particular response in the privacy questionnaire and/or the user provided one or more unsubmitted inputs prior to providing the submitted input for the particular response, the system may calculate a low confidence factor score for the particular response.

(327) Further, the system may calculate a confidence score for the computerized privacy questionnaire based at least in part on the confidence factor score for each of the one or more responses to one or more questions in the computerized privacy questionnaire. Upon calculating the confidence score, the system can use the confidence score to determine whether user abnormal behavior occurred in providing the one or more responses. In some implementations, a low confidence factor score for a single response may cause the confidence score of the privacy questionnaire to automatically indicate user abnormal behavior occurred in providing the privacy questionnaire. However, in other embodiments, this is not the case. For example, if only two out of twenty confidence factor scores are very low (i.e., indicate a higher likelihood of user abnormal behavior in providing the particular response), the system may determine, based on the calculated confidence score for the privacy questionnaire, that user abnormal behavior did not occur in completing the privacy questionnaire.

(328) Privacy Assessment Monitoring Module

(329) In particular embodiments, a Privacy Assessment Monitoring Module **2000** is configured to: (1) monitor user inputs when the user is providing information related to a privacy campaign or completing a privacy impact assessment; and (2) determine, based at least in part on the user inputs, whether the user has provided one or more abnormal inputs or responses. In various embodiments, the Privacy Assessment Monitoring Module **2000** is configured to determine whether the user is, or may be, attempting to provide incomplete, false, or misleading information or responses related to the creation of a particular privacy campaign, a privacy impact assessment associated with a particular privacy campaign, etc.

(330) Turning to FIG. **20**, in particular embodiments, when executing the Privacy Assessment Monitoring Module **2000**, the system begins, at Step **2010**, by receiving an indication that a user is submitting one or more responses to one or more questions related to a particular privacy campaign. In various embodiments, the system is configured to receive the indication in response to a user initiating a new privacy campaign (e.g., on behalf of a particular organization, sub-group within the organization, or other suitable business unit). In other embodiments, the system is configured to receive the indication while a particular user is completing a privacy impact assessment for a particular privacy campaign, where the privacy impact assessment provides oversight into various aspects of the particular privacy campaign such as, for example: (1) what personal data is collected as part of the privacy campaign; (2) where the personal data is stored; (3) who has access to the stored personal data; (4) for what purpose the personal data is collected, etc.

(331) In various embodiments, the system is configured to receive the indication in response to determining that a user has accessed a privacy

campaign initiation system (e.g., or other privacy system) and is providing one or more pieces of information related to a particular privacy campaign. In particular embodiments, the system is configured to receive the indication in response to the provision, by the user, of one or more responses as part of a privacy impact assessment. In various embodiments, the system is configured to receive the indication in response to any suitable stimulus in any situation in which a user may provide one or more potentially abnormal responses to one or more questions related to the collection, storage or use of personal data.

(332) In various embodiments, the privacy campaign may be associated with an electronic record (e.g., or any suitable data structure) comprising privacy campaign data. In particular embodiments, the privacy campaign data comprises a description of the privacy campaign, one or more types of personal data related to the campaign, a subject from which the personal data is collected as part of the privacy campaign, a storage location of the personal data (e.g., including a physical location of physical memory on which the personal data is stored), one or more access permissions associated with the personal data, and/or any other suitable data associated with the privacy campaign. In various embodiments, the privacy campaign data is provided by a user of the system.

(333) An exemplary privacy campaign, project, or other activity may include, for example: (1) a new IT system for storing and accessing personal data (e.g., include new hardware and/or software that makes up the new IT system); (2) a data sharing initiative where two or more organizations seek to pool or link one or more sets of personal data; (3) a proposal to identify people in a particular group or demographic and initiate a course of action; (4) using existing data for a new and unexpected or more intrusive purpose; and/or (5) one or more new databases which consolidate information held by separate parts of the organization. In still other embodiments, the particular privacy campaign, project or other activity may include any other privacy campaign, project, or other activity discussed herein, or any other suitable privacy campaign, project, or activity.

(334) During a privacy impact assessment for a particular privacy campaign, a privacy impact assessment system may ask one or more users (e.g., one or more individuals associated with the particular organization or sub-group that is undertaking the privacy campaign) a series of privacy impact assessment questions regarding the particular privacy campaign and then store the answers to these questions in the system's memory, or in memory of another system, such as a third-party computer server.

(335) Such privacy impact assessment questions may include questions regarding, for example: (1) what type of data is to be collected as part of the campaign; (2) who the data is to be collected from; (3) where the data is to be stored; (4) who will have access to the data; (5) how long the data will be kept before being deleted from the system's memory or archived; and/or (6) any other relevant information regarding the campaign. In various embodiments a privacy impact assessment system may determine a relative risk or potential issues with a particular privacy campaign as it related to the collection and storage of personal data. For example, the system may be configured to identify a privacy campaign as being "High" risk, "Medium" risk, or "Low" risk based at least in part on answers submitted to the questions listed above. For example, a Privacy Impact Assessment that revealed that credit card numbers would be stored without encryption for a privacy campaign would likely cause the system to determine that the privacy campaign was high risk.

(336) As may be understood in light of this disclosure, a particular organization may implement operational policies and processes that strive to comply with industry best practices and legal requirements in the handling of personal data. In various embodiments, the operational policies and processes may include performing privacy impact assessments (e.g., such as those described above) by the organization and/or one or more sub-groups within the organization. In particular embodiments, one or more individuals responsible for completing a privacy impact assessment or providing privacy campaign data for a particular privacy campaign may attempt to provide abnormal, misleading, or otherwise incorrect information as part of the privacy impact assessment. In such embodiments, the system may be configured to receive the indication in response to receiving an indication that a user has initiated or is performing a privacy impact assessment.

(337) Returning to Step **2020**, the system is configured to, in response to receiving the indication at Step **2010**, monitor (e.g., actively monitor) the user's system inputs. In particular embodiments, actively monitoring the user's system inputs may include, for example, monitoring, recording, tracking, and/or otherwise taking account of the user's system inputs. These system inputs may include, for example: (1) one or more mouse inputs; (2) one or more keyboard (e.g., text) inputs; (3) one or more touch inputs; and/or (4) any other suitable inputs (e.g., such as one or more vocal inputs, etc.). In various embodiments, the system is configured to actively monitor the user's system inputs, for example: (1) while the user is viewing one or more graphical user interfaces for providing information regarding or responses to questions regarding one or more privacy campaigns; (2) while the user is logged into a privacy portal; and/or (3) in any other suitable situation related to the user providing information related to the collection or storage of personal data (e.g., in the context of a privacy campaign). In other embodiments, the system is configured to monitor one or more biometric indicators associated with the user such as, for example, heart rate, pupil dilation, perspiration rate, etc.

(338) In particular embodiments, the system is configured to monitor a user's inputs, for example, by substantially automatically tracking a location of the user's mouse pointer with respect to one or more selectable objects on a display screen of a computing device. In particular embodiments, the one or more selectable objects are one or more selectable objects (e.g., indicia) that make up part of a particular privacy impact assessment, privacy campaign initiation system, etc. In still other embodiments, the system is configured to monitor a user's selection of any of the one or more selectable objects, which may include, for example, an initial selection of one or more selectable objects that the user subsequently changes to selection of a different one of the one or more selectable objects.

(339) In any embodiment described herein, the system may be configured to monitor one or more keyboard inputs (e.g., text inputs) by the user that may include, for example, one or more keyboard inputs that the user enters or one or more keyboard inputs that the user enters but deletes without submitting. For example, a user may type an entry relating to the creation of a new privacy campaign in response to a prompt that asks what reason a particular piece of personal data is being collected for. The user may, for example, initially begin typing a first response, but delete the first response and enter a second response that the user ultimately submits. In various embodiments of the system described herein, the system is configured to monitor the un-submitted first response in addition to the submitted second response.

(340) In still other embodiments, the system is configured to monitor a user's lack of input. For example, a user may mouse over a particular input indicia (e.g., a selection from a drop-down menu, a radio button or other selectable indicia) without selecting the selection or indicia. In particular embodiments, the system is configured to monitor such inputs. As may be understood in light of this disclosure, a user that mouses over a particular selection and lingers over the selection without actually selecting it may be contemplating whether to: (1) provide a misleading response; (2) avoid providing a response that they likely should provide in order to avoid additional follow up questions; and/or (3) etc.

(341) In other embodiments, the system is configured to monitor any other suitable input by the user. In various embodiments, this may include, for example: (1) monitoring one or more changes to an input by a user; (2) monitoring one or more inputs that the user later removes or deletes; (3) monitoring an amount of time that the user spends providing a particular input; and/or (4) monitoring or otherwise tracking any other suitable information related to the user's response to a particular question and/or provision of a particular input to the system.

(342) Returning to Step **2030**, the system is configured to store, in memory, a record of the user's submitted and un-submitted system inputs. As discussed above, the system may be configured to actively monitor both submitted and un-submitted inputs by the user. In particular embodiments, the system is configured to store a record of those inputs in computer memory (e.g., in the One or More Storage Devices **130** shown in FIG. **1**). In particular embodiments, storing the user's submitted and un-submitted system inputs may include, for example, storing a record of: (1) each system input made by the user; (2) an amount of time spent by the user in making each particular input; (3) one or more changes to one or more inputs made by the user; (4) an amount of time spent by the user to complete a particular form or particular series of questions prior to submission; and/or (5) any other suitable information related to the user's inputs as they may relate to the provision of information related to one or more privacy campaigns.

(343) Continuing to Step **2040**, the system is configured to analyze the user's submitted and un-submitted inputs to determine one or more changes to the user's inputs prior to submission. In particular embodiments, the system may, for example: (1) compare a first text input with a second text input

to determine one or more differences, where the first text input is an unsubmitted input and the second text input is a submitted input; (2) determine one or more changes in selection, by the user, of a user-selectable input indicia (e.g., including a number of times the user changed a selection); and/or (3) compare any other system inputs by the user to determine one or more changes to the user's responses to one or more questions prior to submission. In various embodiments, the system is configured to determine whether the one or more changes include one or more changes that alter a meaning of the submitted and unsubmitted inputs.

(344) In various embodiments, the system is configured to compare first, unsubmitted text input with second, submitted text input to determine whether the content of the second text input differs from the first text input in a meaningful way. For example, a user may modify the wording of their text input without substantially modifying the meaning of the input (e.g., to correct spelling, utilize one or more synonyms, correct punctuation, etc.). In this example, the system may determine that the user has not made meaningful changes to their provided input.

(345) In another example, the system may determine that the user has changed the first input to the second input where the second input has a meaning that differs from a meaning of the first input. For example, the first and second text inputs may: (1) list one or more different individuals; (2) list one or more different storage locations; (3) include one or more words with opposing meanings (e.g., positive vs. negative, short vs. long, store vs. delete, etc.); and/or (4) include any other differing text that may indicate that the responses provided (e.g., the first text input and the second text input) do not have essentially the same meaning. In this example, the system may determine that the user has made one or more changes to the user's inputs prior to submission.

(346) Returning to Step **2050**, the system continues by determining, based at least in part on the user's system inputs and the one or more changes to the user's inputs, whether the user has provided one or more abnormal responses to the one or more questions. In various embodiments, the system is configured to determine whether the user has provided one or more abnormal responses to the one or more questions based on determining, at Step **2040**, that the user has made one or more changes to a response prior to submitting the response (e.g., where the one or more changes alter a meaning of the response).

(347) In other embodiments, the system is configured to determine that the user has provided one or more abnormal responses based on determining that the user took longer than a particular amount of time to provide a particular response. For example, the system may determine that the user has provided an abnormal response in response to the user taking longer than a particular amount of time (e.g., longer than thirty seconds, longer than one minute, longer than two minutes, etc.) to answer a simple multiple choice question (e.g., "Will the privacy campaign collect personal data for customers or employees?").

(348) In particular embodiments, the system is configured to determine that the user has provided one or more abnormal responses based on a number of times that the user has changed a response to a particular question. For example, the system may determine a number of different selections made by the user when selecting one or more choices from a drop down menu prior to ultimately submitting a response. In another example, the system may determine a number of times the user changed their free-form text entry response to a particular question. In various embodiments, the system is configured to determine that the user provided one or more abnormal responses in response to determining that the user changed their response to a particular question more than a threshold number of times (e.g., one time, two times, three times, four times, five times, etc.).

(349) In still other embodiments, the system is configured to determine that the user has provided one or more abnormal responses based at least in part on whether a particular question (e.g., threshold question) is a "critical" question. In particular embodiments, a critical question may include a question that, if answered in a certain way, would cause the system's conditional logic trigger to present the user with one or more follow-up questions. For example, the system may, in response to receiving the user's full set of answers to the threshold questions, automatically identify any individual question within the series of threshold questions that, if answered in a particular way (e.g., differently than the user answered the question) would have caused the system to display one or more follow up questions.

(350) In various embodiments, the system is configured, for any questions that are deemed "critical" (e.g., either by the system, or manually) to determine whether the user exhibited any abnormal behavior when answering the question. For example, the system may check to see whether the user changed their answer once, or multiple times, before submitting their answer to the question (e.g., by tracking the user's keystrokes or other system inputs while they are answering the threshold question, as described above). As another example, the system may determine whether it took the user longer than a pre-determined threshold amount of time (e.g., 5 minutes, 3 minutes, etc.) to answer the critical threshold question.

(351) In particular embodiments, the system is configured to determine whether the user provided one or more abnormal responses based on any suitable combination of factors described herein including, for example: (1) one or more changes to a particular response; (2) a number of changes to a particular response; (3) an amount of time it took to provide the particular response; (4) whether the response is a response to a critical question; and/or (5) any other suitable factor.

(352) Continuing to Step **2060**, the system, in response to determining that the user has provided one or more abnormal responses, automatically flags the one or more questions in memory. In particular embodiments, the system is configured to automatically flag the one or more questions in memory by associating the one or more questions in memory with a listing or index of flagged questions. In other embodiments, the system, in response to flagging the one or more questions, is further configured to generate a notification and transmit the notification to any suitable individual. For example, the system may transmit a notification that one or more question have been flagged by a particular privacy officer or other individual responsible ensuring that a particular organization's collection and storage of personal data meets one or more legal or industry standards.

(353) In particular embodiments, the system is configured to generate a report of flagged questions related to a particular privacy campaign. In various embodiments, flagging the one or more questions is configured to initiate a follow up by a designated individual or team (e.g., a member of the organization's privacy team) regarding the one or more questions. In particular embodiments, the system may also, or alternatively, be adapted to automatically generate and transmit a message to one or more individuals (e.g., the organization's chief privacy officer) indicating that the threshold question may have been answered incorrectly and that follow-up regarding the question may be advisable. After receiving the message, the individual may, in particular embodiments, follow up with the individual who answered the question, or conduct other additional research, to determine whether the question was answered accurately.

(354) Privacy Assessment Modification Module

(355) In particular embodiments, a Privacy Assessment Modification Module is configured to modify a questionnaire to include at least one additional question in response to determining that a user has provided one or more abnormal inputs or responses regarding a particular privacy campaign. For example, the system may, as discussed above, prompt the user to answer one or more follow up questions in response to determining that the user gave an abnormal response to a critical question. In particular embodiments, modifying the questionnaire to include one or more additional questions may prompt the user to provide more accurate responses which may, for example, limit a likelihood that a particular privacy campaign may run afoul of legal or industry-imposed restrictions on the collection and storage of personal data.

(356) Turning to FIG. **21**, in particular embodiments, when executing the Privacy Assessment Modification Module, the system begins, at Step **2110**, by receiving an indication that a user has provided one or more abnormal inputs or responses to one or more questions during a computerized privacy assessment questionnaire. In particular embodiments, the system is configured to receive the indication in response to determining that the user has provided one or more abnormal responses to one or more questions as part of Step **2050** of the Privacy Assessment Monitoring Module **2000** described above.

(357) Continuing to Step **2120**, in response to receiving the indication, the system is configured to flag the one or more questions and modify the questionnaire to include at least one additional question based at least in part on the one or more questions. In various embodiments, the system is configured to modify the questionnaire to include at least one follow up question that relates to the one or more questions for which the user provided

one or more abnormal responses. For example, the system may modify the questions that the system would have prompted the user to answer if the user had submitted a response that the user had initially provided but not submitted. For example, a user may have initially provided a response that social security numbers would be collected as part of a privacy campaign but deleted that response prior to submitting what sort of personal data would be collected. The system may, in response to determining that the user had provided an abnormal response to that question, modify the questionnaire to include one or more additional questions related to why social security numbers would need to be collected (or to double check that they, in fact, would not be).

(358) In other embodiments, the system is configured to take any other suitable action in response to determining that a user has provided one or more abnormal responses. The system may, for example: (1) automatically modify a privacy campaign; (2) flag a privacy campaign for review by one or more third party regulators; and/or (3) perform any other suitable action.

(359) Automated Vendor Risk Compliance Assessment Systems and Related Methods

(360) In particular embodiments, a vendor risk scanning system is configured to scan one or more webpages associated with a particular vendor (e.g., provider of particular software, particular entity, etc.) in order to identify one or more vendor attributes. In particular embodiments, the system may be configured to scan the one or more web pages to identify one or more vendor attributes such as, for example: (1) one or more security certifications that the vendor does or does not have (e.g., ISO 27001, SOC II Type 2, etc.); (2) one or more awards and/or recognitions that the vendor has received (e.g., one or more security awards); (3) one or more security policies and/or 3.sup.rd party vendor parties; (4) one or more privacy policies and/or cookie policies for the one or more webpages; (5) one or more key partners or potential sub processors of one or more services associated with the vendor; and/or (6) any other suitable vendor attribute. Other suitable vendor attributes may include, for example, membership in a Privacy Shield, use of Standardized Information Gathering (SIG), etc.

(361) In various embodiments, the system is configured to scan the one or more webpages by: (1) scanning one or more pieces of computer code associated with the one or more webpages (e.g., HTML, Java, etc.); (2) scanning one or more contents of the one or more webpages (e.g., using one or more natural language processing techniques); (3) scanning for one or more particular images on the one or more webpages (e.g., one or more images that indicate membership in a particular organization, receipt of a particular award etc.; and/or (4) using any other suitable scanning technique. The system may, for example, identify one or more image hosts of one or more images identified on the website, analyze the contents of a particular identified privacy or cookie policy that is displayed on the one or more webpages, etc. The system may, for example, be configured to automatically detect the one or more vendor attributes described above.

(362) In various embodiments, the system may, for example: (1) analyze the one or more vendor attributes; and (2) calculate a risk rating for the vendor based at least in part on the one or more vendor attributes. In particular embodiments, the system is configured to automatically assign a suitable weighting factor to each of the one or more vendor attributes when calculating the risk rating. In particular embodiments, the system is configured to analyze one or more pieces of the vendor's published applications of software available to one or more customers for download via the one or more webpages to detect one or more privacy disclaimers associated with the published applications. The system may then, for example, be configured to use one or more text matching techniques to determine whether the one or more privacy disclaimers contain one or more pieces of language required by one or more prevailing industry or legal requirements related to data privacy. The system may, for example, be configured to assign a relatively low risk score to a vendor whose software (e.g., and/or webpages) includes required privacy disclaimers, and configured to assign a relatively high risk score to a vendor whose one or more webpages do not include such disclaimers.

(363) In another example, the system may be configured to analyze one or more websites associated with a particular vendor for one or more privacy notices, one or more blog posts, one or more preference centers, and/or one or more control centers. The system may, for example, calculate the vendor risk score based at least in part on a presence of one or more suitable privacy notices, one or more contents of one or more blog posts on the vendor site (e.g., whether the vendor site has one or more blog posts directed toward user privacy), a presence of one or more preference or control centers that enable visitors to the site to opt in or out of certain data collection policies (e.g., cookie policies, etc.), etc.

(364) In particular other embodiments, the system may be configured to determine whether the particular vendor holds one or more security certifications. The one or more security certifications may include, for example: (1) system and organization control (SOC); (2) International Organization for Standardization (ISO); (3) Health Insurance Portability and Accountability ACT (HIPPA); (4) etc. In various embodiments, the system is configured to access one or more public databases of security certifications to determine whether the particular vendor holds any particular certification. The system may then determine the privacy awareness score based on whether the vendor holds one or more security certifications (e.g., the system may calculate a relatively higher score depending on one or more particular security certifications held by the vendor). The system may be further configured to scan a vendor website for an indication of the one or more security certifications. The system may, for example, be configured to identify one or more images indicated receipt of the one or more security certifications, etc.

(365) In still other embodiments, the system is configured to analyze one or more social networking sites (e.g., LinkedIn, Facebook, etc.) and/or one or more business related job sites (e.g., one or more job-posting sites, one or more corporate websites, etc.) or other third-party websites that are associated with the vendor (e.g., but not maintained by the vendor). The system may, for example, use social networking and other data to identify one or more employee titles of the vendor, one or more job roles for one or more employees of the vendor, one or more job postings for the vendor, etc. The system may then analyze the one or more job titles, postings, listings, roles, etc. to determine whether the vendor has or is seeking one or more employees that have a role associated with data privacy or other privacy concerns. In this way, the system may determine whether the vendor is particularly focused on privacy or other related activities. The system may then calculate a privacy awareness score and/or risk rating based on such a determination (e.g., a vendor that has one or more employees whose roles or titles are related to privacy may receive a relatively higher privacy awareness score).

(366) In particular embodiments, the system may be configured to calculate the privacy awareness score using one or more additional factors such as, for example: (1) public information associated with one or more events that the vendor is attending; (2) public information associated with one or more conferences that the vendor has participated in or is planning to participate in; (3) etc. In some embodiments, the system may calculate a privacy awareness score based at least in part on one or more government relationships with the vendor. For example, the system may be configured to calculate a relatively high privacy awareness score for a vendor that has one or more contracts with one or more government entities (e.g., because an existence of such a contract may indicate that the vendor has passed one or more vetting requirements imposed by the one or more government entities).

(367) In any embodiment described herein, the system may be configured to assign, identify, and/or determine a weighting factor for each of a plurality of factors used to determine a risk rating score for a particular vendor. For example, when calculating the rating, the system may assign a first weighting factor to whether the vendor has one or more suitable privacy notices posted on the vendor website, a second weighting factor to whether the vendor has one or more particular security certifications, etc. The system may, for example, assign one or more weighting factors using any suitable technique described herein with relation to risk rating determination. In some embodiments, the system may be configured to receive the one or more weighting factors (e.g., from a user). In other embodiments, the system may be configured to determine the one or more weighting factors based at least in part on a type of the factor.

(368) In any embodiment described herein, the system may be configured to determine an overall risk rating for a particular vendor (e.g., particular piece of vendor software) based in part on the privacy awareness score. In other embodiments, the system may be configured to determine an overall risk rating for a particular vendor based on the privacy awareness rating in combination with one or more additional factors (e.g., one or more additional risk factors described herein). In any such embodiment, the system may assign one or more weighting factors or relative risk ratings to each of the privacy awareness score and other risk factors when calculating an overall risk rating. The system may then be configured to provide the

risk score for the vendor, software, and/or service for use in calculating a risk of undertaking a particular processing activity that utilizes the vendor, software, and/or service (e.g., in any suitable manner described herein).

(369) In a particular example, the system may be configured to identify whether the vendor is part of a Privacy Shield arrangement. In particular, a privacy shield arrangement may facilitate monitoring of an entity's compliance with one or more commitments and enforcement of those commitments under the privacy shield. In particular, an entity entering a privacy shield arrangement may, for example: (1) be obligated to publicly commit to robust protection of any personal data that it handles; (2) be required to establish a clear set of safeguards and transparency mechanisms on who can access the personal data it handles; and/or (3) be required to establish a redress right to address complaints about improper access to the personal data.

(370) In a particular example of a privacy shield, a privacy shield between the United States and Europe may involve, for example: (1) establishment of responsibility by the U.S. Department of Commerce to monitor an entity's compliance (e.g., a company's compliance) with its commitments under the privacy shield; and (2) establishment of responsibility of the Federal Trade Commission having enforcement authority over the commitments. In a further example, the U.S. Department of Commerce may designate an ombudsman to hear complaints from Europeans regarding U.S. surveillance that affects personal data of Europeans.

(371) In some embodiments, the one or more regulations may include a regulation that allows data transfer to a country or entity that participates in a safe harbor and/or privacy shield as discussed herein. The system may, for example, be configured to automatically identify a transfer that is subject to a privacy shield and/or safe harbor as 'low risk.' In this example, U.S. Privacy Shield members may be maintained in a database of privacy shield members (e.g., on one or more particular webpages such as at [www.privacyshield.gov](http://www.privacyshield.gov)). The system may be configured to scan such webpages to identify whether the vendor is part of the privacy shield.

(372) In particular embodiments, the system may be configured to monitor the one or more websites (e.g., one or more webpages) to identify one or more changes to the one or more vendor attributes. For example, a vendor may update a privacy policy for the website (e.g., to comply with one or more legal or policy changes). In some embodiments, a change in a privacy policy may modify a relationship between a website and its users. In such embodiments, the system may be configured to: (1) determine that a particular website has changed its privacy policy; and (2) perform a new scan of the website in response to determining the change. The system may, for example, scan a website's privacy policy at a first time and a second time to determine whether a change has occurred. The system may be configured to analyze the change in privacy policy to determine whether to modify the calculated risk rating for the vendor (e.g., based on the change).

(373) The system may, for example, be configured to continuously monitor for one or more changes. In other embodiments, the system may be configured to scan for one or more changes according to a particular schedule (e.g., hourly, daily, weekly, or any other suitable schedule.). For example, the system may be configured to scan the one or more webpages on an ongoing basis to determine whether the one or more vendor attributes have changed (e.g., if the vendor did not renew its Privacy Shield membership, lost its ISO certification, etc.).

(374) In particular embodiments, any entity (e.g., organization, company, etc.) that collects, stores, processes, or otherwise handles personal data (e.g., on behalf of its customers, employees, or other suitable data subjects) may be subject to various privacy and security policies (e.g., such as the European Union's General Data Protection Regulation (GDPR), the California Consumer Privacy Act (CCPA), Nevada Senate Bill 220 (SB-220), and other such policies) that relate to the handling of such personal data. An entity may, for example, be required to both comply with one or more legal or industry standards related to the collection and/or storage of private information (e.g., such as personal data or personal information) and demonstrate such compliance. One or more systems described herein may be configured to at least partially automate such compliance (e.g., and at least partially automate one or more activities that would support a demonstration of such compliance through use of the one or more systems).

(375) In addition to personal data that an entity (e.g., or other organization) may collect, store, and/or process on its own behalf, an entity may utilize (e.g., contract with) data obtained from and/or collected by one or more third-party vendors that also collect, store, and/or process personal data from one or more data subjects. These third-party vendors may further rely on one or more sub-processors to provide, collect, store, etc. data that those third-party vendors use, and so on. An entity may have agreements and/or contracts (e.g., written agreements) with each third-party vendor that set out the obligations of each party, including obligations to take certain actions in response to privacy-related occurrences, such as a data breach or incident that may affect one or both of the parties. Similarly, third-party vendors may have agreements and/or contracts (e.g., written agreements) with sub-processors that set out the obligations of the third-party vendor and a sub-processor.

(376) Under prevailing legal and industry standards related to the processing of personal data, an entity may be found to be in violation of one or more laws or regulations if the entity utilizes a vendor (e.g., and/or such a vendor utilizes a sub-processor) that mishandles personal data. Accordingly, as may be understood in light of this disclosure, an entity may desire to thoroughly vet (e.g., using one or more risk analysis techniques and/or vendor scoring techniques, such as any suitable technique described herein) any third-party vendors and/or sub-processors: (1) with which the entity contracts; (2) from which the entity receives personal data; (3) that store personal data on behalf of the entity; and/or (4) that otherwise collect, store, process, and/or handle personal data on behalf of the entity, or in association with any activity undertaken by the vendor or sub-processor on behalf of, or for the benefit of, the entity.

(377) Third-party vendors that provide software applications and systems that handle or access the personal data of others may, for example, provide such software to large numbers of different customers (e.g., hundreds or thousands of different customers). This may add an additional level of complexity to complying with one or more prevailing legal or industry standards related to the handling of personal data, because an entity may be required to ensure that any vendor that the entity utilizes is also in compliance with such policies and regulations. As part of ensuring compliance with such regulations, an entity may conduct one or more privacy audits (e.g., of activities undertaken by the entity, of vendors utilized by and/or contracted with the entity, etc.).

(378) Various embodiments of a vendor risk management system described herein may be configured to automate one or more processes related to the risk assessment, scoring, and/or analysis of particular vendors with which an entity may contract (e.g., new vendors that the entity would like to start working with—e.g., by entering into a new contract, or existing vendors that the entity would like to continue working with—e.g., by renewing an existing contract), or whose services an entity may utilize as part of one or more business and/or data processing activities. Various embodiments may also be configured for use in assessing the risk associated with one or more vendors before an entity pays the vendor. Further various embodiments of a vendor risk management system described herein may be configured to determine obligations between an entity and a third-party vendor and/or a sub-processor and perform tasks (e.g., automatically) to comply with such obligations. Particular embodiments of a vendor risk management system are described more fully below.

(379) Exemplary Technical Platforms

(380) As will be appreciated by one skilled in the relevant field, the present invention may be, for example, embodied as a computer system, a method, or a computer program product. Accordingly, various embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment, or an embodiment combining software and hardware aspects. Furthermore, particular embodiments may take the form of a computer program product stored on a computer-readable storage medium having computer-readable instructions (e.g., software) embodied in the storage medium. Various embodiments may take the form of web-implemented computer software. Any suitable computer-readable storage medium may be utilized including, for example, hard disks, compact disks, DVDs, optical storage devices, and/or magnetic storage devices.

(381) Various embodiments are described below with reference to block diagrams and flowchart illustrations of methods, apparatuses (e.g., systems), and computer program products. It should be understood that each block of the block diagrams and flowchart illustrations, and combinations of blocks in the block diagrams and flowchart illustrations, respectively, can be implemented by a computer executing computer program instructions. These computer program instructions may be loaded onto a general-purpose computer, special purpose computer, or other programmable data

processing apparatus to produce a machine, such that the instructions which execute on the computer or other programmable data processing apparatus to create means for implementing the functions specified in the flowchart block or blocks.

(382) These computer program instructions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner such that the instructions stored in the computer-readable memory produce an article of manufacture that is configured for implementing the function specified in the flowchart block or blocks. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer implemented process such that the instructions that execute on the computer or other programmable apparatus provide steps for implementing the functions specified in the flowchart block or blocks.

(383) Accordingly, blocks of the block diagrams and flowchart illustrations support combinations of mechanisms for performing the specified functions, combinations of steps for performing the specified functions, and program instructions for performing the specified functions. It should also be understood that each block of the block diagrams and flowchart illustrations, and combinations of blocks in the block diagrams and flowchart illustrations, can be implemented by special purpose hardware-based computer systems that perform the specified functions or steps, or combinations of special purpose hardware and other hardware executing appropriate computer instructions.

(384) Example System Architecture

(385) FIG. 22 is a block diagram of a Vendor Risk Management System 2200 according to a particular embodiment. In some embodiments, the Vendor Risk Management System 2200 is configured to scan one or more websites associated with a particular vendor to identify and analyze one or more security certifications, privacy and/or cookie policies, etc. The system may, for example, initiate a virtual browsing session on any of the one or more servers and/or computers described below in order to facilitate the scanning of the one or more webpages (e.g., in order to access and then scan the one or more websites).

(386) As may be understood from FIG. 22, the Vendor Risk Management System 2200 includes one or more computer networks 2215, a Vendor Risk Scanning Server 2210, a Vendor Risk Analysis Server 2220 (e.g., which may be configured to analyze data identified during a scan of the vendor's website(s)), One or More Third Party Servers 2260, one or more databases 2240 (e.g., which may be used to store data used as part of the analysis, results of the analysis, etc.), and one or more remote computing devices 2250 (e.g., a desktop computer, laptop computer, tablet computer, etc.). In particular embodiments, the one or more computer networks 2215 facilitate communication between the Vendor Risk Scanning Server 2210, a Vendor Risk Analysis Server 2220, One or More Third Party Servers 2260, one or more databases 2240, and one or more remote computing devices 2250. The Vendor Risk Analysis Server 2220, the Vendor Risk Management System 2200, or a vendor risk management server described herein may be configured to perform any of the functions and processes set forth herein.

(387) The one or more computer networks 2215 may include any of a variety of types of wired or wireless computer networks such as the Internet, a private intranet, a public switch telephone network (PSTN), or any other type of network. The communication link between Vendor Risk Scanning Server 2210 and Vendor Risk Analysis Server 2220 may be, for example, implemented via a Local Area Network (LAN) or via the Internet.

(388) Vendor Management Overview

(389) In particular embodiments, any entity (e.g., organization, company, etc.) that collects, stores, processes, or otherwise handles personal data (e.g., on behalf of its customers, employees, or other suitable data subjects) may be subject to various privacy and security policies (such as the European Union's General Data Protection Regulation (GDPR), the California Consumer Privacy Act (CCPA), Nevada Senate Bill 220 (SB-220), and other such policies) that relate to the handling of such personal data. An entity may, for example, be required to both comply with one or more legal or industry standards related to the collection and/or storage of private information (e.g., such as personal data or personal information) and demonstrate such compliance. One aspect of such compliance may be disclosing data breaches to one or more regulating parties, such as one or more supervisory authorities. One or more systems described herein may be configured to at least partially automate such compliance (e.g., and at least partially automate one or more activities that would support a demonstration of such compliance through the use of the one or more systems).

(390) In addition to personal data that an entity (e.g., a company or other organization) may collect, store, and/or process on its own behalf, an entity may utilize data obtained from and/or collected by one or more third-party vendors that also collect, store, and/or process personal data from one or more data subjects. These third-party vendors may further rely on one or more sub-processors to provide, collect, process, and/or store data that those third-party vendors use, and so on.

(391) Within the context of such business relationships, it is common for an entity to have contractual obligations to disclose privacy-related occurrences, such as a data breach or other privacy or security-related incident, to its business partners. For example, an entity may have one or more verbal or written agreements (e.g., contracts) in place with each of the entity's third-party vendors that set out the obligations of each party, including one or more obligations to take certain actions in response to specified privacy-related occurrences, such as a data security-related incident that may affect any of the parties to the agreement. Similarly, third-party vendors may have respective agreements and/or contracts (e.g., written agreements) with sub-processors that set out respective privacy-related obligations of the third-party vendor and one or more of its sub-processors. One or more systems described herein may be configured to at least partially facilitate and/or automate such compliance with such contractual obligations.

(392) It is noted that under prevailing legal and industry standards related to the processing of personal data, an entity may be found to be in violation of one or more laws or regulations if the entity utilizes a vendor (e.g., and/or such a vendor utilizes a sub-processor) that mishandles personal data. Accordingly, as may be understood in light of this disclosure, an entity may desire to thoroughly vet (e.g., using one or more risk analysis techniques and/or vendor scoring techniques, such as any suitable technique described herein) any third-party vendors and/or sub-processors: (1) with which the entity contracts; (2) from which the entity receives personal data; (3) that store personal data on behalf of the entity; and/or (4) that otherwise collect, store, process, and/or handle personal data on behalf of the entity, or in association with any activity undertaken by the vendor or sub-processor on behalf of, or for the benefit of, the entity.

(393) Third-party vendors that provide software applications and/or systems that handle and/or access the personal data of others may, for example, provide such software to large numbers of different customers (e.g., hundreds or thousands of different customers). This may add an additional level of complexity to complying with one or more prevailing legal or industry standards related to the handling of personal data, because an entity may be required to ensure that any vendor that the entity utilizes is also in compliance with such policies and regulations. As part of ensuring compliance with such regulations, an entity may conduct one or more privacy audits (e.g., of activities undertaken by the entity, of vendors utilized by and/or contracted with the entity, etc.).

(394) Various embodiments of a vendor risk management system described herein may be configured to automate one or more processes related to the risk assessment, scoring, and/or analysis of particular vendors with which an entity may contract, or whose services an entity may utilize as part of one or more business and/or data processing activities. Further various embodiments of vendor risk management systems described herein may be configured to determine obligations between an entity and a third-party vendor and/or a sub-processor and perform tasks (e.g., automatically) to comply with such obligations. Particular embodiments of a vendor risk management system are described more fully below.

(395) Vendor Incident Management

(396) In various embodiments, the system may be configured to automatically facilitate a response to one or more incidents (e.g., security-related incidents, privacy-related incidents, data breaches, etc.). In particular, the system may be configured to: (1) identify a particular incident; (2) determine a method by which the incident was reported (e.g., via webform); (3) identify a country of origin of the incident; (4) generate one or more tasks related to the incident (e.g., one or more reporting tasks and/or notification tasks that should be completed in order to properly respond to the identified incident); (5) communicate the one or more tasks to one or more users; and/or (6) take any other suitable action related to the breach.

(397) The system may, for example, be configured to generate one or more tasks based at least in part on one or more contractual and/or legal

obligations (e.g., with respect to one or more entities, such as one or more vendors of the entity). For example, the system may determine that, based at least in part on one or more contract terms derived, for example, using one or more techniques described herein, the entity is obligated to notify a particular vendor, regulator, sub-processor, or other entity within a specified timeframe of any material data breach. The system may, at least partially in response to identifying such a data breach, be configured to generate a task to notify one or more particular vendors, regulators, and/or other entities (e.g., within the prescribed timeframe). The system may determine such contract terms, for example, by using one or more natural language processing techniques to analyze the text of one or more relevant contracts, such as one or more relevant contracts between an entity and a third-party vendor. The system may be configured to receive any such contracts and agreements as uploaded documents for analysis (e.g., for use by the system in determining, from the documents, one or more key terms, obligations, penalties, etc. that the entity and/or one or more third parties, such as one or more of the entity's vendors are subject to in regard to disclosing, for example, one or more specified types of relevant privacy-related events, such as a data breach).

(398) In various embodiments, the system is configured to automate the submission of notifications of one or more data breaches and/or other privacy-related incidents to one or more entities for which a contractual obligation to notify exists (e.g., a vendor). In particular embodiments, the system is configured to determine one or more attributes of a security-related incident in order to determine whether an obligation to a vendor has arisen, and, if so, what responsive actions should be performed. For example, the system may be configured to determine attributes such as: (1) a geographical region or country in which the incident occurred; (2) a scope of the security-related incident; (3) a date and time of occurrence of the security-related incident; (4) one or more systems, assets, processes, vendors, etc. that were affected by the security-related incident; and/or (5) one or more applicable regulatory or legal schemes.

(399) The system may further be configured to analyze a security-related incident using such attributes to determine additional information. For example, the system may analyze security-related incident attributes to determine a risk level of the security-related incident. The system may then use such determined attributes and optionally additional information to determine the obligations implicated by the security-related incident (e.g., to a particular vendor). Based on such determined obligations, the system may generate one or more tasks (e.g., automatically) to be performed to satisfy the entity's obligations associated with the security-related incident. In various embodiments, the system may recommend a remediation for determined risks in response to the security-related incident with respect to one or more contractual commitments or privacy regulations. In various embodiments, the system may perform such tasks, for example, automatically, or upon receipt of an instruction from a user (e.g., received via an activation of a control on a graphical user interface).

(400) The system may, for example, be configured to: (1) capture, investigate, and/or analyze the risk, liability, and/or obligations of an entity stemming from a security-related incident such as a data breach; (2) parse one or more contracts to identify one or more notification obligations and/or regulatory/jurisdictional obligations to determine one or more required and/or desirable subsequent actions based on a type of incident and/or one or more details about the incident; (3) identify one or more assets, vendors, processes, etc. that are affected by the incident (e.g., based on one or more identified contractual obligations); (4) capture the scope of the incident (e.g., use a mobile application to take a picture relevant to the incident, scan an asset tag of a computing device involved in the incident, etc.); and/or (5) maintain a master database of privacy-related incidents (e.g., based on case law, incident reports, etc.) in order to determine a risk level of a particular incident; etc.

(401) FIG. 23 shows an example process that may be performed by a Vendor Incident Notification Module 2300. In executing the Vendor Incident Notification Module 2300, the system begins at Step 2310, where it receives an indication of a security-related incident. The system may automatically receive this indication, for example, in response to the creation and/or detection, by the system, of an incident report. In various embodiments, such incident reports may be generated, for example: (1) by a user through use of a graphical user interface provided by the system; and/or (2) automatically by a breach detection and/or reporting system, which may be part of the present system.

(402) At Step 2320, the system may determine one or more attributes of the indicated security-related incident. Such attributes may be provided when the incident report was created, for example by a user via a graphical user interface, or as determined by an automated incident report generation system. Such attributes may be stored in or otherwise associated with a record of the incident in the system's memory. Attributes can be any type of information associated with a security-related incident, including, but not limited to (1) a geographical region or country in which the incident occurred; (2) a scope of the incident; (3) a date and time of occurrence of the incident; (4) one or more affected systems, assets, processes, vendors, etc.; and/or (5) one or more controlling regulatory or legal schemes.

(403) At Step 2330, based on the information available about the security-related incident (e.g., attributes as determined at Step 2320), the system may determine additional information for the security-related incident. For example, the system may determine a risk level and/or regulatory regime for an incident based, at least in part, on the location and/or scope of the incident and/or the affected systems. The system may determine any other additional information associated with the incident using any available resources at Step 2330.

(404) At Step 2340, the system may determine one or more third-party entities (e.g., third party vendors) that may be involved and/or associated with the security-related incident using one or more of the attributes of the security-related incident and/or any additional information determined for the security-related incident. For example, the system may determine, in some embodiments based at least in part on one or more attributes of a particular data breach, that the data breach has affected one or more email systems in Germany. The system may then determine that the applicable email systems in Germany are hosted by one or more particular vendors. Accordingly, the system may conclude that the one or more particular vendors have been affected by the data breach.

(405) The system may next, at Step 2350, analyze one or more contracts with the one or more determined entities (e.g., as determined at Step 2340) to determine whether one or more notification obligations to such entities exist and, if so, the particular requirements of such obligations. For example, the system may determine that a particular vendor contract includes an obligation of an entity to alert the particular vendor of any data breach affecting a particular service involving that vendor within 48 hours of the entity learning of the data breach. It should be understood that notification obligations may specify, for example, any particular requirements related to the required notification, such as the form of the notification (e.g., email, phone call, letter, etc.), timeframe of the notification (24 hours, 48 hours, five business days, etc.), information to be included in the notification, etc. The system may be configured to analyze such contracts using natural language processing techniques to scan the language of the contracts in order to determine the particular obligations and associated requirements.

(406) Based on the determined obligations, at Step 2360 the system may generate one or more tasks that should be performed to satisfy such obligations. The system may then present such tasks to a user for completion, for example, in a suitable graphical user interface on a display screen associated with the system. The system may present one or more such tasks to the user along with any related information, as described in more detail herein. The system may also, or instead, automatically perform one or more of such tasks and may notify a user of the system's automatic performance and/or completion of such tasks, for example, via a suitable user interface.

(407) Vendor Risk Scanning and Scoring Systems

(408) A vendor risk management system may be configured to perform any one or more of several functions related to managing vendors and/or other third-party entities. In various embodiments, a vendor management system may be a centralized system providing the functions of vendor compliance demonstration, vendor compliance verification, vendor scoring (e.g., vendor risk rating, vendor privacy compliance scoring, etc.), and/or vendor information collection. The system may use various sources of information to facilitate vendor-related functions, such as, but not limited to: (1) publicly available vendor information (e.g., from websites, regulator bodies, industry associations, etc.); (2) non-publicly available information (e.g., private information, contracts, etc.); and/or (3) internally-generated information (e.g., internally-generated scoring information, internally-generated ranking information, one or more internally-maintained records of interactions with the vendor, one or more internal records of privacy-related incidents, etc.).



(409) In particular embodiments, a vendor risk management system may be configured to scan one or more systems and/or publicly available information associated with a particular vendor. The system may extract vendor information from such sources and/or use the extracted information to determine one or more vendor risk scores for the particular vendor. The system may, for example, be configured to define particular scoring criteria for one or more privacy programs (e.g., associated with a particular vendor of the entity) and use the scoring criteria to determine one or more vendor risk scores for the particular vendor (e.g., a vendor or sub-processor that processes data on behalf of the entity) based on the particular scoring criteria. The system may also, or instead, be configured to define particular scoring criteria for one or more privacy programs (e.g., associated with a particular vendor of the entity and/or a particular product or service of the particular vendor) and use the scoring criteria to determine respective risk scores for one or more products (services, offerings, etc.) provided by the particular vendor based on the particular scoring criteria. In various embodiments, suitable scoring criteria may be based on any suitable vendor information (e.g., any suitable information associated with the vendor), including, but not limited to, publicly available information and non-publicly available information.

(410) Suitable vendor information may include, for example: (1) one or more security certifications that the vendor may or may not have (e.g., ISO 27001, SOC II Type 2, etc.); (2) one or more awards and/or recognitions that the vendor has received (e.g., one or more security awards); (3) one or more security policies the vendor may have in place, (4) one or more third parties (e.g., sub-processors, third-party vendors, etc.) with which the vendor may do business or otherwise interact; (5) one or more privacy policies and/or cookie policies for one or more vendor webpages (e.g., one or more webpages associated with the vendor, operated by the vendor, etc.); (6) one or more partners and/or potential sub-processors associated with one or more products offered by the vendor; (7) one or more typical vendor response times to one or more particular types of incidents; (8) one or more typical vendor response times to one or more particular types of requests for information from the vendor; (9) vendor financial information (e.g., publicly available financial information for the vendor such as revenue, stock price, trends in stock price, etc.); (10) news related to the vendor (e.g., one or more news articles, magazine articles, blog posts, etc.); (11) one or more data breaches experienced by the vendor (e.g., one or more announced breaches) and/or the vendor's response to such breaches; and/or (12) any other suitable vendor information. Other suitable vendor information may include, for example, membership in a Privacy Shield and/or participation in one or more treaties and/or organizations related to a demonstration of meeting certain privacy standards, use of Standardized Information Gathering (SIG), etc. Particular exemplary vendor information is discussed more fully below.

(411) In particular embodiments, the system may, for example, be configured to scan one or more webpages associated with a particular vendor (e.g., one or more webpages operated by the particular vendor, one or more webpages operated on behalf of the particular vendor, one or more webpages comprising information associated with the particular vendor, etc.) in order to identify one or more pieces of vendor information that may serve as a basis for calculating and/or otherwise determining one or more vendor risk scores (e.g., one or more vendor compliance scores, one or more vendor privacy risk scores, one or more vendor security risk scores, etc.). In various embodiments, the system may be configured to scan the one or more webpages by: (1) scanning one or more pieces of computer code associated with the one or more webpages (e.g., HTML, Java, etc.); (2) scanning one or more contents (e.g., text content) of the one or more webpages (e.g., using one or more natural language processing techniques); (3) scanning for one or more particular images on the one or more webpages (e.g., one or more images that indicate membership in a particular organization, receipt of a particular award, etc.); and/or (4) using any other suitable scanning technique to scan the one or more webpages. When scanning a particular webpage or multiple webpages, the system may, for example, perform one or more functions such as identifying one or more hosts of one or more images identified on the particular webpage or multiple webpages, analyzing the contents of one or more particular identified privacy and/or cookie policies that are displayed on the one or more webpages, identify one or more particular terms, policies, and/or other privacy-related language included in the text of the particular webpage or multiple webpages, etc. The system may, for example, be configured to automatically detect any of the one or more pieces of vendor information described above. The system may also, or instead, be configured to detect any of the one or more pieces of vendor information at least partially in response to a detection and/or receipt of a user input, such as the selection of a user-selectable control (e.g., user-selectable indicia, webform button, webpage control, etc.) in a graphical user interface presented to a user. The system may also, or instead, be configured to initiate detection of any of the one or more pieces of vendor information in response to any other type of input or condition.

(412) In various embodiments, the system may, for example analyze the one or more pieces of vendor information and calculate or otherwise determine a risk score for the vendor based at least in part on the one or more pieces of vendor information. The system may also use other information in conjunction with the one or more pieces of vendor information to calculate or otherwise determine a vendor risk score. In particular embodiments, the system is configured to automatically assign one or more weighting factors to each of the one or more pieces of vendor information and/or to each of one or more pieces of other information when calculating the risk score.

(413) In particular embodiments, the system is configured to analyze one or more pieces of a vendor's published software applications of software and/or documentation associated with vendor software (e.g., that may be available to one or more customers for download via one or more webpages) to detect one or more privacy disclaimers associated with such software. The system may then, for example, be configured to use one or more text matching techniques to determine whether the one or more privacy disclaimers contain one or more pieces of language required by one or more prevailing industry and/or legal standards and/or requirements related to data privacy and/or security. The system may, for example, be configured to assign a relatively low risk score to a vendor whose products (e.g., software, services, webpages, other offerings, etc.) include one or more required privacy disclaimers. Likewise, the system may, for example, be configured to assign a relatively high risk score to a vendor whose products do not include such disclaimers.

(414) In various embodiments, the system may be configured to analyze one or more webpages associated with a particular vendor for one or more privacy notices, one or more blog posts, one or more preference centers, and/or one or more control centers. The system may then, for example, calculate a vendor privacy risk score based, at least in part, on a presence of one or more of: (1) one or more suitable privacy notices; (2) contents of one or more blog posts on one or more vendor sites (e.g., whether the vendor site has one or more blog posts directed toward user privacy); (3) a presence of one or more preference centers and/or control centers that enable visitors to the site to opt-in or opt-out of certain data collection policies (e.g., cookie policies, etc.); and/or (4) any other security-related information, privacy-related information etc. that may be present on one or more webpages associated with the particular vendor.

(415) In particular embodiments, the system may be configured to determine whether the particular vendor holds one or more certifications (e.g., one or more security certifications, one or more privacy certifications, one or more industry certifications etc.) such as one or more system and organization controls (SOC) or International Organization for Standardization (ISO) certifications or one or more certifications related to Health Insurance Portability and Accountability Act (HIPAA). In various embodiments, the system is configured to access one or more public databases of certifications to determine whether the particular vendor holds any particular certification. The system may then determine a risk score based, at least in part, on whether the vendor holds one or more certifications (e.g., the system may calculate a relatively higher score if the vendor holds one or more particular certifications). The system may be further configured to scan a vendor website for an indication of one or more certifications. The system may, for example, be configured to identify one or more images that indicate receipt of one or more certifications. In various embodiments, the system may be configured to calculate a vendor risk score based on one or more certifications that the system determines that the vendor does or does not hold.

(416) In a particular embodiment, the system may first scan one or more vendor websites for one or more indications that the vendor has one or more certifications as discussed above. Next, in response to determining that the vendor has indicated that they have one or more certifications (e.g., via their website or otherwise), the system may be adapted to verify whether the vendor actually has the indicated one or more security certifications by automatically confirming this with one or more independent data sources, such as a public database of entities that hold security certifications.

(417) In still other embodiments, the system is configured to analyze one or more social networking sites (e.g., LinkedIn, Facebook, etc.), one or



more business related job sites (e.g., one or more job-posting sites, one or more corporate websites, etc.), and/or one or more other third-party websites that may be associated with and/or contain information pertaining to the vendor (e.g., that are not operated by, or on behalf of, the vendor). The system may, for example, use social networking data (e.g., obtained from one or more social network websites) and/or other data to identify one or more titles of employees of the vendor, one or more job roles for one or more employees of the vendor, one or more job postings for the vendor, etc. The system may then analyze the one or more job titles, postings, listings, roles, etc. to determine whether the vendor has and/or is seeking one or more employees that have a role associated with addressing data privacy, data security, and/or other privacy or security concerns (e.g., a role that requires data privacy experience). In this way, the system may determine whether the vendor is particularly focused on privacy, security, and/or other related activities. The system may then calculate a risk score for the vendor based, at least in part, on such a determination (e.g., a vendor that has one or more employees whose roles and/or titles are related to security may receive a relatively higher risk score as compared to a vendor who does not).

(418) In particular embodiments, the system may be configured to calculate the risk score using one or more additional factors such as, for example: (1) public information associated with one or more events that the vendor is attending; (2) public information associated with one or more conferences that the vendor has participated in and/or is planning to participate in; (3) one or more publications and/or articles written by authors associated with and/or sponsored by the vendor; (4) public relations material issued by the vendor; (5) one or more news articles and/or reports about the vendor; and/or (6) any other public information about and/or associated with the vendor. In some embodiments, the system may calculate a risk score for the vendor based, at least in part, on one or more governmental relationships of the vendor (e.g., relationships that the vendor has with one or more particular government entities). For example, the system may be configured to calculate a relatively low risk score for a vendor that has one or more contracts with one or more government entities (e.g., because an existence of such a contract may indicate that the vendor has passed one or more vetting requirements imposed by the one or more government entities).

(419) In particular embodiments, the system may be configured to determine a vendor risk score based, at least in part, on one or more pieces of information contained in one or more documents that define a relationship between the vendor and the entity (e.g., one or more contracts, one or more agreements, one or more licenses, etc.). The system may be configured to receive one or more such documents as uploaded documents, for example, provided via a suitable user interface. For example, for one or more such documents, the system may be configured to: (1) receive a copy of a particular document; (2) scan the particular document to identify particular language (e.g., one or more particular terms, clauses, etc.) contained in the document; (3) categorize the particular language based on one or more pre-defined term language categories; and/or (4) modify and/or calculate a risk score for the vendor based on the presence and/or absence of the particular language.

(420) In particular embodiments, the system may be configured to analyze (e.g., using natural language processing) one or more such documents to identify key terms. The system may, for example, be automatically configured to identify one or more: (1) term limits; (2) breach notification timeline obligations; (3) sub-processor change notification requirements; (4) liability caps/obligations; (5) data breach liability terms; (6) indemnification terms; (7) required data transfer mechanisms; (8) notification time periods for a data breach; (9) notification requirements for sub-processor changes; (10) terms requiring one or more security certifications; (11) terms requiring compliance with one or more regulatory regimes; and/or (12) any other privacy or security related terms within the one or more documents.

(421) In particular embodiments, as described herein, the system may be configured to generate one or more vendor risk assessment questionnaires and transmit the one or more questionnaires to a particular vendor for completion. The system may later receive the completed questionnaire and use one or more pieces of vendor information (as obtained from the vendor's responses to the various questions within the questionnaire) in calculating the vendor risk score.

(422) In various embodiments, the system may be configured to automatically generate an expiration date for any particular piece of information used in the determination of a vendor risk score (e.g., one or more pieces of vendor information derived from a questionnaire and/or assessment related to the vendor, determined from one or more webpage scans, identified in one or more uploaded documents, etc.). Such an expiration date may, for example, be based on an explicit characteristic of the piece of information, such as the date on which a security certification expires. Alternatively, or in addition, an expiration date may be determined based on one or more system configurations (e.g., privacy-related data may be set to expire six months after the system identifies/determines the information, which may help ensure that the system maintains current information).

(423) The system may use any other criteria to set information expiration dates. Any piece of information may have an expiration date that may be distinct and/or independent from the expiration date associated with any other piece of information. Alternatively, or in addition, a piece of information may have an expiration date tied to and/or associated with an expiration date of another piece of information.

(424) In various embodiments, the system may be configured for, in response to determining that a particular piece of vendor-related information used by the system has expired, automatically requesting and/or attempting to obtain an updated version of the expired information. In various embodiments, automatically requesting and/or obtaining updated information may comprise, for example: (1) generating an updated risk assessment questionnaire for completion by the vendor and facilitating completion of the questionnaire by the vendor; (2) competing an updated scan of one or more pieces of publicly available information associated with the vendor; (3) completing an updated scan of one or more vendor systems; (4) analyzing one or more new versions of one or more particular vendor documents; and/or (5) performing other suitable activities to obtain updated information, etc. In particular embodiments, the system may then be configured to calculate an updated vendor risk score based, at least in part, on one or more pieces of the updated information. In any embodiment described herein, the system may be configured to determine whether the one or more pieces of updated information are sufficient to demonstrate continued compliance, by the vendor, with one or more obligations under one or more privacy laws, standards and/or regulations, one or more obligations under one or more vendor contracts, etc.

(425) In any embodiment described herein, the system may be configured to assign, identify, and/or determine a weighting factor for each of a plurality of factors used to determine a risk score for a particular vendor. For example, when calculating a risk score for a particular vendor, the system may assign a first weighting factor to whether the vendor has one or more suitable privacy notices posted on a website associated with the vendor, a second weighting factor to whether the vendor has one or more particular security certifications, etc. The system may, for example, assign one or more weighting factors using any suitable technique described herein with relation to risk rating determination. In various embodiments, the system may be configured to receive the one or more weighting factors (e.g., from a user). In various embodiments, the system may also, or instead, be configured to determine the one or more weighting factors based at least in part on a type of the factor.

(426) In any embodiment described herein, the system may be configured to determine an overall risk score for a particular vendor (e.g., applicable to all pieces of the vendor's software) based at least in part on a risk score associated with a subset of the vendor's products. In various embodiments, the system may be configured to determine an overall risk score for a particular vendor based at least in part on a risk score associated with a subset of the vendor's products in combination with one or more additional factors (e.g., one or more additional risk factors described herein). In various embodiments, the system may be configured to determine an overall risk rating for a product of a particular vendor based, at least in part, on a risk score associated with one or more of the vendor's other products in combination with one or more additional factors (e.g., one or more additional risk factors described herein). In various embodiments, the system may assign one or more weighting factors to each of one or more risk scores and/or other risk factors that may be used when calculating an overall risk score. The system may then be configured to provide a risk score (e.g., an overall risk score) for the vendor and/or a vendor product for use in calculating a risk of undertaking a particular processing activity that utilizes the vendor and/or a particular product of the vendor (e.g., in any suitable manner described herein).

(427) In a particular example, the system may be configured to determine whether the vendor is part of a Privacy Shield arrangement. In various embodiments, a privacy shield arrangement may facilitate monitoring of a vendor's compliance with one or more commitments and may facilitate enforcement of those commitments under the privacy shield. In particular, a vendor entering a privacy shield arrangement may, for example: (1) be obligated to publicly commit to robust protection of any personal data that it handles; (2) be required to establish a clear set of safeguards and

transparency mechanisms regarding who can access the personal data of the vendor handles; and/or (3) be required to establish a redress right to address complaints about improper access to the personal data. The system may then be configured to use the determinization of the vendor's participation and/or membership in a privacy shield and/or one or more similar arrangement to determine a risk score for that vendor.

(428) In a particular example of a privacy shield arrangement between the United States and Europe, the U.S. Department of Commerce may be responsible for monitoring a vendor's compliance (e.g., a company's compliance) with its commitments under the privacy shield and the Federal Trade Commission may be responsible for enforcement authority over such commitments. In a further example, the U.S. Department of Commerce may designate an ombudsman to hear complaints from Europeans regarding U.S. surveillance that affects personal data of Europeans.

(429) In various embodiments, regulations related to data privacy and/or data security may include one or more regulations that allow data transfer to a country or entity that participates in a safe harbor and/or a privacy shield as discussed herein. The system may, for example, be configured to automatically identify a transfer that is subject to a privacy shield and/or safe harbor as "low risk." For example, U.S. Privacy Shield members may be maintained in a database of privacy shield members (e.g., on one or more particular webpages such as [www.privacyshield.gov](http://www.privacyshield.gov)). The system may be configured to scan one or more webpages reflecting information stored in such databases to determine whether the vendor is part of the privacy shield and/or to otherwise obtain information associated with the vendor.

(430) In particular embodiments, the system may be configured to monitor the one or more websites (e.g., one or more webpages) and/or other systems to identify one or more changes to one or more pieces of vendor information. For example, a vendor may update a privacy policy for one of its websites (e.g., to comply with one or more legal or policy changes). In various embodiments, a change in a privacy policy may modify a relationship between a website and its users. In particular embodiments, the system may be configured to determine that a particular website has changed its privacy policy and responsively perform a new scan of the website to obtain updated privacy-related information for the vendor. The system may, for example, scan a website's privacy policy at a first time and at a second, later time and compare such scans to determine whether a change has occurred. The system may be configured to perform scanning of websites and/or other sources of vendor information routinely and/or automatically. The system may be configured to analyze any changes (e.g., a change in a privacy policy for the vendor posted on a particular web page of the web site) to determine whether and how to modify a calculated risk score for a vendor (e.g., based on the change).

(431) The system may, for example, be configured to continuously monitor a particular web site and/or web page for one or more changes. In various embodiments, the system may be configured to scan for one or more changes according to a particular schedule (e.g., hourly, daily, weekly, or any other suitable schedule.). For example, the system may be configured to scan one or more webpages and/or other sources of vendor information on an ongoing basis to determine whether any pieces of vendor information have changed (e.g., whether the vendor has not renewed its Privacy Shield membership, lost its ISO certification, etc.).

(432) FIG. 24 shows an example process that may be performed by a Vendor Compliance Demonstration Module 2400. In executing the Vendor Compliance Demonstration Module 2400, the system begins at Step 2410, where it determines vendor information. The Vendor Compliance Demonstration Module 2400 may determine vendor information based on a selection of a control on a graphical user interface, such as a control or indicia on an interface associated with a vendor. In various embodiments, the Vendor Compliance Demonstration Module 2400 may determine vendor information from user input such as text input on a graphical user interface, for example, when a user inputs information for a new vendor to be analyzed for compliance as described herein. In various embodiments, the Vendor Compliance Demonstration Module 2400 may determine vendor information using information (e.g., a vendor name) received from a user and/or associated with an interface activity (e.g., selection of a control) to query a database of vendor information.

(433) At Step 2410, determining vendor information may include performing analysis on one or more documents to determine the vendor information. For example, the system may be configured to retrieve one or more contracts that an entity has entered into with a vendor from a database using a vendor's name. The system may then analyze such one or more contracts (e.g., using natural language processing) to identify one or more particular terms used in the one or more contract that may be useful in calculating a vendor risk score for the vendor. The system may be configured to also, or instead, obtain and/or determine any other internally sourced data associated with the vendor at Step 2410, such as internal records of interactions with the vendor, business relationship information for the vendor, service provided by the vendor, length of relationship with vendor, expiration of vendor service agreements, etc.

(434) At Step 2420, the system may obtain publicly available vendor information. In doing so, the system may be configured to scan one or more webpages operated by or on behalf of the vendor and perform analysis of such webpages to determine, for example, any of the various factors related to privacy and/or security described herein. The system may also be configured to scan one or more webpages that are not operated by, or on behalf of, the vendor and perform analysis of such sites to determine any of the various factors related to privacy and/or security described herein. For example, the system may scan and analyze websites of one or more privacy certification organizations and/or industry groups to extract one or more factors related to privacy and/or security associated with the vendor. The system may perform such analysis using natural language processing and/or metadata analysis to extract data from one or more websites and/or other sources of information.

(435) The system may also verify one or more factors at Step 2420. For example, the system may determine that a vendor's webpage indicates that the vendor holds a particular privacy certification and may then analyze the webpage of the organization that issues the particular privacy certification to verify that the vendor does indeed hold the claimed privacy certification or to determine that the vendor does not hold the privacy certification as claimed. At Step 2420, the system may access and/or analyze information from one or more other publicly available sources of information, such as databases, publications, libraries, etc.

(436) At Step 2430, the system may calculate a vendor risk score, as described in more detail herein. In various embodiments, this calculation may be performed based at least in part on the vendor information determined at Step 2410 and/or the publicly available information obtained at Step 2420. In determining the vendor's risk score, the system may use any one or more factors, each of which may be weighted according to any criteria as described herein.

(437) At Step 2440, the system may use any of the vendor information (e.g., as determined at Step 2410), publicly available vendor information (e.g., as determined at Step 2420), and/or a calculated vendor risk score (e.g., as determined at Step 2430) to determine any additional vendor information. For example, the system may calculate a supplemental score for the vendor (e.g., based at least in part on the score determined at Step 2430 in combination with another score associated with the particular vendor). Such a supplemental score may relate to any one or more security attributes of the particular vendor, one or more privacy attributes of the particular vendor, and/or one or more privacy or security attributes of one or more products provided by the particular vendor.

(438) In various examples, the system may perform analysis of vendor information, publicly available vendor information, and/or one or more vendor risk scores at Step 2440 to determine the additional information. For example, the system may analyze one or more news reports retrieved at Step 2420 to identify a data breach involving the particular vendor and determine, as additional vendor information, that the breach was a high risk incident. In another example, the system may analyze the status of a privacy certification held by the particular vendor and determine that the certification expires within a short time period. In response, as additional vendor information, the system may determine at Step 2440 (e.g., based on one or more additional pieces of information) that the particular vendor is at high risk of losing the privacy certification. In another example, the system may analyze a number of and/or one or more descriptions of privacy-related officers in the particular vendor's organization (e.g., their respective job titles and/or backgrounds) and determine, as additional vendor information, that the particular vendor treats privacy issues as a high priority, and therefore has lower relative privacy risk as opposed to other organizations. In yet another example, the system may determine one or more additional scores and/or rankings beyond a vendor risk score reflecting calculations based on other criteria at Step 2440, such as a compliance score reflecting the particular vendor's compliance with a particular privacy standard and/or regulatory regime. The system may use any information

available for the particular vendor to determine any additional vendor information.

(439) At Step **2450**, the system may generate a graphical user interface and present, to a user, all or any subset of the vendor information, the publicly-available vendor information, the vendor privacy risk score, and/or the additional vendor information.

(440) As noted herein, each piece of information associated with a vendor, regardless of how obtained or used by the presently disclosed systems, may have an associated expiration date. FIG. **25** shows an example process that may be performed by a Vendor Information Update Module **2500** that may utilize such expiration dates. In executing the Vendor Information Update Module **2500**, the system begins at Step **2510**, where it determines a piece of vendor information. This may be suitable any piece of vendor information, such as, but not limited to, a piece of non-publicly available vendor information, a piece of publicly available vendor information, a vendor risk score, and/or a piece of additional vendor information (e.g., as described herein). Such a piece of vendor information may be retrieved from a database and/or otherwise obtained using any suitable means. (441) At Step **2520**, an expiration date associated with the retrieved piece of vendor information may be evaluated and determined to have passed. This expiration date may have been set based on an intrinsic characteristic of the piece of information (e.g., a date of expiration of privacy certification) and/or on one or more criteria associated with the acquisition, determination, and/or storage of the piece of information (e.g., six months after a date of acquisition, determination, and/or storage of the piece of information).

(442) At Step **2530**, responsive to determining that the expiration date has passed, the system may initiate a process to obtain and/or determine an updated piece of information. For example, the system may generate and transmit another assessment to the particular vendor associated with the expired piece of information to acquire an updated corresponding piece of information. In another example, the system may recalculate a risk score for the particular vendor associated with an expired risk score using current information. In another example, the system may scan one or more webpages for updates in order to determine an updated piece of information.

(443) At Step **2540**, the system may determine whether a valid updated piece of vendor information was obtained (e.g., determined, received). If an updated piece of information was successfully obtained (e.g., one or more responses to an updated assessment sent to a vendor were received, an updated privacy risk score was calculated, updated information was determined from analyzed webpages, etc.), at Step **2550** the system may store this updated piece of information and a new expiration date, associating the updated piece of information and the new expiration date with the appropriate vendor. Alternatively, if the system was unable to update an expired piece of information (e.g., no response was received to an updated assessment questionnaire sent to a vendor, an updated privacy risk score could not be calculated due to a lack of sufficient current information, no updated information is currently available from current webpages, etc.), at Step **2560**, the system may store an indication that the piece of information is expired, invalid, and/or otherwise should not be relied upon (e.g., store such an indication in a database and associate the indication with the piece of information and/or the vendor).

(444) FIG. **26** shows an example process that may be performed by a Vendor Risk Score Calculation Module **2600**. In executing the Vendor Risk Score Calculation Module **2600**, the system begins at Step **2610**, where it determines and/or otherwise obtains non-publicly available vendor information (e.g., non-publicly available vendor information, information determined from one or more documents, etc.), publicly available vendor information, and/or vendor assessment information (e.g., as described herein). Such information may be any information and criteria as described herein.

(445) At Step **2620**, for each piece of non-publicly available vendor information, publicly available vendor information, and/or vendor assessment information, the system may be configured to determine whether the piece of information is valid. In various embodiments, to determine whether a piece of information is valid, the system may determine whether an expiration date associated with the piece of information has passed. If the expiration date has passed (e.g., the information has expired), the system may be configured to request updated information corresponding to the expired piece of information using, for example, means described herein (e.g., one or more processes such as those described in regard to FIG. **25**). Other verification criteria may also, or instead, be used. For example, the system may analyze a piece of vendor information to determine whether it matches known information (e.g., a vendor name on a security certification matches a known vendor name, a vendor address on an industry membership roll matches a known vendor address, a name of vendor representative in a particular position listed in a contract matches a known vendor representative in that position, etc.). Any invalid information may be addressed in any effective manner, such as those described herein. (446) At Step **2630**, the system may determine a value for each piece of non-publicly available vendor information, publicly available vendor information, and/or vendor assessment information that is to be used in calculating a vendor risk score (e.g., a vendor privacy risk score, a vendor security risk score, a vendor privacy risk rating, a vendor security risk rating, etc.). For example, in order to calculate a numerical vendor risk score, the system may determine a numerical value for each piece of non-publicly available vendor information, publicly available vendor information, and/or vendor assessment information. The system may be configured to assign a numerical value to each respective piece of non-publicly available vendor information, publicly available vendor information, and/or vendor assessment information using any criteria, including those described herein and/or any other suitable process, algorithm, etc.

(447) At Step **2640**, the system may be configured to apply a weighting factor to each respective value determined for each respective piece of non-publicly available vendor information, publicly available vendor information, and/or vendor assessment information. In various embodiments, some pieces of such information may be considered more important in determining a vendor risk score than others. The system may be configured to assign a greater weight to such information of elevated importance when calculating a vendor risk score. For example, a vendor's current one or more security certifications may be considered to be of greater importance than a vendor's attendance at one or more privacy-related events. In such an example, the system may apply a weighting factor to the value associated with the vendor's security certifications that is greater than the weighting factor applied to the value associated with the vendor's attendance at privacy events. Various means of determining suitable weighting factors may be used, including as described herein.

(448) At Step **2650**, the system may calculate the vendor risk score using the respective weighted values of each piece of non-publicly available vendor information, publicly available vendor information, and/or vendor assessment information. The system may, for example, be configured to perform a calculation to determine the score, such as averaging the weighted values of each piece of information. Alternatively, or in addition, the system may be configured to employ more detailed calculations and/or algorithms using the weighted values of each piece of information to determine the vendor privacy risk score. At Step **2660**, the system may generate a graphical user interface and present the vendor risk score to a user. In various embodiments, the system may present the vendor privacy risk score on a graphical user interface that displays other information as well, including any interface described herein.

(449) In particular embodiments, the system may be configured to generate and maintain a database of vendor information (e.g., including a risk analysis for each of a plurality of particular vendors). Any information associated with a vendor in any way (e.g., any vendor-related information described herein) may be stored in and/or retrieved from such a vendor information database. Such information may be acquired and/or determined by the system via any means described herein (e.g., scanning of webpages, analyzing vendor privacy risk assessments, analyzing contractual terms, analyzing one or more documents associated with the vendor, etc.). The system may provide access to, or provide information retrieved from, such a vendor information database to entities that may wish to contract with (e.g., in a new contract or by renewing an existing contract), pay, or otherwise utilize or interact with one or more vendors that are in the database. The system may also provide access to, or provide information retrieved from, such a vendor information database to entities that already have an existing relationship with one or more vendors that are in the database. In this way, the system may enable such entities to assess the risk of, for example, integrating new vendors into a new or existing processing activity, a risk associated with paying the vendor, and/or the risk of continuing a relationship with one or more vendors.

(450) In various embodiments, vendor information (of any type) may be retrieved using one or more data models. A data model may be stored in a vendor information database and/or in any other storage means available to the disclosed systems. A data model may be associated with a vendor and

may map one or more relationships between and/or among a plurality of data assets utilized by a vendor (e.g., alone or in combination with another entity). In particular embodiments, each of the plurality of data assets (e.g., data systems) may include, for example, any asset that collects, processes, contains, and/or transfers data (e.g., such as a software application, “internet of things” computerized device, database, website, data-center, server, etc.). For example, a first data asset may include any software or device (e.g., server or servers) utilized by a particular vendor for such data collection, processing, transfer, storage, etc. A data model may store any of the following information: (1) the vendor that owns and/or uses a particular data asset; (2) one or more departments within the vendor responsible for the data asset; (3) one or more software applications that collect data (e.g., personal data) for storage in and/or use by the data asset (e.g., or one or more other suitable collection assets from which the personal data that is collected, processed, stored, etc. by the primary data asset is sourced); (4) one or more particular data subjects and/or categories of data subjects that information is collected from for use by the data asset; (5) one or more particular types of data that are collected by each of the particular applications for storage in and/or use by the data asset; (6) one or more individuals (e.g., particular individuals or types of individuals) that are permitted to access and/or use the data stored in, or used by, the data asset; (7) which particular types of data each of those individuals are allowed to access and use; and/or (8) one or more data assets (destination assets) that the data is transferred to for other use, and which particular data is transferred to each of those data assets. In particular embodiments, the data model stores this information for each of a plurality of different data assets and may include links between, for example, a portion of the model that provides information for a first particular data asset and a second portion of the model that provides information for a second particular data asset.

(451) In various embodiments, vendor information (of any type) may be retrieved using one or more data maps (e.g., privacy-related data maps). A data map may include a visual and/or computer-readable representation of one or more data models that may include one or more data assets, one or more connections between the one or more data assets, one or more inventory attributes, one or more vendor attributes, etc. For example, a data map may include one or more of: (1) a visual or other indication of a first data asset (e.g., a storage asset), a second data asset (e.g., a collection asset), and a third data asset (e.g., a transfer asset); (2) a visual or other indication of a flow of data (e.g., personal data) from the second data asset to the first data asset (e.g., from the collection asset to the storage asset); (3) a visual or other indication of a flow of data (e.g., personal data) from the first data asset to the third data asset (e.g., from the storage asset to the transfer asset); (4) one or more visual or other indications of a risk level associated with the transfer of personal data; and/or (5) any other suitable information related to the one or more data assets, the transfer of data between/among the one or more data assets, access to data stored or collected by the one or more data assets, etc.

(452) In particular embodiments, the data map identifies one or more electronic associations between at least two data assets within a data model comprising a respective digital inventory for each of the two or more data assets, each respective digital inventory comprising one or more respective inventory attributes selected from a group consisting of: (A) one or more processing activities associated with each of the respective data assets; (B) transfer data associated with each of the respective data assets; and (C) respective identifiers of one or more pieces of personal data associated with each of the respective data assets.

(453) The system may be configured to provide a user-accessible “dashboard” (e.g., a graphical user interface) through which a user (e.g., on behalf of an entity) may initiate a process of requesting information for a vendor (a current or new vendor to the entity). The system may, for example, perform a risk assessment (e.g., privacy risk assessment, security risk assessment, privacy impact assessment, etc.) for a specified particular vendor, which may include: (1) determining whether a current risk assessment exists for the particular vendor within the system (e.g., whether a current risk assessment is stored within a data structure (e.g., a database) associated with the system); (2) determining how long the particular vendor (e.g., a business entity) has been in business; (3) identifying one or more privacy and/or security related incidents (e.g., data breaches) associated with the particular vendor and/or one or more sub-processors utilized by the particular vendor; and/or (4) analyzing any other available data related to the particular vendor. Based at least in part on the analyzed vendor data, the system may determine whether to: (1) automatically trigger a new or updated risk assessment for the vendor; (2) automatically approve the particular vendor (e.g., as a business partner for a particular entity and/or for involvement in a particular processing activity); and/or (3) automatically reject the particular vendor (e.g., as a business partner for a particular entity and/or for involvement in a particular processing activity).

(454) For example, at least partially in response to determining that the particular vendor has an existing, older vendor risk assessment stored within a database stored within a data structure associated with the system (e.g., a vendor risk assessment that is past a particular age, such as six months), the system may be configured to trigger a new vendor risk assessment for the particular vendor (e.g., using any suitable technique described herein). In another example, the system may be configured to trigger a new vendor risk assessment for the particular vendor in response to determining that the particular vendor has experienced one or more privacy-related incidents and/or a security-related incidents (e.g., a data breach) after the most recent vendor risk assessment was completed for the particular vendor. In yet another example, the system may be configured to automatically approve the particular vendor in response to determining that the system currently stores a recent vendor risk assessment for the particular vendor, and/or that the particular vendor has had no recent privacy and/or security incidents. Any such approvals or rejections may also be based, at least in part, on other information associated with the particular vendor, including, but not limited to: (1) one or more vendor risk scores; (2) one or more terms contained in one or more documents (e.g., contracts, licenses, agreements, etc.) involving the vendor; (3) one or more privacy and/or security certifications held by the vendor; (4) any other public information about the vendor (e.g., retrieved by scanning webpages or accessing databases); and/or (5) any other suitable vendor-related information, described herein or otherwise.

(455) In particular embodiments, the system is configured to maintain a database of vendor privacy-specific information (e.g., scoring criteria) for use in such assessments. The system may be configured to periodically (e.g., every month, every week, annually, every six months, or at any other suitable interval) update such privacy-specific information and/or to monitor for one or more changes to such privacy-specific information (e.g., vendor privacy information) and update the database in response to identifying any such changes. Any information in such a database may have an associated expiration date, the passing of which may trigger the system to (e.g., substantially automatically) attempt to obtain updated information for the vendor.

(456) FIG. 27 shows an example process that may be performed by a Vendor Privacy Risk Determination Module 2700. In executing the Vendor Privacy Risk Determination Module 2700, the system begins at Step 2710, where it receives a request assess the risk associated with a particular vendor. The system may receive such a request via a graphical user interface where a user has selected the vendor from a prepopulated listing or otherwise specified the particular vendor for which information is desired (e.g., as described herein).

(457) At Step 2720, the system may attempt to retrieve any currently available information for the particular vendor (e.g., a completed risk assessment (e.g., a privacy risk assessment, a security risk assessment, etc.) for the vendor, a summary of such a risk assessment, and/or any other suitable information regarding the vendor), for example, from a vendor information database.

(458) At Step 2730, the system may determine whether a current risk assessment was retrieved from the vendor information database for the particular vendor. In various embodiments, if no current, valid vendor risk assessment for the vendor exists in the database (e.g., an existing assessment has expired, is invalid, or is not present), the system may be configured to responsively obtain an updated (e.g., new) vendor risk assessment from the particular vendor at Step 2731 (e.g., as described herein). At least partially in response to obtaining an updated vendor risk assessment for the vendor and/or determining that a current, valid vendor risk assessment was retrieved from the vendor information database, the system may proceed to Step 2740.

(459) At Step 2740, the system may determine whether other vendor information (e.g., any vendor information described herein beyond a vendor risk assessment) retrieved from the vendor information database for the particular vendor is present, current, and valid. In various embodiments, if the system retrieves expired or otherwise invalid vendor information at this step, and/or any required vendor information is not present in the vendor information database, the system may be configured to responsively obtain updated (e.g., new) information (e.g., using any means described herein)

at Step **2741**. At least partially in response to obtaining any needed vendor information and/or determining that all required vendor information retrieved from the vendor database is current and valid, the system may proceed to Step **2750**.

(460) At Step **2750**, the system may determine whether a current vendor risk score retrieved from the vendor information database for the particular vendor is available to the system (e.g., saved to a database associated with the system) and current. If the system retrieves an expired vendor risk score or there is no vendor risk score present in the vendor information database for the particular vendor, the system may be configured to responsively calculate an updated (e.g., new) vendor risk score (e.g., using any means described herein) at Step **2751**. At least partially in response to calculating an updated vendor risk score and/or determining that the vendor risk score retrieved from the vendor database is current, the system may proceed to Step **2760**.

(461) At Step **2760**, the system may be configured to determine whether to approve the use (e.g., new or continued) of the particular vendor based at least in part on the information retrieved and/or otherwise determined previously (e.g., in prior steps). In various embodiments, any or all of the information described in regard to FIG. 27, or elsewhere herein, may be used, at least in part, by the system to make this determination. If, at Step **2770**, the system determines that the particular vendor is approved for new or continued use with the entity, then, at Step **2771**, the system may present an indication of such approval to a user. The system may present such an indication on a graphical user interface (or via any other suitable communications mechanism—e.g., a paper report, an audio signal, etc.) that may also include a presentation of any of the vendor information described herein. If, at Step **2770**, the system determines that the particular vendor is rejected from new or continued use with the entity, then, at Step **2772**, the system may instead present an indication of such rejection to a user. Here again, the system may present such an indication on a graphical user interface (or via any other suitable communications mechanism—e.g., a paper report, an audio signal, etc.) that may also include presentation of any of the vendor information described herein.

(462) It should be understood that various alternative embodiments of the system may function differently than described above. For example, while the system is described above as using three different types of information to determine whether to approve or reject a particular vendor, other embodiments may use only one or two of these three types of information or may use different or other information when making this determination.

(463) Dynamic Vendor Training Material Generation

(464) In particular embodiments, the system may be configured to generate training material associated with a particular vendor based at least in part on privacy information associated with that particular vendor, such as the vendor's privacy risk score, any privacy-related information for the vendor, any publicly available information for the vendor, sub-processors used by the vendor, privacy and/or security incidents involving the vendor, etc. (e.g., any information described herein that may be associated with a vendor). In various embodiments, such training material may be intended for use by an entity to train employees on how to evaluate, interact, and/or otherwise operate with the particular vendor with whom the training is associated. In various embodiments, such training material may be intended for use by the particular vendor itself, for example as training recommended and/or required by the entity engaging the particular vendor. Any other use of such training material is contemplated in various embodiments.

(465) The system may generate vendor-specific training material on-demand, for example, at least partially in response to the detection of a selection of a user-selectable control on a graphical user interface, where the control is associated with requesting the generation of such material.

(466) The system may also, or instead, generate vendor-specific training material at least partially in response to detection of an occurrence associated with the particular vendor. For example, the system may be configured to detect (e.g., using any suitable technique described herein) a change in any vendor information described herein (e.g., a change in a vendor risk score, a change in a vendor sub-processor, etc.) and/or detect an incident or other event involving the vendor (e.g., a privacy breach, a security incident, etc.). In response to detection of such an occurrence, the system may be configured to dynamically (e.g., substantially automatically) update training material associated with the involved vendor to reflect the detected occurrence. The system may be configured to adjust existing training material in an appropriate manner, update existing training material, and/or generate new training material based at least in part on the occurrence. In various embodiments, the generated training material may also include one or more training assessments that may be used to gauge how well the recipients of the training material have absorbed the material. The system may be configured to store training material in a vendor database as described herein or in any appropriate system.

(467) FIG. 28 shows an example process that may be performed by a Dynamic Vendor Privacy Training Material Generation Module **2800**. In executing the Dynamic Vendor Privacy Training Material Generation Module **2800**, the system begins at Step **2810**, where a request to generate vendor-related training may be received by the module. Such a request may be received via a graphical user interface where a user has selected the vendor from a prepopulated listing of vendors and/or otherwise specified the particular vendor for which training is desired (e.g., as described herein).

(468) At Step **2820**, the system may retrieve any currently available information for the particular vendor, for example, from a vendor information database. This information may include any vendor information described herein (e.g., vendor privacy risk assessment, vendor risk score, vendor incident history, publicly available vendor information, etc.). This information may also include any other suitable information that may be of use in generating training material associated with a particular vendor, such as: (1) one or more training material templates; (2) general information to be included in any vendor training; (3) background on applicable privacy and/or security laws and regulations; (4) one or more standard procedures for interacting with vendors; and/or (5) any other generally applicable vendor training material.

(469) At Step **2830**, the system may generate the training material associated with the particular vendor using any of the information obtained at Step **2820**. The generated training material may take any suitable form (e.g., one or more manuals, slide decks, audio files, video files, etc.). At Step **2840**, the system may present an indication on a graphical user interface that the training material associated with the particular vendor has been generated and/or may include a user-selectable control on such an interface that allows a user to download or otherwise access such training material. Such a graphical user interface may also include presentation of any of the vendor information described herein. At Step **2840**, the system may also store the generated training material, for example, in a vendor database as described herein and/or in any appropriate system.

(470) FIG. 29 shows an example process that may be performed by a Dynamic Vendor Privacy Training Material Update Module **2900**. In executing the Dynamic Vendor Privacy Training Material Update Module **2900**, the system begins at Step **2910**, where the system may detect an occurrence associated with a particular vendor. For example, the system may detect a change in any vendor information and/or an incident involving the vendor (e.g., any information or occurrence as described herein).

(471) At Step **2920**, in response to detecting the change or occurrence associated with the particular vendor, the system may retrieve any updated information for the particular vendor (e.g., from a vendor information database) and/or any other information relevant to the detected change or occurrence. This information may include any information described herein. As with the process of FIG. 29, this information may also include any other information that may be of use in generating training material associated with a particular vendor.

(472) At Step **2930**, the system may generate the training material associated with the particular vendor using any of the updated and/or occurrence information obtained at Step **2920**. At Step **2940**, the system may present an indication on a graphical user interface that the updated training material associated with the particular vendor has been generated. Such a graphical user interface may include a user-selectable control that allows a user to download or otherwise access such updated training material. Such a graphical user interface may also include presentation of any of the vendor information described herein. At Step **2940**, the system may also store the generated training material in a vendor database as described herein or in any appropriate system.

(473) It should be understood that various alternative embodiments of the system may function differently than described above. For example, while the system is described above as using three different types of information to determine whether to approve or reject a particular vendor, other embodiments may use only one or two of these three types of information or may use different or other information when making this determination.

(474) Exemplary User Experience

(475) Exemplary Vendor Incident Management User Experience

(476) FIGS. **30-34** depict exemplary screen displays that a user may encounter when utilizing an exemplary system configured to provide notifications of a security-related incident to one or more vendors of a particular entity. For example, a vendor list page **3010** illustrated in FIG. **30** presents a listing of vendors and associated vendor attributes (e.g., vendor name, service products provided by each respective vendor, vendor score (which may, for example, indicate a privacy rating and/or security rating for the vendor), criticality of each respective vendor to the particular entity, associated business unit for each respective vendor (e.g., that the entity does direct business with), privacy impact assessment status for each respective vendor, status of each respective vendor with respect to the entity, etc.). The vendor list page **3010** may be represented in a graphical user interface, or in any other suitable format.

(477) At least partially in response to an occurrence and/or detection of an incident, the system may generate and/or present an incident alert **3020** on the vendor list page **3010**. Incident alert **3020** may include a summary and/or brief description of the incident and may be, or include, a user-selectable object that instructs the system to generate an incident detail page, such as incident detail page **3110** of FIG. **31**.

(478) Turning now to FIG. **31**, at least partially in response to an occurrence and/or detection, by the system, of an incident and/or in response to selection of a control requesting incident details, the system may generate a page presenting the details of a security-related incident, such as incident detail page **3110**. The incident detail page **3110** may be represented in a graphical user interface, such as a webpage.

(479) The incident detail page **3110** may include various attributes **3120** of a security-related incident. For example, as may be understood from FIG. **31**, incident detail page **3110** may display: (1) the method used to report the incident; (2) a date that the incident was reported (e.g., May 12, 2018); (3) a geographical location of occurrence of the incident (e.g., USA); and/or (4) a description of the incident. Additional information may also be presented, such as potentially impacted processing activities and/or contracts **3130** (e.g., processing activities and/or contracts that may be affected by the particular incident). The system may receive additional information, such as the potentially impacted processing activities and/or contracts **3130**, when receiving information about the incident and/or the system may determine such additional information based on information received about the incident and/or one or more attributes of the incident (e.g., attributes **3120**) and/or the system's analysis of such information and/or attributes.

(480) As noted herein, at least partially in response to receiving and/or analyzing incident information and/or one or more attributes of the incident, the system may determine one or more vendors associated with the incident and/or the notification obligations for each such vendor.

(481) Turning now to FIG. **32**, the system may generate a page presenting the details of a security-related incident and associated vendor notification tasks, such as incident detail page **3210**. The incident detail page **3210** may be presented in a graphical user interface. Similar to the incident detail page **3110**, the incident detail page **3210** may include various attributes **3220** of security-related incident. For example, as seen on the incident detail page **3210**, a method of reporting the incident may be presented (e.g., web form), as well as a date reported (e.g., May 12, 2018), a geographical location of occurrence of the incident (e.g., USA), and a description of the incident.

(482) The system may also include, on incident detail page **3210**, a listing of tasks **3230** to be performed to satisfy one or more of the entity's incident notification obligations to the vendor. As noted herein, the system may determine one or more affected vendors and associated obligations, and any information associated therewith, by analyzing one or more vendor contracts and/or one or more attributes of the incident. The listing of tasks **3230** may include a title for each respective task (e.g., "Notify Amazon Web Services"), a status for each respective task (e.g., "New"), a timeframe for completion of each respective task (e.g., "48 Hrs"), whether each respective task is required (e.g., "Yes"), a user to whom each respective task is assigned (e.g., "UserName Here"), and/or a deadline for completion of each respective task (e.g., "4/25/2018").

(483) One or more sections of each task listing presented in listing of tasks **3230** may be user selectable. At least partially in response to activating (e.g., "hovering" or moving a cursor onto) such a section, the system may generate a pop-up window **3240** providing a brief description of the task to be performed. In response to clicking on, or otherwise selecting, a task from the listing of tasks **3230**, the system may generate a task details page, such as the task detail page **3310** of FIG. **33**.

(484) Turning now to FIG. **33**, the system may generate a page presenting the details of a vendor notification task, such as task detail page **3310**. The task detail page **3310** may include a reason section **3320** that may provide a brief explanation for why this vendor incident notification task should be performed. A detailed explanation section **3330** may provide additional information, such as one or more excerpts from the applicable contract, agreement, regulation, law, etc. A task information section **3340** may list the task to be performed and any responses that may have been received to the task received (e.g., from the vendor, from those asked to perform the task, etc.). A user may provide any additional information associated with the task by uploading one or more files to the system in upload section **3350**. For example, the communication (e.g., email, letter, documentation of a phone call) used to satisfy the task may be uploaded or otherwise recorded here. Upon completion of the task, the task may be marked as complete by a user at completion control **3360**. Any other changes to the task, such as status change, indication of actions taken, partial completion of the task, changes made to the task details, etc., may be saved by the user (e.g., via task detail page **3310**). The system may store any such task details and changes, including an indication of satisfaction of a vendor incident notification task, in a suitable database or elsewhere.

(485) The system may provide a summary of incidents that includes one or more incidents associated with one or more vendors for ease of evaluation. Turning now to FIG. **34**, the system may generate a page, such as incident summary page **3410**, presenting a listing of incident-related tasks, including vendor notification tasks. The incident summary page **3410** may include an incident summary listing **3420** that may include a listing of tasks (e.g., to be performed, in progress, and/or completed). The incident summary listing **3420** may indicate a type of each respective task (e.g., "Data Leak", "Vendor Incident"), a severity of each respective task (e.g., "Very High", "Medium"), a status of each respective task (e.g., "Notify—New", "Complete"), a contact person for each respective task (e.g., "Steve", "Carrie"), and a date of creation of each respective task (e.g., "12/20/17", "11/15/17", "10/20/17").

(486) Exemplary Vendor Risk Scanning and Scoring Experience

(487) FIGS. **35-46** depict exemplary screen displays that a user may encounter when utilizing any suitable system described herein to view and/or determine a vendor's compliance, privacy, and/or security scoring and/or other attributes. These exemplary screen displays may also, or instead, be encountered by a user when onboarding a new vendor on behalf of an entity utilizing any suitable system described herein. For example, these exemplary screen displays may be encountered by a user associated with an entity in evaluating a vendor according to the disclosed embodiments. These exemplary screen displays may also, or instead, be encountered by a vendor in completing an evaluation requested by an entity, as part of one or more processing activities.

(488) FIG. **35** depicts the exemplary listing **3520** of one or more vendors in a database as represented in the exemplary interface **3510**. The listing **3520** may include one or more vendors with which an entity is already engaging in one or more contracts. Each item listed in the listing **3520** may include vendor information, which may include: (1) the vendor's name; (2) a product provided by the vendor; (3) a risk score for the vendor or the vendor's product(s); (4) a criticality rating for the vendor (or vendor's product); (5) a business unit for which the vendor provides services; (6) an privacy impact assessment status for the vendor (or vendor's product) (e.g., does the entity have a current privacy impact assessment for the vendor); and (7) a current status of the vendor. Some portion of the listing for each vendor shown in the listing **3520** may be a user-selectable control (e.g., a user-selectable indicia, a webpage control, etc.) that, when selected and/or otherwise activated, presents the user with additional vendor information as described herein.

(489) The exemplary interface **3510** may also include a user-selectable control **3530** for adding a new vendor to the database of vendor information. In response to the user selecting the control **3530**, the system may be configured to generate the interface **3610** shown in FIG. **36** which may facilitate the creation of a new database entry for the new vendor. The system may access a prepopulated database of potential vendor information and use

such information to provide a listing of one or more potential vendors **3630** from which a user may select a vendor. The system may also allow a user of the interface **3610** to search for a particular vendor from among those available in a database of potential vendor using a search field **3620**. In some examples, the system may populate a drop-down box **3621** based on the user's input to the search field **3620**, allowing the user to select a vendor from the drop-down box **3621**. Should the user not locate the desired vendor from the listing of vendors provided by the interface **3610**, the user may select the control **3640** to add a new vendor without using prepopulated information.

(490) Upon selection of a vendor from the prepopulated listing on the interface **3610** or selection of the control **3640** to add a new vendor without using predetermined information, the system may generate an exemplary interface **3710** of FIG. **37**. Where the user has selected a particular vendor as the vendor to be added to a database of vendor information (e.g., by selecting a vendor on the interface **3610** of FIG. **36**), the system may prepopulate some or all of the field and information shown in the interface **3710**. Where the user has chosen to add a new vendor without using predetermined information, some or all of the field and information shown in the interface **3710** may be left blank.

(491) The fields available in the interface **3710** may include the vendor information fields **3720** (e.g., in the example of FIG. **37**, for ABC, Inc., an audit and financial advisory firm). The vendor information fields **3720** may include respective fields for: (1) a vendor name; (2) a vendor description; (3) one or more vendor addresses or locations (e.g., a vendor headquarters address, a location within which the vendor operates, a jurisdiction to which the vendor is subject, etc.); (4) one or more vendor contacts; (5) contact information for the one or more vendor contacts; (6) respective roles and/or responsibilities of the one or more vendor contacts; and/or (7) any other suitable vendor information. Some or all of the vendor information fields **3720** may be prepopulated based on known vendor information (e.g., in response to a user selecting a vendor on the interface **3610** of FIG. **36**). The fields available in the interface **3710** may include a services field **3730** that may allow a user to select or view one or more of the services, products, software, offerings, etc. that the vendor may provide to the entity. The user may select and/or deselect such services as appropriate. Some or all of the services shown in the services field **3730** may be preselected and/or prepopulated based on known vendor services information (e.g., in response to a user selecting a vendor on the interface **3610** of FIG. **36**). The system may be configured to enable a user to update any information (e.g., that may be incorrect or non-current) that may have been prepopulated.

(492) Upon entry or receipt of vendor information (e.g., as described in regard to FIG. **37**), the system may be configured to enable a user to upload one or more documents associated with the vendor (e.g., one or more licenses, agreements, contracts, etc. that an entity may be entering into and/or engaged in with the vendor). To facilitate this document uploading, the system may generate an interface such as the exemplary interface **3810** shown in FIG. **38**. The interface **3810** may be configured to receive one or more documents for uploading and analysis, for example using the upload field **3820**. The interface **3810** may also display a listing **3830** of documents that have already been uploaded for this particular vendor. Such a listing may be prepopulated based on an earlier selection of the particular vendor (as described in regard to FIG. **36**) and/or may reflect documents already uploaded using the interface **3810**.

(493) Upon receipt of one or more documents associated with the vendor, the system may be configured to analyze such one or more documents using any suitable analysis technique (e.g., natural language processing) to identify key language and/or terms in the documents. The system may, for example, be automatically configured to identify, from such documents, one or more of: (1) term limits; (2) breach notification timeline obligations; (3) sub-processor change notifications; (4) liability caps and/or obligations; (5) data breach liability information; (6) indemnification information; (7) data transfer mechanisms; (8) notification time periods for a breach; (9) notification requirements for sub-processor changes; and/or (10) any other suitable information that may be included in any documents associated with a vendor.

(494) FIG. **39** depicts the exemplary interface **3910** showing results of such analysis. The system may be configured to indicate one or more particular identified features and/or terms of the documents in the critical data section **3920**, which may list such features and/or terms as one or more respective user-selectable controls associated with one or more respective locations in the uploaded document where the particular identified features and/or terms may be found. Upon selection of a control for a particular feature or term, the system may be configured to display the document section from which the particular feature or term was derived in the document display section **3930**. For example, as shown in the interface **3910**, the system has identified breach notification requirements, liability obligations, and data transfer obligations in the critical data section **3920**. When the highlighted breach notification requirements indicia in the critical data section **3920** is selected, the system is configured to display the corresponding text from the document from which such requirements were derived in the document display section **3930**.

(495) As described herein, the system may be configured to determine and/or analyze publicly available information sources and/or shared information sources that may have data associated with the vendor. Such information sources may include one or more webpages (e.g., operated by the vendor and/or operated by third parties), databases to which the entity may have access, news sources, governmental bodies, regulatory agencies, industry groups, etc. FIG. **40** depicts the exemplary interface **4010** that may indicate to a user the information sources that are being analyzed in the listing **4020**. In this analysis, the system may be configured to use any suitable analysis technique (e.g., natural language processing) to determine the desired vendor-related information. Among the analysis performed by the system, the system may be configured to: (1) analyze one or more local/privacy/jurisdiction laws associated with the vendor; (2) analyze shared data with the vendor; (3) analyze one or more consent withdrawal obligations from one or more vendor documents; (4) analyze one or more data subject requests associated with the vendor; and (5) analyze one or more sub-processors associated with the vendor.

(496) FIG. **41** depicts the exemplary interface **4110** showing a vendor overview. The system may be configured to generate and display the vendor overview interface **4110** based on any vendor information the system has determined, including information determined based on the vendor analyses described herein. The interface **4110** may include a description of the vendor (e.g., "ADB, Inc." in FIG. **41**) in the vendor description section **4120** that may include the vendor's name, location, description, etc.

(497) The system may be configured to determine additional information for the vendor based on one or more of: (1) information gathered from the vendor (e.g., assessment responses from the vendor); (2) information about the vendor gathered from public or shared sources (e.g., webpages, databases, etc.); documents associated with the vendor (e.g., contracts, licenses, agreements, etc.); and/or (3) and other vendor information (e.g., known vendor data, historical information about the vendor, etc.). Such additional information may be displayed on the interface **4110**.

(498) In various embodiments, as part of additional vendor information, the system may calculate a vendor risk score for the vendor, shown as "Vendor Score" in the vendor score section **4170** of the interface **4110**. As described herein, the system may, for example, calculate the vendor risk score based on any factor(s) and/or criteria described herein or that may be suitable (e.g., information transfer, contract terms, assessments performed, etc.). The system may also calculate one or more other scores (e.g., as one or more internal vendor-related scores based on criteria different than that used to determine a vendor risk score) and display such scores in the vendor score section **4170**.

(499) In various embodiments, as part of additional vendor information, the system may determine and/or highlight one or more vendor risks (e.g., data encryption incidents, personal information compromises, 3.sup.rd party breaches, etc.) and display such risks in the vendor risk section **4130**. In various embodiments, as part of additional vendor information, the system may determine and display third-party vendors utilized by the vendor in the third-party vendor section **4140**. In various embodiments, as part of additional vendor information, the system may determine and display historical incidents associated with the vendor in the historical incident section **4150**. In various embodiments, as part of additional vendor information, the system may determine and display a listing of services provided by the vendor in the services listing **4160**. The system may be configured to determine and display any other information relevant to risks associated with the vendor.

(500) FIG. **42** depicts the exemplary interface **4210** showing vendor details. The system may be configured to generate and display the vendor details interface **4210** based on any vendor information the system has determined, including information determined based on the vendor analyses described herein. The interface **4210** may include any vendor information described herein, including the vendor information shown in the section **4240** of the interface **4210**, and vendor information such as: (1) a number of security and/or privacy officers (e.g., as shown in the section **4220** of the



interface **4210**); (2) one or more certifications, verifications, and/or awards obtained by the vendor (e.g., as shown in the section **4230** of the interface **4210**); (3) one or more vendor contacts and their respective roles at the vendor organization (e.g., as shown in the section **4250** of the interface **4210**); (4) entity personnel responsible for interacting with the vendor and their respective roles at the entity organization (e.g., as shown in the section **4260** of the interface **4210**); (5) notes regarding interactions with the vendor and related information (e.g., as shown in the section **4270** of the interface **4210**); and/or (6) any other information that may be of use in evaluating and interacting with the vendor.

(501) As described herein, a vendor may complete one or more privacy and/or security-related assessments (e.g., that may include question/answer pairings), the responses to which the system may use in calculating one or more vendor risk scores and/or determining other vendor information. FIG. **43** depicts the exemplary interface **4310** for requesting that an assessment be sent to a vendor. The system may be configured to detect the selection of a vendor from the listing of vendors **4320** and/or the selection of the assessment control **4330**. Responsive to such detection, the system may be configured to request desired assessment information, for example using the assessment information window **4340**. The assessment information window **4340** may include fields or selections that allow a user to specify a template for the assessment (e.g., as shown in the field **4341**), a name for the assessment (e.g., as shown in the field **4342**), and a recipient of the assessment, such as a particular vendor employee or representative to designated to receive such an assessment (e.g., as shown in the field **4343**).

(502) After completion of an assessment request (e.g., as described in regard to FIG. **43**), a designated vendor representative may receive an indication that a new assessment has arrived. FIG. **44** depicts the exemplary interface **4410** that may include a notification **4420** of a new assessment. Note that the system may be configured to generate such an interface in response a user requesting that such an assessment be sent because vendor information queried by the assessment has expired, as described herein. The assessment notification **4420** may include a control that allows the recipient vendor representative to initiate the assessment.

(503) At least partially in response to initiating the assessment, the system may be configured to present the exemplary interface **4510** as shown in FIG. **45** that may request information using, for example, one or more question and answer pairs (e.g., as described herein). For example, the first question and answer section **4520** may be presented to the vendor representative completing the assessment, followed by the second question and answer section **4530** that may, in some examples, not be active until the preceding question and answer section is complete. Upon completing the required one or more question and answer sections of the assessment, the vendor representative may activate the assessment submission control **4540** to submit the completed assessment to the entity requesting the assessment.

(504) In various embodiments, answers to one or more questions within a vendor assessment may be pre-populated based on known and/or previously provided information. This may be especially helpful where a subset of information acquired via an assessment has expired but the remaining information remains valid. In such embodiments, the system may be configured to generate and present an interface that includes prepopulated information, such as the exemplary interface **4610** shown in FIG. **46**. In this example, the system may generate a window including the section of prepopulated information **4620** that the vendor representative may then evaluate and update as needed.

(505) The system may be configured to detect a change in a vendor's information and responsively inquire of a user whether the vendor should be sent an updated assessment. In various embodiments, the system may be configured to substantially automatically identify a change in a sub-processor by one or more vendors. The system may, for example, be configured to monitor one or more RSS feeds to identify one or more changes to one or more sub-processors utilized by a particular vendor. In response to identifying that a vendor has changed (e.g., been added or removed) one or more sub-processors, the system may be configured to substantially automatically generate and/or transmit a privacy assessment and/or a security assessment to the vendor based at least in part on the detected change. Alternatively, the system may be configured to prompt a user to send a new assessment.

(506) FIG. **47** depicts the exemplary interface **4710** that includes the notification **4720** of a detected vendor change. The notification **4720** includes a user-selectable control that may initiate creation and/or transmission of a new vendor assessment (e.g., as described herein). Note that any detected vendor changes may initiate a new vendor assessment and/or generate a prompt to a user inquiring of the need to send a new assessment to the vendor.

(507) FIGS. **48-50** depict exemplary screen displays that a user may encounter when utilizing any suitable system described herein to determine the risk (e.g., privacy risk, security risk, etc.) that a particular vendor may present, as well as to view other attributes and information about the particular vendor. For example, these exemplary screen displays may be encountered by a user associated with an entity in evaluating a vendor to determine whether to begin or continue a relationship (e.g., business relationship) with such a vendor according to various disclosed embodiments.

(508) FIG. **48** depicts an exemplary listing of vendors **4830** in a database as represented in an exemplary user interface **4810**. The system may access a prepopulated database of vendor information and use such information to provide the listing of vendors **4830** from which a user may select a vendor. The system may also allow a user of the interface **4810** to search for a particular vendor from among those available in a database of vendor information using a search field **4820**. In some examples, the system may populate a drop-down box **4821** based at least in part on the user's input to the search field **4820**, allowing the user to select a vendor from the drop-down box **4821**. Should the user not locate the desired vendor from the listing of vendors provided by the interface **4810**, the user may select a control **4840** to add, or request to have added, a new vendor to the vendor information database. The user may then take the necessary steps to add or request to add the new vendor.

(509) Upon selection of a particular vendor on interface **4810**, the system may generate exemplary interface **4910** as depicted in FIG. **49** on a display screen. The exemplary interface **4910** may show a vendor overview for the particular vendor. The system may be configured to generate and display the vendor overview interface **4910** based at least in part on any vendor information the system has determined, including information determined based at least in part on the vendor analyses described herein. The interface **4910** may include a description of the vendor (e.g., "ABC, Inc." in FIG. **49**) in a vendor description section **4920**, which may include the vendor's name, location, description, etc.

(510) The system may be configured to determine additional information for the vendor as described herein, including based at least in part on one or more of: (1) information gathered from the vendor (e.g., assessment responses from the vendor); (2) information about the vendor gathered from public and/or shared sources (e.g., webpages, databases, etc.); documents associated with the vendor (e.g., contracts, licenses, agreements, etc.); and/or (3) and other vendor information (e.g., publicly known vendor data, historical information about the vendor, etc.). Such additional information may be displayed on interface **4910**.

(511) In various embodiments, as part of the additional vendor information, the system may calculate a vendor risk score (e.g., vendor security risk score, vendor privacy risk score, etc.) for the vendor, shown as "Vendor Score" in a vendor score section **4970** of interface **4910**. As described herein, the system may, for example, calculate the vendor risk score based at least in part on any factor or criteria described herein or any other suitable information (e.g., information transfer information, one or more contract terms, assessments previously performed for the vendor, etc.). The system may also calculate one or more other scores of any type (e.g., as one or more internal vendor-related scores based at least in part on criteria that differs from criteria used to determine one or more other vendor risk scores) and display such scores in the vendor score section **4970**.

(512) In various embodiments, as part of additional vendor information, the system may determine and/or highlight one or more vendor risks (e.g., data encryption incidents, personal information compromises, third-party breaches, etc.) and display such risks in the vendor risk section **4930**. In various embodiments, as part of the additional vendor information, the system may determine and display third-party vendors utilized by the vendor in the third-party vendor section **4940**. In various embodiments, as part of the additional vendor information, the system may determine and display one or more historical incidents associated with the vendor in the historical incident section **4950**. In various embodiments, as part of the additional vendor information, the system may determine and display a listing of services provided by the vendor in a services listing **4960**. The system may be configured to determine and display any other information relevant to one or more privacy risks associated with the vendor. The system may be configured to determine whether, based, for example, on any vendor information described herein, the particular vendor is approved or rejected for



use by, and/or interaction with, the entity requesting the assessment of the vendor's risk. Based at least in part on this determination, the system may present an approval indication or a rejection indication in an approval section **4980** of the user interface.

(513) FIG. **50** depicts an exemplary interface **5010** showing vendor details. The system may be configured to generate and display the vendor details interface **5010** in response to a selection, by a user, of a particular vendor on interface **4810** of FIG. **48**, for example, as an alternative to displaying interface **4910** of FIG. **49**, or in response to a selection, by a user, of a control on interface **4910** of FIG. **49** requesting further vendor details. In various embodiments, the system may generate interface **5010** based at least in part on any vendor information the system has determined, including information determined based at least in part on the vendor analyses described herein. The interface **5010** may include any additional detailed vendor information described herein, including the vendor information shown in the section **5040** of the interface **5010**, and vendor information such as: (1) a number of security and/or privacy officers associated with the vendor (e.g., as shown in section **5020**); (2) one or more certifications, verifications, and/or awards obtained by the vendor (e.g., as shown in section **5030**); (3) vendor employees (e.g., employees who serve as contacts with the requesting entity) and their roles at the vendor organization (e.g., as shown in section **5050**); (4) entity personnel responsible for interacting with the vendor and their roles at the entity organization (e.g., as shown in section **5060**); (5) notes regarding one or more interactions with the vendor and related information (e.g., as shown in section **5070**); and (6) any other information that may be of use in evaluating and interacting with the vendor. As noted above, in various embodiments, the system may be configured to determine whether, based at least in part on any vendor information described herein, the particular vendor is approved or rejected for use by, and/or for interaction with, the entity requesting the assessment of the vendor's privacy risk. Based at least in part on this determination, the system may present an approval indication or a rejection indication in approval section **5080**.

(514) Exemplary Vendor Training Material Generation Experience

(515) FIGS. **51-53** depict exemplary screen displays that a user may encounter when utilizing any suitable system described herein to generate and/or update training material associated with a particular vendor, as well as to view other attributes and/or information about the particular vendor. For example, these exemplary screen displays may be encountered by a user associated with an entity who may be operating the disclosed system to obtain privacy-related training material and/or security-related training material that may assist the user in understanding how to interact with a particular vendor. In another example, these exemplary screen displays may be encountered by a user associated with a vendor who may be operating the disclosed system to obtain privacy-related training material and/or security-related training material provided by an entity with which the vendor interacts.

(516) FIG. **51** depicts the exemplary listing of vendors **5130** in a database as represented in the exemplary interface **5110**. The system may access a prepopulated database of vendor information and use such information to provide the listing of vendors **5130** from which a user may select a vendor. The system may also allow a user of the interface **5110** to search for a particular vendor from among those available in a database of vendor information using the search field **5120**. In some examples, the system may populate the drop-down box **5121** based at least in part on the user's input to the search field **5120**, allowing the user to select a vendor from the drop-down box **5121**.

(517) Upon selection of a particular vendor on the interface **5110**, the system may generate the exemplary interface **5210** showing a vendor overview for the particular vendor, as depicted in FIG. **52**. The interface **5210** may include the user-selectable control **5280** that may indicate that training material has been generated for the particular vendor. The user-selectable control **5280** may allow a user to download or otherwise access (e.g., via a subsequent interface) the training material generated by the system.

(518) In various embodiments, the interface **5210** may also provide a date of generation of such training material (e.g., on or proximate to the user-selectable control **5280**). The system may also be configured to generate and/or display the vendor overview interface **5210** based at least in part on any vendor information the system has determined, including information determined based at least in part on the vendor analyses described herein. The interface **5210** may include a description of the vendor (e.g., "ABC, Inc." in FIG. **52**) in vendor description section **5220**, a "Vendor Score" in vendor score section **5270**, one or more vendor risks in vendor risk section **5230**, third-party vendors utilized by the vendor in third-party vendor section **5240**, historical incidents associated with the vendor in historical incident section **5250**, a listing of services provided by the vendor in services listing **5260**, etc.

(519) As noted herein, the system may be configured to detect a change in a vendor's information and/or an occurrence involving a vendor and responsively update training material associated with that particular vendor. For example, the system may be configured to substantially automatically identify a change in sub-processor by one or more vendors. FIG. **53** depicts the exemplary interface **5310** that includes a notification **5320** of a detected vendor change of a sub-processor. The notification **5320** includes a user-selectable control that may allow a user to download and/or otherwise access training material that has been updated based at least in part on the detected change or occurrence (e.g., as described herein). Alternatively, in response to selection of the user-selectable control, the system may generate an interface such as interface **5210** of FIG. **52**. The user may then access the updated training material using such an interface. Referring again to FIG. **52**, where the system has generated updated training material in response to some detected change or occurrence, the indication of such training material generation (e.g., control **5280**) may include a date of creation (e.g., updating) of such updated training material.

(520) Mapping of Data Breach Regulation Questions

(521) A large number of regulations govern the actions that are required to be taken in response to a data breach. The particular regulations that apply to a data breach may be defined by the jurisdiction (e.g., country, state, defined geographic area, or other suitable region, such as any defined area sharing at least one common reporting requirement related to one or more data breaches) in which the data breach occurs, the nationality of one or more potential victims (e.g., data subjects) of the data breach, and/or the business sector involved in the data breach (e.g., healthcare, finance, telecommunications, utilities, defense, cybersecurity, etc.). For example, a data breach that results in the improper disclosure of personal health information within the U.S. may trigger the disclosure provisions of the Health Insurance Portability and Accountability Act (HIPAA). Examples of security standards or regulations that may indicate how a data breach is to be managed may include International Organization for Standardization (ISO) 27000 series standards, National Institute of Standards and Technology (NIST) standards, Health Information Technology for Economic and Clinical Health (HITECH) standards, Health Insurance Portability and Accountability Act (HIPAA) standards, American Institute of Certified Public Accountants (AICPA) System and Organization Controls (SOC) standards, the EU General Data Protection Regulation (GDPR), and the California Consumer Privacy Act (CCPA). Jurisdictions may also develop and use their own sets of requirements for handling data breaches. Entities (e.g., corporations, organizations, companies, etc.) may also have their own requirements and policies regarding the management of data breaches.

(522) Therefore, a breach of personal data by a large, multinational company may trigger a need to analyze and comply with (potentially numerous) applicable privacy regulations of a potentially large number of different territories. This can pose a daunting challenge for an organization because, in currently available systems, a privacy officer would typically have to complete a data breach disclosure questionnaire for each affected territory and/or business segment. Each such questionnaire can include a large number of (e.g., 40, 50, or more) questions, making this process very time consuming when there are many different jurisdictions involved.

(523) Systems and methods according to various embodiments may store, in memory, an ontology that maps respective questions from a data breach disclosure questionnaire for a first territory and/or business sector (e.g., an initial, high-level questionnaire that is used to determine whether it is necessary to disclose a particular data breach within the first territory) to: (1) corresponding questions within one or more data breach disclosure questionnaires (e.g., similar threshold questionnaires) for other territories and/or business sectors; and/or (2) corresponding questions within a master questionnaire. For example, the health care sectors of Germany, France, and the United States may all use "The number of data subjects whose data was affected by the breach" as a factor in determining whether a particular breach must be disclosed, who the breach must be disclosed to, and/or how quickly the breach must be disclosed. In various embodiments, however, each jurisdiction may include one or more data breach disclosure

questionnaires related to the number of data subjects that are in a different language, are worded differently, are posed differently (e.g., one questionnaire may require a free-form text entry response, another may include one or more user selectable responses, etc.), etc. As may be understood in light of this disclosure, although each respective questionnaire may include one or more respective questions that have different wording or form, each question may still map back to the same specific question within a data breach master questionnaire.

(524) In an example embodiment, the master questionnaire may include the question “How many data subjects were affected by the breach?” This question may be important because various jurisdictions may have varying threshold of affected numbers of data subject that trigger reporting requirements. The system may map this question, via the ontology (which may map questions, at least in part, based on pattern matching between respective questions), to corresponding questions within the respective threshold data breach questionnaires for Germany, France, and the United States. In a particular example, in response to receiving, from a user, an answer to this question in the master questionnaire, the system may then use the answer in conjunction with the ontology to populate the answer to the corresponding questions within the questionnaires for Germany, France, and the United States. For example, if the user indicated in the answer to this question in the master questionnaire that the personal data of 150 people was affected by the breach, the system may save, in system memory, an answer corresponding to “150 people” to the particular question “How many data subjects were affected by the breach” (or similar questions that may, for example, be worded differently) in the threshold data breach questionnaires for Germany, France, and the United States.

(525) It should be understood that the ontology may vary in complexity based on the circumstances. In particular embodiments, one or more questions from a master questionnaire (e.g., 1, 2, 3, 4, 5, 10, 25, 50, etc. questions) may each be respectively mapped to one or more corresponding questions in a plurality of (e.g., any number between 1 and 500, or more) data breach questionnaires for respective territories and/or business sectors. For example, the question above regarding the number of affected data subjects may be mapped to a respective question in data breach questionnaires for 40 different jurisdictions.

(526) The system may include any number and type of questions in a master questionnaire and any data breach questionnaire for a particular territory and/or business sector. The system may use the answers to any such questions to determine the notification obligations for any particular territory. In this way, the system may determine the notification obligations for various territories that may each have varying disclosure requirements. The questions that the system may include on a master questionnaire and/or a data breach questionnaire for a particular territory may include, but are not limited to, a number of affected data subject and/or consumers, types of data elements involved in the breach, a volume of data involved in the breach, a classification of data involved in the breach, a business sector associated with the breach, questions associated with any type of regulatory trigger that may initiate a requirement for disclosure, etc.

(527) FIG. 54 illustrates an exemplary Data Structure 5400 representing a data breach ontology according to particular embodiments that may be used for determining data breach response requirements and/or gathering data breach reporting information. The Data Structure 5400 may include requirements for each territory and/or business sector regarding, for example, what types of data breaches must be disclosed (e.g., whether a particular type of data breach must be disclosed and to whom), when different types of affected breached need to be disclosed (e.g., one or more reporting deadlines), and/or how different types of data breaches need to be disclosed (e.g., what information needs to be reported, the form of reporting, etc.). The Data Structure 5400 may also facilitate the gathering of data for, and the reporting of, data breaches.

(528) The Data Breach Master Questionnaire 5410 represents data received as answers to a master questionnaire that the system provided to a user. The system may map answers to questions in the master questionnaire to corresponding answers for one or more other questionnaires. For example, the system may map one or more answers for the Master Questionnaire 5410 to one or more answers for the Data Breach Disclosure Questionnaire for Germany 5420 and/or the Data Breach Disclosure Questionnaire for France 5430, as shown in FIG. 54. The system may also, or instead, map answers to questions in any particular questionnaire to corresponding answers for any one or more other questionnaires. For example, the system may map one or more questions for the Data Breach Disclosure Questionnaire for Germany 5420 to one or more questions for the Data Breach Disclosure Questionnaire for France 5430, as shown in FIG. 54.

(529) For example, the system may map data associated with question 5410A of the Data Breach Master Questionnaire 5410, which may provide a number of data subjects affected by a data breach, to question 5420A for the Data Breach Disclosure Questionnaire for Germany 5420 and to question 5430C for the Data Breach Disclosure Questionnaire for France 5430. Also, or instead, the system may map data associated with question 5420A for the Data Breach Disclosure Questionnaire for Germany 5420 to question 5430C for the Data Breach Disclosure Questionnaire for France 5430. The system may also, or instead, map data associated with question 5410B of the Data Breach Master Questionnaire 5410, which may provide a date for the detection of a data breach, to question 5420L for the Data Breach Disclosure Questionnaire for Germany 5420, but not to a question in the Data Breach Disclosure Questionnaire for France 5430. The system may also, or instead, map data associated with question 5410Y of the Data Breach Master Questionnaire 5410 to question 5430FH for the Data Breach Disclosure Questionnaire for France 5430, but not to a question in the Data Breach Disclosure Questionnaire for Germany 5420. In various embodiments, an ontology may map any one or more questions of any questionnaire to any one or more questions in any one or more other questionnaires in the ontology, or to no question in any other questionnaire.

(530) One potential advantage of various embodiments of computer-implemented versions of this ontology is that it may allow a user to effectively complete at least a portion of a large number of data breach questionnaires by only completing a single master questionnaire. In various embodiments, the system may prompt the user to input answers to each respective question in the master questionnaire. The system would then map the answer to each of the questions to also be the answer of any corresponding questions in the data breach questionnaires of any other countries in which the entity was doing business or that were involved in a particular data breach (e.g., as determined by input from a user).

(531) In particular embodiments, the system may be configured to dynamically edit the current master questionnaire for a particular entity so that the master questionnaire includes, for example, at least one question that will provide the answer for each question within a data breach disclosure questionnaire of a plurality of territories in which the entity does business (e.g., all of the territories in which the entity does business) or that were involved in a particular data breach (e.g., all of the territories affected by the particular data breach).

(532) For example, in a particular embodiment, if a data breach disclosure questionnaire includes a question that is unique to Brazil, the master questionnaire will include that question as long as the entity's profile information indicates that the entity is doing business in Brazil or that Brazil is involved in the associated data breach. However, if a user modifies the entity's profile information to indicate that the entity no longer does business in Brazil, the system may automatically modify the master questionnaire to remove the question (since the question will no longer be applicable to the entity). Similarly, if a user even later updates the entity's profile to indicate that the entity has resumed doing business in Brazil, the system may automatically update the master questionnaire to include the Brazil-specific question (and/or questions).

(533) In various embodiments, the system may be configured to generate a master questionnaire at any appropriate time. For example, in a particular embodiment, the system may prompt a user to indicate one or more territories (e.g., regions, jurisdictions, and/or countries) and/or sectors in which an entity is doing business and, at least partially in response to receiving the user's input, generate a threshold list of questions that the system may then use to determine which territories require disclosure of a particular data breach. In another particular embodiment, the system may prompt a user to indicate one or more territories (e.g., regions, jurisdictions, and/or countries) and/or sectors affected (e.g., potentially affected) by a particular data breach and, at least partially in response to receiving the user's input, generate a threshold list of questions that the system may then use to determine which territories affected by the data breach require disclosure of the data breach.

(534) For example, in a particular embodiment, after a user identifies a particular data breach, the system may responsively execute a disclosure compliance module, such as the exemplary Disclosure Compliance Module 5500 shown in FIG. 55. In executing the Disclosure Compliance Module 5500, at Step S510, the system may prompt the user to indicate the territories (e.g., regions, jurisdictions, countries, etc.) in which the entity does

business. Alternatively, or in addition, at Step **S510**, the system may prompt the user to indicate the territories that may be affected by the particular data breach. In various embodiments, the system may ask the user to select territories from a listing of territories. Alternatively, or in addition, the system may prompt the user to indicate the applicable territories using any suitable technique. Further at Step **S510**, the system may receive input from the user indicating the applicable territories. In particular embodiments, the system may facilitate such prompting for territories and receipt of indications of applicable territories by using graphical user interfaces.

(535) Next, at Step **S520**, the system may prompt the user to indicate the business sectors (e.g., healthcare, finance, etc.) in which the entity is doing business. Alternatively, or in addition, at Step **S510**, the system may prompt the user to indicate the business sectors that may be affected by the particular data breach. In various embodiments, the system may ask the user to select business sectors from a listing of business sectors. Alternatively, or in addition, the system may prompt the user to indicate the applicable business sectors using any suitable technique. Further at Step **S520**, the system may receive input from the user indicating the applicable business sectors. In particular embodiments, the system may facilitate such prompting for business sectors and receipt of indications of applicable business sectors by using one or more graphical user interfaces.

(536) In response to the user-indicated applicable territories and/or business, at Step **S530** the system may generate a master questionnaire of threshold questions for the applicable territories and business sectors, e.g., as described above. At Step **S540**, the system may present the master questionnaire to the user and prompt the user for input indicating answers to the threshold questions in the master questionnaire. Further at Step **S540**, the system may receive input from the user indicating answers to the threshold questions in the master questionnaire. The system may prompt the user to indicate the answers to the threshold questions using any suitable techniques. In particular embodiments, the system may facilitate such prompting for answers to the threshold questions and receipt of indications of answers to the threshold questions by using graphical user interfaces.

(537) At Step **S550**, the system may use the ontology to map the user's answers to the threshold questions in the master questionnaire back to the threshold questionnaires for each particular applicable territory and/or business sector. At Step **S560**, the system may determine based on the information mapped from the master questionnaire answers to the threshold questionnaires for each particular applicable territory and/or business sector, whether, under the applicable laws of each particular applicable territory and/or within the particular applicable business sector, the entity must disclose the data breach (e.g., in addition to the matter of any required disclosure, timing of any required disclosure, etc.). In various embodiments, the system may be configured to determine a respective disclosure requirement for each of one or more territories and/or one or more business sectors in which a particular entity operates. In particular embodiments, the system is configured to simultaneously determine, for at least two or more jurisdictions in which the entity operates, a respective disclosure requirement for each of the at least two or more jurisdictions (e.g., the system is configured to determine the respective disclosure requirements for each of the at least two or more jurisdictions in parallel). The system may, for example, utilize one or more parallel processing techniques.

(538) If so, at Step **S570**, the system generates one or more disclosure questionnaires, each of which may reflect questions from a breach notification template for a particular territory and/or business sector, for completion by the user. Alternatively, the system may generate one or more disclosure questionnaires that may each include a consolidated master list of disclosure questions that are respectively mapped (e.g., using the ontology) to any one or more corresponding questions in one or more respective disclosure questionnaires (e.g., breach notification templates) for each of the territories in which the entity is required to disclose the breach (e.g., as determined by the system). Alternatively, or in addition, the system may facilitate the user completing a breach notification template for each territory individually. At Step **S580**, the system may present the one or more disclosure questionnaires to the user and prompt the user for input indicating answers to the questions in each disclosure questionnaire. Further at Step **S580**, the system may receive input from the user indicating answers to the questions in each disclosure questionnaire. The system may prompt the user to indicate the answers to questions in each disclosure questionnaire using any suitable techniques. In particular embodiments, the system may facilitate such prompting for answers to the questions in each disclosure questionnaire and receipt of indications of answers to the questions in each disclosure questionnaire by using graphical user interfaces. The system may then use the answers to the questions in each disclosure questionnaire to generate the applicable disclosure document(s) for each territory.

(539) At Step **S590**, after receiving the user's answers to the questions in each disclosure questionnaire, the system may use the input received from the user (e.g., when completing the master questionnaire and/or when providing answers to the questions in each disclosure questionnaire) to automatically generate a suitable disclosure document disclosing the breach for each territory in which disclosure of the breach is required. The system may then access, from system memory, information regarding how to properly submit the required disclosure document to each territory and display that information to the user. This information may include, for example, a mailing address or email address to which the disclosure document must be submitted, the entity or person to which the disclosure document should be sent, etc. In a particular embodiment, the system may be adapted to auto-submit one or more of the disclosure documents to the entity or person to which the disclosure document should be sent (e.g., via a suitable electronic or paper transmission of the document).

(540) In various embodiments, the system may be adapted to present questions for a particular jurisdiction in the order in which they are presented on the jurisdiction's disclosure form. This may make it easier for the individual to prepare and finalize the disclosure form. In particular embodiments, the system may be further adapted to, based on a user's answers to one or more of the master list of disclosure questions, automatically promote an incident to a breach status.

(541) In various embodiments, the system may be configured to present the results of the disclosure determination using a graphical user interface. FIG. 56 depicts an exemplary interface **5600** showing the results of a disclosure determination as described herein (e.g., by the Disclosure Compliance Module **5500**). The system may indicate on interface **5600** the territories for which the system has determined that disclosure is required. The system may also indicate on such an interface the territories for which the system has determined that disclosure is not required. The interface **5600** may include a graphical representation of one or more territories, such as map **5610**. The system may color code, shade, or otherwise visually indicate which of the territories shown in the map **5610** require notification of a data breach and which do not. The system may also color code, shade, or may otherwise visually indicate which of the territories shown in the map **5610** are not territories in which the entity is conducting business (and therefore were not included in the disclosure analysis performed by the system). The system may generate a legend **5620** in the interface **5600** to illustrate to the user the meaning of the color coding, shading, visual indications, etc. used on the map **5610** to illustrate the disclosure status of each territory and/or whether each territory was included in the disclosure analysis.

(542) The interface **5600** may also include details of the disclosure requirements determined by a data breach disclosure determination as described herein. For example, the system may present disclosure requirements listing **5630** on the interface **5600** listing data breach notification requirements for the various jurisdictions in which disclosure is required. The interface **5600** may also include details of each particular disclosure requirement for a territory in which disclosure is required. For example, the system may present disclosure requirement subtasks listing **5640** on the interface **5600** listing particular subtasks associated with a particular data breach notification requirement for a particular territory in which disclosure is required, such as the territory highlighted in the disclosure requirements listing **5630**.

(543) The system may also present further detailed information regarding the disclosure requirements for a particular territory for which the system has determined that disclosure of the data breach is required. FIG. 57 depicts an exemplary interface **5700** showing detailed results of a disclosure determination as described herein (e.g., by the Disclosure Compliance Module **5500**) for a particular territory. The interface **5700** may include a graphical representation of one or more territories, such as map **5710**. Upon selection of one of these territories, the system may highlight the selected territory, for example, the selected territory **5715** on the interface **5700**. The system may then, in response to user selection of the selected territory **5715**, generate detailed information regarding the selected territory **5715** in the detailed information section **5720**. The detailed information section **5720** may include detailed information regarding the reporting requirements for the selected territory **5715**, such as the particular laws or regulation that require disclosure, the regulating body, contact information for the regulators, etc.

(544) As in FIG. 56, the interface 5700 of FIG. 57 also includes details of the disclosure requirements determined by a data breach disclosure determination as described herein, such as disclosure requirements listing 5730 listing data breach notification requirements for the various jurisdictions in which disclosure is required and disclosure requirement subtasks listing 5740 on listing particular subtasks associated with a particular data breach notification requirement for the selected territory 5715.

(545) In any embodiment described herein, they system may be configured to at least partially automatically determine and populate one or more responses to one or more questions in the master questionnaire (e.g., prior to mapping the one or more responses to a corresponding questionnaire for a particular jurisdiction and/or business unit). The system may, for example, use one or more data mapping techniques (such as any data mapping technique described herein), for example, to determine particular data subjects involved, particular data assets involved, a location of those data assets, a type of data elements involved in the data breach, a volume of data subjects affected by the data breach, a classification of data involved in the breach, and/or any other suitable data related to the breach that may be relevant to one or more reporting and/or disclosure requirements. The system may, in various embodiments, at least partially automatically populate one or more responses to a master questionnaire and: (1) optionally prompt a user to confirm the automatically populated responses; and (2) prompt a user to provide any additional responses that the system did not automatically populate. In a particular example, in response to a data breach involving a payroll processing database utilized by an entity, the system may be configured to access a data model for the entity to determine, for example: (1) a number of employees whose personal data (e.g., name, mailing address, banking information, etc.) may have been affected by the breach; (2) a type of data potentially exposed by the breach (e.g., routing numbers, names, social security numbers, etc.); (3) a number of other entity data assets that may have been affected (e.g., by virtue of interfacing with the payroll processing database, sending or receiving data to the databased, etc.); and/or (4) any other data related to the payroll processing database that may be relevant to determine what disclosure requirements may need to be met by the entity in response to the data breach. The system may then use the determined data to at least partially automatically populate one or more master questionnaires (e.g., one or more responses in the one or more master questionnaires) for use in one or more breach disclosure assessments.

(546) Assessing Entity and/or Vendor Compliance with Privacy Standards

(547) Systems and methods according to various embodiments may store, in memory, an ontology that maps respective controls that are required for compliance with a first privacy standard (e.g., HIPAA, NIST, HITECH, GDPR, CCPA, etc.) to: (1) corresponding controls required for compliance with one or more other privacy standards; and/or (2) respective corresponding questions within a master questionnaire. For example, each of the HIPAA, NIST, and HITECH privacy standards may all require multi-factor authentication of employees before allowing the employees to access sensitive data. Accordingly, the ontology may map, to each other, respective controls listed in the HIPAA, NIST and HITECH privacy standards that each involve multi-factor authentication of employees.

(548) The ontology may also, or alternatively, map each of the respective controls listed in a privacy standard or required by a privacy regulation (e.g., HIPAA, NIST, HITECH, GDPR, CCPA, etc.) to a question in a master list of questions that is used to determine compliance with the one or more privacy standards and/or regulations. For example, the master questionnaire may include a question regarding the use of multi-factor authentication of employees that maps to a requirement of one or more privacy standards. Such a question may be, for example, “Does your organization require multi-factor authentication of employees before they access sensitive data?”. In a particular example, in response to receiving the answer to this question in the master questionnaire from a user, the system may use the answer in conjunction with the ontology to populate the answer to the corresponding questions within particular questionnaires that are used to assess an entity's level of compliance with a plurality of privacy standards and/or regulations, where each particular questionnaire is specific to a particular privacy standard or regulation (e.g., HIPAA, NIST, HITECH, CSA, GDPR, CCPA, etc.). For example, if the user indicated in the answer to this question in the master questionnaire that the user's organization does require multi-factor authentication of employees before they access sensitive data, the system may save, in system memory using the ontology, an answer corresponding to “Yes” to that particular question (or similar questions that may, for example, be worded differently) in the particular privacy standard compliance questionnaires for HIPAA, NIST, and HITECH.

(549) It should be understood that the ontology may vary in complexity based on the circumstances. In particular embodiments, one or more questions from the master list a master questionnaire (e.g., 1, 2, 3, 4, 5, 10, 25, 50, etc. questions) may each be respectively mapped to one or more corresponding questions in a plurality of (e.g., any number between 1 and 500, or more) respective compliance questionnaires for other privacy standards. For example, the question above regarding multi-factor authentication may be mapped to a respective question in compliance questionnaires for 20 different privacy standards.

(550) The system may include any number and type of questions in a master questionnaire and any compliance questionnaire for a particular privacy regulation and/or privacy standard. The system may use the answers to any such questions to determine whether and to what extent an entity and/or a vendor complies with a particular privacy regulation and/or privacy standard. In this way, the system may determine vendor and/or entity compliance with various privacy regulations and/or privacy standards that may each have varying requirements. The questions that the system may include on a master questionnaire and/or a compliance questionnaire for a particular privacy regulation and/or privacy standard may include, but are not limited to, controls on access to sensitive data, controls on modification and storage of sensitive data, required employee certifications, required security controls on devices/websites/systems, and any other questions associated with any type of control or requirement needed to comply with any privacy standard or privacy regulation.

(551) FIG. 58 illustrates an exemplary Data Structure 5800 representing a compliance ontology according to particular embodiments that may be used for determining particular privacy standard/regulation compliance and/or gathering privacy standard/regulation compliance information. The Data Structure 5800 may include requirements for each particular privacy standard and regulation, for example, what types of controls must be in place, what types of security measures are required, employee requirements (e.g., training, certifications, background checks, etc.), physical requirements, software requirements, etc. The Data Structure 5800 may also facilitate the gathering of data for, and the determination of, compliance with any one or more privacy standards and privacy regulations.

(552) The Compliance Master Questionnaire 5810 represents data received as answers to a master questionnaire that the system provided to a user. The system may map answers to questions in the master questionnaire to corresponding answers for one or more other questionnaires. For example, the system may map one or more answers for the Master Questionnaire 5810 to one or more answers for the Privacy Standard Compliance Questionnaire for HIPAA 5820 and/or the Privacy Standard Compliance Questionnaire for NIST 5830, as shown in FIG. 58. The system may also, or instead, map answers to questions in any particular questionnaire to corresponding answers for any one or more other questionnaires. For example, the system may map one or more questions for the Privacy Standard Compliance Questionnaire for HIPAA 5820 to one or more questions for the Privacy Standard Compliance Questionnaire for NIST 5830, as shown in FIG. 58.

(553) For example, the system may map data associated with question 5810A of the Compliance Master Questionnaire 5810, which may indicate whether multi-factor authentication is required, to question 5820A for the Privacy Standard Compliance Questionnaire for HIPAA 5820 and to question 5830C for the Privacy Standard Compliance Questionnaire for NIST 5830. Also, or instead, the system may map data associated with question 5820A for the Privacy Standard Compliance Questionnaire for HIPAA 5820 to question 5830C for the Privacy Standard Compliance Questionnaire for NIST 5830. The system may also, or instead, map data associated with question 5810B of the Compliance Master Questionnaire 5810, which may provide an indication as to whether a particular certification is required for employees, to question 5820L for the Privacy Standard Compliance Questionnaire for HIPAA 5820, but not to a question in the Privacy Standard Compliance Questionnaire for NIST 5830. The system may also, or instead, map data associated with question 5810Y of the Compliance Master Questionnaire 5810 to question 5830FH for the Privacy Standard Compliance Questionnaire for NIST 5830, but not to a question in the Privacy Standard Compliance Questionnaire for HIPAA 5820. In various embodiments, an ontology may map any one or more questions of any questionnaire to any one or more questions in any one or more other

questionnaires in the ontology, or to no question in any other questionnaire.

(554) One potential advantage of various embodiments of computer implemented versions of this ontology is that it may allow a user to effectively complete at least a portion of a large number of privacy standard and/or regulation compliance questionnaires by only completing a single, master questionnaire. In various embodiments, the system may prompt the user to input answers to each respective question in the master questionnaire. The system would then, using the ontology, map the answer to each of the questions to also be the answer of any corresponding questions in the respective compliance questionnaires for any suitable privacy standards.

(555) In particular embodiments, the system may be configured to dynamically edit the current master questionnaire for a particular entity or vendor so that the master questionnaire includes, for example, at least one question that will provide the answer for each question within a privacy standard compliance questionnaire of a plurality of data standards. For example, if a privacy standard compliance questionnaire includes a question that is unique to HIPAA, the master questionnaire will include that question if a user indicates that they would like to assess an entity's compliance with HIPAA. However, if a user indicates that the entity (or the user) no longer wishes to assess the entity's compliance with HIPAA, the system may automatically modify the master questionnaire to remove the question (since the question will no longer be applicable to the entity). Similarly, if a user later updates the entity's profile to indicate that the entity (or user) again wishes to evaluate the entity's compliance with HIPAA, the system may automatically update the master questionnaire to include the HIPAA-specific question.

(556) In various embodiments, the system may be configured to generate the master questionnaire at any appropriate time. For example, in a particular embodiment, the system may prompt the user to indicate the privacy standards and/or regulations that the user would like to have an entity or vendor evaluated for compliance with before generating a master list of questions that the system then uses to determine the extent to which the entity or vendor complies with the indicated privacy standards.

(557) After a user provides answers to the questions in a master list, the system may use the ontology to map the user's answers to the questions back to the compliance questionnaires for each specified privacy standard and regulation to determine the extent to which the entity or vendor complies with each respective privacy standard and regulation. In various embodiments, the results of this determination may be selectively communicated to the user in any suitable way. For example, the system may generate and present to the user a report showing the degree to which (e.g., in percentages) an entity complies with each specified privacy standard and regulation.

(558) In particular embodiments, the system may be adapted to not re-present questions that the system already has answers for. In such embodiments, the system may only present, to the user, compliance questions for selected privacy standards that the system doesn't already have an analogous answer for (e.g., based on an earlier-answered question from a master list of questions and/or an earlier-answered question from a compliance question for another privacy standard or regulation.)

(559) In particular embodiments, the system may be adapted to automatically determine that a particular entity complies, fully or partially (e.g., in regard to consent) with one or more particular standards (e.g., the HITECH standard) based on the entity's compliance with one or more other standards and/or the answers to various questions within a master questionnaire.

(560) In various embodiments, the questions presented to a user (e.g., as part of a master questionnaire) may be answered based on different types of information that may be associated with different levels of confidence. For example, each particular question may be answered with: (1) unsubstantiated data provided by the entity or vendor; (2) data that is substantiated via a remote interview; or (3) data that is substantiated by an on-site audit. In particular embodiments, the system is adapted to store an indication of the confidence level of the answer to each compliance question in memory (e.g., along with answer data associated with the question in a master questionnaire and/or a compliance questionnaire for a particular standard or regulation) and to selectively provide this information to a user (e.g., in the form of a report). In this way, the system may provide the user with an indication of the confidence level that the entity actually complies with the standard. For example, the system may generate an aggregate confidence score for an entity's compliance with a particular privacy standard based on the individual confidence levels associated with each answer to each question in the compliance questionnaire for that particular privacy standard.

(561) In particular embodiments, the entity being assessed in the manner described above may be a vendor. The system may be adapted to allow the vendor to allow other entities to access the vendor's compliance data (e.g., as described herein) and to use such data to independently assess whether the vendor complies with any of a plurality of privacy standards and/or regulations. For example, if a particular potential customer of a vendor wishes to determine whether the vendor complies with the GDPR, the system may execute a privacy standard compliance module, such as those described herein, to assess whether the vendor complies with the GDPR. If the system doesn't have answers to all of the questions within a GDPR compliance assessment questionnaire, the system may prompt the user to provide answers to those questions as discussed above. The system may then optionally save the provided answers for later use by the vendor, or other potential customers of the vendor.

(562) A potential advantage of various such embodiments is that they may allow a vendor to complete a single master questionnaire (e.g., a master Privacy Impact Assessment) that may be used by the vendor and/or a plurality of the vendor's customers to assess the vendor's current compliance with various applicable privacy standards and/or regulations. This may alleviate the need for the vendor to provide this data to multiple parties individually. Another advantage is that such embodiments may allow an entity, such a vendor, to use a single privacy impact assessment questionnaire when assessing each of the entity's business processes.

(563) In various embodiments, the system may execute a privacy standard and/or privacy regulation compliance module, such as the exemplary Privacy Standard Compliance Module **5900** shown in FIG. **59**. In particular embodiments, the system may execute the Privacy Standard Compliance Module **5900** in response to user input requesting the evaluation of an entity's (e.g., company, organization, vendor, etc.) compliance with one or more privacy standards and/or privacy regulations. In executing the Privacy Standard Compliance Module **5900**, at Step **S910**, the system may prompt the user to indicate one or more particular privacy standards and/or regulations. In various embodiments, the system may ask the user to select one or more standards and/or regulations from a listing of standards and/or regulations. Alternatively, or in addition, the system may prompt the user to indicate the applicable standards/regulations using any suitable means. Further at Step **S910**, the system may receive input from the user indicating the applicable standards/regulations. In particular embodiments, the system may facilitate such prompting for standards and/or regulations and receipt of indications of applicable standards and/or regulations by using graphical user interfaces.

(564) At Step **S920**, in response to receiving the specified standards and/or regulations, the system may generate or otherwise obtain a particular compliance questionnaire for each specified standard or regulation. At Step **S930**, the system may generate a master questionnaire of compliance questions based on the specified standards and/or regulations. In various embodiments, the system may generate the ontology mapping questions in each particular compliance questionnaire to questions in the master questionnaire and/or to questions in other particular compliance questionnaires at Step **S930**. In particular embodiments, for example as described above, the system may generate a master questionnaire that includes every question from each particular compliance questionnaire for each specified standard or regulation, while eliminating questions that represent substantially duplicative data. For example, the system may use pattern matching, machine learning techniques, or any other means to determine which questions from a particular privacy standard compliance questionnaire are the same or similar to another question in another privacy standard compliance questionnaire and include just one such question in the master questionnaire, reducing the total number of questions presented to the user.

(565) Further at Step **S930**, questions in the master questionnaire may be customized in any suitable manner. For example, questions may be presented in natural language form to solicit the corresponding information for respective privacy standard compliance questionnaires. Questions may also be presented in a language appropriate for a particular vendor or user, translated from another language used in one or more of the privacy standard compliance questionnaires if need be. The system may use machine learning, machine translation, neural networking, and/or any other suitable means of preparing and mapping questions in a master questionnaire so that the responsive data provided by a user can be used in one or more privacy standard and/or privacy regulation compliance questionnaires.

(566) At Step **S940**, the system may present the master questionnaires to the user and prompt the user to input answers to the compliance questions in the master questionnaire. Further at Step **S940**, the system may receive input from the user indicating answers to the compliance questions in the master questionnaire. Also at Step **S940**, the system may determine a confidence level for each question, for example, based on the form of substantiation for the respective question as described above. The system may prompt the user to indicate the answers to the compliance questions using any suitable means. In particular embodiments, the system may facilitate such prompting for answers to the compliance questions and receipt of indications of answers to the compliance questions by using graphical user interfaces.

(567) At Step **S950**, the system may use the ontology to map the user's answers to the compliance questions in the master questionnaire back to the compliance questionnaires for each particular privacy standard or privacy regulation. At Step **S960**, the system may determine, based on the information mapped from the master questionnaire answers to the compliance questionnaires for each particular privacy standard or privacy regulation, whether and/or to what extent the entity is in compliance with the particular privacy standard or privacy regulation. At Step **S970**, the system may determine a confidence score for each particular privacy standard or privacy regulation compliance determination, for example, based on the confidence level for each question in the compliance questionnaire for that particular privacy standard or privacy regulation as described above. At Step **S980**, the system may present the results of the compliance determinations to the user. In various embodiments, these determinations may be presented on a graphical user interface or in a report of any form. The system may also, or instead, present the results of any compliance determination and/or associated confidence determination using any suitable means.

(568) Assessing Entity and/or Vendor Readiness to Comply with Privacy Regulations

(569) Systems and methods according to various embodiments may store, in memory, an ontology that maps respective data privacy requirements for a particular jurisdiction or set of regulations (e.g., GDPR, CCPA, French privacy regulations, German privacy regulations, etc.) to: (1) corresponding data privacy requirements required for compliance with one or more other particular jurisdictions or sets of regulations; and/or (2) respective corresponding questions within a master questionnaire. For example, the GDPR and the CCPA regulations may each require a particular privacy policy to be in compliance with the respective set of regulations. Accordingly, the ontology may map, to each other, corresponding privacy policies listed in the GDPR and the CCPA regulations. By gathering answers to questions in a single master questionnaire, the system can map the answers to data privacy requirements required for compliance with the regulations in various jurisdictions and/or regions and assess the readiness of an entity to be in compliance with the regulations for such jurisdictions and/or regions.

(570) In various embodiments, an ontology generated and/or stored by the system may also, or instead, include respective requirements for sectoral laws (e.g., laws related or applicable to particular business sectors, such as health, finance, etc., in some instances, in a particular jurisdiction) to: (1) corresponding requirements required for compliance in another particular business sector (e.g., in a particular jurisdiction); (2) corresponding data privacy requirements required for compliance with one or more other particular jurisdictions or sets of regulations; and/or (3) respective corresponding questions within a master questionnaire. For example, the healthcare information regulations (e.g., HIPAA) in a particular jurisdiction may require a particular privacy policy to be in compliance. Accordingly, the ontology may map, to each other, corresponding healthcare information regulations. By gathering answers to questions in a single master questionnaire, the system can map the answers to sectoral requirements required for compliance with sectoral regulations (e.g., healthcare information regulations, financial information regulations, etc.) for various jurisdictions and/or regions and assess the readiness of an entity to be in compliance with the sectoral requirements for such jurisdictions and/or regions.

(571) The ontology may map each of the respective controls listed in a set of regulations for a particular region or territory (e.g., GDPR, CCPA, etc.) to a question in a master list of questions that is used to assess the entity's compliance with the set of regulations for that particular region or territory. For example, the master questionnaire may include a question regarding the use of a particular privacy data control or the implementation of a particular privacy policy. The system may map this question in the ontology to a requirement of one or more privacy regulations for particular jurisdictions and/or regions. Examples of such a question may include "Does your organization require multi-factor authentication of employees before they access sensitive data?" and "Do you prominently display a link to your privacy policy on your homepage?". In a particular example, in response to receiving the answer to this question in the master questionnaire from a user, the system may use the answer in conjunction with the ontology to populate the data associated with corresponding requirements within particular questionnaires that are used to assess an entity's readiness to comply with a plurality of privacy regulations for particular jurisdictions and/or regions, where each particular questionnaire is specific to a particular set of privacy regulations for a particular jurisdiction and/or region (e.g., GDPR, CCPA, etc.). For example, if the user indicated in the answer to this question in the master questionnaire that the user's organization does not prominently display a link to its privacy policy on its homepage, the system may save, in a computer memory using the ontology, an answer corresponding to "entity does not prominently display link to privacy policy on homepage" to that particular requirement (or similar requirements that may, for example, be worded differently) as represented in a questionnaire for the particular privacy regulations for a particular region.

(572) It should be understood that the ontology may vary in complexity based on the circumstances. In particular embodiments, one or more questions from a master questionnaire (e.g., 1, 2, 3, 4, 5, 10, 25, 50, etc. questions) may each be respectively mapped to one or more corresponding questions in a plurality of (e.g., any number between 1 and 500, or more) respective questionnaires for particular sets of regulations for particular regions or territories. For example, the question above regarding displaying a link to a privacy policy on a homepage may be mapped to a respective question in questionnaires for 20 different sets of regulations, each associated with a different territory or region.

(573) The system may include any number and type of questions in a master questionnaire and any readiness questionnaire for a particular set of privacy regulations for any particular territory or region. The system may use the answers to any such questions to determine whether and to what extent an entity (or a vendor) is ready to comply with a particular set of privacy regulations for any particular territory or region. Note that any of the particular sets of privacy regulations for any particular territory or region described herein may be currently in force or may be prospective (e.g., planned but not yet in force). In this way, the system may determine entity readiness for compliance with various sets of privacy regulations that may each have varying requirements and may each be currently in force or anticipated to be implemented in the future. The questions that the system may include on a master questionnaire and/or a readiness questionnaire for a particular territory or region may include, but are not limited to, controls on access to sensitive data, controls on modification and storage of sensitive data, required disclosures, required security controls on devices/websites/systems, require policies, required contact information, require consent modifications, and any other questions associated with any type of control or requirement needed to comply with any set of regulations for any territory, jurisdiction, or region.

(574) FIG. 60 illustrates an exemplary Data Structure **6000** representing a global readiness assessment ontology according to particular embodiments that may be used for determining an entity's readiness to comply with one or more particular sets of privacy regulations compliance and/or for gathering regulatory compliance information. The Data Structure **6000** may include requirements for each particular set of regulations for a particular territory or region (and/or for particular sectors in a particular territory or region), for example, what types of controls must be in place, what types of policies are required, physical requirements, software requirements, data handling requirements, etc. The Data Structure **6000** may also facilitate the gathering of data for, and the determination of, compliance (or readiness to comply) with any one or more sets of privacy regulations.

(575) The Global Readiness Master Questionnaire **6010** represents data received as answers to a master questionnaire that the system provided to a user. The system may map answers to questions in the master questionnaire to corresponding answers for one or more other questionnaires. For example, the system may map one or more answers for the Master Questionnaire **6010** to one or more answers for the GDPR Readiness Questionnaire **6020** and/or the CCPA Readiness Questionnaire **6030**, as shown in FIG. 60. The system may also, or instead, map answers to questions in any particular questionnaire to corresponding answers for any one or more other questionnaires. For example, the system may map one or more questions for the GDPR Readiness Questionnaire **6020** to one or more questions for the CCPA Readiness Questionnaire **6030**, as shown in FIG. 60.

(576) For example, the system may map data associated with question **6010A** of the Global Readiness Master Questionnaire **6010**, which may indicate whether a link to a privacy policy is prominently displayed on the entity's homepage, to question **6020A** for the GDPR Readiness Questionnaire **6020** and to question **6030C** for the CCPA Readiness Questionnaire **6030**. Also, or instead, the system may map data associated with question **6020A** for the GDPR Readiness Questionnaire **6020** to question **6030C** for the CCPA Readiness Questionnaire **6030**. The system may also, or instead, map data associated with question **6010B** of the Global Readiness Master Questionnaire **6010**, which may provide an indication as to whether a link is provided to allow a data subject to request a consent modification, to question **6020L** for the GDPR Readiness Questionnaire **6020**, but not to a question in the CCPA Readiness Questionnaire **6030**. The system may also, or instead, map data associated with question **6010Y** of the Global Readiness Master Questionnaire **6010** to question **6030FH** for the CCPA Readiness Questionnaire **6030**, but not to a question in the GDPR Readiness Questionnaire **6020**. In various embodiments, an ontology may map any one or more questions of any questionnaire to any one or more questions in any one or more other questionnaires, or to no question in any other questionnaire.

(577) One potential advantage of various embodiments of computer implemented versions of this ontology is that it may allow a user to effectively complete at least a portion of a large number of regulatory readiness questionnaires by only completing a single, master questionnaire. In various embodiments, the system may prompt the user to input answers to each respective question in the master questionnaire. The system may then, using the ontology, map the answer to each of the questions to also be the answer of any corresponding questions in the respective regulatory readiness questionnaires for any suitable set of regulations.

(578) In particular embodiments, the system may be configured to dynamically generate and/or edit the current master questionnaire so that the master questionnaire includes, for example, at least one question that will provide the answer for each question within each readiness questionnaire of a plurality of readiness questionnaires for a plurality of respective sets of regulations (e.g., jurisdictional, sectoral, etc.). For example, if a readiness questionnaire for the GDPR includes a question that is unique to the GDPR (e.g., among the possible or available sets of regulations for which readiness may be assessed), the master questionnaire will include that question if a user indicates that they would like to assess the entity's compliance with the GDPR. However, if a user indicates that the entity (or the user) no longer wishes to assess the entity's readiness to comply with the GDPR, the system may automatically modify the master questionnaire to remove the question (since the question will no longer be applicable to any relevant set of regulations). Similarly, if a user later updates the entity's profile to indicate that the entity (or user) again wishes to evaluate the entity's readiness to comply with the GDPR, the system may automatically update the master questionnaire to include the GDPR-specific question.

(579) In various embodiments, the system may be configured to generate the global readiness master questionnaire at any appropriate time. For example, in a particular embodiment, the system may prompt the user to indicate the regions and territories for which the user would like to have the entity evaluated for readiness to comply with the applicable privacy regulations. In response to receiving this information from the user, the system may generate a master list of questions that the system then uses to assess the readiness of the entity to comply with the applicable privacy regulations.

(580) After a user provides answers to the questions in a master list, the system may use the ontology to map the user's answers to the questions back to the readiness questionnaires for each specified set of regulations for each particular region/territory to determine the extent to which the entity is ready to comply with each respective set of regulations. In various embodiments, the results of this assessment may be selectively communicated to the user in any suitable way. For example, the system may generate and present to the user a report showing the degree of readiness (e.g., in percentages) the entity has to comply with each specified set of privacy regulations.

(581) In particular embodiments, the system may be adapted to not re-present questions that the system already has answers for. In such embodiments, the system may only present, to the user, readiness questions for selected sets of privacy regulations that the system doesn't already have analogous data for (e.g., based on an earlier-answered question from a master list of questions and/or an earlier-answered question from a readiness questionnaire for another set of privacy regulations or an earlier completed readiness questionnaire for this particular set of privacy regulations.)

(582) In particular embodiments, the system may be adapted to automatically determine to what extent the entity is ready to comply with one or more particular sets of privacy regulations for one or more particular regions or territories (e.g., GDPR, CCPA, etc.), and/or for particular sectors in one or more particular regions or territories, based on data provided for the entity in response to various questions within a readiness questionnaire associated with one or more other sets of privacy regulations and/or in response to various questions within a master questionnaire.

(583) In particular embodiments, the entity being assessed in the manner described above may be a vendor. The system may be adapted to allow the vendor to allow other entities to access the vendor's readiness assessment data (e.g., as described herein) and to use such data to independently determine the readiness of the vendor to comply with any of a plurality of set of privacy regulations. For example, if a particular potential customer of a vendor wishes to determine whether the vendor complies with the GDPR, the system may execute a readiness assessment module, such as those described herein, to assess the extent to which the vendor is prepared to comply with the GDPR. If the system doesn't have answers to all of the questions within a GDPR readiness assessment questionnaire, the system may prompt the user to provide answers to those questions as discussed herein. The system may then optionally save the provided answers for later use by the vendor or other potential customers of the vendor in future readiness assessments.

(584) A potential advantage of various such embodiments is that they may allow a vendor to complete a single master questionnaire (e.g., a master global readiness questionnaire) that may be used by the vendor and/or a plurality of the vendor's customers to assess the vendor's readiness to comply with various sets of privacy regulations. This may alleviate the need for the vendor to provide this data to multiple parties individually. Another advantage is that such embodiments may allow an entity, such a vendor, to use a single master questionnaire when assessing its readiness to comply with multiple sets of privacy regulations.

(585) In various embodiments, the system may execute a global readiness assessment module, such as the exemplary Global Readiness Assessment Module **6100** shown in FIG. **61**. In particular embodiments, the system may execute the Global Readiness Assessment Module **6100** in response to user input requesting the evaluation of an entity's (e.g., company, organization, vendor, etc.) readiness to comply with one or more particular sets of privacy regulations for one or more regions or territories and/or with one or more particular sets of privacy regulations for one or more particular sectors in one or more particular regions or territories. In executing the Global Readiness Assessment Module **6100**, at Step **6110**, the system may prompt the user to indicate one or more particular regions, territories, and/or sectors, for example, in which the entity conducts business or has customers. In various embodiments, the system may ask the user to select one or more regions and/or territories from a map of regions and/or territories or from a listing of regions, territories, and/or sectors. Alternatively, or in addition, the system may prompt the user to indicate the applicable regions, territories, and/or sectors using any suitable means. Further at Step **6110**, the system may receive input from the user indicating the applicable regions, territories, and/or sectors. In particular embodiments, the system may facilitate such prompting for regions, territories, and/or sectors and receipt of indications of applicable regions, territories, and/or sectors using one or more graphical user interfaces.

(586) In various embodiments, the system may allow a user to specify or select the particular sets of regulations rather than, or in addition to, selecting regions, territories, and/or sectors. At Step **6120**, the system may prompt the user to indicate one or more particular sets of regulations (e.g., GDPR, CCPA, etc.), for example, governing the entity's conduct in various regions, territories, and/or sectors. In various embodiments, the system may ask the user to select one or more sets of regulations using a map indicating the regions and/or territories where such sets of regulations are in force or from a listing of sets of regulations. Alternatively, or in addition, the system may prompt the user to indicate the applicable sets of regulations using any suitable means. Further at Step **6120**, the system may receive input from the user indicating the applicable sets of regulations. In particular embodiments, the system may facilitate such prompting for sets of regulations and receipt of indications of applicable sets of regulations using one or more graphical user interfaces.



(587) At Step **6130**, the system may generate a master questionnaire of global readiness questions based on the specified regions, territories, sectors, and/or sets of regulations. In various embodiments, the system may generate the ontology mapping questions in each particular compliance questionnaire to questions in the master questionnaire and/or to questions in other particular compliance questionnaires at Step **6130**. In particular embodiments, for example as described above, the system may generate a master questionnaire that includes every question from each particular readiness questionnaire for each specified set of regulations, while eliminating questions that represent substantially duplicative data. For example, the system may use pattern matching, machine learning techniques, or any other means to determine which questions from a particular readiness questionnaire for a particular set of regulations are the same or similar to another question in another readiness questionnaire for a different particular set of regulations and include just one such question in the global readiness master questionnaire, reducing the total number of questions presented to the user.

(588) Further at Step **6130**, questions in the global readiness master questionnaire may be customized in any suitable manner. For example, questions may be presented in natural language form to solicit the corresponding information for respective readiness questionnaires. Questions may also be presented in a language appropriate for a particular user, translated from another language used in one or more of the readiness questionnaire if need be. The system may use machine learning, machine translation, neural networking, and/or any other suitable means of preparing and mapping questions in a master questionnaire so that the responsive data provided by a user can be used in one or more readiness questionnaires.

(589) At Step **6140**, the system may present the global readiness master questionnaire to the user and prompt the user for input indicating answers to the compliance readiness questions in the master questionnaire. Further at Step **6140**, the system may receive input from the user indicating answers to the questions in the global readiness master questionnaire. The system may prompt the user to indicate the answers to the compliance readiness questions using any suitable means. In particular embodiments, the system may facilitate such prompting for answers to the compliance readiness questions and receipt of indications of answers to the compliance readiness questions using one or more graphical user interfaces.

(590) At Step **6150**, the system may use the ontology to map the user's answers to the compliance readiness questions in the master questionnaire back to the readiness questionnaires for each particular set of privacy regulations. At Step **6160**, the system may determine, based on the information mapped from the master questionnaire answers to the readiness questionnaires for each particular set of privacy regulations, whether and/or to what extent the entity is prepared to comply with each particular set of privacy regulations. In particular embodiments, the system may determine a percentage of readiness to comply with a particular set of privacy regulations based on the percentage of answers to questions in a respective questionnaire for that particular set of privacy regulations that indicate compliance. For example, if the user's answers to 25% of the questions in a questionnaire for a particular set of regulations indicate that the entity complies with the respective requirements represented by those questions, the system may determine that the entity is at 25% readiness to comply with that particular set of regulations. Alternatively, or in addition, the system may employ an algorithm or other means of calculating a readiness level or score (e.g., weighting particular questions) that may be represented in any suitable manner (e.g., percentage, raw score, relative score, etc.). The system may use any other suitable means of determining an extent of the entity's readiness to comply with the regulations associated with any particular region or territory.

(591) At Step **6170**, the system may present the results of the compliance readiness determination to the user. In various embodiments, these results may be presented on a graphical user interface or in a report of any form. The system may also, or instead, present the results of any readiness determination using any suitable means.

(592) In various embodiments, the system may be configured to solicit input regarding territories, regions, sectors, and/or sets of regulations for which readiness is to be assessed and/or to present the results of such readiness assessments using a graphical user interface. FIG. **62** depicts an exemplary interface **6200** showing a map **6210** of regions and territories that allows a user to select one or more territories for a global readiness assessment (e.g., by the Global Readiness Assessment Module **6100**). The system may indicate on interface **6200** the territories selected and the associated regulation for a selected territory. For example, territory **6215** may be highlighted or otherwise emphasized as a selected territory, and the system may, in response to selecting the territory **6215**, present a summary **6220** of the privacy regulations that are applicable to the territory **6215**. The system may color code, shade, or otherwise visually indicate which of the territories shown in the map **6210** are associated with which regulations. The system may also present a listing of regulations **6230** that may be applicable to one or more territories shown in map **6210**. By detecting a user selection of any of the regions or territories shown in the map **6210** and/or the listing of regulations **6230**, the system may responsively add the selected regions and territories to a listing of regions and territories that the system will evaluate for compliance readiness.

(593) FIG. **63** depicts an exemplary interface **6300** showing a listing of privacy regulations **6320**. This listing may represent the regulations implicated when a user selected one or more regions or territories, such as on interface **6200** of FIG. **62**. The listing of privacy regulations **6320** may also, or instead, allow the user to select additional sets of regulations for which the entity's readiness is to be evaluated and/or may allow the user to deselect sets of regulations, thereby removing such regulations from those for which the entity's readiness is to be evaluated. The listing of privacy regulations **6320** may be filtered or sorted based on regions and territories, for example using the region listing **6310**.

(594) As selection of one of the sets of regulations presented in the listing of privacy regulations **6320** may generate another interface (e.g., a pop-up window) providing further details regarding that set of privacy regulations, such as interface **6400** shown in FIG. **64**. The interface **6400** may include a user-interactive listing of the various requirements of the selected set of regulations, allowing a user to view the details of complying with that particular set of regulations.

(595) FIG. **65** depicts an exemplary interface **6500** showing the results of compliance readiness assessments. The interface **6500** may include a map **6510** that may indicate the regions, territories, and/or sectors for which the entity's readiness was evaluated. The system may generate a listing of the results of the readiness analysis **6520** for each applicable set of regulations. Each entry in the listing **6520** may include specific results for the respective set of regulations. For example, the entry **6522** may indicate that the entity is 79% ready to comply with the EU-U.S. PrivacyShield regulations, while the entry **6524** may indicate that the entity is 68% ready to comply with the GDPR. Each such entry may also provide options that a user may select to view more details about the results and/or the associated set of regulations. As noted above, the system may provide the results of a compliance readiness assessment in any suitable form.

(596) Generation of an Intelligent Data Breach Response Plan

(597) Because of the large number of regulations that must be followed across various jurisdictions in order to remain in compliance such regulations and to properly respond in the event of a data breach or other incident, it can be very difficult for an entity to develop proper response and compliance plans. In some instances, various requirements and regulations (e.g., jurisdictional, sectoral, standards-based, etc.) may be in conflict with one another, making the planning and response process even more complex. In particular embodiments, the system may be configured to automatically develop a plan for responding to a particular data breach or other incident based upon various criteria that take into account requirements and regulations for various regions, territories, and/or sectors. The system may, for example, use one or more of the follow criteria in developing a response plan for a data breach: (1) the respective disclosure requirements of each regions, territories, and/or sectors (e.g., whether and how quickly the region/territory/sector requires disclosure of the data breach); (2) how frequently each region, territory, and/or sector enforces its data breach disclosure requirements; (3) any penalty (e.g., applicable fine) for not properly satisfying the disclosure requirements of each region, territory, and/or sector; (4) how important each region, territory, and/or sector is to the entity's business (e.g., how much business the entity does in the region, territory, and/or sector); and/or (5) any other suitable factor. Such a plan may be particularly helpful in situations where there are conflicts (e.g., irreconcilable conflicts) between the laws or regulations regarding how and when a particular breach must be disclosed. For example, where there are conflicts between the regulations of two or more regions, territories, and/or sectors, the system may be configured to determine the particular region, territory, or sector for which violation of a regulation is less (or more) impactful and develop a response plan based on that determination.



(598) In various embodiments, the system may generate and/or store one or more ontologies in a suitable data structure, for example as described herein. In exemplary embodiments, such a data structure (or any data structure configured to organize the data disclosed herein) may include, for example, the requirements of each territory and/or business sector, such as the types of data breaches need to be disclosed in a particular territory, when and how different types of data breaches need to be disclosed in a particular territory, etc. In particular embodiments, the data structure may also include information regarding, for each particular region, territory, and/or sector, one or more of: (1) how often the regulations (e.g., breach-related regulations) of the particular region, territory, or sector are enforced; (2) the fine(s) for not disclosing a breach as required by the particular region, territory, or sector; (3) how other privacy officers within the entity (or other, similar entities) typically handle data breaches within the particular region, territory, or sector (e.g., do they routinely comply with a territory's applicable breach disclosure requirements?); and (4) other applicable information that may be useful in developing a decision as to how to best handle a privacy breach that impacts one or more of the regions, territories, and/or sectors in which the entity conducts business.

(599) In various embodiments, the system may enable a user to execute a regulatory disclosure compliance module that prompts the user to input, in addition to the information described above, information regarding the importance of each particular region, territory, or sector to the entity's business and any other business information that may be helpful in prioritizing efforts in responding to the disclosure requirements of multiple different regions, territories, and/or sectors.

(600) After receiving this information, the system may then use any suitable algorithm to create an ordered list of regions, territories, and/or sectors in which the entity needs to disclose the breach. Particular territories may be listed, for example, in order of the urgency with which the disclosure must be filed in the respective territories (e.g., based on how soon from the current date the disclosure must be filed in each territory and/or the importance of the territory to the entity's business). In particular embodiments, the system may, for example, generate a disclosure urgency score for each territory and order the list based on the determined respective disclosure urgency scores for each of the countries.

(601) In various embodiments, the system may communicate this information via a heat map display of a plurality of territories, where the heat map visually indicates (e.g., by displaying the territories in different respective colors) which territories require the most immediate disclosure. In other embodiments, the system may present to a user a listing of affected regions, territories, and/or sectors ordered by their relative urgency. In various embodiments, the system is configured to display detailed information regarding a particular region's, territory's, or sector's disclosure requirements in response to a user selecting the territory on the heat map or from a listing of affected regions, territories, and/or sectors.

(602) In addition, or instead, the system may be configured to generate a list of recommended steps (e.g., an ordered checklist of steps) that the user (or entity) should complete to satisfy data breach reporting requirements and recommendations according to the system's logic. The system may present questions to a user soliciting information required to satisfy each step and may automatically generate reporting communications that may be required by the affected jurisdictions and/or sectors. This may be advantageous because it may allow a user to satisfy multiple different jurisdictions' and/or sectors' respective disclosure obligations, for example, by providing answers to a single questionnaire (e.g., as described herein in regard to the Data Structure 5400). This may further be advantageous because it may allow a user to satisfy multiple different jurisdictions' (or different business sectors') respective disclosure obligations according to a particular protocol that takes into account internal conflict-of-laws logic by completing each step in the list in the specified order.

(603) It should be understood, based on the discussion above, that a list of compliance or disclosure steps may omit one or more steps that are necessary to comply with the regulations of one or more territories regarding the data breach. For example, the system may have determined that, since the penalty for non-compliance in a particular territory is below a particular monetary threshold, and since the company needs to allocate resources to disclosing the data breach to many other territories that have relatively high monetary fines for non-disclosure, it is recommended not to comply, in the particular instance, with the disclosure regulations of the particular territory.

(604) It should also be understood that the list of steps may be in any suitable order. For example, steps for complying with a particular jurisdiction's disclosure laws may be listed in consecutive order or intermixed with one or more steps for steps for complying with the disclosure laws of one or more other jurisdictions. This may be useful, for example, in situations where a particular jurisdiction requires the disclosure requirement to be completed in two stages, with a first stage to be completed before the due date of a particular action that is due in another jurisdiction, and a second stage to be completed after the due date of that particular action.

(605) Also, in various embodiments, the system may allow a user to modify the list of action items (e.g., by deleting certain action items, adding additional action items, or by reordering the list of action items so that, for example, at least one of the actions is performed sooner than it would have been in the original ordered list. In particular embodiments, such manual modifications of the original list may be used by one or more machine learning modules within the system to adjust the logic used to present future lists of action items for the entity or for other entities.

(606) In various embodiments, the system may automate one or more of the steps described herein, for example, as part of a workflow. The system may automatically route one or more of the tasks generated to particular recipients for completion as part of such a workflow. Upon determining the particular type of breach or incident and details relating thereto, the system may automatically generate or select a suitable workflow that may include such tasks. The system may also use a determined workflow as a template and integrate details of required tasks based on specific information related to the particular breach or incident. In particular embodiments, the system may automatically route any of the subtasks and/or any items in any of the checklists described herein to one or more suitable recipients based on the parameters or details of the associated incident and or the type of incident.

(607) FIG. 66 depicts a Disclosure Prioritization Module 6600 according to a particular embodiment, which may be executed, for example, on any of the servers, devices, or computing devices described herein, or on any combination thereof. The Disclosure Prioritization Module 6600 may also generate, modify, otherwise interoperate with one or more ontologies as described herein. Note that the steps that the Disclosure Prioritization Module 6600 may perform are described here in an exemplary order. The Disclosure Prioritization Module 6600 according to various embodiments may perform any subset of these steps in any order and/or in conjunction with any one or more other functions and activities.

(608) When executing the Disclosure Prioritization Module 6600, the system may begin, at Step 6610, by generating and presenting an interface to a user prompting the user to provide data breach information. This interface may take any form capable of presenting and collecting information from a user. In a particular embodiment, the system may generate a data breach information interface as a GUI presented on one or more computer display devices. The Disclosure Prioritization Module 6600 may use the data breach information interface to solicit any useful information about the data breach. For example, the data breach information interface may ask the user to provide an incident name, type of data involved (e.g., personal data, particular type of personal data, etc.), an amount of data involved, a number of data subjects affected, a date on which the breach was discovered (and, in some examples, a time of discovery), the jurisdictions affected, the method used to detect the data breach (e.g., manually, automatically), a name of user reporting breach, a sector affected by the breach, and/or any other information that may be of use in generating a data breach response plan. The data breach information interface may request information regarding the importance of each affected territory to the entity's business and/or any other business information that may be helpful in prioritizing efforts in responding to the disclosure requirements of multiple different territories. Further at Step 6610, the Disclosure Prioritization Module 6600 may receive the data breach information from the user via the interface.

(609) At Step 6620, according to various embodiments, the system may store the received data breach information in a data structure that may incorporate an ontology for future use. For example, after determining the affected jurisdictions, the Disclosure Prioritization Module 6600 may generate an ontology (e.g., similar to that described in regard to the Data Structure 5400) that maps respective requirements and recommendations for compliance with a first privacy law, regulation, standard, and/or policy in a first jurisdiction to corresponding requirements and recommendations for compliance with one or more other privacy laws, regulations, standards and/or policies. The ontology generated by the Disclosure Prioritization Module 6600 may also, or alternatively, map each of the requirements and recommendations for compliance with each privacy law, regulation,

standard, and/or policy in each affected jurisdiction (and, in particular embodiments, sector) to a question in a master list of questions in a master questionnaire that may be used to request information to address such requirements and recommendations (e.g., as described above). The Disclosure Prioritization Module **6600** may store the answers received at Step **6610** as answers to a master questionnaire and subsequently map those answers to the respective requirements and recommendations for compliance with for each affected jurisdiction.

(610) At Step **6630**, the Disclosure Prioritization Module **6600** may begin generating a plan for responding to the breach by first determining the data breach disclosure requirements, if any, for each applicable jurisdiction and/or sector. The Disclosure Prioritization Module **6600** may also, at step **6630**, determine the consequences, if any, of failures to address these requirements. The Disclosure Prioritization Module **6600** may also, at step **6630**, determine one or more recommended (e.g., but not required) actions associated with responding to the data breach in each particular jurisdiction or sector. For example, for a breach of the type indicated by the information provided by the user for each affected jurisdiction, the Disclosure Prioritization Module **6600** may determine whether disclosing the breach is required, any deadlines associated with disclosing the breach, any penalties associated with a failure to timely disclose the breach, the form of notification required in disclosing the breach, one or more recommended internal notifications (e.g., notify the entity's legal department, notify one or more particular privacy officers, etc.), and/or any other information that may be specified as required or recommended for a territory or region for data breach reporting. Such information may be obtained from one or more data structures, including one or more data structures having, or associated with, one or more ontologies as described herein.

(611) At Step **6640**, the Disclosure Prioritization Module **6600** may continue generating a plan for responding to the breach by determining one or more enforcement characteristics for each affected jurisdiction and/or sector. For example, for a breach of the type indicated by the user, the Disclosure Prioritization Module **6600** may determine, for each affected jurisdiction and/or sector, how often regulations associated with that type of breach are enforced, how often fines are imposed for not disclosing a such a breach as required, the potential liability to data subjects and/or consumers for such a breach, how other privacy officers within this and/or one or more other entities typically handle similar data breaches, and/or any other applicable information that may be useful in developing a data breach response plan. Here again, such information may be obtained from one or more data structures, including one or more data structures having, or associated with, one or more ontologies as described herein.

(612) At Step **6650**, the Disclosure Prioritization Module **6600** may determine or assign a score or grade to each region, territory, and/or sector implicated in the data breach based on the information available. For example, the Disclosure Prioritization Module **6600** may assign one or more points or a score for each of several attributes for each jurisdiction and/or sector. Such attributes may include a business importance of a jurisdiction and/or sector, a penalty associated with not satisfying requirements for a jurisdiction and/or sector, a difficulty of satisfying requirements for a jurisdiction and/or sector, the temporal proximity of a deadline for satisfying requirements for a jurisdiction and/or sector, an availability of a cure period, and/or any other criteria or attributes that may be associated with a region, territory, and/or sector and its respective data breach response requirements. The Disclosure Prioritization Module **6600** may determine a sum of such points associated with respective attributes for a particular jurisdiction and/or sector, in some embodiments applying a weight to one or more particular attributes, as a total score for that jurisdiction or sector. The Disclosure Prioritization Module **6600** may instead, or in conjunction, use other any other algorithm or method to determine a score or other indicator of the importance of each jurisdiction and/or sector relative to the other affected jurisdictions and/or sectors at Step **6650**.

(613) At Step **6660**, the Disclosure Prioritization Module **6600** may rank the affected jurisdictions and/or sectors based on the scoring determined for each jurisdiction and/or sector at Step **6650**. The system may generate this ranking based solely on scores or grades assigned to each affected jurisdiction/sector or may use a combination of factors that may or may not include such scoring. In particular embodiments, at Step **6660**, the Disclosure Prioritization Module **6600** may determine that one or more jurisdictions and/or sectors have a score, grade, or other associated attribute(s) that indicates that the one or more jurisdictions and/or sectors should not be included in a representation of affected jurisdictions at all. For example, the Disclosure Prioritization Module **6600** may determine that, because the penalty for non-compliance in a particular territory is below a particular monetary threshold, a penalty score for that jurisdiction may be very low, zero, or even negative (e.g., to reduce the importance of an otherwise important territory due to the very low penalty for non-compliance). The Disclosure Prioritization Module **6600** may also, or instead, weight a penalty score for each jurisdiction and/or sector so that any very low or zero penalty removes the jurisdiction from a list of affected jurisdictions and/or sectors requiring a data breach report (e.g., by using a penalty score as a multiplier such that a score for the jurisdiction or sector will be zero when other scores for the jurisdiction or sector are multiplied by the penalty score). This may allow an entity to allocate its limited resources to disclosing the data breach to other territories and/or sectors that may have relatively higher monetary fines for non-disclosure by not complying in a particular jurisdiction or sector where the penalty for non-compliance is relatively inconsequential.

(614) At Step **6670**, the Disclosure Prioritization Module **6600** may generate a data representation of the requirements for each jurisdiction and/or sector and/or the ranking of the affected jurisdictions and/or sectors. Note that, at Step **6670**, the Disclosure Prioritization Module **6600** may not present all such data in a single data representation. The Disclosure Prioritization Module **6600** may generate a ranked list, a heat map, or other visual representation indicating all, or a subset, of the affected jurisdictions and/or sectors. The system may allow a user to manipulate an indicator of each jurisdiction in such a representation and may, in response to detecting such manipulation, present the requirements and/or recommendations for that jurisdiction and/or sector. For example, a user may click or tap on a country represented in a heat map and the system may, in response, generate another visual representation that shows the data breach response requirements and/or recommendations for that country. Such requirements and/or recommendations may be presented in an interactive list format that allows a user to provide data indicating whether each item in such a list has been performed or to otherwise provide data and input associated with the item (e.g., a checklist).

(615) The Disclosure Prioritization Module **6600** may present scores, rankings, data breach response requirements, and/or any other data in any of various formats. For example, the Disclosure Prioritization Module **6600** may generate visual interface presented on one or more computer monitors or display devices indicating scores, rankings, data breach response requirements, and/or any other data. In addition, or instead, the Disclosure Prioritization Module **6600** may generate one or more printed reports indicating scores, rankings, data breach response requirements, and/or any other data. In addition, or instead, the Disclosure Prioritization Module **6600** may generate one or more audible indications of scores, rankings, data breach response requirements, and/or any other data. The Disclosure Prioritization Module **6600** may generate and/or provide any other form of report or provision of scores, rankings, data breach response requirements, and/or any other data, and any combinations thereof.

(616) FIG. **67** depicts a Data Breach Reporting Module **6700** according to a particular embodiment, which may be executed, for example, on any of the servers, devices, or computing devices described herein, or on any combination thereof. The Data Breach Reporting Module **6700** may also generate, modify, otherwise interoperate with one or more ontologies as described herein. Note that the steps that the Data Breach Reporting Module **6700** may perform are described here in an exemplary order. The Data Breach Reporting Module **6700** according to various embodiments may perform any subset of these steps in any order and/or in conjunction with any one or more other functions and activities.

(617) When executing the Data Breach Reporting Module **6700**, the system may begin, at Step **6710**, by determining one or more jurisdictions affected by a data breach. The Data Breach Reporting Module **6700** may determine such one or more jurisdictions using a data map, questionnaire, received user input (e.g., as described herein), or any other source of information. At Step **6720**, the Data Breach Reporting Module **6700** may determine one or more business sectors affected by the data breach. The Data Breach Reporting Module **6700** may determine such one or more business sectors using a data map, questionnaire, received user input (e.g., as described herein), or any other source of information. The affected business sector may be important because a jurisdiction may have different reporting requirements for data breaches in different business sectors.

(618) At Step **6730**, the Data Breach Reporting Module **6700** may determine whether the data breach should be reported in each of the one or more affected jurisdictions and business sectors. For example, the system may determine, at Step **6730**, whether to include each particular jurisdiction in an ontology used to generate a master questionnaire soliciting information for reporting the data breach. In particular embodiments, the Data Breach Reporting Module **6700** may determine that the entity should not allocate limited resources to disclosing the data breach in a relatively

(e.g., based on applicable penalties for not reporting the breach) jurisdiction. For example, using one or more particular embodiments described herein, the system may determine that, for a particular territory, the penalty for non-compliance is below a particular monetary threshold (e.g., based on a penalty score assigned to that jurisdiction of zero or negative as described above). In response, the Data Breach Reporting Module **6700** may determine, at Step **6730**, to not report the data breach in that particular jurisdiction. In this way, the system may avoid requesting user responses to questions in a disclosure or master questionnaire that are specific to that jurisdiction, thereby saving valuable user and entity resources. (619) In various embodiments, the Data Breach Reporting Module **6700** may receive or obtain a listing of jurisdictions in which reporting should be performed from a module such as the Disclosure Compliance Module **5500** or the Disclosure Prioritization Module **6600**, either of which may have taken into account the relative importance of each jurisdiction and may therefore have already removed one or more affected jurisdictions based on its analysis of their consequence to the entity.

(620) At Step **6740**, the Data Breach Reporting Module **6700** may determine the particular data breach reporting requirements and recommendations, if any, for each applicable jurisdiction. For example, the Data Breach Reporting Module **6700** may determine that a letter to a regulatory agency that includes a number of affected data subjects and date of discovery of the data breach must be generated for a particular jurisdiction. The Data Breach Reporting Module **6700** may also, or instead, determine that an internal report to the entity's privacy officer that includes the amount of personal data compromised and name of the user handling the data breach is recommended to be prepared. The Data Breach Reporting Module **6700** may also, or instead, determine that a notification of the data breach must be sent to affected data subjects or consumers.

(621) Based on the data breach reporting requirements and recommendations, at Step **6750**, the Data Breach Reporting Module **6700** may generate an ontology that maps respective requirements and recommendations for compliance with the regulations in a first jurisdiction to corresponding requirements and recommendations for compliance in one or more other jurisdictions. The Data Breach Reporting Module **6700** may also, or instead, generate an ontology at Step **6750** that maps each of the requirements and recommendations for compliance with a particular regulation in a particular jurisdiction to a question in a master list of questions in a master questionnaire that may be used to request information needed to satisfy disclosure requirements in several jurisdictions.

(622) Once a master questionnaire is generated, at Step **6760**, the Data Breach Reporting Module **6700** may present the questionnaire to a user prompting the user to answer questions with information needed to properly disclose the data breach. For example, the Data Breach Reporting Module **6700** may generate an interactive graphical user interface on a computer display device that allows a user to view the questionnaire and submit data, information, and/or documentation as answers to questions in the questionnaire. In response to receiving data, information, and/or documentation for a question in the master questionnaire at Step **6760**, the Data Breach Reporting Module **6700** may use the data, information, and/or documentation and the ontology to populate the data, information, and/or documentation of a corresponding question associated with a jurisdiction and required for compliance with the particular applicable regulations in that jurisdiction. In this way, the Data Breach Reporting Module **6700** may gather the required information for a reporting a data breach in several jurisdictions according to their applicable laws, and regulations using a single master questionnaire rather than a different questionnaire per jurisdiction. For example, the Data Breach Reporting Module **6700** may prompt the user to input answers (e.g., number of data subject affected, date of breach discovery, amount of personal data compromised, etc.) to each respective question in the master questionnaire. The Data Breach Reporting Module **6700** may then map the answer to each of these questions to the respective answer of any corresponding questions in the questionnaires for any jurisdiction as appropriate.

(623) At Step **6770**, using the data collected and organized using an ontology at Step **6760**, the Data Breach Reporting Module **6700** may generate the communications (e.g., a regulatory report or a report to a regulatory body) required for data breach reporting for a particular jurisdiction. The Data Breach Reporting Module **6700** may format, and/or transmit such reports based on the requirements of the particular jurisdiction for which the report is generated. These communications may be presented to a user for approval or further modification before transmission to a regulatory agency or may be transmitted (e.g., automatically) to a regulatory agency.

(624) FIG. **68** depicts a Regulatory Conflict Resolution Module **6800** according to a particular embodiment, which may be executed, for example, on any of the servers, devices, or computing devices described herein, or on any combination thereof. The Regulatory Conflict Resolution Module **6800** may also generate, modify, otherwise interoperate with one or more ontologies as described herein. Note that the steps that the Regulatory Conflict Resolution Module **6800** may perform are described here in an exemplary order. The Regulatory Conflict Resolution Module **6800** according to various embodiments may perform any subset of these steps in any order and/or in conjunction with any one or more other functions and activities.

(625) When executing the Regulatory Conflict Resolution Module **6800**, the system may begin, at Step **6810**, by determining, receiving, or otherwise obtaining requirements (e.g., regulations, standards, laws, other requirements, etc.) for multiple jurisdictions (e.g., territories, regions, etc.) and/or sectors. For example, the Regulatory Conflict Resolution Module **6800** may determine such one or more requirements using a data map, questionnaire, received user input (e.g., as described herein), or any other source of information (e.g., as part of collecting data breach requirements; as part of determining compliance for a particular jurisdiction or standard, etc.) At Step **6820**, the Regulatory Conflict Resolution Module **6800** may determine a requirement for a first jurisdiction and/or sector conflicts with a similar requirement in a second jurisdiction and/or sector. For example, the Regulatory Conflict Resolution Module **6800** may determine that a first territory requires that the entity stores collected personal data for no longer than 90 days while a second territory requires that the entity stores collected personal data for at least 90 days. In another example, the Regulatory Conflict Resolution Module **6800** may determine that a first sector in a particular territory requires that the entity report a data breach in a first time and manner that is incompatible with the data breach time and manner reporting requirements for a second sector in that particular territory. The system may detect any type of conflict and number of conflicts between regulations, requirements, etc. of any set of regulations or standards.

(626) At Step **6830**, the Regulatory Conflict Resolution Module **6800** may determine a risk of non-compliance with each of the regulations that is in conflict with another regulations. For example, the system may determine that failure to delete collected personal data after 90 days in a first territory that requires it incurs only a small yearly monetary fine if such a failure is detected in an audit that is rarely performed. The system may further determine that failure to retain collected personal data beyond 90 days in a second territory that requires it incurs an immediate suspension of the entity's business license and a large monetary fine if such a failure is detected in routinely performed monthly audits. In this example, the system may determine that the risk in the first territory is much less than the risk in the second territory.

(627) In particular embodiments, the system may also, or instead, take into account the business risk involved in non-compliance of conflicting requirements. For example, the system may determine that the risk of non-compliance is much lower in jurisdictions and/or sectors where the entity has few customers (e.g., below a threshold number of customers, such as 10, 50, 100, etc.) and/or much higher in jurisdictions and/or sectors where the entity has many customers (e.g., above a threshold number of customers, such as 100,000, 1,000,000 etc.). In particular embodiments, the system may use a scoring method to determine risk that takes into account several attributes or factors, each of which may be weighted based on various criteria. For example, at Step **6830**, the Regulatory Conflict Resolution Module **6800** may use the scores generated by the Disclosure Prioritization Module **6600** to determine, at least in part, the risk of non-compliance with conflicting data breach reporting requirements. The system may use any other methods and algorithms to determine risk, including those dedicated to such risk determination. The system may also use any criteria for determining risk, including, but not limited to, a risk of audit, a past history in a particular jurisdiction and/or sector, a history of how an entity has addressed similar conflicts in the past, how similar entities have addressed similar conflicts, a volume of data processed in a particular jurisdiction and/or sector, types of services offered in a particular jurisdiction and/or sector, business goals in a particular jurisdiction and/or sector, etc.

(628) At Step **6840**, the Regulatory Conflict Resolution Module **6800** may determine a particular recommended course of action based on the risk determinations of Step **6830**. For example, the Regulatory Conflict Resolution Module **6800** may compare the risks of non-compliance determined at Step **6830** and determine to recommend complying with the least risky requirement. Alternatively, the system may determine to report the conflict and seek user input regarding the course of action to be taken.

(629) At Step **6850**, the Regulatory Conflict Resolution Module **6800** may provide the recommended course of action to a user, for example, via a graphical user interface. Alternatively, the Regulatory Conflict Resolution Module **6800** may proceed with the course of action automatically, for example, if configured to do so. Such courses of action may include any activity or function described herein, including those relating to complying with data breach disclosure requirements or requirements for compliance with any regulation, requirements, rules, standards, etc.

(630) The disclosed systems may generate GUIs that may facilitate implementation of the disclosed subject matter, examples of which will now be described in greater detail. FIG. **69** illustrates an exemplary interface **6900**. A system may generate the interface **6900** on a computing device and may present the interface **6900** on a display device. In some embodiments, the system may generate the interface **6900** as a webpage presented within a web browser. The system may generate the interface **6900** in response to detecting the activation of a control indicating that a data breach has been discovered.

(631) The interface **6900** may include data entry area **6910** that allow a user to input details about the data breach. The interface **6900** may allow the entry, in data entry area **6910**, of any data breach information described herein, and any other data breach information. For example, the interface **6900** may allow the entry of a number of data subjects affected, a volume or quantity of data compromised, a type of personal data compromised, a data breach discovery date and/or time, a data breach occurrence date and/or time, a data breach reporting date and/or time, a name of the data breach discovering user or organization, a method of receiving a report of the data breach, a description of the data breach, one or more business sectors affected by the data breach, and/or a name of the particular data breach. The interface **6900** may also allow submission of one or more affected jurisdictions, but in other embodiments jurisdictions may be provided at a different interface, such as interface **7000** of FIG. **70**.

(632) FIG. **70** illustrates an exemplary interface **7000**. A system may generate the interface **7000** on a computing device and may present the interface **7000** on a display device. In some embodiments, the system may generate the interface **7000** as a webpage presented within a web browser. The system may generate the interface **7000** in response to detecting the activation of a control indicating that a data breach has been discovered or in response to detecting an indication that information has been received from an earlier presented interface, such as the interface **6900** of FIG. **69**.

(633) The interface **7000** may include a data entry area **7010** that allow a user to input details about one or more jurisdictions and/or sectors affected by the data breach. The interface **7000** may allow a user to indicate one or more affected jurisdictions, in the data entry area **7010**, by selection of jurisdictions from a map that may include all or a subset of the jurisdictions in which the entity conducts business. In another example, the interface **7000** may allow a user to indicate one or more affected jurisdictions and/or sectors by selecting jurisdictions and/or sectors from a list of jurisdictions and/or sectors in which the entity conducts business. In another example, the interface **7000** may allow a user to indicate one or more affected jurisdictions and/or sectors by entry of the jurisdictions and/or sectors into a text box. In various other embodiments, any method of collecting affected jurisdiction and/or sector information may be used.

(634) As described herein, once jurisdiction, sector, and/or other data breach information has been collected, the system may determine data breach disclosure and reporting requirement for each affected jurisdiction and/or sector (e.g., as performed by the Disclosure Compliance Module **5500**, the Disclosure Prioritization Module **6600**, the Data Breach Reporting Module **6700**, and/or in any other suitable manner). The system may also determine a score or urgency value for each affected jurisdiction and may rank the affected jurisdictions and/or sectors, in some embodiments, removing those for which there are no consequential penalties for failing to report the data breach. In particular embodiments, the system may also, or instead, remove particular jurisdictions and/or sectors from a ranking for which a regulatory conflict analysis has determined that those particular jurisdictions and/or sectors have a lower risk of non-compliance than others that may be left in the ranking. In various embodiments, the system may present affected jurisdictions in a heat map, with various colors and/or textures used to indicate the relative urgency of data breach reporting for each jurisdiction. In other embodiments, the system may generate a listing in order of urgency of the affected jurisdictions and/or sectors. In still other embodiments, other methods may be used to present the affected jurisdictions and/or sectors and their respective data breach reporting urgency.

(635) Also as described herein, the system may generate an interactive list of items that should be addressed in the event of a data breach. For example, the system may generate a listing of actions required by the laws, regulations, standards, and/or policies associated with a respective jurisdiction and/or sector. The listing may include inputs that allow a user to “check off” items as they are completed, or to otherwise provide information related to that item. Any such listing may be ordered based on the urgency, ranking, or other priority as described herein. For example, the system may place items required to be completed sooner and/or subject to a higher non-compliance penalty than other items earlier in a list, for example, based on a score assigned to each item and/or to its respective jurisdiction or sector. In another example, the system may place items that do not have an associated cure period earlier in a list, for example, based on a score assigned to each item and/or to its respective jurisdiction or sector.

(636) In the example shown in FIG. **71**, the system may generate an exemplary interface **7100** that may include a heat map **7110**. The heat map **7110** may indicate various jurisdictions, at least a subset of which may include one or more jurisdictions affected by the data breach. The system may color code and/or generate texture for each affected jurisdiction as shown in the heat map **7110**. The interface **7100** may include legend **7120** that may indicate the values or descriptions of the urgency associated with each color shown in the heat map **7110**. The system may also, or instead, use coloring and/or texture to indicate the affected business sector in each affected jurisdiction.

(637) The interface **7100** may also include one or more listings of tasks to be performed and/or recommended next steps, each of which may be presented in order of importance or urgency. For example, the listing **7130** may provide a list of steps that are recommended and/or required to be performed in response to a data breach. The listing **7130** may include items that are generally required and/or applicable to more than one affected jurisdiction and/or sectors (e.g., instead of items associated with only one jurisdiction). The listing **7130** may include items ordered by urgency, which the system may have determined based on a score or other value assigned to each item. The system may provide a check box for each of the items in the listing **7130**. Upon completion of an item, a user may select the check box for that item. In various embodiments, the system may remove that item from the listing **7130** and/or make a record of item completion and no longer present that item to a user as part of a list of incomplete data breach response activities. The system may also provide a mechanism allowing the assignment of each item in the listing **7130** to a particular user or to an organization. Upon assignment to a particular user or organization, the system may remove that item from the listing **7130** and/or make a record of item completion and no longer present that item to a user as part of a list of incomplete data breach response activities. Alternatively, the system may leave any assigned items on the listing **7130** until the assigned user or organization provides an indication or confirmation that the item has been completed.

(638) Each of the items in the listing **7130** may have one or more associated tasks to be performed. For example, for the highlighted first item in the listing **7130**, the system may generate a listing of tasks associated with the item may be provided in the subtask listing **7140**. The subtask listing **7140** may include tasks ordered by urgency, which, as for items in the listing **7130**, the system may have determined based on a score or other value assigned to each task. The system may provide a check box for each of the tasks in the subtask listing **7140**. Upon completion of a task, a user may select the check box for that task. In various embodiments, the system may remove that task from the subtask listing **7140** and/or make a record of task completion and no longer present that task to a user as part of a list of incomplete data breach response activities. The system may also provide a mechanism allowing the assignment of each task in the subtask listing **7140** to a particular user or to an organization. Upon assignment to a particular user or organization, the system may remove that task from the subtask listing **7140** and/or make a record of task completion and no longer present that task to a user as part of a list of incomplete data breach response activities. Alternatively, the system may leave any assigned tasks on the subtask listing **7140** until the assigned user or organization provides an indication or confirmation that the task has been completed.

(639) As described herein, the system may be configured to display detailed information regarding a particular jurisdiction's disclosure requirements in response to a user selecting the jurisdiction on a heat map or from a listing of affected jurisdictions. In the example shown in FIG. **72**, the system may generate an exemplary interface **7200** that may include a heat map **7210**. The heat map **7210** may indicate various jurisdictions (e.g., geographical territories, regions), at least a subset of which may include one or more jurisdictions affected by the data breach. The system may color

code and/or add texture to each affected jurisdiction as shown in the heat map **7210**. Upon selection of an affected jurisdiction of the United Kingdom in the particular example of FIG. 72), the interface **7200** may generate data breach response details **7220** that may provide details about the recommended and/or required data breach response actions for the selected jurisdiction.

(640) The interface **7200** may also include listings of tasks to be performed and/or recommended next steps, each of which may be presented in order of importance or urgency. For example, the listing **7230** may provide a list of steps recommended and/or required to be performed in response to a data breach. The listing **7230** may include items that are particularly required and/or applicable to the selected affected jurisdiction or sector (the United Kingdom in the particular example of FIG. 72). Alternatively, the listing **7230** may include items that are generally required and/or applicable to more than one affected jurisdiction or sector, while data breach response details **7220** may provide details about the recommended and/or required data breach response actions for the selected jurisdiction or sector (e.g., in the particular example of FIG. 72, the listing **7230** may show items that are generally required and/or applicable to multiple jurisdictions and/or sectors, while data breach response details **7220** may show items particularly relevant to the United Kingdom). The listing **7230** may include items ordered by urgency, which the system may have determined based on a score or other value assigned to each item. The system may provide a check box for each of the items in the listing **7230**. Upon completion of an item, a user may select the check box for that item. In various embodiments, the system may remove that item from the listing **7230** and/or make a record of item completion and no longer present that item to a user as part of a list of incomplete data breach response activities. The system may also provide a mechanism allowing the assignment of each item in the listing **7230** to a particular user or to an organization. Upon assignment to a particular user or organization, the system may remove that item from the listing **7230** and/or make a record of item completion and no longer present that item to a user as part of a list of incomplete data breach response activities. Alternatively, the system may leave any assigned items on the listing **7230** until the assigned user or organization provides an indication or confirmation that the item has been completed.

(641) The system may determine one or more associated tasks to be performed for each of the items in the listing **7230**. For example, for the highlighted first item in the listing **7230**, a listing of tasks associated with that particular item may be provided in the subtask listing **7240**. The subtask listing **7240** may include tasks ordered by urgency, which, as for items in the listing **7230**, the system may have determined based on a score or other value assigned to each task. The system may provide a check box for each of the tasks in the subtask listing **7240**. Upon completion of a task, a user may select the check box for that task. In various embodiments, the system may remove that task from the subtask listing **7240** and/or make a record of task completion and no longer present that task to a user as part of a list of incomplete data breach response activities. The system may also provide a mechanism allowing the assignment of each task in the subtask listing **7240** to a particular user or organization. Upon assignment to a particular user or organization, the system may remove that task from the subtask listing **7240** and/or make a record of task completion and no longer present that item to a user as part of a list of incomplete data breach response activities. Alternatively, the system may leave any assigned tasks on the subtask listing **7240** until the assigned user or organization provides an indication or confirmation that the task has been completed.

(642) In the example shown in FIG. 73, the system may generate an exemplary interface **7300** that may include a listing **7310** of one or more items required to be performed in response to a data breach. The listing **7310** may include items **7320**, **7330**, and **7340** that may be ordered by urgency or otherwise ranked based on a score or other value determined by the system and assigned to each item, for example, as described herein. For example, the item **7320** may have the highest urgency score, and therefore is listed first, followed by the item **7330**, which may have the second highest urgency score, and then followed by the item **7340**, which may have the third highest urgency score. Each of the items **7320**, **7330**, and **7340** may include a summary or a detailed description of its requirements and associated characteristics, such as the jurisdiction and/or sector to which the item corresponds. Items that may typically be required for compliance may be removed from a list such as the listing **7310** due to conflict-of-laws decisions made earlier, as described above.

(643) The system may present a check box for each of the items **7320**, **7330**, and **7340** in the interface **7300**. Upon completion of an item, a user may select the check box for that item. In various embodiments, the system may remove that item from its listing of required items and/or make a record of item completion and no longer present that item to a user as part of a list of incomplete data breach response activities. The system may also provide a mechanism allowing the assignment of each of the items **7320**, **7330**, and **7340** in interface **7300** to a particular user or organization. Upon assignment to a particular user or organization, the system may remove that item from the listing **7310** and/or make a record of item completion and no longer present that item to a user as part of a list of incomplete data breach response activities. Alternatively, the system may leave any assigned items on the listing **7310** until the assigned user or organization provides an indication or confirmation that the item has been completed.

(644) As described herein, the system may determine which affected jurisdictions and/or sectors require reporting of data breaches. The system may use information collected via a master questionnaire to populate a data structure that uses an ontology to map answers to questions in the master questionnaire to questions associated with particular jurisdictions and/or sectors. In the example shown in FIG. 74, an exemplary interface **7400** may include questions **7410** from a master questionnaire that allow a user to input answers to each question in the master questionnaire. The interface **7400** may allow the entry, via questions **7410** from the master questionnaire, of any data breach information described herein or otherwise and/or that may be needed to complete the data breach reporting requirements for one or more jurisdictions. For example, questions **7410** may include questions soliciting a number of data subjects affected, a volume or quantity of data compromised, a type of personal data compromised, a data breach discovery date and/or time, a data breach occurrence date and/or time, a data breach reporting date and/or time, a method of receiving a report of the data breach, a business sector affected by the breach, and/or a description of the data breach. In response to receiving the data breach information as answers to the questions **7410**, the system may map the answers to respective questions in particular questionnaires for particular jurisdictions as described herein.

(645) In various embodiments, the system may present questions in a master questionnaire, such questions **7410** from a master questionnaire, in an order that corresponds to the order of such questions in corresponding reporting documents or other communications. This may make it easier for a user to prepare and finalize the reporting communications or documentation for each jurisdiction and/or sector. Alternatively, or in addition, the system may present questions in an order that allows the system to take into account internal conflict-of-laws logic by addressing such conflicts in turn.

(646) To further illustrate the disclosed embodiments, an example will now be provided. This example is only intended to further illustrate exemplary aspects of the various embodiments and is not intended to provide any limitations to any embodiments of the disclosed subject matter.

(647) In an example, a business may determine that a breach of personal data or personal information has occurred. The business may determine that 500,000 user accounts having personal data or personal information for users in the U.S. and Canada have been accessed by an unauthorized system. Each such user account may include a user's first name and last name and at least one credit card number. In response, an employee of the business may operate a system, such as those described herein, to interact with one or more interfaces (e.g., as described in regard to interface **6900**, interface **7000**, etc.) to provide incident information, such as the type of data compromised (here, names and credit card numbers), the affected jurisdictions (in this example, the U.S. and Canada), a number of compromised accounts (in this example, 500,000), and a date of discovery of the breach. The employee may provide any other useful information to the system. The system may then process the information (e.g., as performed by the Disclosure Compliance Module **5500**, the Disclosure Prioritization Module **6600**, the Data Breach Reporting Module **6700**, and/or in any other suitable manner) and present the next steps to the employee regarding reporting requirements, for example, in a prioritized listing (e.g., as described in regard to interfaces **7100**, **7200**, **7300**, **7400**). For example, the system may provide a listing that includes supplying a notification to the business's legal department, supplying a notification to a California regulatory agency, and supplying a notification to a Canadian regulatory agency, in that order. The system may also include penalties associated with each step, such as the potential civil penalties for failure to provide the notifications to the California regulatory agency and the Canadian regulatory agency. Alternatively, the system may substantially automatically take actions to report or otherwise address the breach as described herein. As the user completes the steps provided by the system, the user may provide information via an

interface (e.g., as described in regard to interfaces **7100**, **7200**, **7300**, **7400**, **7500**) may be used to track the completion of steps (e.g., the system may then, automatically or upon demand, update the listing of steps to remove completed steps and/or add additional steps based on newly received information).

#### (648) Systems and Methods for Bundling Privacy Policies

(649) Currently, in order to comply with various privacy laws, it is frequently necessary for entities (e.g., companies or other organizations) to display an applicable privacy policy to users, such as users who interact with a computer interface associated with the company (e.g., a website or software application, such as an app running on a mobile device). This can be difficult since different privacy laws may require different language to be included in the terms of the privacy policy and since it may be difficult to determine which privacy laws/regulations apply to a particular situation.

(650) For example, a first situation where a resident of Europe accesses a French-language version of a website of a particular multinational corporation regarding a first service provided by the corporation may require a first set of one or more of privacy terms to be included in a controlling privacy policy. Similarly, a second situation where a resident of California accesses an English-language version of the same website of the particular multinational corporation regarding a second service provided by the corporation may require a second set of one or more of privacy terms to be included in a controlling privacy policy.

(651) An entity wishing to comply with the privacy policy requirements of various jurisdictions may provide individual links to different web pages that each respectively include the text of an applicable privacy policy for a particular situation. Taking this approach may be labor-intensive for an entity developing the website, since it would typically require the entity to know the applicable privacy laws that control each of many different types of situations and the applicable privacy policy language required for each situation. Accordingly, there is currently a need for a solution that makes it easier for entities to display the appropriate privacy policy to a particular individual in any of a variety of different situations.

(652) In various embodiments, the system may be configured to address this issue by providing and/or executing computer code (e.g., for inclusion on a web page or within a particular software application) on a computing device that sends a request (and optionally relevant information) to a remote computer system (e.g., one or more remote servers) to have software running on the server: (1) determine which one or more privacy policies of a plurality of privacy policies apply to a particular situation (e.g., combination of user, services, and jurisdiction); and (2) in response to determining that a particular privacy policy should control the particular situation, providing text for the privacy policy or bundle of privacy policies for display to a user (e.g., on the web page or within the particular software application). Alternatively, or in addition, the system may be configured to receive a request (and optionally relevant information) from a remote computer system (e.g., one or more remote servers) and, in response: (1) determine which one or more privacy policies of a plurality of privacy policies apply to a particular situation (e.g., combination of user, services, and jurisdiction); and (2) in response to determining that a particular privacy policy or bundle of privacy policies should control the particular situation, transmit text for a privacy policy or bundle of privacy policies (or information allowing retrieval of such text) for display to a user (e.g., on the web page or within the particular software application).

(653) In various embodiments, the system may be configured to substantially automatically determine which of a plurality of potentially applicable privacy policies (e.g., which bundle of a bundle of privacy policies) applies to a particular situation based on one or more criteria that apply to the particular situation. Such criteria may include one or more criteria associated with one or more particular products or services, such as, for example, one or more of: (1) one or more particular products or services being provided/offered in a current situation (e.g., one or more particular products or services that are being offered via a particular website or app); (2) an entity offering and/or providing the one or more particular products or services; (3) one or more geographical and/or jurisdictional locations of the entity providing the one or more particular products or services; (4) the respective type of each of the one or more particular products or services; (5) one or more subgroups (divisions, subsidiaries, etc.) of the entity providing the one or more particular products or services; (6) one or more geographical and/or jurisdictional locations of one or more computing systems hosting, providing, or otherwise facilitating access to one or more websites associated with the one or more particular products or services or the entity providing the one or more particular products or services; and/or (7) any other criteria associated with one or more particular products or services that may relate to a determination of one or more privacy policies for a particular situation.

(654) Such criteria may also, or instead, include criteria associated with a user and/or potential purchaser of the one or more particular products and/or services, such as, for example, one or more of: (1) a geographical and/or jurisdictional location of the user; (2) a language of the user; (3) a territory of residence (e.g., country or other territory) of the user; (4) an originating location or access means of the user (e.g., a website from which the user is accessing one or more services or an advertisement for one or more products); (5) a citizenship of the user; and/or (6) any other criteria associated with the user that may relate to a determination of one or more privacy policies for a particular situation. Any such criteria and/or related information may be determined using, for example, any suitable process described herein, including, but not limited to, using one or more data maps and/or data asset information to make the determination.

(655) In various embodiments, the system may be configured for, at least partially in response to receiving a request for a privacy policy or a bundle of privacy policies in a particular situation (e.g., in response to receiving a request, from a particular user of a particular website, for the system to display a controlling privacy policy or bundle of privacy policies): (1) evaluating criteria, such as the criteria discussed above, that is associated with the current situation (e.g., that is associated with one or more particular products or services offered and/or provided via the particular website and/or the requesting user); and (2) determining the one or more particular privacy policies applicable to the particular situation. The system may then retrieve, generate, and/or display information associated with such one or more particular privacy policies to the user. For example, a user may select a privacy policy control on a webpage associated with one or more services to which the user subscribes, and the system may, at least partially in response to such a selection, determine a particular privacy policy or bundle of privacy policies that apply to the situation (e.g., associated with the combination of one or more particular products or services offered and/or provided via the particular website and/or the requesting user) from a plurality of associated privacy policies. The system may then display some or all of the contents of the applicable particular privacy policy or bundle of privacy policies (e.g., the full or abbreviated text of the applicable policy/policies) to the user.

(656) In various embodiments, when setting up and/or operating the system, an operator of the system (e.g., an employee of the entity, an agent of the entity, a privacy officer of the entity, an employee of a third-party, etc.) may configure the system to allow a particular individual (e.g., a privacy specialist) to associate one or more particular privacy policies with one or more combinations of user criteria and/or product or service criteria. Such associations may be rule-based. Accordingly, in particular embodiments, by evaluating one or more specified rules, the system may determine an applicable privacy policy or bundle of policies that apply to a particular situation by applying the specified rules to criteria associated with the particular situation.

#### (657) Privacy Policy Bundle Determination Module

(658) FIG. 75 depicts a Privacy Policy Bundle Determination Module **7500** according to a particular embodiment, which may be executed, for example, on any of the servers, devices, or computing devices described herein, or on any combination thereof. The Privacy Policy Bundle Determination Module **7500** may also generate, modify, otherwise interoperate with one or more data models, data maps, data inventories, ontologies, and/or any other data structures and computing systems described herein. Note that the steps that the Privacy Policy Bundle Determination Module **7500** may perform are described here in an exemplary order. The Privacy Policy Bundle Determination Module **7500**, according to various embodiments, may perform any subset of these steps in any order and/or in conjunction with any one or more other functions and activities.

(659) When executing the Privacy Policy Bundle Determination Module **7500**, the system may begin, at Step **7510**, by receiving a request to determine one or more applicable privacy policies for a particular situation. In particular embodiments, a user may select a privacy policy control on a particular website that may generate such a request. Alternatively, or in addition, a remote system may generate such a request in response to

detecting user activity on a website, at the remote system, and/or at a system communicatively connected to the remote system. The request may include and/or reference criteria associated with a particular situation in which the request was generated. The request may also, or instead, include and/or reference other information that will enable the system to determine criteria that may be used to determine the or more applicable privacy policies for the particular situation.

(660) At Step **7520**, the system may begin the process of determining the appropriate navigational elements to present to a user that, when activated, may provide the applicable privacy policy dataset to the user for the user's particular situation. For example, at Step **7520** the system may determine product or service criteria associated with the request. In particular embodiments, the system may evaluate criteria such as those detailed above that may be associated with one or more particular products or services. The system may determine such criteria from the request and/or based on the request. For example, product or service criteria may be included in the request. Alternatively, or in addition, information indicated by the request may be used to determine one or more applicable products or services. Such criteria may be included in the request automatically by a browser and/or any other system involved in generating the request. Alternatively, or in addition, the criteria may be detected from browser data, for example, from a state of a browser application executing on the user's device. Such criteria may also, or instead, be received from the user and/or included in the request based on user input.

(661) In various embodiments, the system may use a data model, map, and/or inventory to identify and/or determine product or service criteria. For example, the system may use information contained or indicated in the request (e.g., user identifier) to evaluate a data model and/or one or more associated data inventories to determine a product or service associated with the user. In another example, the system may use information contained or indicated in the request (e.g., product identifier, service identifier) to evaluate a data model and/or one or more associated data inventories to determine product or service criteria (e.g., a jurisdiction or geographical location associated with the product or service, the entity providing the product or service, location of one or more data assets associated with the product or service, etc.).

(662) At Step **7530**, the system may determine user criteria associated with the request. In particular embodiments, the system may evaluate user criteria such as those detailed above that may be associated with the user originating or otherwise associated with the request. The system may determine such criteria from the request and/or based on the request. For example, user criteria may be included in the request. Alternatively, or in addition, information indicated by the request may be used to determine one or more applicable user criteria. Such criteria may be included in the request automatically by a browser (e.g., stored username) and/or any other system involved in generating the request. Alternatively, or in addition, such criteria may be detected from browser data, for example, from a state of a browser application executing on the user's device. Such criteria may also, or instead, be received from the user and/or included in the request based on user input.

(663) In various embodiments, the system may use a data model, map, and/or inventory to identify and/or determine user criteria. For example, the system may use information contained or indicated in the request (e.g., user identifier) to evaluate a data model and/or one or more associated data inventories to determine criteria associated with the user (e.g., a jurisdiction, age, address, language, location, associated products or services, etc.).

(664) At Step **7540**, the system may execute a privacy policy bundle determination rules engine using the product/service criteria and/or the user criteria to determine one or more navigation elements that, when selected by a user, may be configured to cause a user device (e.g., a browser executing on the user device) to present one or more applicable privacy policies (e.g., one or more applicable privacy policy datasets). In particular embodiments, the system may evaluate rules configured by an operator using the criteria to determine the particular navigational elements and/or privacy policies that apply to the situation associated with the request. For example, a particular privacy policy in a particular language may be applicable to a user located in a particular EU country and using a particular product. In various embodiments, the system may determine more than one applicable policy (a "bundle" of such policies). In particular embodiments, two or more policies may be determined, but may be in conflict. In such embodiments, the system may use a ranking or score for such policies to select the most applicable policy.

(665) In various embodiments, the system may use a ranking or score to select from multiple potentially applicable policies, regardless of whether there is a conflict. For example, a product may have a general privacy policy associated with it and a particular privacy policy applicable to users of the product in a particular jurisdiction. The system may determine a score for each policy, for example by assigning a numeric value to each matching criterion. For these two example policies, the system may determine that a score for the jurisdiction-specific policy is higher than a score for the general policy because the jurisdiction-specific policy matches an additional criterion (jurisdiction). Thus, because the score is higher for the jurisdiction-specific policy, the system may select the jurisdiction-specific policy over the general policy and therefore determine to include the navigational element associated with the jurisdiction-specific policy in include in a graphical user interface for presentation to the user in response to the request.

(666) In various embodiments, the request and/or the criteria may not allow the system to determine a specific policy. In such cases, the system may use a navigational element associated with a default privacy policy applicable to all users. In this way, the system is configured to include the navigational element associated with affirmative privacy policy information (e.g., a default privacy policy dataset) in a graphical user interface presented to a requesting user regardless of the criteria available to the system to evaluate applicable policies.

(667) At Step **7550**, in response to detecting the user selection of a navigational element associated with a privacy policy dataset, the system may instruct the user device (e.g., a browser executing on the user device) to retrieve or otherwise obtain some or all of the privacy policy dataset associated with the selected navigational element and present the dataset to the user. For example, the system may determine one or more particular sections of a privacy policy dataset that are applicable to individual users. The system may transmit an instruction to the user device causing the user device to retrieve those sections for presentation to the user, present the sections to user, and/or otherwise facilitate the presentation of those sections to the user (e.g., by sending data that can be used to obtain the relevant sections to the user device).

(668) Example User Interfaces for Privacy Policy Bundle Determination and Additional Embodiments

(669) As noted above, in a particular example, the system may include one or more "setup" user interfaces that a first individual may use to define one or more privacy policy rule definitions that may be used by the system and/or one or more second individuals (e.g., privacy officers) to define one or more rules that are used to determine which particular one or more privacy policies and/or associated navigational elements may be associated with a particular situation. FIG. **76** illustrates an exemplary graphical user interface **7600** that allows a user to define one or more various parameters that may be used in evaluating criteria, for example, included in or indicated by a request for applicable privacy policy information. A user may select a type of parameter in the section **7610** (e.g., entity division, region (geographical and/or jurisdiction), product, etc.) and then provide specifics of the parameter in the details section **7620** (e.g., particular product name, applicable product type (movie, TV show, game, etc.)).

(670) In various embodiments, the user (or another, second user) may use a suitable privacy policy rule definition screen to define one or more rules that map one or more privacy policies to particular criteria. For example, as shown in FIG. **77** illustrating exemplary graphical user interface **7700**, the user (or one or more second individuals, such as a privacy officer) may define a first rule (indicated by the rule name **7722**, that may be user-defined in this interface) specifying that the system should display a navigational element associated with a first particular privacy policy **7721** when a first set **7720** of one or more criteria are satisfied. The selection of this navigational element by a user on a user device may then cause the user device to retrieve and/or present to the user a privacy policy dataset associated with the first particular privacy policy **7721**. The first set of criteria may, for example, be that: (1) the user is from the United States, France, or Germany; (2) the user is seeking information regarding products offered by Warner Home Video; and (3) the user is seeking information regarding TV Shows (See FIG. **37**). The rule summary **7730** may also be provided and reflect updates to the rule via the interface **7700** in real-time.

(671) Similarly, a user may define a second rule specifying that the system should display a second navigational element associated with a second particular privacy policy when a second set of one or more criteria are satisfied. The second set of criteria may, for example, include the following criteria: (1) the user is from Australia; (2) the user is seeking information regarding products offered by Warner Home Video; and (3) the user is



seeking information regarding TV Shows.

(672) In various embodiments, the user may add one or more of the various rules to a rule group that applies, for example, to a particular web site or software application. As shown in FIG. 78, the graphical user interface **7800** may provide a listing of rule groups **7810** from which the user may select the rule group **7820**. Each of the rule groups in the listing of rule groups **7810** may have a unique rule group name and description, as well as particular privacy policies that it applies to and various combinations of criteria with which each is associated. A set of rules **7821** may be associated with the rule group **7820**. One or more controls **7830** may be available that allow the user to add, move, delete, and/or copy a rule in any particular rule group.

(673) FIG. 79 illustrates an exemplary graphical user interface **7900** showing rule group details. As seen in this figure, such details may include the rule group name **7910** and the rule group description **7930**. Each rule group may also have a default, or “fall back,” privacy policy that the user may define, such as fallback privacy policy **7920**. As noted above, a default or fallback privacy policy may be the policy that will apply as a default in situations that aren't specifically covered by any rule within the rule group or where a particular privacy policy cannot be determined based on the available criteria.

(674) In various embodiments, the system may include instructions defining relative prioritizations (e.g., numeric scores or values) of various rules within a rule group. In such embodiments, if two rules within a particular group of rules specify different privacy policies for a particular situation, the system may use the relative priorities of the rules to determine which rule's privacy policy should control in the particular situation. For example, the system may determine that the privacy policy associated with the rule having the higher priority (e.g., higher score) should control. In another example, the system may determine a score for each privacy policy based on the number of matching criteria, the importance of the criteria that match (e.g., one criteria may be given more weight than another or may be weighted due to importance), or any other basis. The system may then configure the navigational element on a graphical user interface that is associated with the applicable privacy policy based on determining the policy with the highest score.

(675) In particular embodiments, a particular rule group may be associated with a particular entity, website, or software application. The system may use the rule group to determine which particular privacy policy to display in various situations (e.g., when a particular visitor to a particular website satisfies certain criteria, such as any of those covered by rules within the rule group).

(676) In various embodiments, when a user visits a particular website or a particular software application, the user may be presented with an option (e.g., a navigational element) to request information regarding a privacy policy that applies to the user's current situation. In a particular embodiment, in response to receiving this request, the system may provide a suitable user interface (e.g., a graphical user interface) that prompts the user for information regarding their current situation. FIG. 80 illustrates an example graphical user interface **8000** that shows the applicable privacy policy text **8010** that has been presented to a user in response to receiving an instruction from the system to retrieve and present such privacy policy information based on the user selection of a navigational element as described above. The GUI **8000** further allows a user to specify additional criteria to further refine the displayed privacy policy text. For example, in the privacy policy filters sections **8020**, a user may select or enter a location **8021** of the user, a division **8022** of a particular entity that provides products or services that the user uses or in which the user may be interested, and one or more particular products or services **8023** that the user uses or in which the user may be interested. As described above, these may be included or indicated in a request as criteria that the system may then use to determine the appropriate navigational element to configure in a graphical user interface that, when selected by a user, will cause the user device to retrieve and present the appropriate policy or bundle of policies. Such a request may be generated in response to the user selecting or otherwise providing information that can be used as criteria to determine one or more applicable privacy policies. The system may use this information (with or without any other criteria) in conjunction with a rule group that is associated with the particular website to determine a particular navigational element to configure in a graphical user interface that, when selected by a user, will cause the user device to retrieve and present the privacy policy that controls the situation specified by the user (e.g., as defined by the parameters entered by the user and/or based on other criteria).

(677) In various embodiments, rather than prompting a user to enter one or more of the particular parameters, the system may be adapted to automatically determine the one or more particular parameters in any suitable manner. In particular embodiments, the system may detect and use state data from a browser executing on the user's device to determine one or more parameters. For example, if the website is a site associated with a particular product or service (e.g., a website for the Spotify music subscription service), the system may automatically determine from browser data that the user is a user of, or is interested in, the Spotify service. In another example, code executed by a web browser operated by the user may collect one or more particular parameters that may be included or indicated in a request for privacy policy information, for example, from metadata.

(678) In various embodiments, the system may be able to automatically determine some or all of the parameters needed to determine which particular privacy policy or bundle of policies applies to a particular situation. In such embodiments, the system may, for example, upon receiving a request from a user to display the privacy policy associated with the current situation, automatically display the controlling privacy policy to the user.

(679) It should be understood that the system may take any suitable form and that the various steps outlined above may be implemented by any suitable computer system. For example, in various embodiments, computer code is provided to be included on a particular web page that executes a call to a remote server (which may be, for example, operated by an entity that is not associated with the web page or related website; such an entity may be, for example, a third party privacy management company). The remote server may then determine which of a plurality of privacy policies applies to the particular situation and, for example, pass the privacy policy back to the web page for display to the user. In alternative embodiments, the remote server may communicate information regarding the privacy policy (e.g., a partial or entire text of the policy) directly back to the user. In other embodiments, the functionality described above may be executed locally, for example, on a computing device that is executing a particular software application to which the privacy policy applies.

## CONCLUSION

(680) Although embodiments above are described in reference to various systems and methods for assessing the risk associated with particular vendors, it should be understood that any applicable concept described herein could be done with entities other than vendors—for example business partners other than vendors, tenants in the context of landlord/tenant relationships, etc.

(681) Also, although embodiments above are described in reference to various systems and methods for creating and managing data flows related to individual privacy campaigns, it should be understood that various aspects of the system described above may be applicable to other privacy-related systems, or to other types of systems, in general. For example, the functionality described above for obtaining the answers to various questions (e.g., assigning individual questions or sections of questions to multiple different users, facilitating collaboration between the users as they complete the questions, automatically reminding users to complete their assigned questions, and other aspects of the systems and methods described above) may be used within the context of Privacy Impact Assessments (e.g., in having users answer certain questions to determine whether a certain project complies with an organization's privacy policies).

(682) While this specification contains many specific embodiment details, these should not be construed as limitations on the scope of any invention or of what may be claimed, but rather as descriptions of features that may be specific to particular embodiments of particular inventions. Certain features that are described in this specification in the context of separate embodiments may also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment may also be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

(683) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be



performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems may generally be integrated together in a single software product or packaged into multiple software products.

(684) Many modifications and other embodiments of the invention will come to mind to one skilled in the art to which this invention pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. While examples discussed above cover the use of various embodiments in the context of operationalizing privacy compliance and assessing risk of privacy campaigns, various embodiments may be used in any other suitable context. Therefore, it is to be understood that the invention is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for the purposes of limitation.

## Claims

1. A method comprising: analyzing, by computing hardware, browser data associated with a browser application displaying a website to determine a user parameter identifying a geographical location of a user device executing the browser application; receiving, by the computing hardware, product or service information provided via a product or service parameter input displayed on the website; determining, by the computing hardware and based on the product or service information, a product or service parameter, wherein the product or service parameter comprises at least one of a particular product or service being provided or offered via the website, an entity offering or providing the particular product or service, a geographical location of the entity offering or providing the particular product or service, a type of the particular product or service, a subgroup of the entity offering or providing the particular product or service, or a geographical location of a computing system configured for at least one of hosting, providing, or facilitating access to the website; identifying, by the computing hardware and based on the website, a plurality of privacy policy rules; analyzing, by the computing hardware, the plurality of privacy policy rules using the user parameter and the product or service parameter to identify an applicable privacy policy rule from the plurality of privacy policy rules; identifying, by the computing hardware and based on the applicable privacy policy rule, an applicable privacy policy; configuring, by the computing hardware, a navigation element displayed on the website to navigate to a display element that presents the applicable privacy policy; receiving, by the computing hardware, an indication of a selection of the navigation element; and responsive to receiving the indication, transmitting, by the computing hardware, an instruction to the browser application causing the browser application to retrieve and present the display element.
2. The method of claim 1, wherein analyzing the plurality of privacy policy rules using the user parameter and the product or service parameter to identify the applicable privacy policy rule from the plurality of privacy policy rules is based on the applicable privacy policy rule having a priority higher than a second privacy policy rule of the plurality of privacy policy rules that is also applicable based on the user parameter and the product or service parameter.
3. The method of claim 2 further comprising assigning, by the computing hardware based on the user parameter, a first numeric value to the applicable privacy policy and a second numeric value to the second privacy policy rule, wherein the applicable privacy policy rule has the priority higher than the second privacy policy rule due to the first numeric value being higher than the second numeric value.
4. The method of claim 2, wherein analyzing the plurality of privacy policy rules is based on the applicable privacy policy rule having the priority higher than the second privacy policy rule is performed due to a conflict between the applicable privacy policy rule and the second privacy policy rule.
5. The method of claim 1, wherein receiving the product or service information comprises receiving a request sent by computing code included in a web page of the website and executing on the user device.
6. The method of claim 1, wherein the user parameter further comprises at least one of a language of a user of the user device, a territory of residence of the user, or a citizenship of the user.
7. The method of claim 1, wherein the user parameter further comprises a language of a user of the user device and the applicable privacy policy is provided in the language.
8. A system comprising: a non-transitory computer-readable medium storing instructions; and a processing device communicatively coupled to the non-transitory computer-readable medium, wherein, the processing device is configured to execute the instructions and thereby perform operations comprising: receiving a user parameter identifying a geographical location of a user device executing a user interface; determining product or service information based on user activity on the user interface; determining, based on the product or service information, a product or service parameter, wherein the product or service parameter comprises at least one of a particular product or service being provided or offered via the user interface, an entity offering or providing the particular product or service, a geographical location of the entity offering or providing the particular product or service, a type of the particular product or service, a subgroup of the entity offering or providing the particular product or service, or a geographical location of a computing system configured for at least one of hosting, providing, or facilitating access to the user interface; identifying a plurality of privacy policy rules; analyzing the plurality of privacy policy rules using the user parameter and the product or service parameter to identify an applicable privacy policy rule from the plurality of privacy policy rules; identifying, based on the applicable privacy policy rule, an applicable privacy policy; configuring a navigation element displayed on the user interface to navigate to a display element that presents the applicable privacy policy; receiving an indication of a selection of the navigation element; and responsive to receiving the indication, causing the user interface to present the display element.
9. The system of claim 8, wherein the user interface comprises at least one of a web page associated with a website or a display interface provided in a software application.
10. The system of claim 8, wherein identifying the plurality of privacy policy rules is based on at least one of the user parameter, the product or service parameter, the user interface, or the entity.
11. The system of claim 8, wherein analyzing the plurality of privacy policy rules using the user parameter and the product or service parameter to identify the applicable privacy policy rule from the plurality of privacy policy rules is based on the applicable privacy policy rule having a priority higher than a second privacy policy rule of the plurality of privacy policy rules that is also applicable based on the user parameter and the product or service parameter.
12. The system of claim 11, wherein the operations further comprise assigning, based on the user parameter, a first numeric value to the applicable privacy policy and a second numeric value to the second privacy policy rule, wherein the applicable privacy policy rule has the priority higher than the second privacy policy rule due to the first numeric value being higher than the second numeric value.
13. The system of claim 11, wherein analyzing the plurality of privacy policy rules is based on the applicable privacy policy rule having the priority higher than the second privacy policy rule is performed due to a conflict between the applicable privacy policy rule and the second privacy policy rule.
14. The system of claim 8, wherein the user parameter further comprises a language of a user of the user device and the applicable privacy policy is provided in the language.
15. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by one or more processing devices, configure the one or more processing devices to perform operations comprising: receiving a user parameter identifying a geographical location of a

user device executing a user interface; determining product or service information based on user activity on the user interface; determining, based on the product or service information, a product or service parameter, wherein the product or service parameter comprises at least one of a particular product or service being provided or offered via the user interface, an entity offering or providing the particular product or service, a geographical location of the entity offering or providing the particular product or service, a type of the particular product or service, a subgroup of the entity offering or providing the particular product or service, or a geographical location of a computing system configured for at least one of hosting, providing, or facilitating access to the user interface; identifying a plurality of policy rules; analyzing the plurality of policy rules using the user parameter and the product or service parameter to identify an applicable policy rule from the plurality of policy rules; identifying, based on the applicable policy rule, an applicable policy; configuring a navigation element displayed on the user interface to navigate to a display element that presents the applicable policy; receiving an indication of a selection of the navigation element; and responsive to receiving the indication, causing the user interface to present the display element.

16. The non-transitory computer-readable medium of claim 15, wherein the user interface comprises at least one of a web page associated with a website or a display interface provided in a software application.

17. The non-transitory computer-readable medium of claim 15, wherein identifying the plurality of policy rules is based on at least one of the user parameter, the product or service parameter, the user interface, or the entity.

18. The non-transitory computer-readable medium of claim 15, wherein analyzing the plurality of policy rules using the user parameter and the product or service parameter to identify the applicable policy rule from the plurality of policy rules is based on the applicable policy rule having a priority higher than a second policy rule of the plurality of policy rules that is also applicable based on the user parameter and the product or service parameter.

19. The non-transitory computer-readable medium of claim 18, wherein the operations further comprise assigning, based on the user parameter, a first numeric value to the applicable policy and a second numeric value to the second policy rule, wherein the applicable policy rule has the priority higher than the second policy rule due to the first numeric value being higher than the second numeric value.

20. The non-transitory computer-readable medium of claim 18, wherein analyzing the plurality of policy rules is based on the applicable policy rule having the priority higher than the second policy rule is performed due to a conflict between the applicable policy rule and the second policy rule.

---